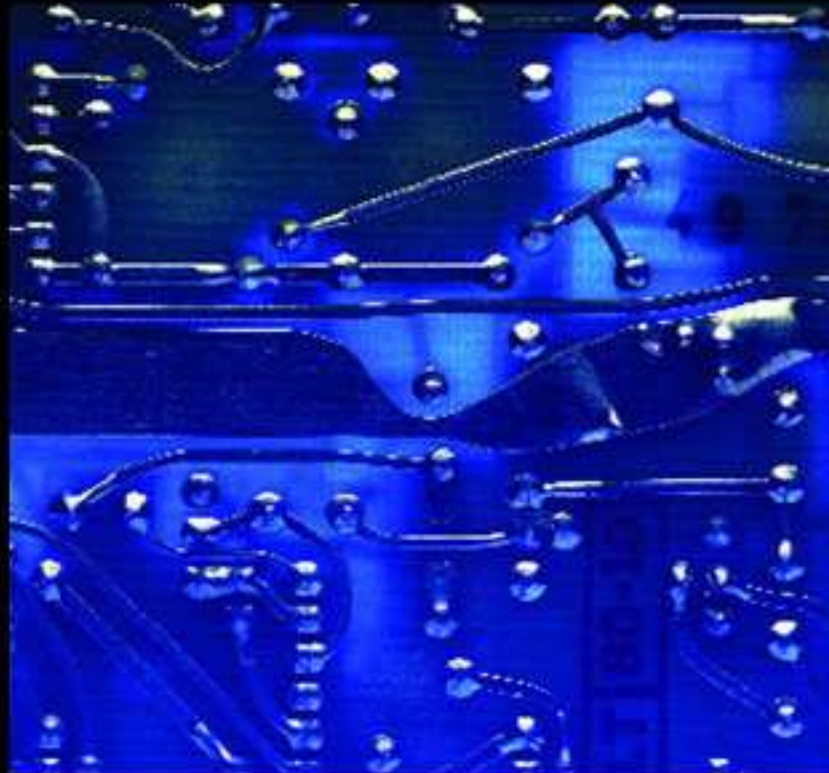


ELECTRONIC DEVICES AND CIRCUIT THEORY

TENTH EDITION

BOYLESTAD

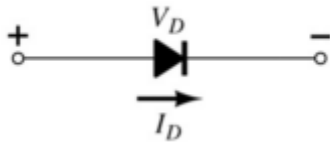


PEARSON

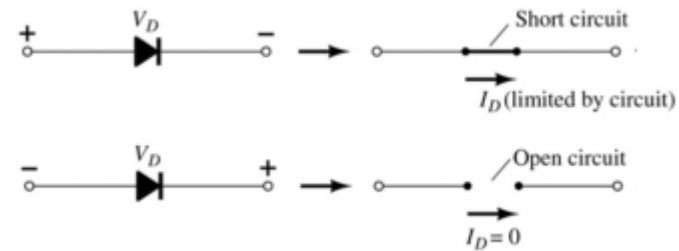
Chapter 1: Semiconductor Diodes

Diodes

The diode is a 2-terminal device.

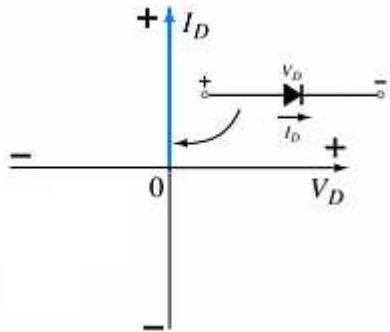


A diode ideally conducts in only one direction.



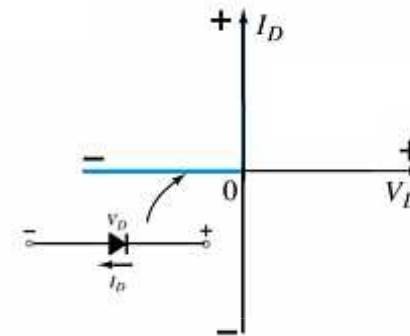
Diode Characteristics

Conduction Region



- The voltage across the diode is 0 V
- The current is infinite
- The forward resistance is defined as $R_F = V_F / I_F$
- The diode acts like a short

Non-Conduction Region



- All of the voltage is across the diode
- The current is 0 A
- The reverse resistance is defined as $R_R = V_R / I_R$
- The diode acts like open

Semiconductor Materials

Materials commonly used in the development of semiconductor devices:

- **Silicon (Si)**
- **Germanium (Ge)**
- **Gallium Arsenide (GaAs)**

Doping

The electrical characteristics of silicon and germanium are improved by adding materials in a process called doping.

There are just two types of doped semiconductor materials:

***n*-type**

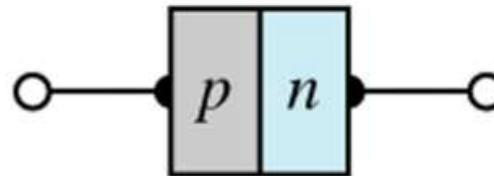
***p*-type**

- ***n*-type materials contain an excess of conduction band electrons.**
- ***p*-type materials contain an excess of valence band holes.**

p-n Junctions

One end of a silicon or germanium crystal can be doped as a *p*-type material and the other end as an *n*-type material.

The result is a *p-n junction*.

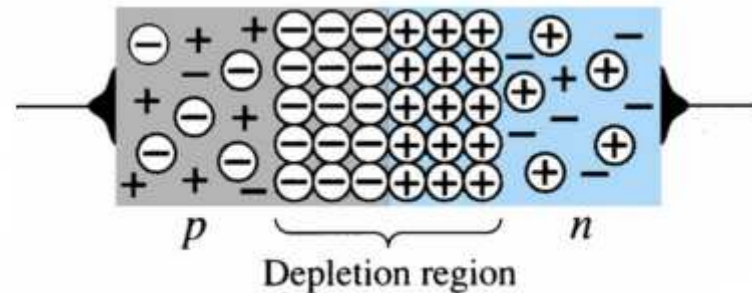


p-n Junctions

At the *p-n* junction, the excess conduction-band electrons on the *n*-type side are attracted to the valence-band holes on the *p*-type side.

The electrons in the *n*-type material migrate across the junction to the *p*-type material (electron flow).

The electron migration results in a **negative** charge on the *p*-type side of the junction and a **positive** charge on the *n*-type side of the junction.



The result is the formation of a **depletion region** around the junction.

Diode Operating Conditions

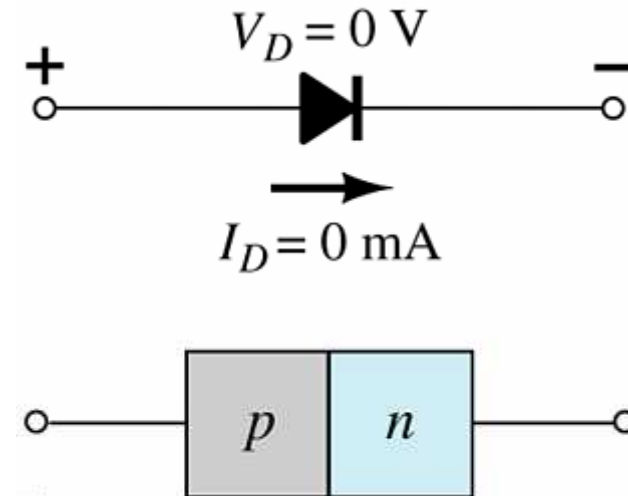
A diode has three operating conditions:

- **No bias**
- **Forward bias**
- **Reverse bias**

Diode Operating Conditions

No Bias

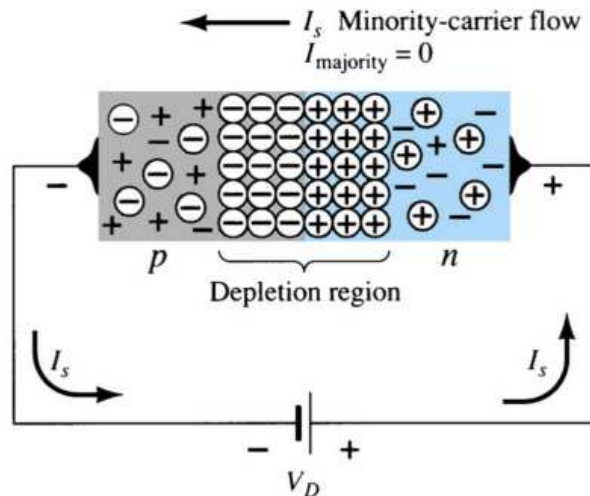
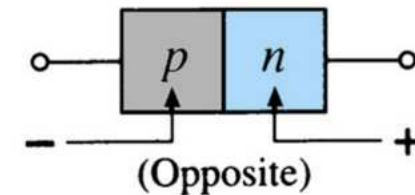
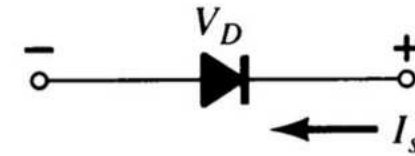
- No external voltage is applied: $V_D = 0 \text{ V}$
- No current is flowing: $I_D = 0 \text{ A}$
- Only a modest depletion region exists



Diode Operating Conditions

Reverse Bias

External voltage is applied across the p - n junction in the opposite polarity of the p - and n -type materials.

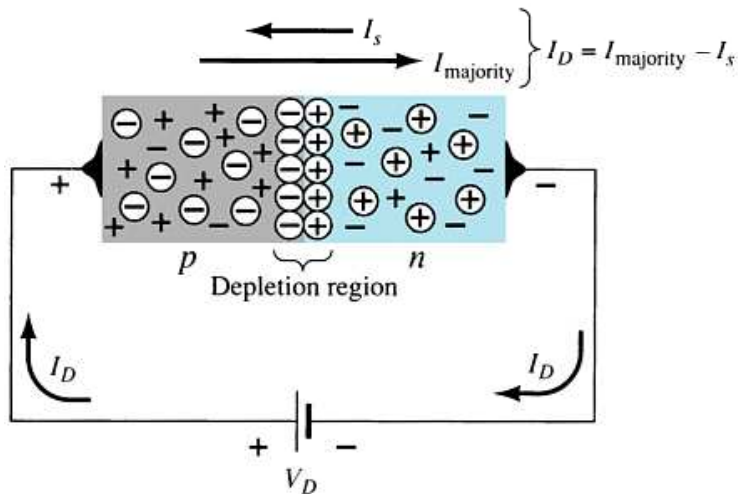
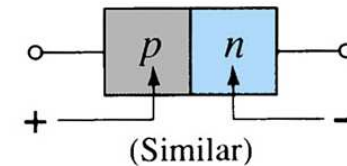
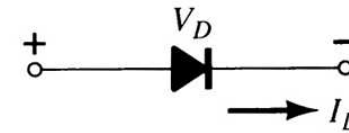


- The reverse voltage causes the depletion region to widen.
- The electrons in the n -type material are attracted toward the positive terminal of the voltage source.
- The holes in the p -type material are attracted toward the negative terminal of the voltage source.

Diode Operating Conditions

Forward Bias

External voltage is applied across the p - n junction in the same polarity as the p - and n -type materials.

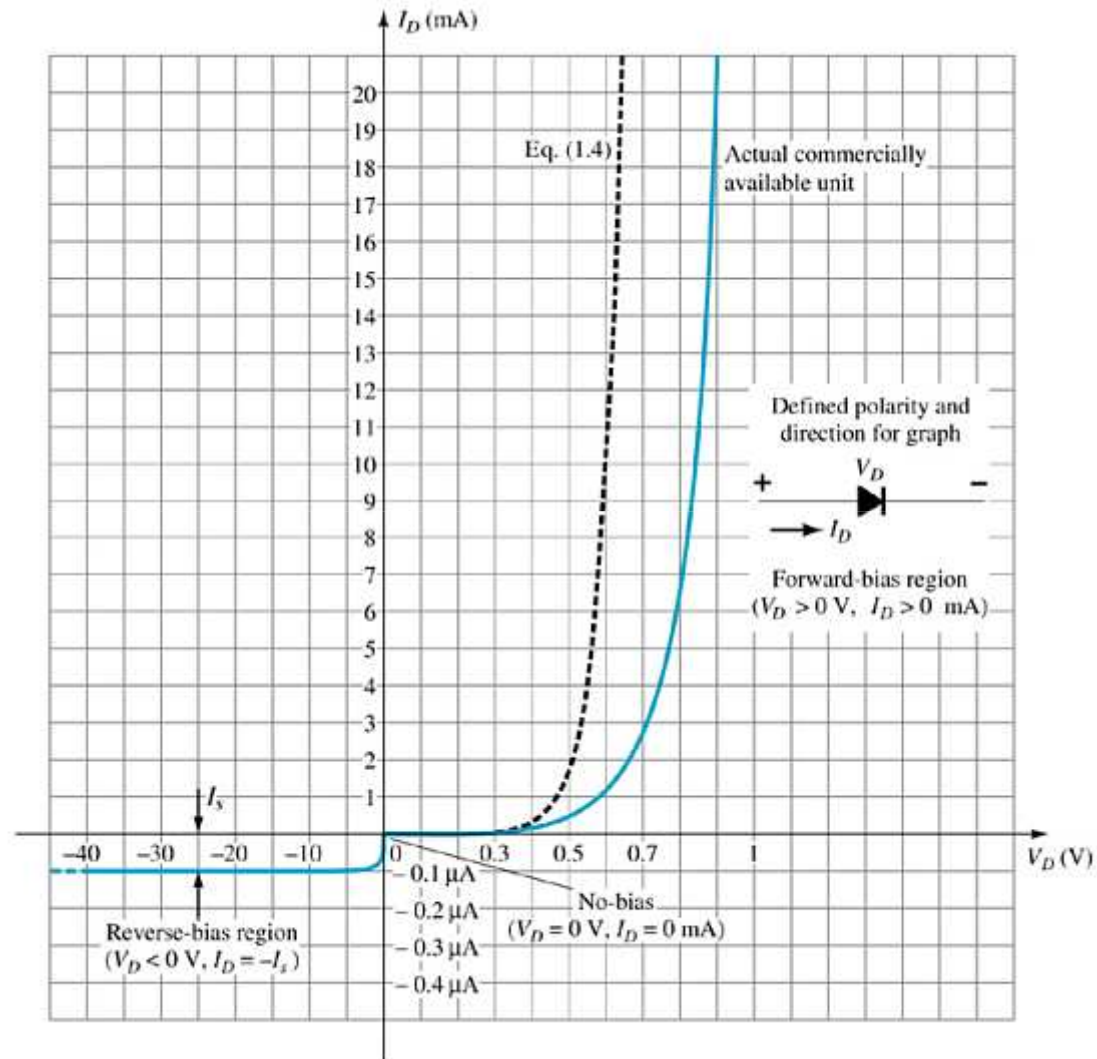


- The forward voltage causes the depletion region to narrow.
- The electrons and holes are pushed toward the p - n junction.
- The electrons and holes have sufficient energy to cross the p - n junction.

Actual Diode Characteristics

Note the regions for no bias, reverse bias, and forward bias conditions.

Carefully note the scale for each of these conditions.



Majority and Minority Carriers

Two currents through a diode:

Majority Carriers

- The majority carriers in *n*-type materials are electrons.
- The majority carriers in *p*-type materials are holes.

Minority Carriers

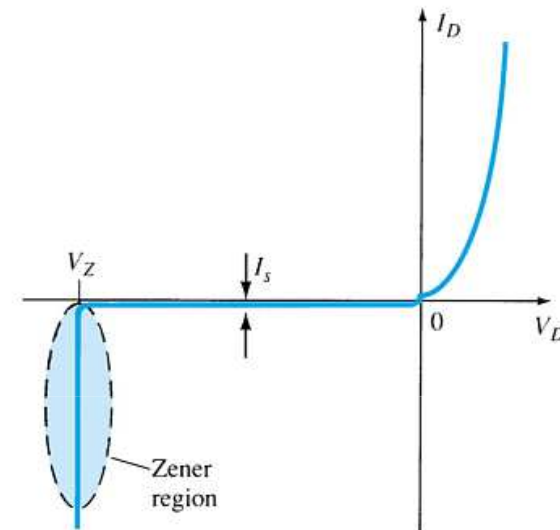
- The minority carriers in *n*-type materials are holes.
- The minority carriers in *p*-type materials are electrons.

Zener Region

The Zener region is in the diode's reverse-bias region.

At some point the reverse bias voltage is so large the diode breaks down and the reverse current increases dramatically.

- The maximum reverse voltage that won't take a diode into the zener region is called the **peak inverse voltage** or **peak reverse voltage**.
- The voltage that causes a diode to enter the zener region of operation is called the **zener voltage (V_Z)**.



Forward Bias Voltage

The point at which the diode changes from no-bias condition to forward-bias condition occurs when the electrons and holes are given sufficient energy to cross the p - n junction. This energy comes from the external voltage applied across the diode.

The forward bias voltage required for a:

- **gallium arsenide diode $\cong 1.2$ V**
- **silicon diode $\cong 0.7$ V**
- **germanium diode $\cong 0.3$ V**

Temperature Effects

As temperature increases it adds energy to the diode.

- **It reduces the required forward bias voltage for forward-bias conduction.**
- **It increases the amount of reverse current in the reverse-bias condition.**
- **It increases maximum reverse bias avalanche voltage.**

Germanium diodes are more sensitive to temperature variations than silicon or gallium arsenide diodes.

Resistance Levels

Semiconductors react differently to DC and AC currents.

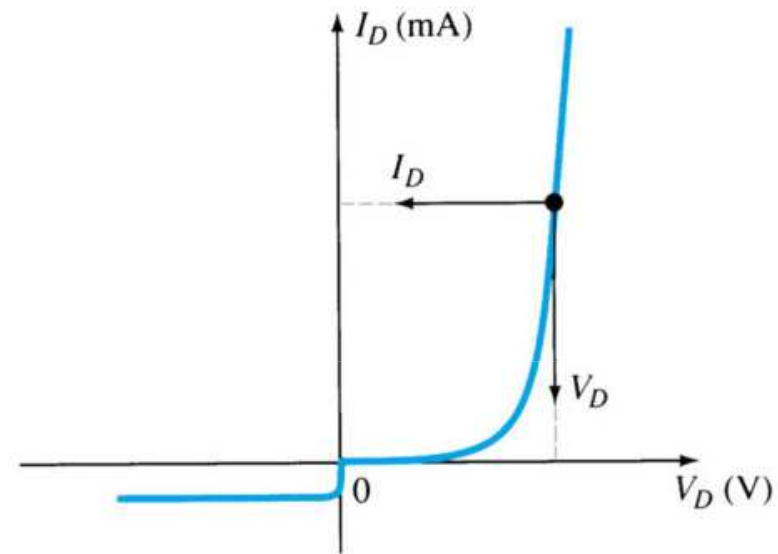
There are three types of resistance:

- **DC (static) resistance**
- **AC (dynamic) resistance**
- **Average AC resistance**

DC (Static) Resistance

For a specific applied DC voltage V_D , the diode has a specific current I_D , and a specific resistance R_D .

$$R_D = \frac{V_D}{I_D}$$



AC (Dynamic) Resistance

In the forward bias region:

$$r'_d = \frac{26\text{mV}}{I_D} + r_B$$

- The resistance depends on the amount of current (I_D) in the diode.
- The voltage across the diode is fairly constant (26 mV for 25°C).
- r_B ranges from a typical 0.1 Ω for high power devices to 2 Ω for low power, general purpose diodes. In some cases r_B can be ignored.

In the reverse bias region:

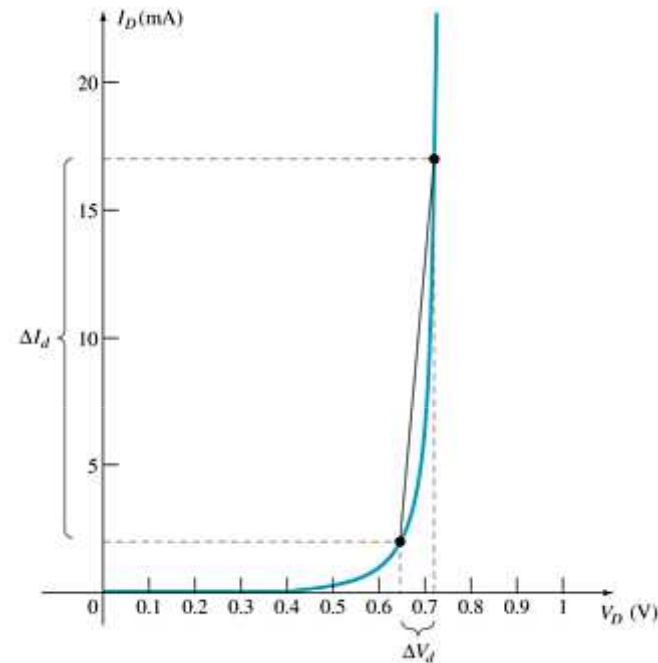
$$r'_d = \infty$$

The resistance is effectively infinite. The diode acts like an open.

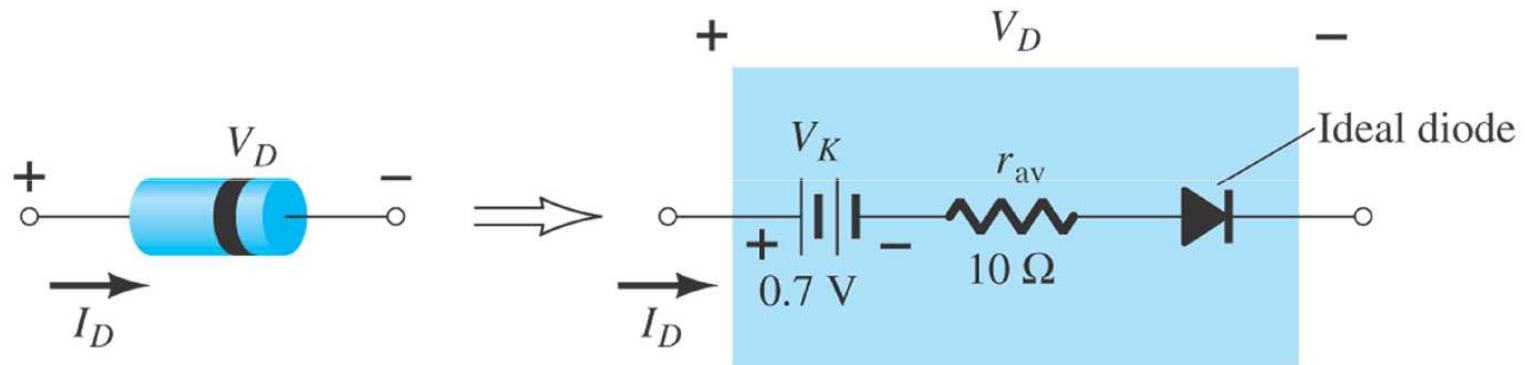
Average AC Resistance

$$r_{av} = \frac{\Delta V_d}{\Delta I_d} \quad \text{pt. to pt.}$$

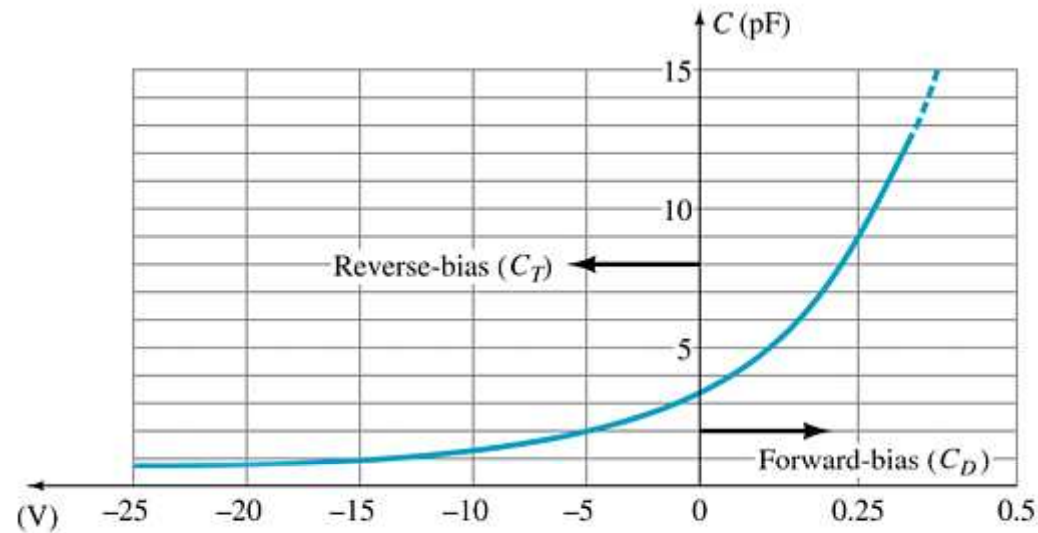
AC resistance can be calculated using the current and voltage values for two points on the diode characteristic curve.



Diode Equivalent Circuit



Diode Capacitance

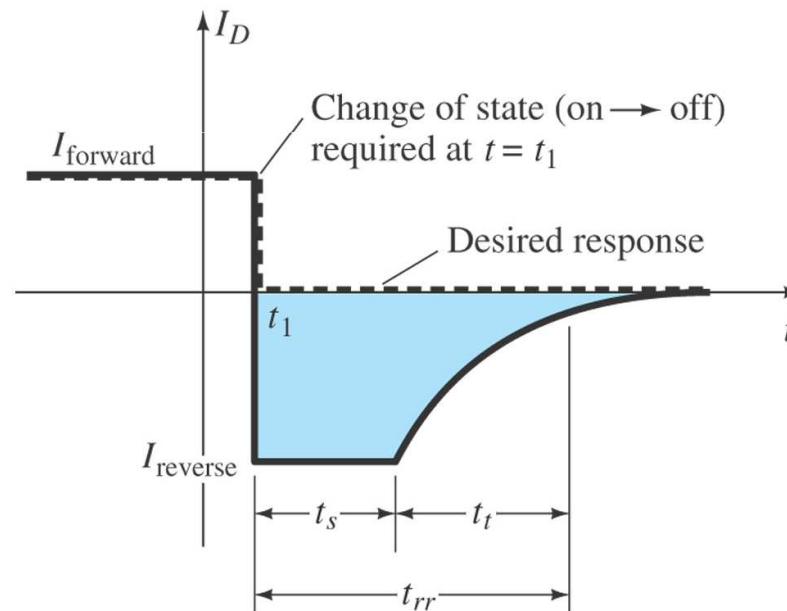


In reverse bias, the depletion layer is very large. The diode's strong positive and negative polarities create capacitance, C_T . The amount of capacitance depends on the reverse voltage applied.

In forward bias storage capacitance or diffusion capacitance (C_D) exists as the diode voltage increases.

Reverse Recovery Time (t_{rr})

Reverse recovery time is the time required for a diode to stop conducting once it is switched from forward bias to reverse bias.

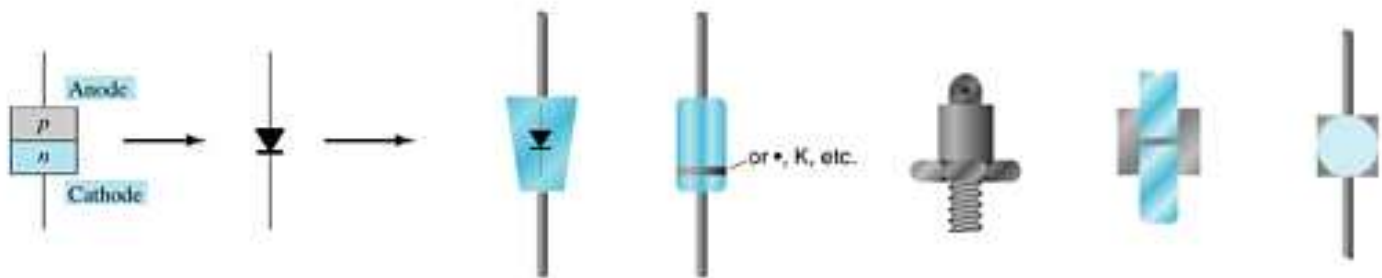


Diode Specification Sheets

Data about a diode is presented uniformly for many different diodes. This makes cross-matching of diodes for replacement or design easier.

1. Forward Voltage (V_F) at a specified current and temperature
2. Maximum forward current (I_F) at a specified temperature
3. Reverse saturation current (I_R) at a specified voltage and temperature
4. Reverse voltage rating, PIV or PRV or $V(BR)$, at a specified temperature
5. Maximum power dissipation at a specified temperature
6. Capacitance levels
7. Reverse recovery time, t_{rr}
8. Operating temperature range

Diode Symbol and Packaging



The anode is abbreviated A
The cathode is abbreviated K

Diode Testing

Diode checker
Ohmmeter
Curve tracer

Diode Checker

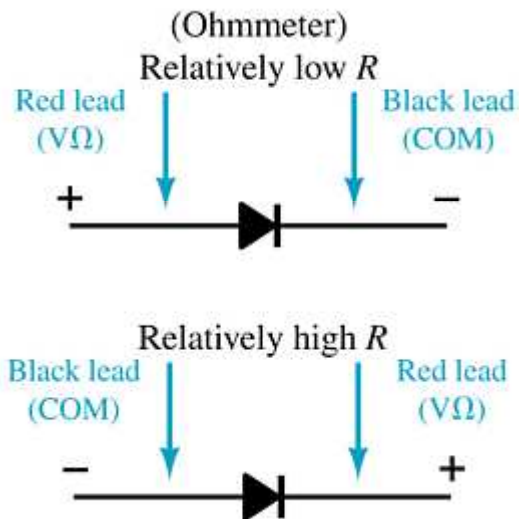
**Many digital multimeters have a diode checking function.
The diode should be tested out of circuit.**

A normal diode exhibits its forward voltage:

- **Gallium arsenide $\cong 1.2$ V**
- **Silicon diode $\cong 0.7$ V**
- **Germanium diode $\cong 0.3$ V**

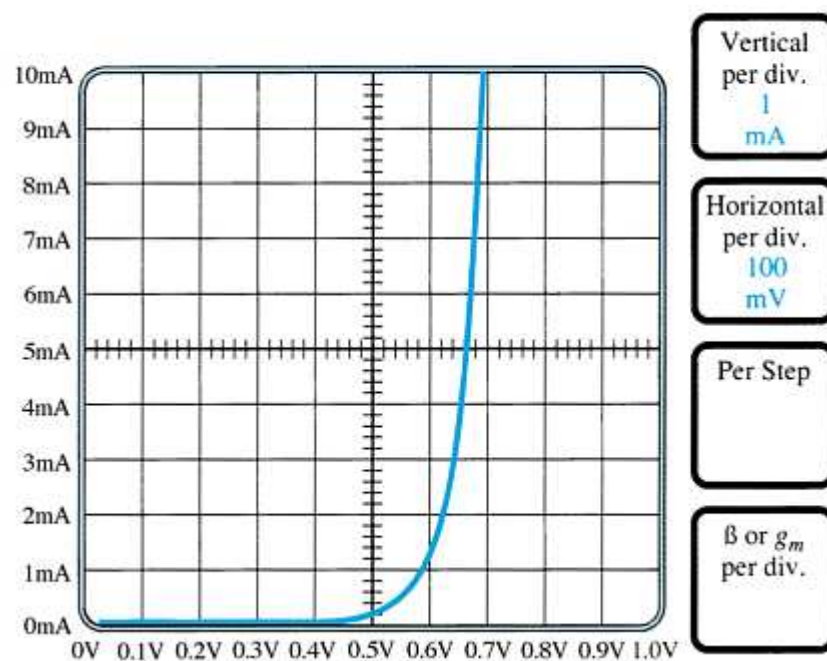
Ohmmeter

An ohmmeter set on a low Ohms scale can be used to test a diode. The diode should be tested out of circuit.



Curve Tracer

A curve tracer displays the characteristic curve of a diode in the test circuit. This curve can be compared to the specifications of the diode from a data sheet.



Other Types of Diodes

Zener diode

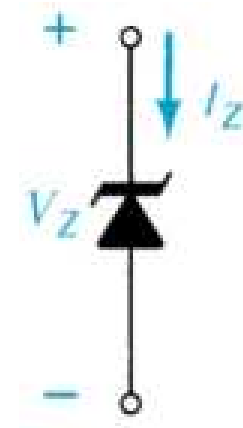
Light-emitting diode

Diode arrays

Zener Diode

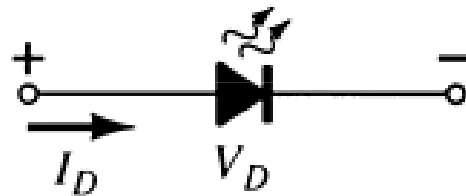
A Zener is a diode operated in reverse bias at the Zener voltage (V_Z).

Common Zener voltages are between 1.8 V and 200 V



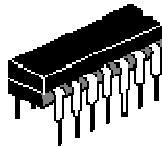
Light-Emitting Diode (LED)

**An LED emits photons when it is forward biased.
These can be in the infrared or visible spectrum.
The forward bias voltage is usually in the range of 2 V to 3 V.**



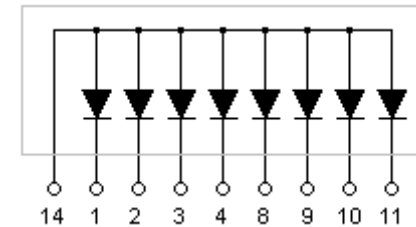
Diode Arrays

Multiple diodes can be packaged together in an integrated circuit (IC).

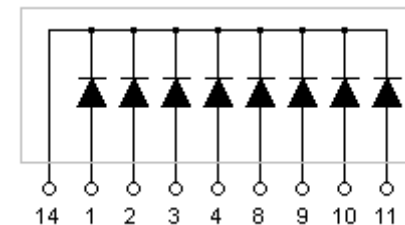


A variety of combinations exist.

Common Anode



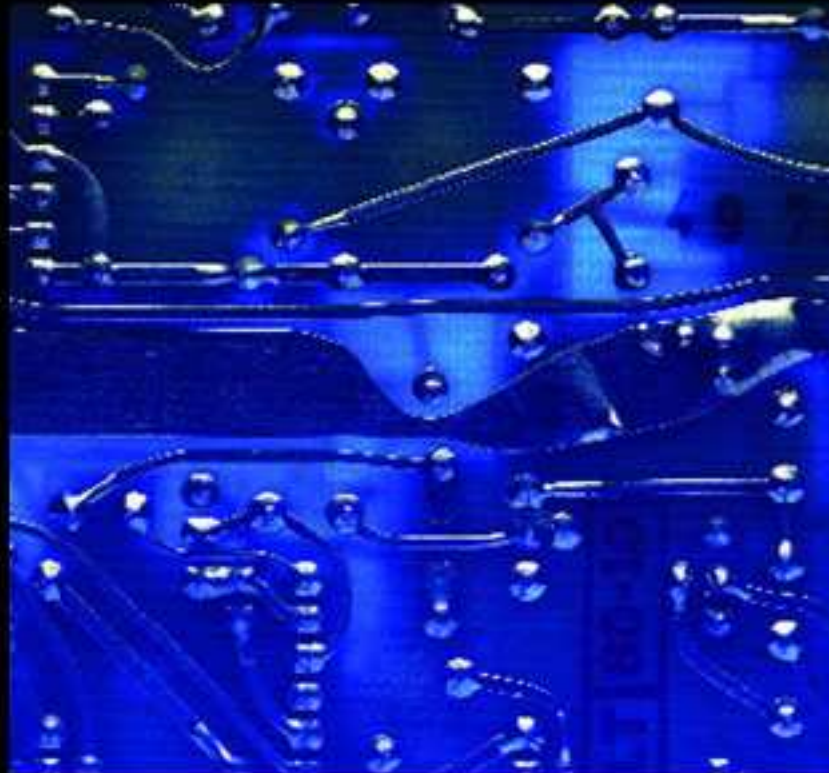
Common Cathode



ELECTRONIC DEVICES AND CIRCUIT THEORY

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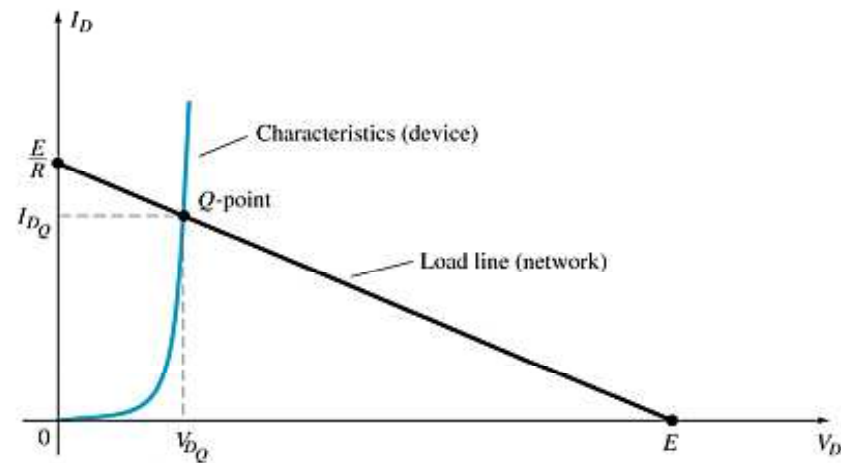
PEARSON

Chapter 2: Diode Applications

Load-Line Analysis

The load line plots all possible combinations of diode current (I_D) and voltage (V_D) for a given circuit. The maximum I_D equals E/R , and the maximum V_D equals E .

The point where the load line and the characteristic curve intersect is the Q-point, which identifies I_D and V_D for a particular diode in a given circuit.



Series Diode Configurations

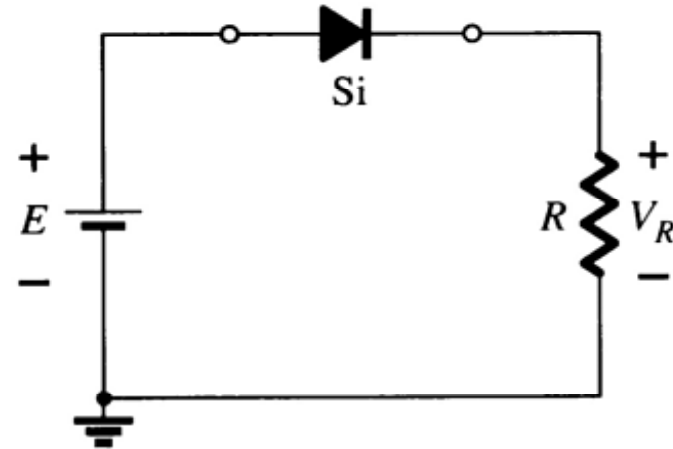
Forward Bias

Constants

- Silicon Diode: $V_D = 0.7 \text{ V}$
- Germanium Diode: $V_D = 0.3 \text{ V}$

Analysis (for silicon)

- $V_D = 0.7 \text{ V}$ (or $V_D = E$ if $E < 0.7 \text{ V}$)
- $V_R = E - V_D$
- $I_D = I_R = I_T = V_R / R$



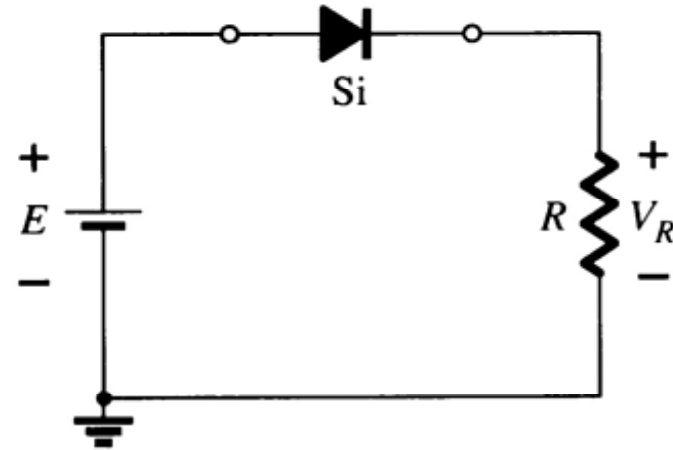
Series Diode Configurations

Reverse Bias

Diodes ideally behave as open circuits

Analysis

- $V_D = E$
- $V_R = 0 \text{ V}$
- $I_D = 0 \text{ A}$



Parallel Configurations

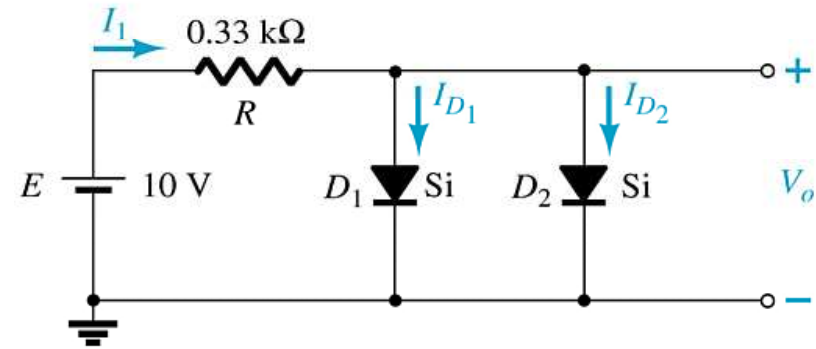
$$V_D = 0.7 \text{ V}$$

$$V_{D1} = V_{D2} = V_O = 0.7 \text{ V}$$

$$V_R = 9.3 \text{ V}$$

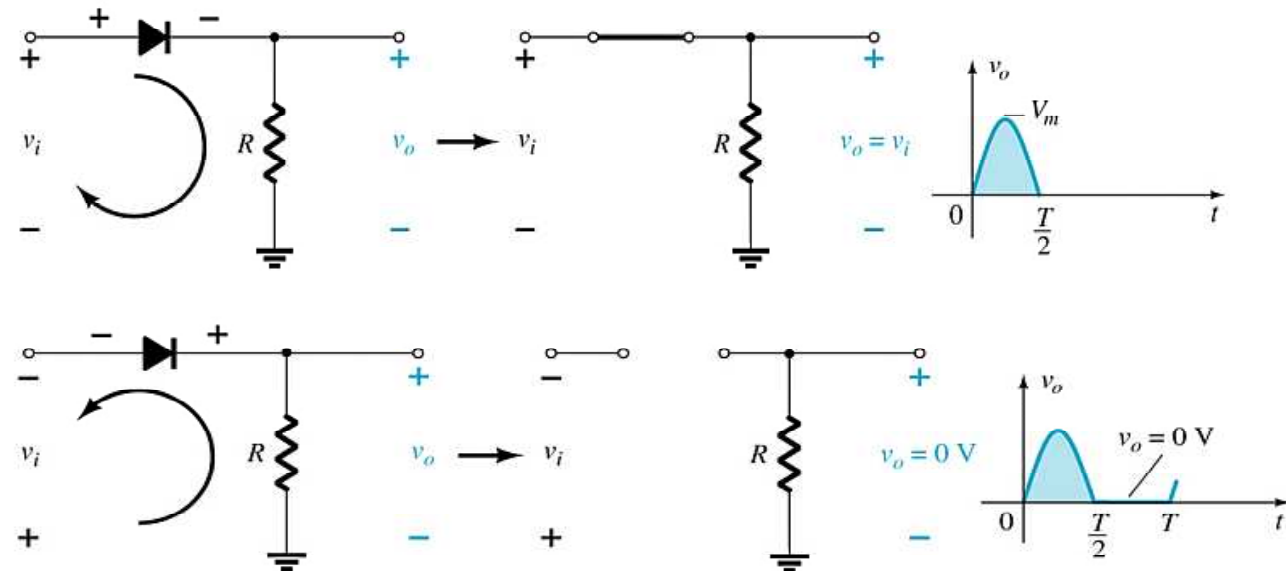
$$I_R = \frac{E - V_D}{R} = \frac{10 \text{ V} - .7 \text{ V}}{.33 \text{ k}\Omega} = 28 \text{ mA}$$

$$I_{D1} = I_{D2} = \frac{28 \text{ mA}}{2} = 14 \text{ mA}$$



Half-Wave Rectification

The diode only conducts when it is forward biased, therefore only half of the AC cycle passes through the diode to the output.



The DC output voltage is $0.318V_m$, where V_m = the peak AC voltage.

PIV (PRV)

Because the diode is only forward biased for one-half of the AC cycle, it is also reverse biased for one-half cycle.

It is important that the reverse breakdown voltage rating of the diode be high enough to withstand the peak, reverse-biasing AC voltage.

$$\text{PIV (or PRV)} > V_m$$

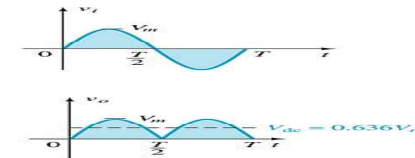
- **PIV = Peak inverse voltage**
- **PRV = Peak reverse voltage**
- **V_m = Peak AC voltage**

Full-Wave Rectification

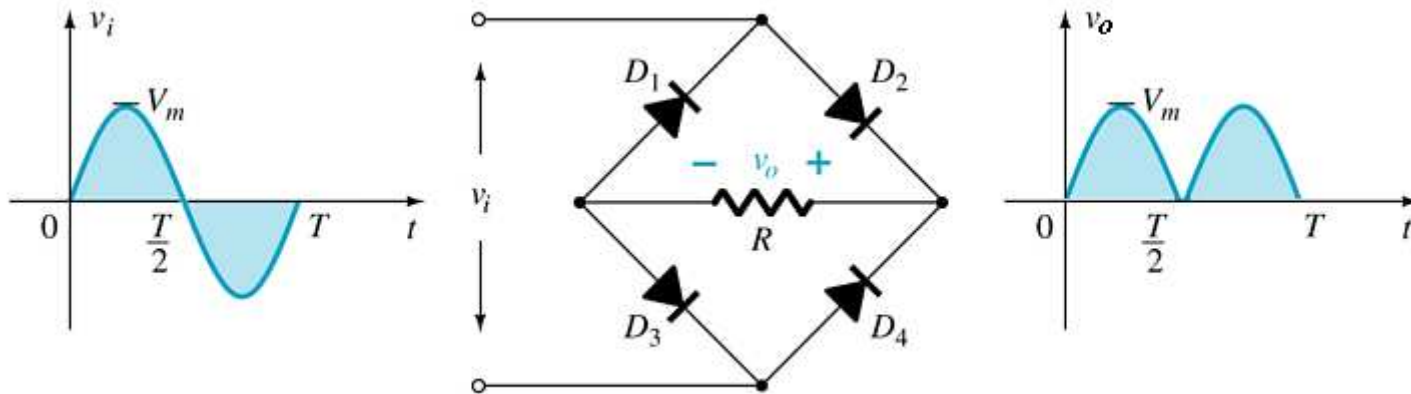
The rectification process can be improved by using a full-wave rectifier circuit.

Full-wave rectification produces a greater DC output:

- Half-wave: $V_{dc} = 0.318V_m$
- Full-wave: $V_{dc} = 0.636V_m$

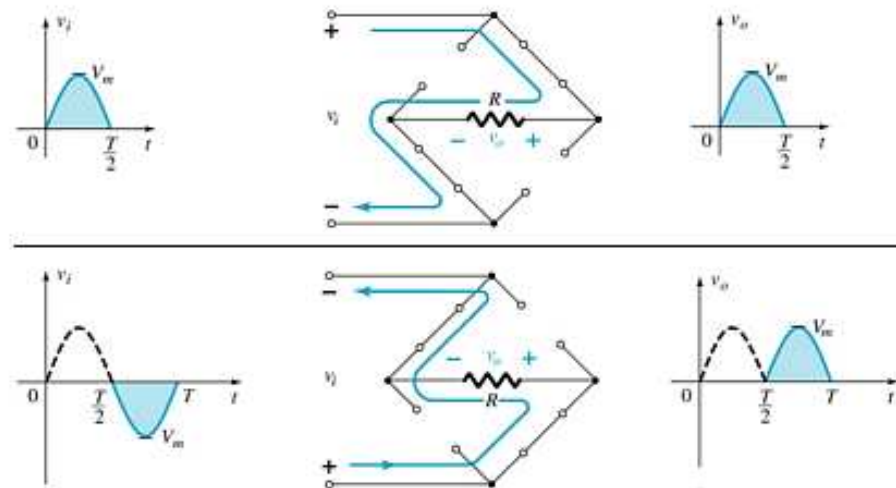


Full-Wave Rectification

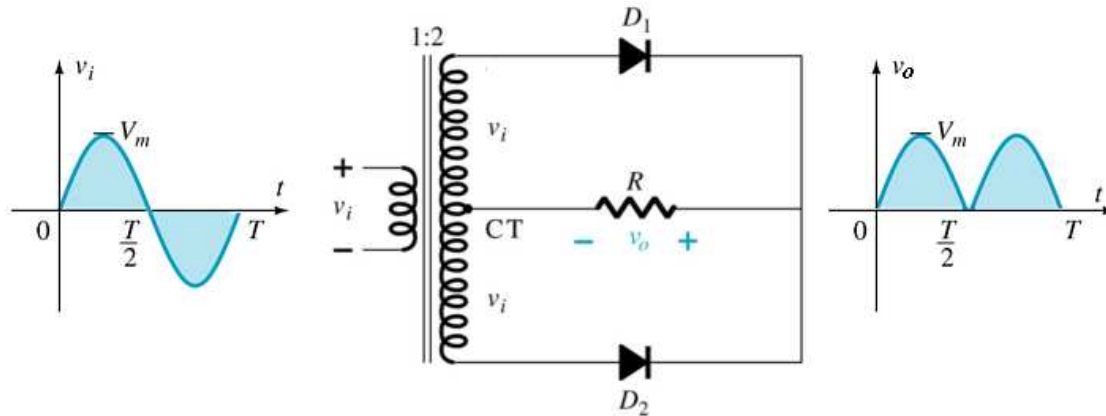


Bridge Rectifier

- Four diodes are connected in a bridge configuration
- $V_{DC} = 0.636V_m$



Full-Wave Rectification

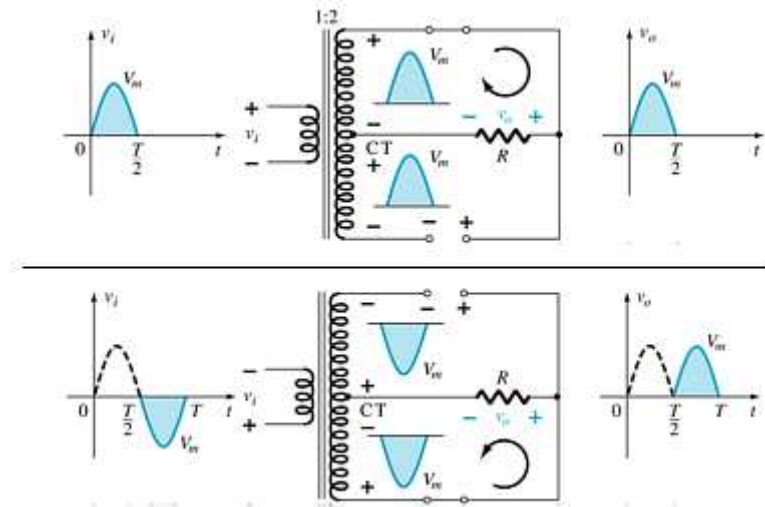


Center-Tapped Transformer Rectifier

Requires

- Two diodes
- Center-tapped transformer

$$V_{DC} = 0.636V_m$$



Summary of Rectifier Circuits

Rectifier	Ideal V_{DC}	Realistic V_{DC}
Half Wave Rectifier	$V_{DC} = 0.318V_m$	$V_{DC} = 0.318V_m - 0.7$
Bridge Rectifier	$V_{DC} = 0.636V_m$	$V_{DC} = 0.636V_m - 2(0.7 \text{ V})$
Center-Tapped Transformer Rectifier	$V_{DC} = 0.636V_m$	$V_{DC} = 0.636V_m - 0.7 \text{ V}$

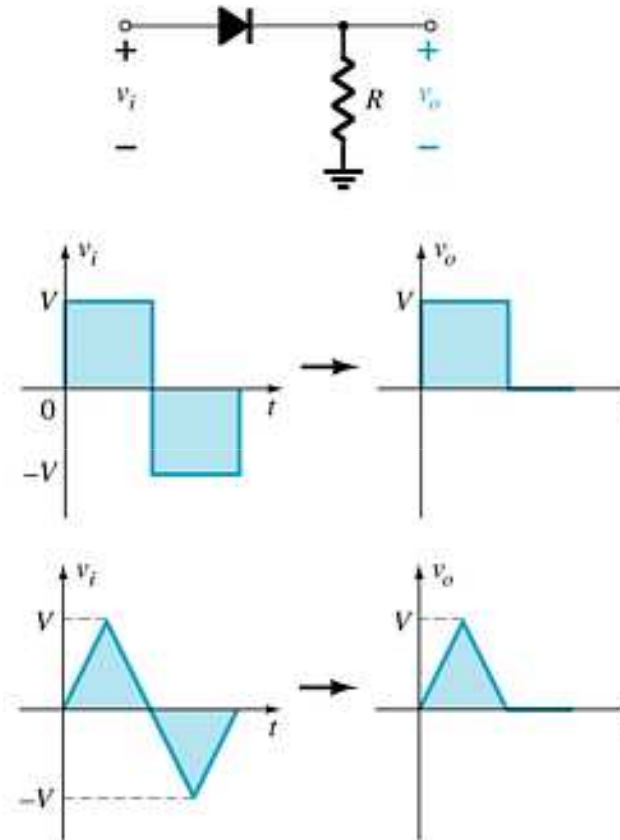
V_m = peak of the AC voltage.

In the center tapped transformer rectifier circuit, the peak AC voltage is the transformer secondary voltage to the tap.

Diode Clippers

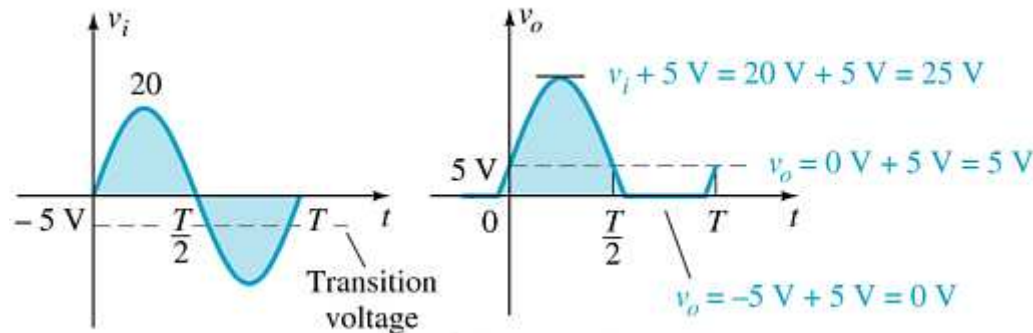
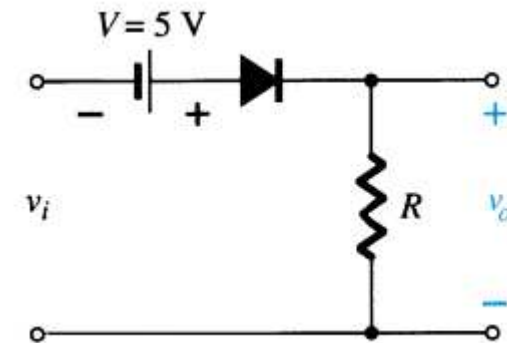
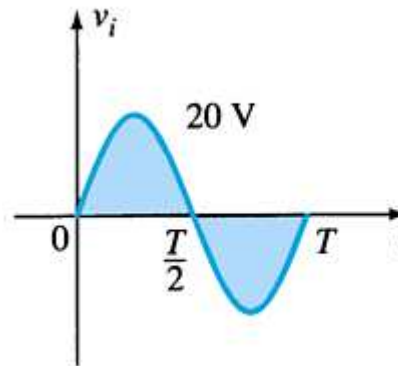
The diode in a **series clipper** “clips” any voltage that does not forward bias it:

- A reverse-biasing polarity
- A forward-biasing polarity less than 0.7 V (for a silicon diode)



Biased Clippers

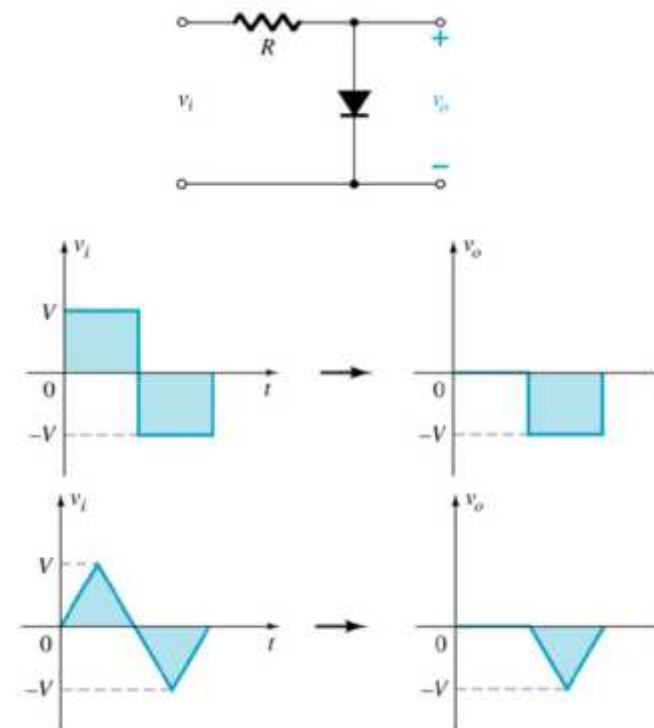
Adding a DC source in series with the clipping diode changes the effective forward bias of the diode.



Parallel Clippers

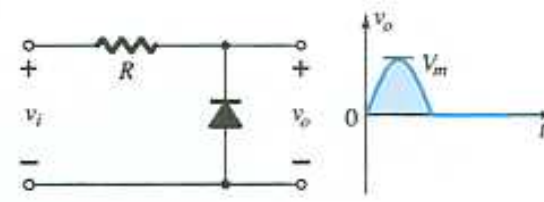
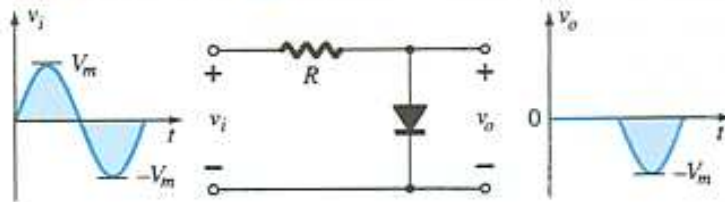
The diode in a **parallel clipper** circuit “clips” any voltage that forward bias it.

DC biasing can be added in series with the diode to change the clipping level.

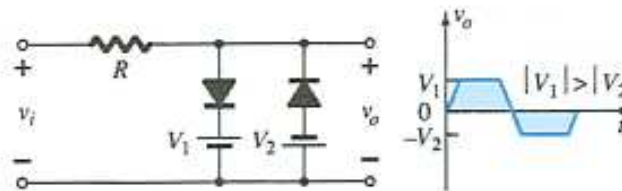
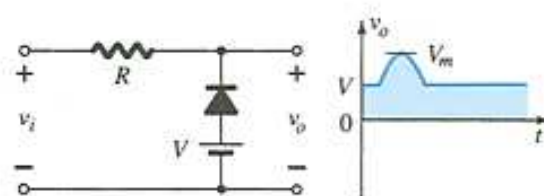
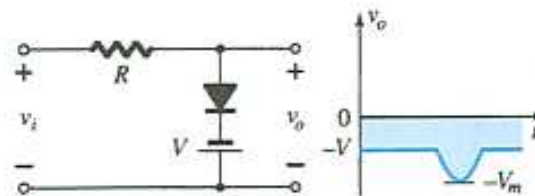
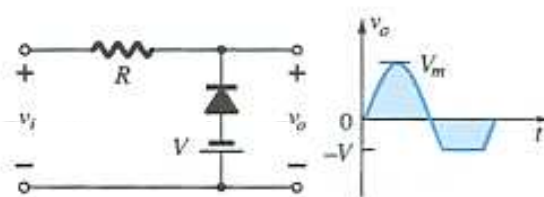
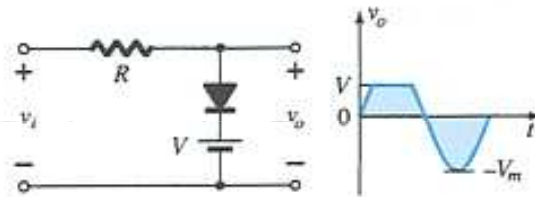


Summary of Clipper Circuits

Simple Parallel Clippers (Ideal Diodes)



Biased Parallel Clippers (Ideal Diodes)

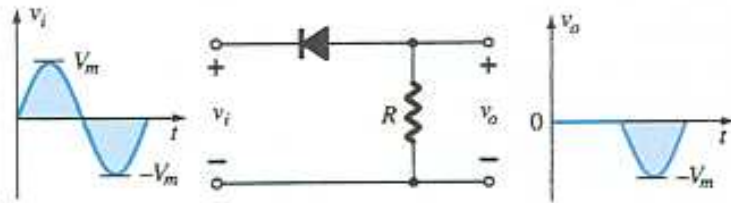


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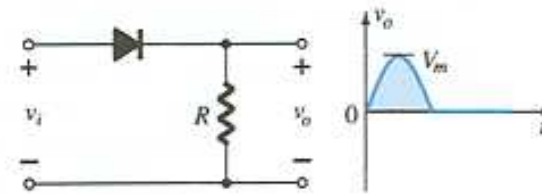
Summary of Clipper Circuits

Simple Series Clippers (Ideal Diodes)

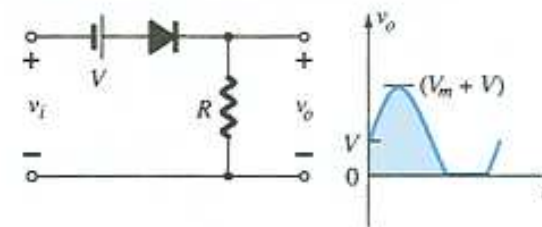
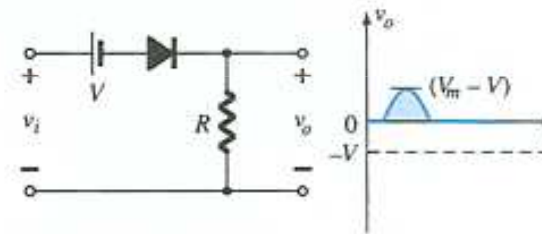
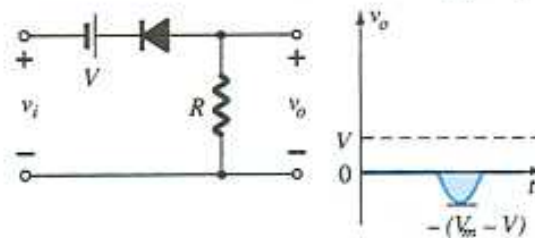
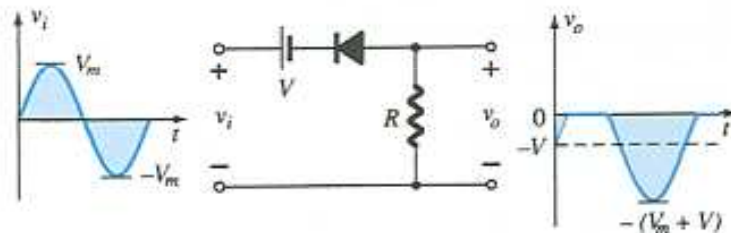
POSITIVE



NEGATIVE

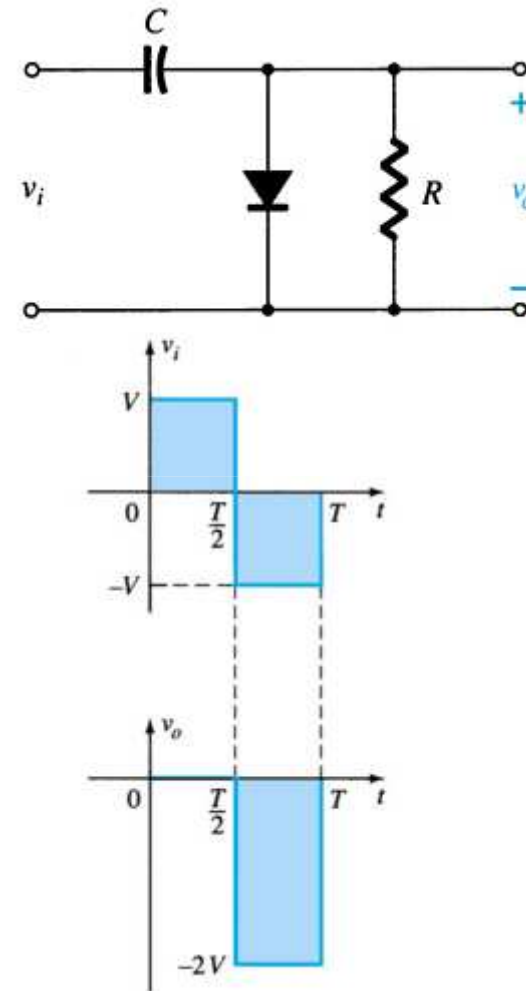


Biased Series Clippers (Ideal Diodes)



Clampers

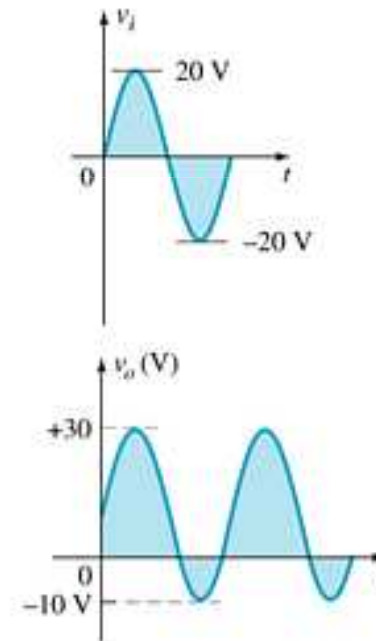
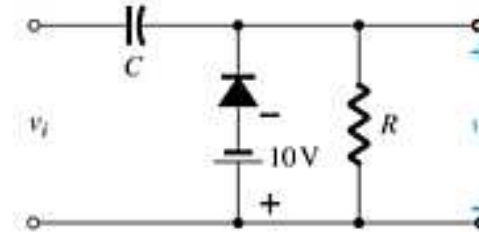
A diode and capacitor can be combined to “clamp” an AC signal to a specific DC level.



Biased Clamper Circuits

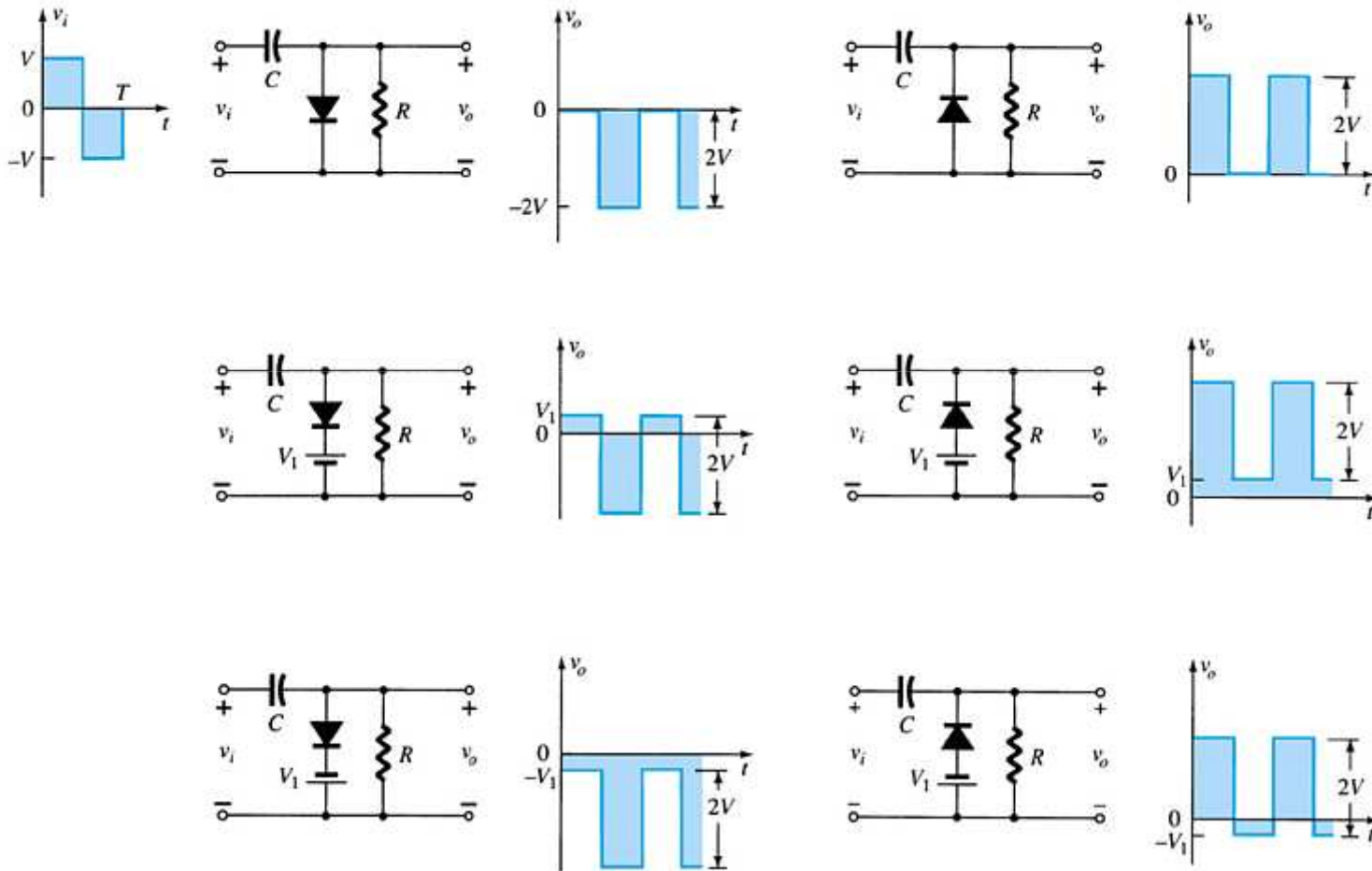
The input signal can be any type of waveform such as sine, square, and triangle waves.

The DC source lets you adjust the DC clamping level.



Summary of Clamper Circuits

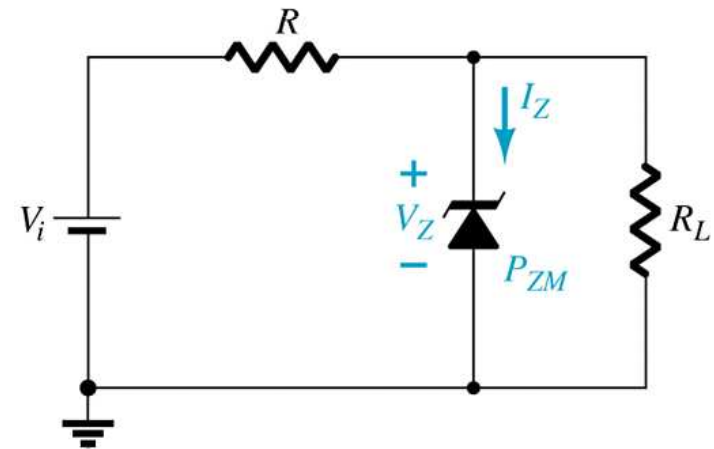
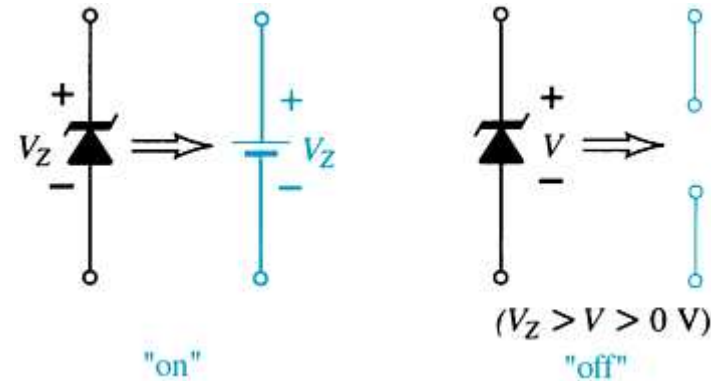
Clamping Networks



Zener Diodes

The Zener is a diode operated in reverse bias at the Zener Voltage (V_Z).

- When $V_i \geq V_Z$
 - The Zener is on
 - Voltage across the Zener is V_Z
 - Zener current: $I_Z = I_R - I_{RL}$
 - The Zener Power: $P_Z = V_Z I_Z$
- When $V_i < V_Z$
 - The Zener is off
 - The Zener acts as an open circuit



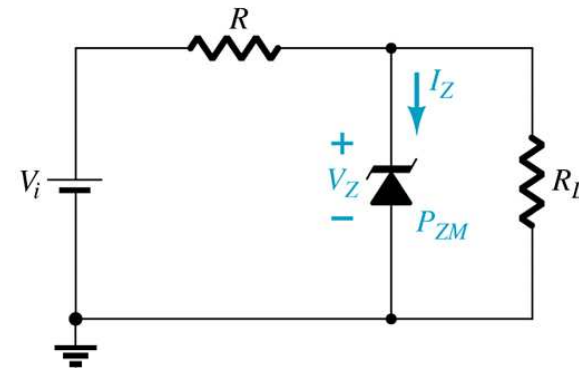
Zener Resistor Values

If R is too large, the Zener diode cannot conduct because the available amount of current is less than the minimum current rating, I_{ZK} . The minimum current is given by:

$$I_{Lmin} = I_R - I_{ZK}$$

The *maximum* value of resistance is:

$$R_{Lmax} = \frac{V_Z}{I_{Lmin}}$$



If R is too small, the Zener current exceeds the maximum current rating, I_{ZM} . The maximum current for the circuit is given by:

$$I_{Lmax} = \frac{V_L}{R_L} = \frac{V_Z}{R_{Lmin}}$$

The *minimum* value of resistance is:

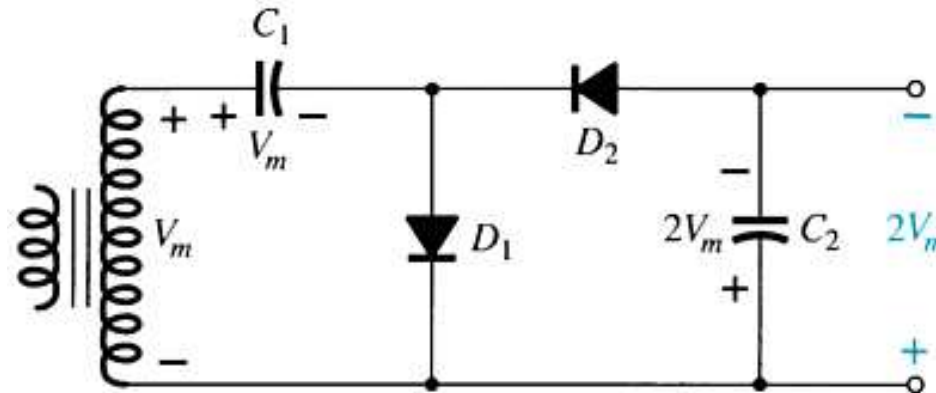
$$R_{Lmin} = \frac{RV_Z}{V_i - V_Z}$$

Voltage-Multiplier Circuits

Voltage multiplier circuits use a combination of diodes and capacitors to step up the output voltage of rectifier circuits.

- **Voltage Doubler**
- **Voltage Tripler**
- **Voltage Quadrupler**

Voltage Doubler



This half-wave voltage doubler's output can be calculated by:

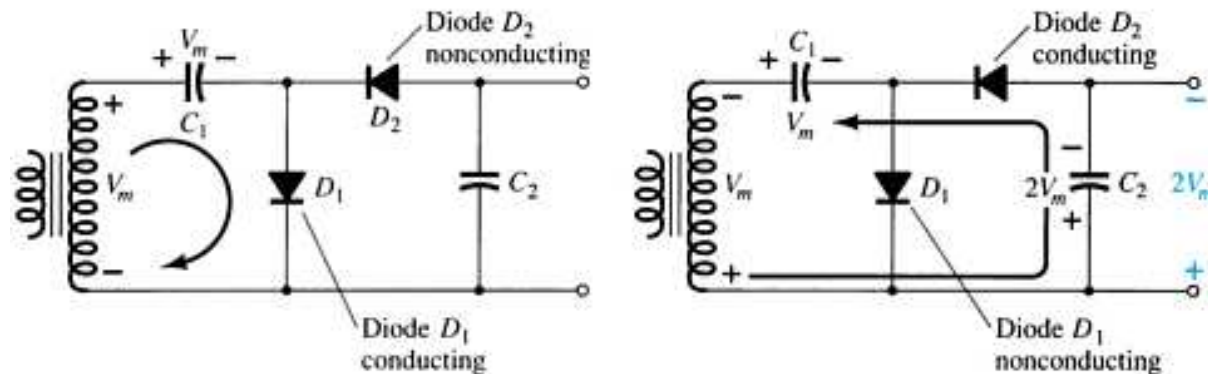
$$V_{\text{out}} = V_{C2} = 2V_m$$

where V_m = peak secondary voltage of the transformer

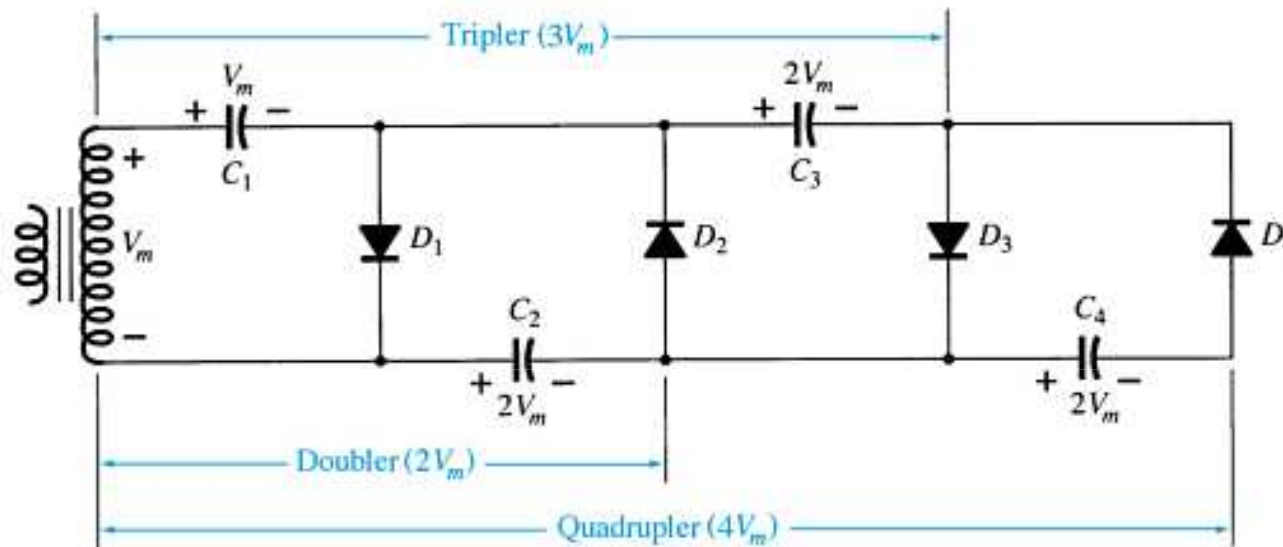
Voltage Doubler

- **Positive Half-Cycle**
 - D_1 conducts
 - D_2 is switched off
 - Capacitor C_1 charges to V_m
- **Negative Half-Cycle**
 - D_1 is switched off
 - D_2 conducts
 - Capacitor C_2 charges to V_m

$$V_{\text{out}} = V_{C2} = 2V_m$$



Voltage Tripler and Quadrupler



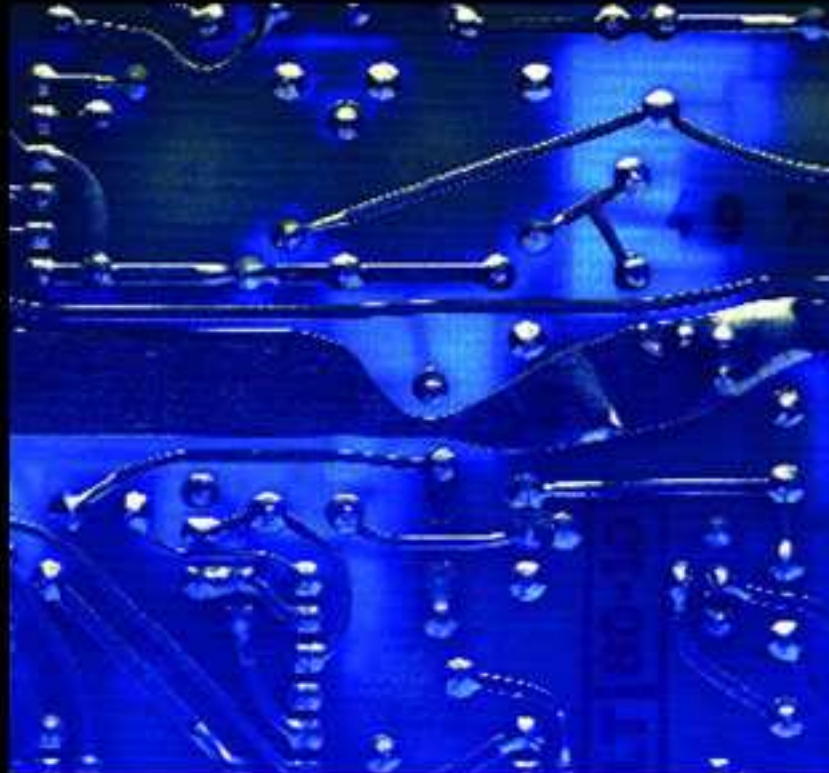
Practical Applications

- **Rectifier Circuits**
 - **Conversions of AC to DC for DC operated circuits**
 - **Battery Charging Circuits**
- **Simple Diode Circuits**
 - **Protective Circuits against**
 - **Overcurrent**
 - **Polarity Reversal**
 - **Currents caused by an inductive kick in a relay circuit**
- **Zener Circuits**
 - **Overvoltage Protection**
 - **Setting Reference Voltages**

ELECTRONIC DEVICES AND CIRCUIT THEORY

TENTH EDITION

BOYLESTAD



PEARSON

Chapter 3: Bipolar Junction Transistors

Transistor Construction

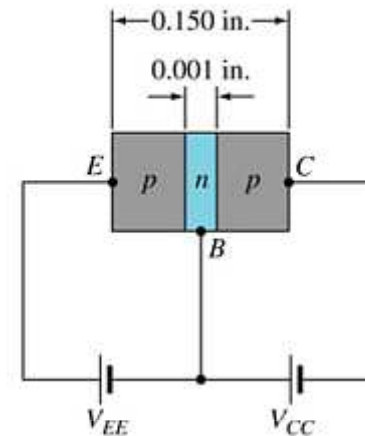
There are two types of transistors:

- *pnp*
- *npn*

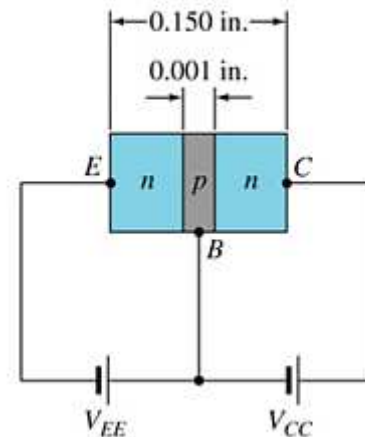
The terminals are labeled:

- **E - Emitter**
- **B - Base**
- **C - Collector**

pnp



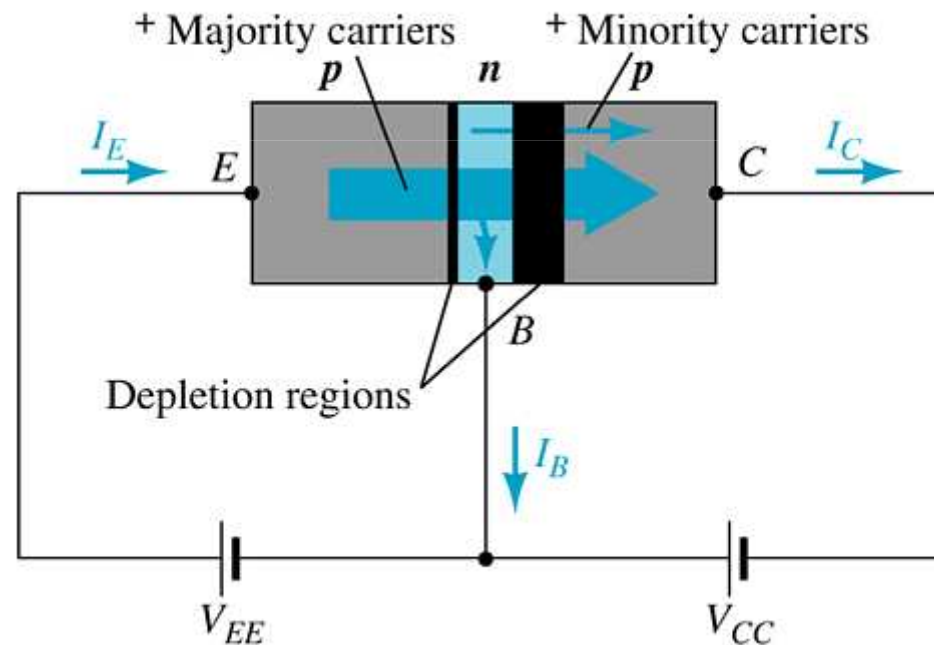
npn



Transistor Operation

With the external sources, V_{EE} and V_{CC} , connected as shown:

- The emitter-base junction is forward biased
- The base-collector junction is reverse biased



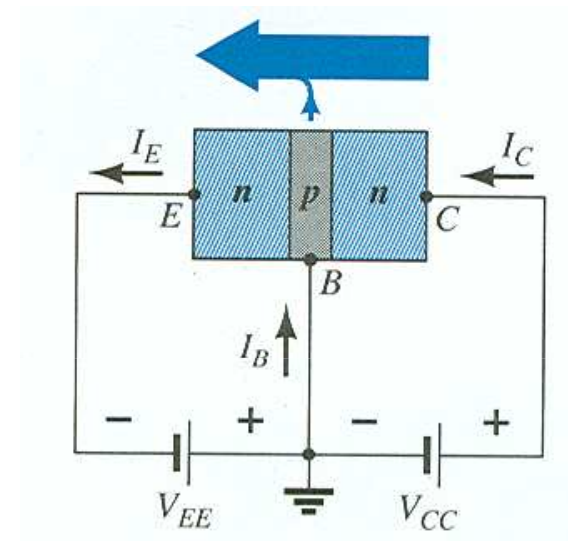
Currents in a Transistor

Emitter current is the sum of the collector and base currents:

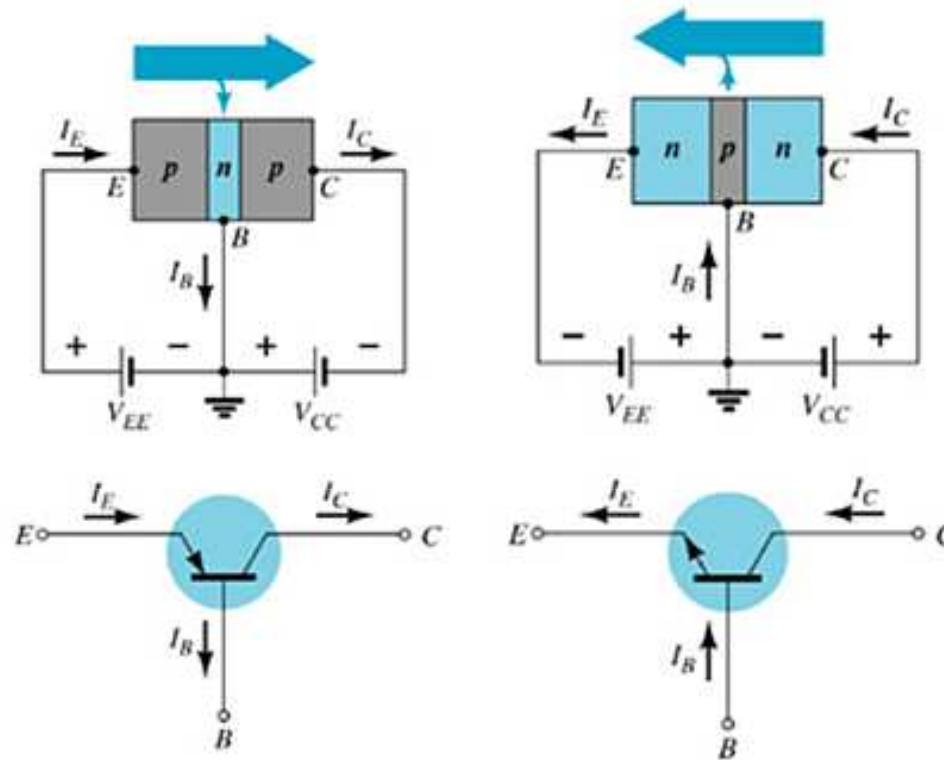
$$I_E = I_C + I_B$$

The collector current is comprised of two currents:

$$I_C = I_{C_{\text{majority}}} + I_{C_{\text{minority}}}$$



Common-Base Configuration

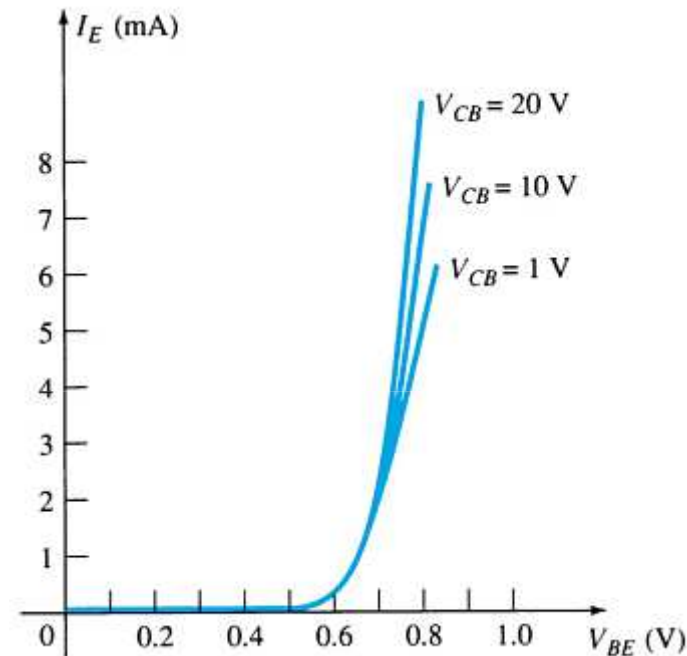


The base is common to both input (emitter–base) and output (collector–base) of the transistor.

Common-Base Amplifier

Input Characteristics

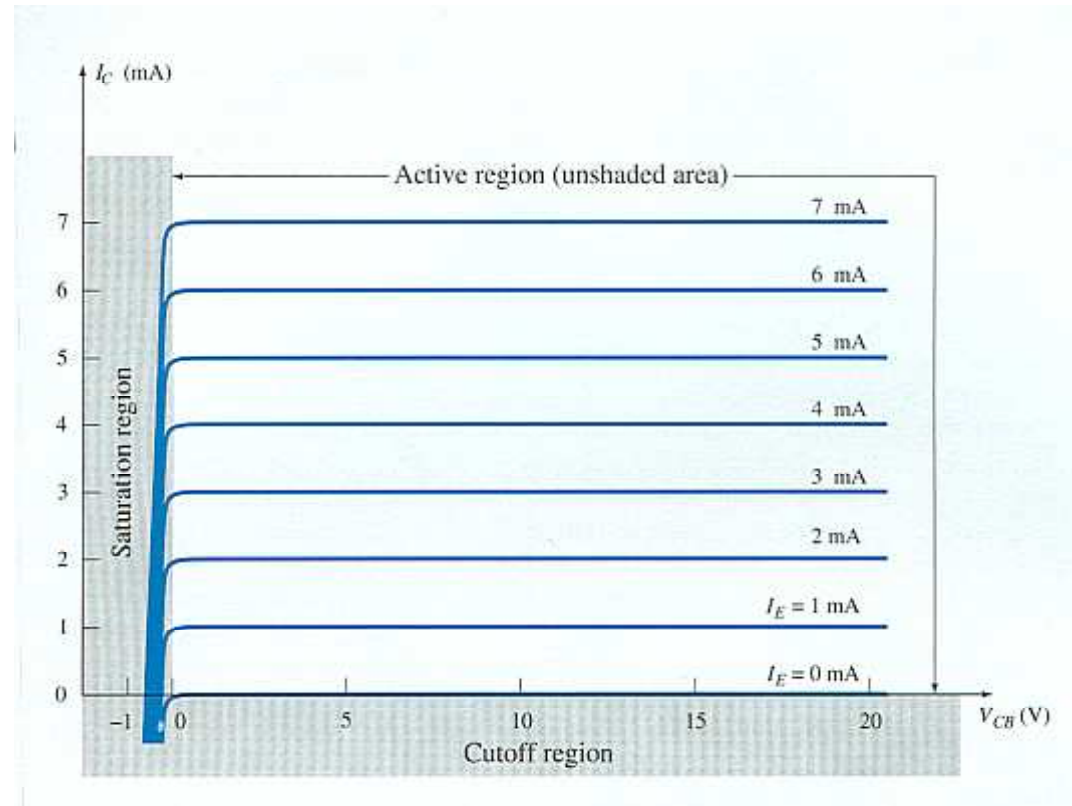
This curve shows the relationship between input current (I_E) to input voltage (V_{BE}) for three output voltage (V_{CB}) levels.



Common-Base Amplifier

Output Characteristics

This graph demonstrates the output current (I_C) to an output voltage (V_{CB}) for various levels of input current (I_E).



Operating Regions

- **Active** – Operating range of the amplifier.
- **Cutoff** – The amplifier is basically off. There is voltage, but little current.
- **Saturation** – The amplifier is full on. There is current, but little voltage.

Approximations

Emitter and collector currents:

$$I_C \cong I_E$$

Base-emitter voltage:

$$V_{BE} = 0.7 \text{ V (for Silicon)}$$

Alpha (α)

Alpha (α) is the ratio of I_C to I_E :

$$\alpha_{dc} = \frac{I_C}{I_E}$$

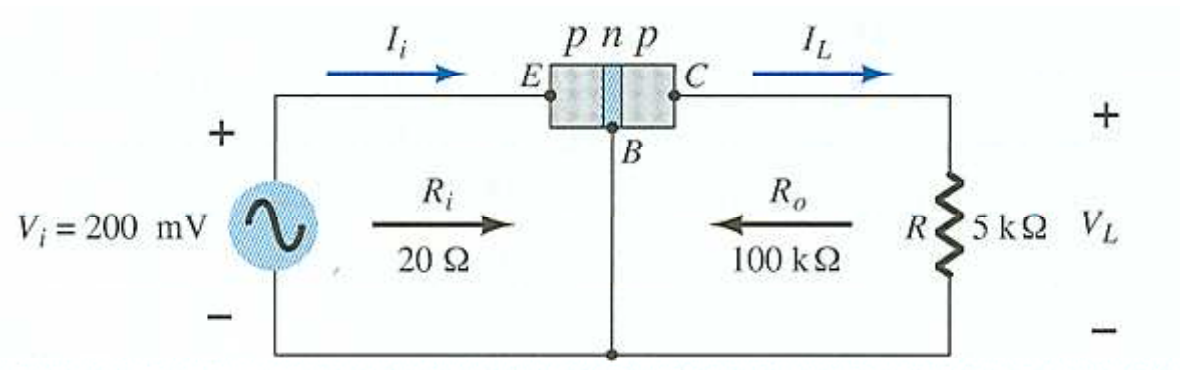
Ideally: $\alpha = 1$

In reality: α is between 0.9 and 0.998

Alpha (α) in the AC mode:

$$\alpha_{ac} = \frac{\Delta I_C}{\Delta I_E}$$

Transistor Amplification



Currents and Voltages:

$$I_E = I_i = \frac{V_i}{R_i} = \frac{200\text{mV}}{20\Omega} = 10\text{mA}$$

$$I_C \cong I_E$$

$$I_L \cong I_i = 10\text{mA}$$

$$V_L = I_L R = (10\text{mA})(5\text{k}\Omega) = 50\text{V}$$

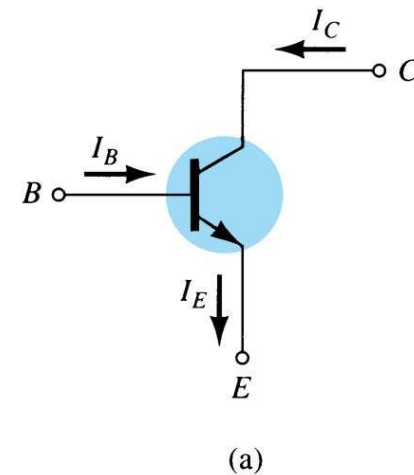
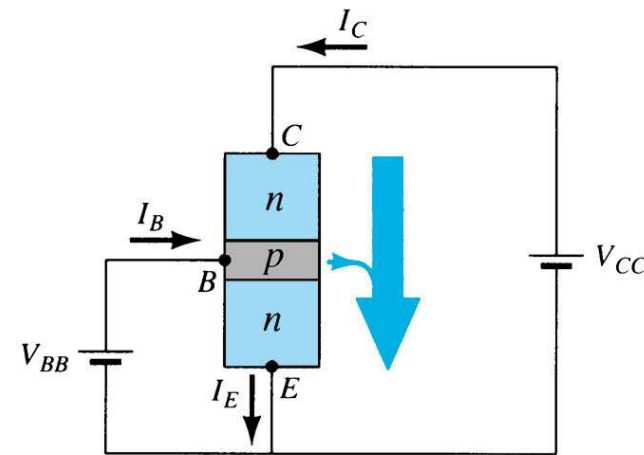
Voltage Gain:

$$A_v = \frac{V_L}{V_i} = \frac{50\text{V}}{200\text{mV}} = 250$$

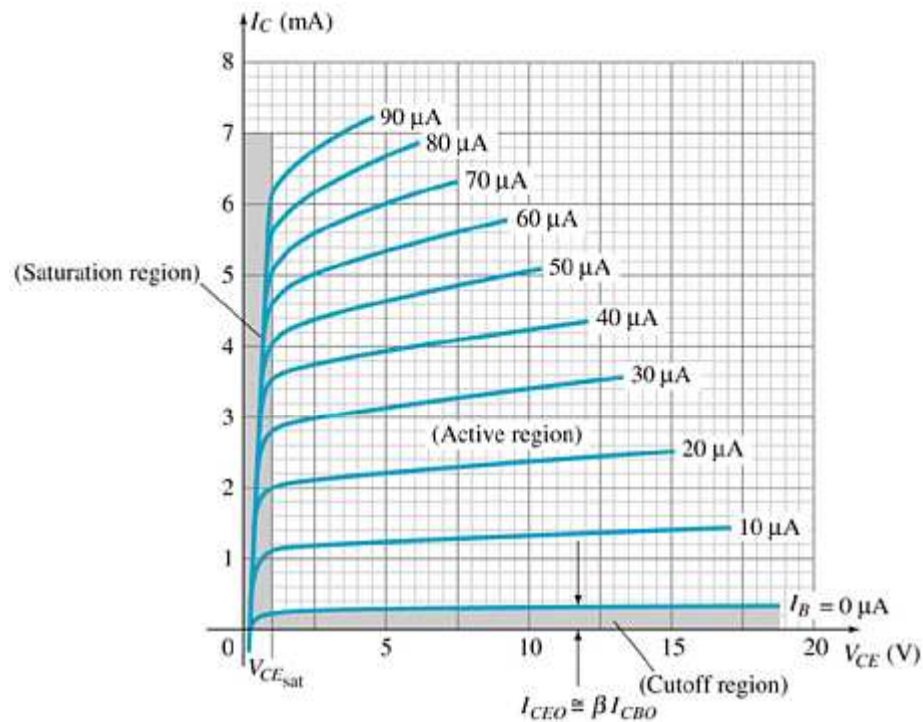
Common-Emitter Configuration

The emitter is common to both input (base-emitter) and output (collector-emitter).

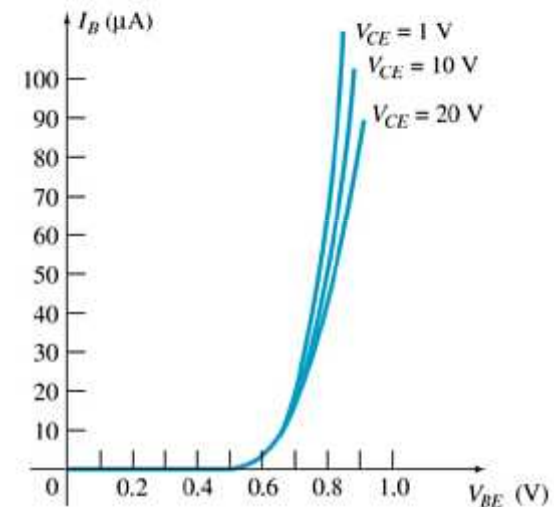
The input is on the base and the output is on the collector.



Common-Emitter Characteristics



Collector Characteristics



Base Characteristics

Common-Emitter Amplifier Currents

Ideal Currents

$$I_E = I_C + I_B$$

$$I_C = \alpha I_E$$

Actual Currents

$$I_C = \alpha I_E + I_{CBO} \quad \text{where } I_{CBO} = \text{minority collector current}$$

I_{CBO} is usually so small that it can be ignored, except in high power transistors and in high temperature environments.

When $I_B = 0 \mu\text{A}$ the transistor is in cutoff, but there is some minority current flowing called I_{CEO} .

$$I_{CEO} = \frac{I_{CBO}}{1 - \alpha} \bigg|_{I_B = 0 \mu\text{A}}$$

Beta (β)

β represents the amplification factor of a transistor. (β is sometimes referred to as h_{fe} , a term used in transistor modeling calculations)

In DC mode:

$$\beta_{dc} = \frac{I_C}{I_B}$$

In AC mode:

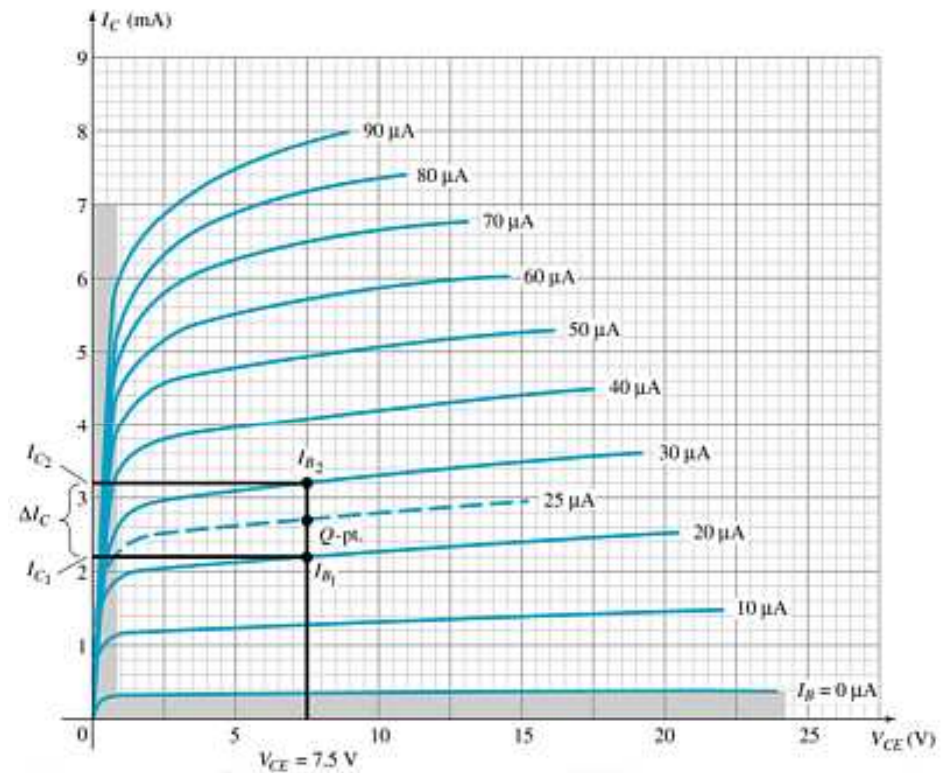
$$\beta_{ac} = \left. \frac{\Delta I_C}{\Delta I_B} \right|_{V_{CE} = \text{constant}}$$

Beta (β)

Determining β from a Graph

$$\begin{aligned}\beta_{AC} &= \frac{(3.2\text{mA} - 2.2\text{mA})}{(30\mu\text{A} - 20\mu\text{A})} \\ &= \frac{1\text{mA}}{10\mu\text{A}} \bigg|_{V_{CE}=7.5} \\ &= 100\end{aligned}$$

$$\begin{aligned}\beta_{DC} &= \frac{2.7\text{mA}}{25\mu\text{A}} \bigg|_{V_{CE}=7.5} \\ &= 108\end{aligned}$$



Beta (β)

Relationship between amplification factors β and α

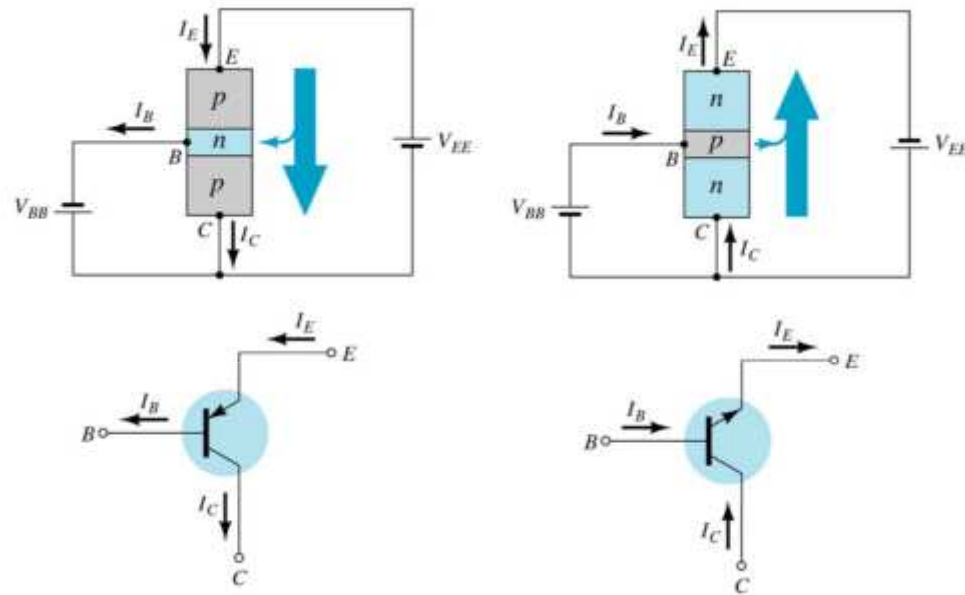
$$\alpha = \frac{\beta}{\beta + 1} \qquad \beta = \frac{\alpha}{\alpha - 1}$$

Relationship Between Currents

$$I_C = \beta I_B \qquad I_E = (\beta + 1)I_B$$

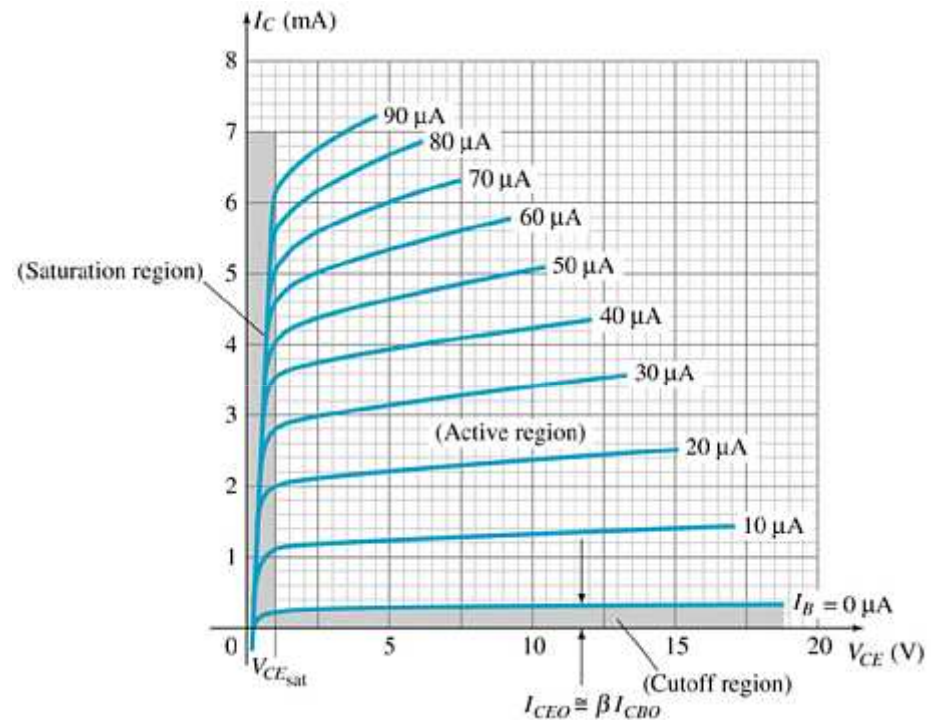
Common-Collector Configuration

The input is on the base and the output is on the emitter.



Common-Collector Configuration

The characteristics are similar to those of the common-emitter configuration, except the vertical axis is I_E .

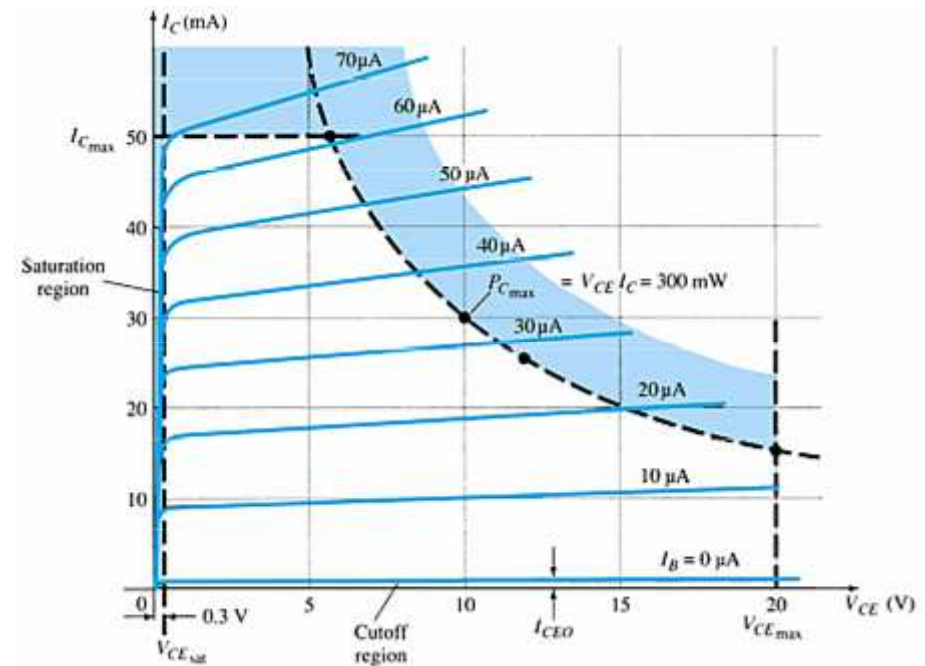


Operating Limits for Each Configuration

V_{CE} is at maximum and I_C is at minimum ($I_{C_{max}} = I_{CEO}$) in the cutoff region.

I_C is at maximum and V_{CE} is at minimum ($V_{CE_{max}} = V_{CE_{sat}} = V_{CEO}$) in the saturation region.

The transistor operates in the active region between saturation and cutoff.



Power Dissipation

Common-base:

$$P_{Cmax} = V_{CB}I_C$$

Common-emitter:

$$P_{Cmax} = V_{CE}I_C$$

Common-collector:

$$P_{Cmax} = V_{CE}I_E$$

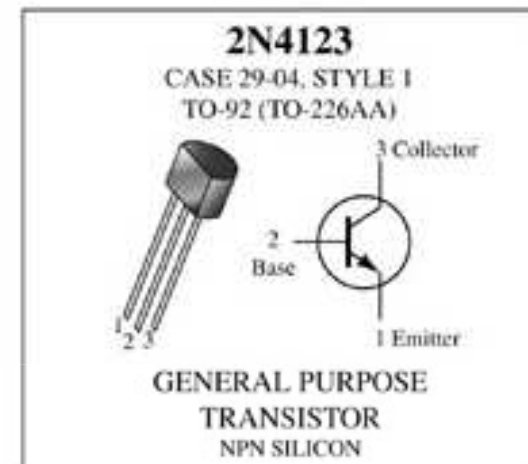
Transistor Specification Sheet

MAXIMUM RATINGS

Rating	Symbol	2N4123	Unit
Collector-Emitter Voltage	V_{CE0}	30	Vdc
Collector-Base Voltage	V_{CB0}	40	Vdc
Emitter-Base Voltage	V_{EB0}	5.0	Vdc
Collector Current – Continuous	I_C	200	mAdc
Total Device Dissipation @ $T_A = 25^\circ\text{C}$ Derate above 25°C	P_D	625 5.0	mW mW/°C
Operating and Storage Junction Temperature Range	T_J, T_{stg}	-55 to +150	°C

THERMAL CHARACTERISTICS

Characteristic	Symbol	Max	Unit
Thermal Resistance, Junction to Case	$R_{\theta JC}$	83.3	°C/W
Thermal Resistance, Junction to Ambient	$R_{\theta JA}$	200	°C/W



Transistor Specification Sheet

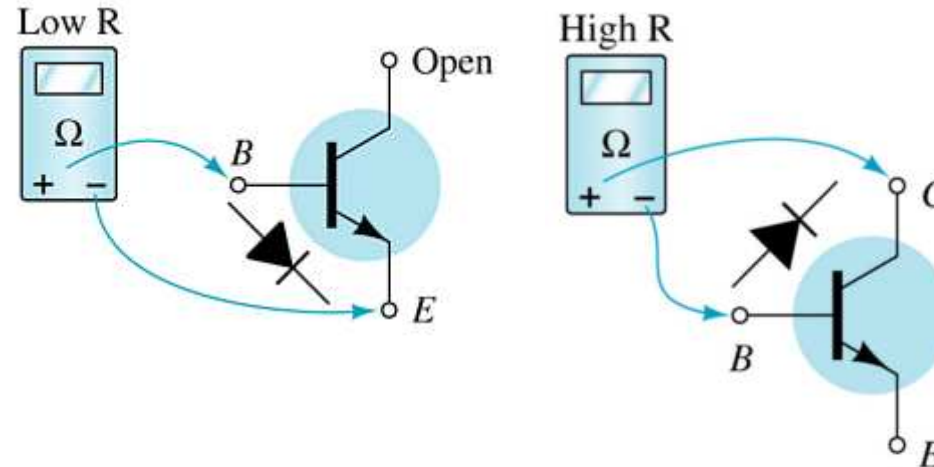
ELECTRICAL CHARACTERISTICS ($T_A = 25^\circ\text{C}$ unless otherwise noted)

Characteristic	Symbol	Min	Max	Unit
OFF CHARACTERISTICS				
Collector-Emitter Breakdown Voltage (1) ($I_C = 1.0\text{ mAdc}$, $I_E = 0$)	$V_{(BR)CEO}$	30		Vdc
Collector-Base Breakdown Voltage ($I_C = 10\text{ }\mu\text{Adc}$, $I_E = 0$)	$V_{(BR)CBO}$	40		Vdc
Emitter-Base Breakdown Voltage ($I_E = 10\text{ }\mu\text{Adc}$, $I_C = 0$)	$V_{(BR)EBC}$	5.0	–	Vdc
Collector Cutoff Current ($V_{CE} = 20\text{ Vdc}$, $I_E = 0$)	I_{CBO}	–	50	nAdc
Emitter Cutoff Current ($V_{BE} = 3.0\text{ Vdc}$, $I_C = 0$)	I_{EBO}	–	50	nAdc
ON CHARACTERISTICS				
DC Current Gain(1) ($I_C = 2.0\text{ mAdc}$, $V_{CE} = 1.0\text{ Vdc}$) ($I_C = 50\text{ mAdc}$, $V_{CE} = 1.0\text{ Vdc}$)	h_{FE}	50 25	150 –	–
Collector-Emitter Saturation Voltage(1) ($I_C = 50\text{ mAdc}$, $I_B = 5.0\text{ mAdc}$)	$V_{CE(sat)}$	–	0.3	Vdc
Base-Emitter Saturation Voltage(1) ($I_C = 50\text{ mAdc}$, $I_B = 5.0\text{ mAdc}$)	$V_{BE(sat)}$	–	0.95	Vdc
SMALL-SIGNAL CHARACTERISTICS				
Current-Gain – Bandwidth Product ($I_C = 10\text{ mAdc}$, $V_{CE} = 20\text{ Vdc}$, $f = 100\text{ MHz}$)	f_T	250		MHz
Output Capacitance ($V_{CE} = 5.0\text{ Vdc}$, $I_E = 0$, $f = 100\text{ MHz}$)	C_{obo}	–	4.0	pF
Input Capacitance ($V_{BE} = 0.5\text{ Vdc}$, $I_C = 0$, $f = 100\text{ kHz}$)	C_{ibo}	–	8.0	pF
Collector-Base Capacitance ($I_E = 0$, $V_{CB} = 5.0\text{ V}$, $f = 100\text{ kHz}$)	C_{cb}	–	4.0	pF
Small-Signal Current Gain ($I_C = 2.0\text{ mAdc}$, $V_{CE} = 10\text{ Vdc}$, $f = 1.0\text{ kHz}$)	h_{ie}	50	200	–
Current Gain – High Frequency ($I_C = 10\text{ mAdc}$, $V_{CE} = 20\text{ Vdc}$, $f = 100\text{ MHz}$) ($I_C = 2.0\text{ mAdc}$, $V_{CE} = 10\text{ V}$, $f = 1.0\text{ kHz}$)	h_{fe}	25 50	– 200	–
Noise Figure ($I_C = 100\text{ }\mu\text{Adc}$, $V_{CE} = 5.0\text{ Vdc}$, $R_S = 1.0\text{ k ohm}$, $f = 1.0\text{ kHz}$)	NF	–	6.0	dB

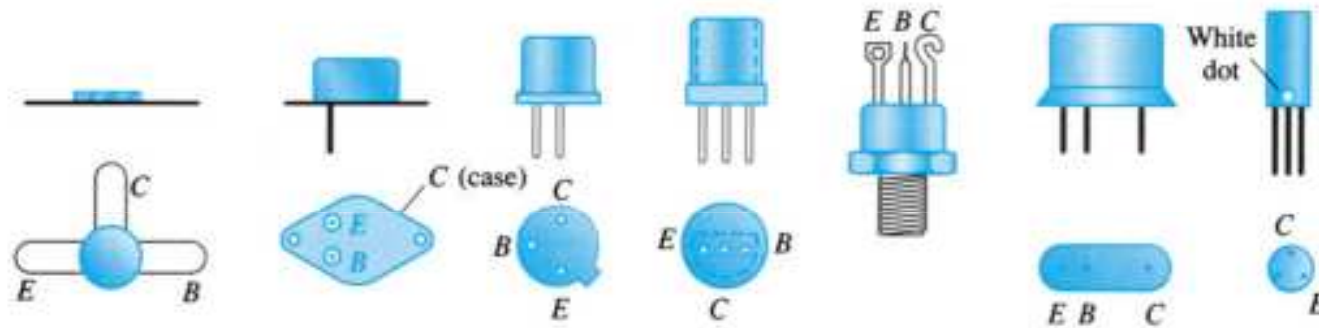
(1) Pulse Test: Pulse Width = 300 μs , Duty Cycle = 2.0%

Transistor Testing

- **Curve Tracer**
Provides a graph of the characteristic curves.
- **DMM**
Some DMMs measure β_{DC} or h_{FE} .
- **Ohmmeter**



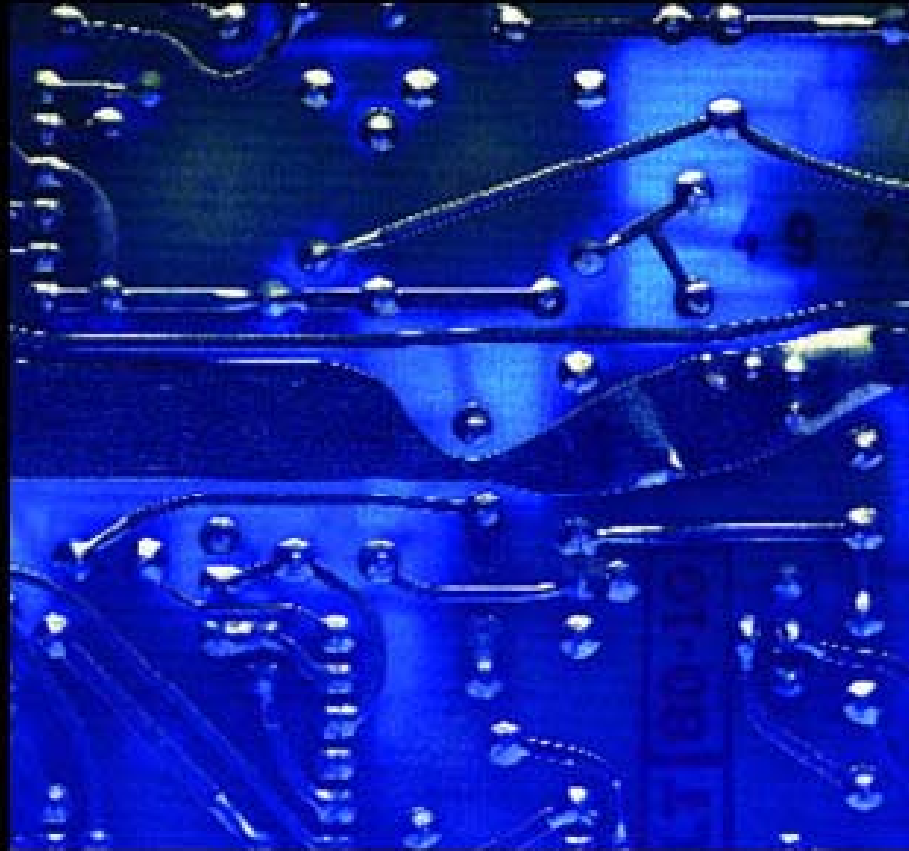
Transistor Terminal Identification



ELECTRONIC DEVICES AND CIRCUIT THEORY

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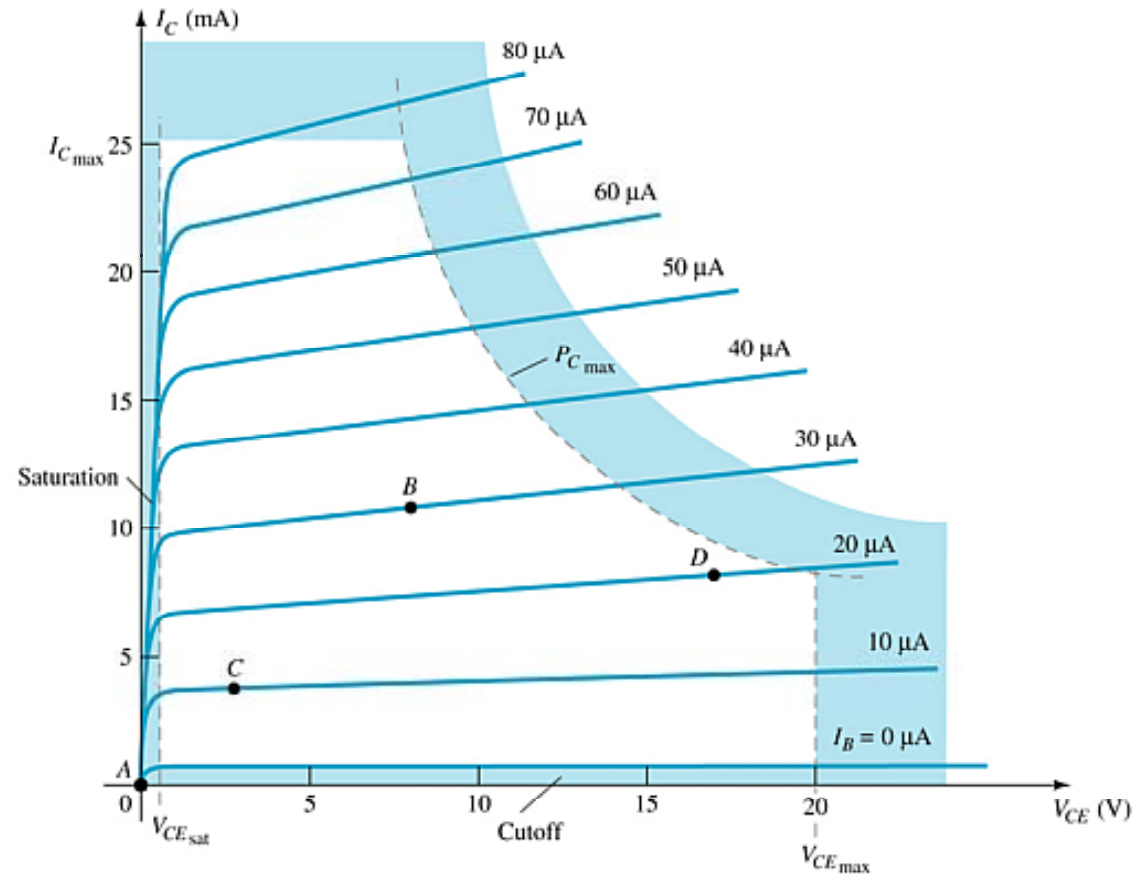
Chapter 4 DC Biasing—BJTs

Biasing

Biasing: The DC voltages applied to a transistor in order to turn it on so that it can amplify the AC signal.

Operating Point

The DC input establishes an operating or *quiescent point* called the ***Q-point***.



The Three States of Operation

- **Active or Linear Region Operation**

Base–Emitter junction is forward biased

Base–Collector junction is reverse biased

- **Cutoff Region Operation**

Base–Emitter junction is reverse biased

- **Saturation Region Operation**

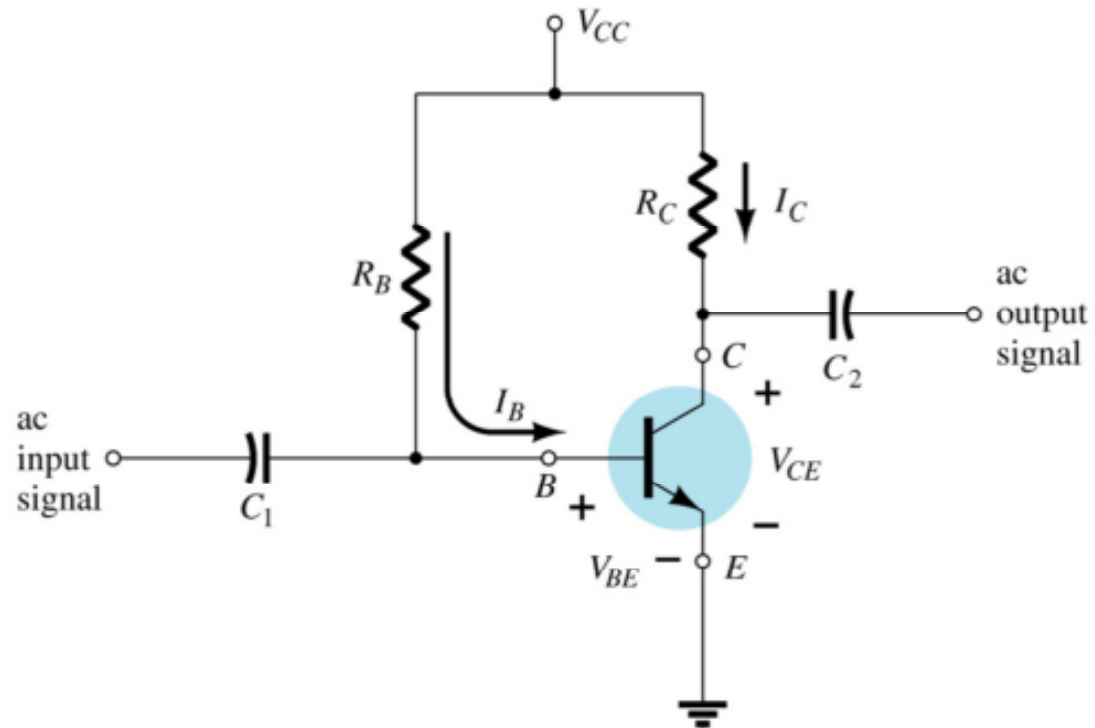
Base–Emitter junction is forward biased

Base–Collector junction is forward biased

DC Biasing Circuits

- Fixed-bias circuit
- Emitter-stabilized bias circuit
- Collector-emitter loop
- Voltage divider bias circuit
- DC bias with voltage feedback

Fixed Bias



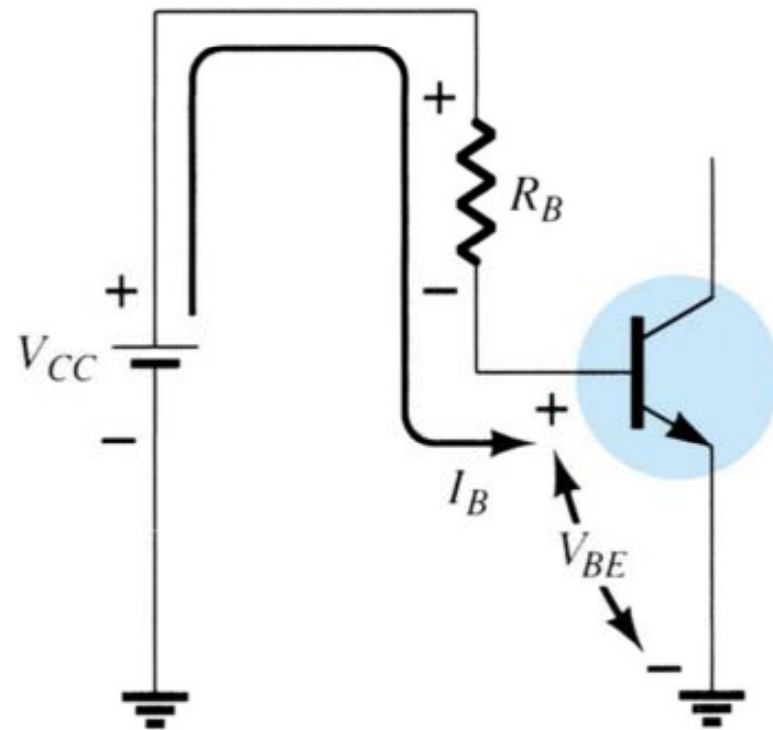
The Base-Emitter Loop

From Kirchhoff's voltage law:

$$+V_{CC} - I_B R_B - V_{BE} = 0$$

Solving for base current:

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$



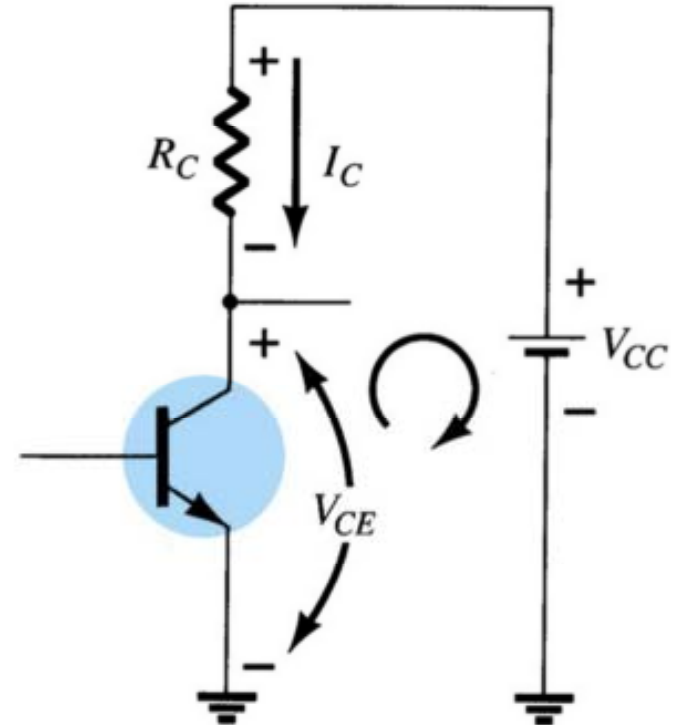
Collector-Emitter Loop

Collector current:

$$I_C = \beta I_B$$

From Kirchhoff's voltage law:

$$V_{CE} = V_{CC} - I_C R_C$$



Saturation

When the transistor is operating in saturation, current through the transistor is at its *maximum* possible value.

$$I_{C\text{sat}} = \frac{V_{CC}}{R_C}$$

$$V_{CE} \cong 0\text{ V}$$

Load Line Analysis

The end points of the load line are:

$I_{C\text{sat}}$

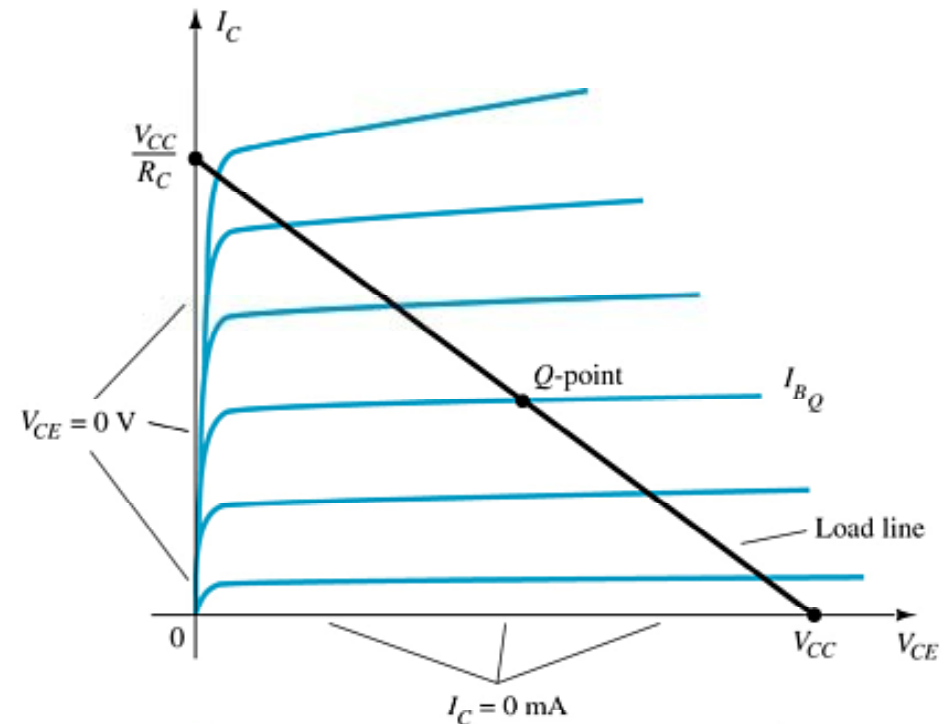
$$I_C = V_{CC} / R_C$$

$$V_{CE} = 0 \text{ V}$$

$V_{CE\text{cutoff}}$

$$V_{CE} = V_{CC}$$

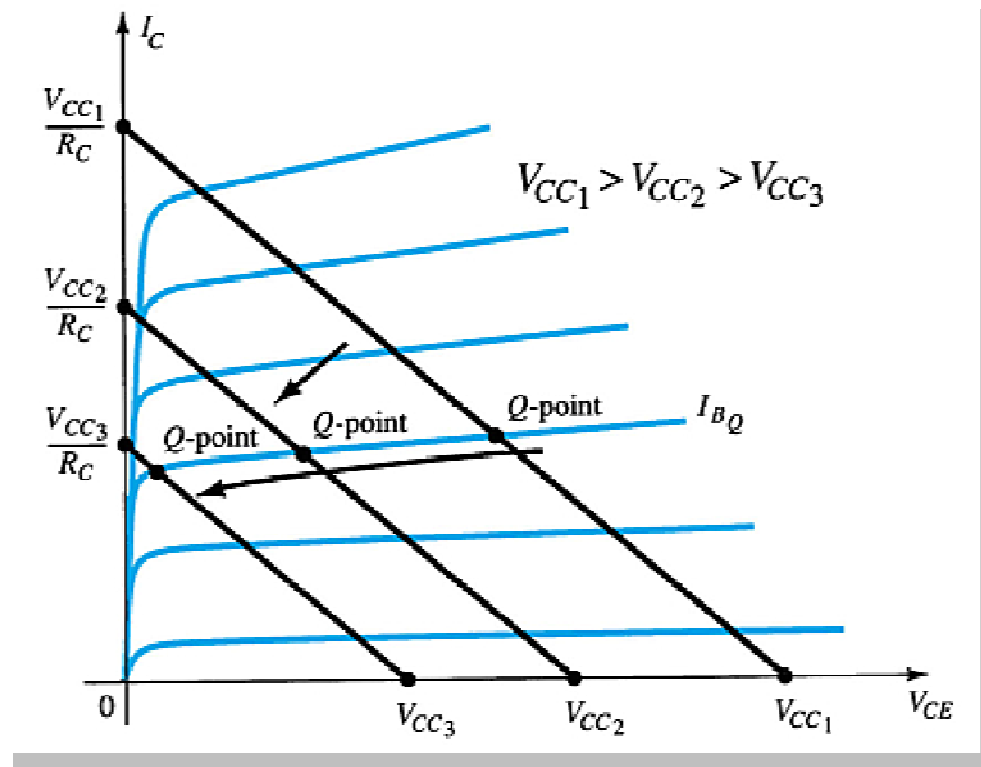
$$I_C = 0 \text{ mA}$$



The Q-point is the operating point:

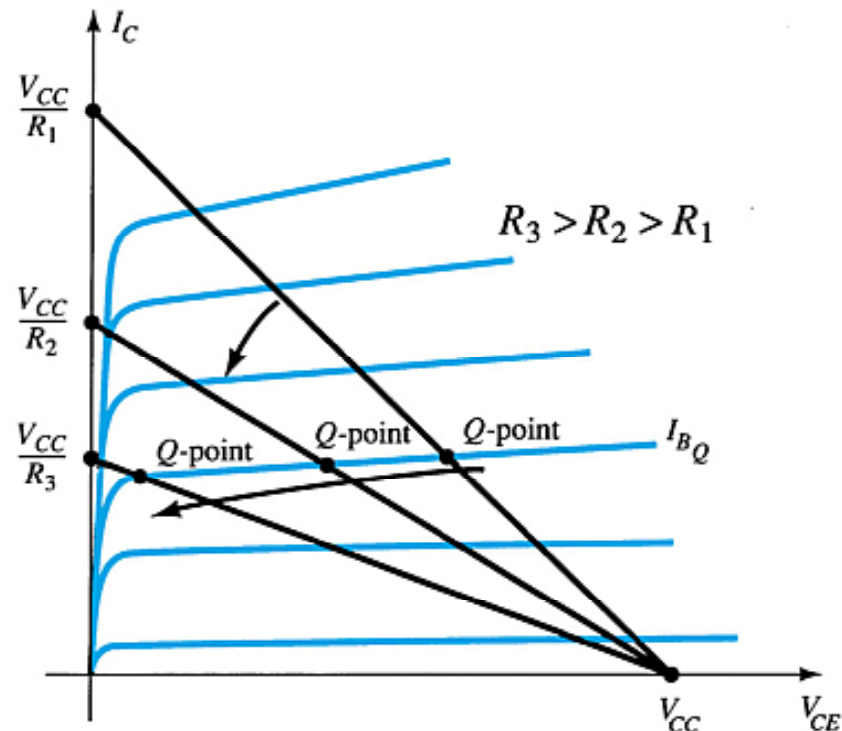
- where the value of R_B sets the value of I_B
- that sets the values of V_{CE} and I_C

Circuit Values Affect the Q-Point



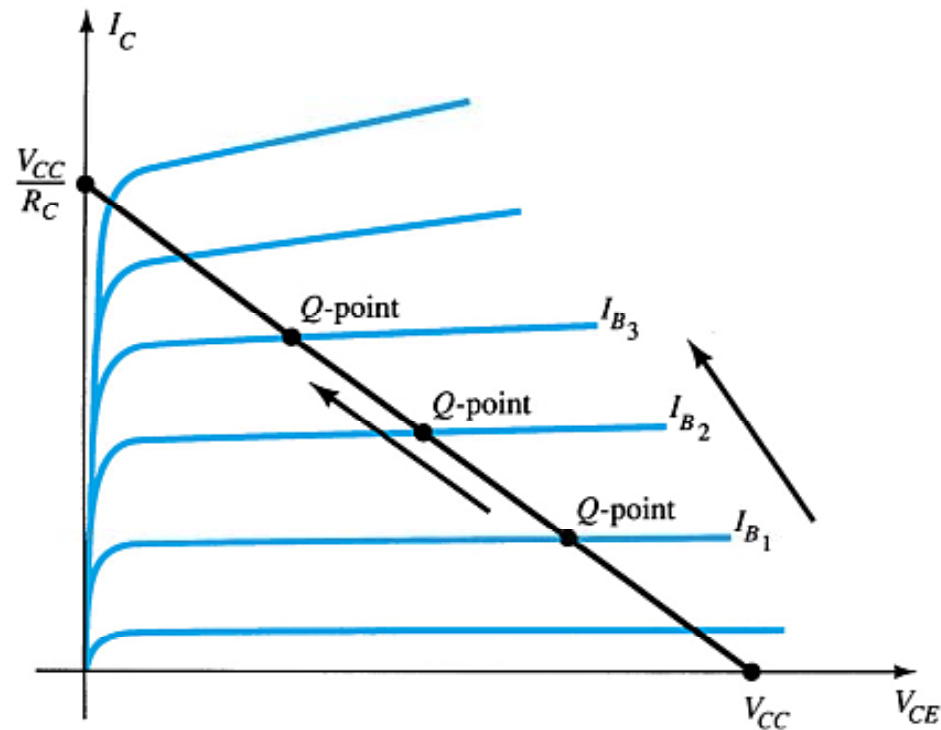
more ...

Circuit Values Affect the Q-Point



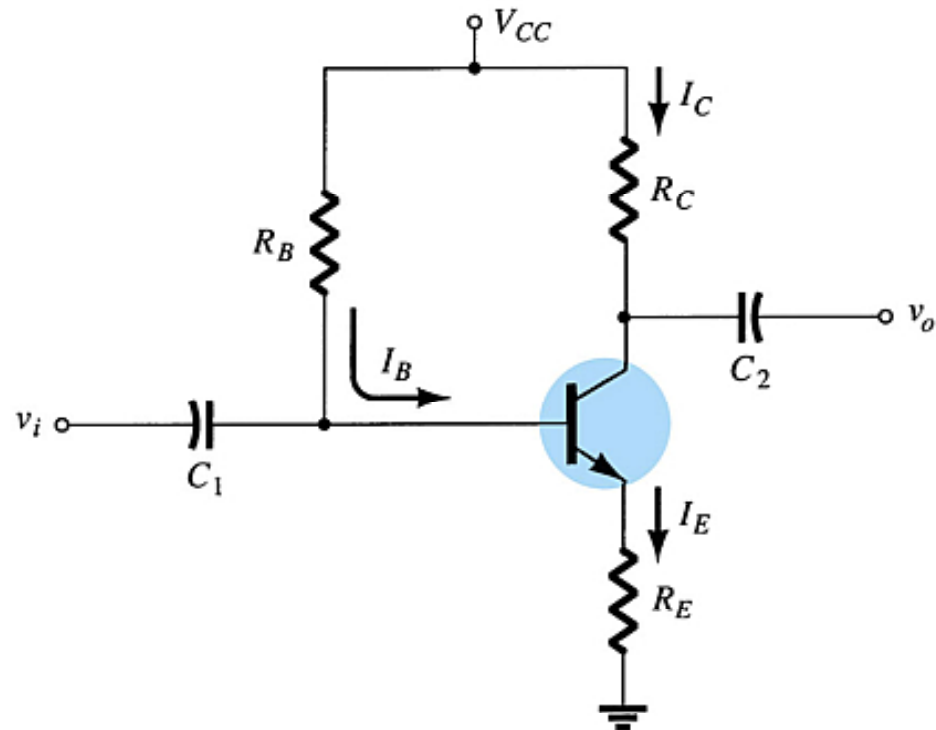
more ...

Circuit Values Affect the Q-Point



Emitter-Stabilized Bias Circuit

Adding a resistor (R_E) to the emitter circuit stabilizes the bias circuit.



Base-Emitter Loop

From Kirchhoff's voltage law:

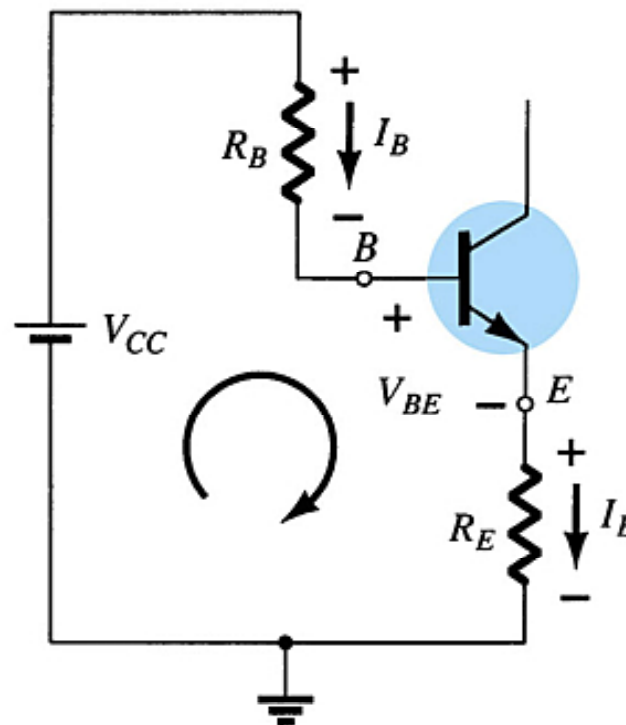
$$+V_{CC} - I_E R_E - V_{BE} - I_E R_E = 0$$

Since $I_E = (\beta + 1)I_B$:

$$V_{CC} - I_B R_B - (\beta + 1)I_B R_E = 0$$

Solving for I_B :

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E}$$



Collector-Emitter Loop

From Kirchhoff's voltage law:

$$I_E R_E + V_{CE} + I_C R_C - V_{CC} = 0$$

Since $I_E \cong I_C$:

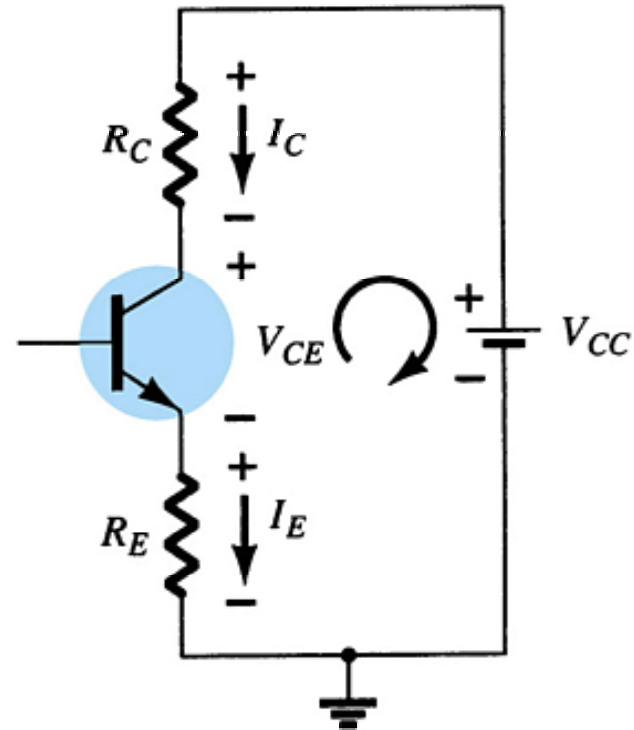
$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

Also:

$$V_E = I_E R_E$$

$$V_C = V_{CE} + V_E = V_{CC} - I_C R_C$$

$$V_B = V_{CC} - I_R R_B = V_{BE} + V_E$$

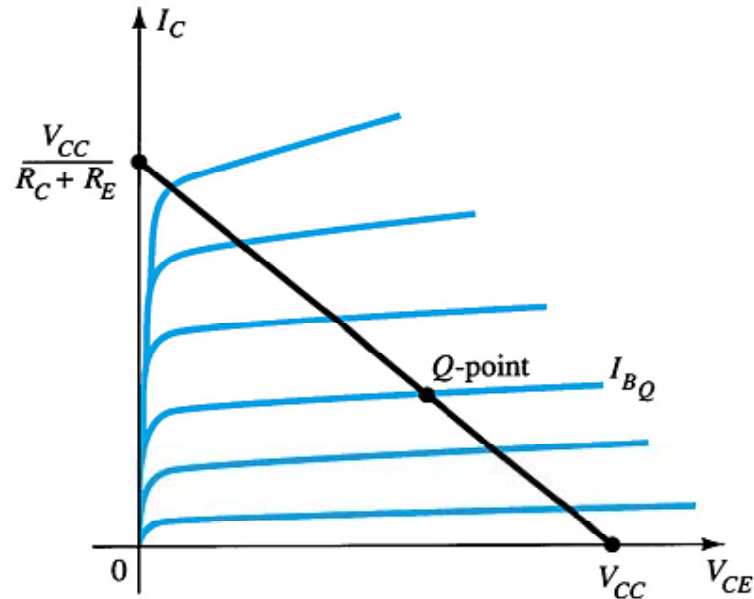
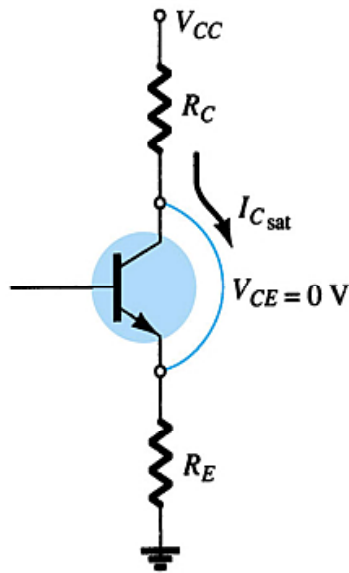


Improved Biased Stability

Stability refers to a circuit condition in which the currents and voltages will remain fairly constant over a wide range of temperatures and transistor Beta (β) values.

Adding RE to the emitter improves the stability of a transistor.

Saturation Level



The endpoints can be determined from the load line.

$V_{CE\text{cutoff}}$:

$$V_{CE} = V_{CC}$$

$$I_C = 0 \text{ mA}$$

$I_{C\text{sat}}$:

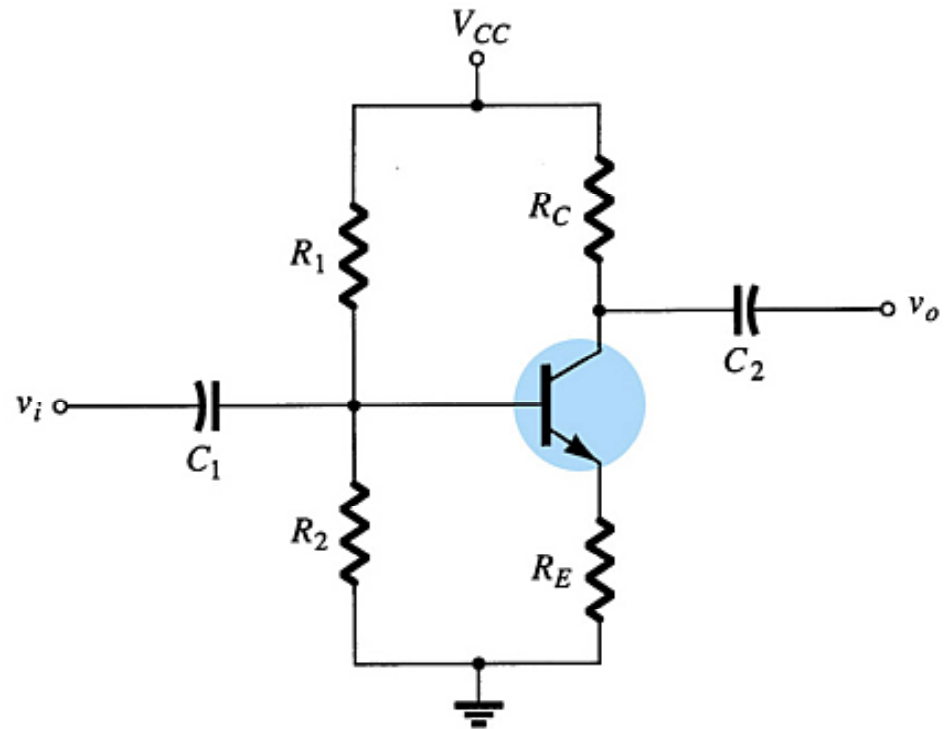
$$V_{CE} = 0 \text{ V}$$

$$I_C = \frac{V_{CC}}{R_C + R_E}$$

Voltage Divider Bias

This is a very stable bias circuit.

The currents and voltages are nearly independent of any variations in β .



Approximate Analysis

Where $I_B \ll I_1$ and $I_1 \cong I_2$:

$$V_B = \frac{R_2 V_{CC}}{R_1 + R_2}$$

Where $\beta R_E > 10R_2$:

$$I_E = \frac{V_E}{R_E}$$

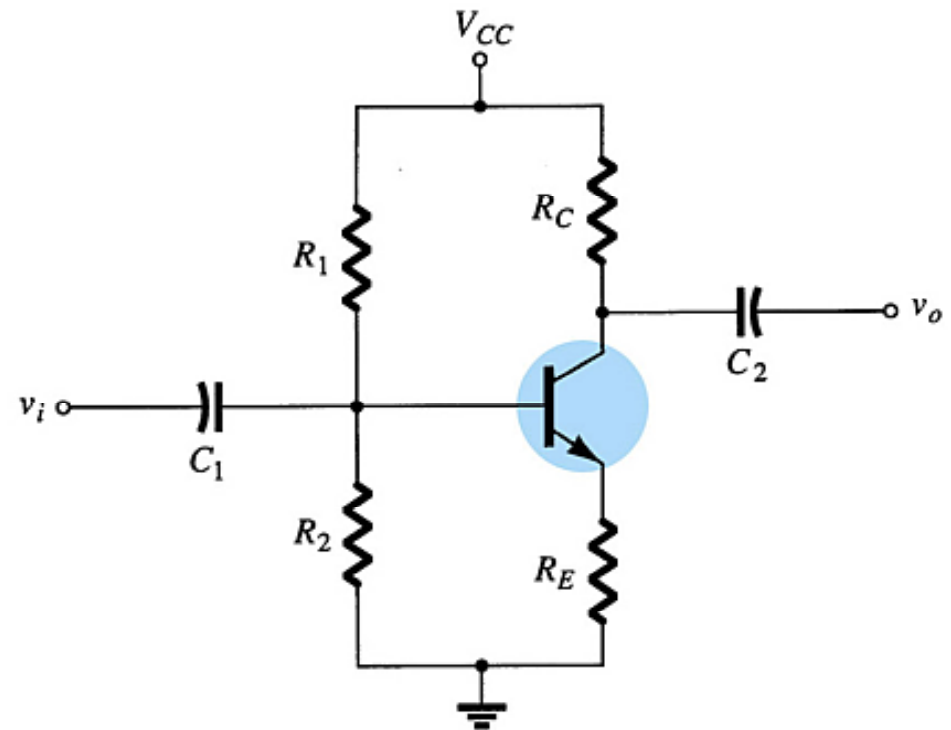
$$V_E = V_B - V_{BE}$$

From Kirchhoff's voltage law:

$$V_{CE} = V_{CC} - I_C R_C - I_E R_E$$

$$I_E \cong I_C$$

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$



Voltage Divider Bias Analysis

Transistor Saturation Level

$$I_{C\text{sat}} = I_{C\text{max}} = \frac{V_{CC}}{R_C + R_E}$$

Load Line Analysis

Cutoff:

$$V_{CE} = V_{CC}$$
$$I_C = 0\text{mA}$$

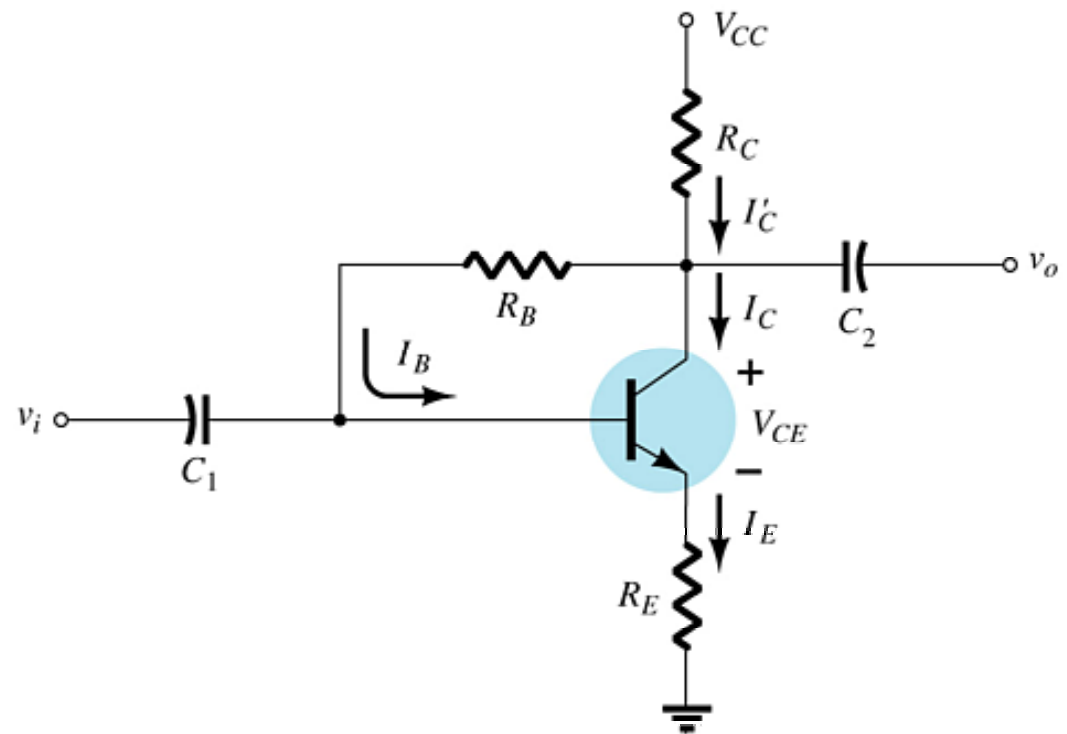
Saturation:

$$I_C = \frac{V_{CC}}{R_C + R_E}$$
$$V_{CE} = 0\text{V}$$

DC Bias with Voltage Feedback

Another way to improve the stability of a bias circuit is to add a feedback path from collector to base.

In this bias circuit the Q-point is only slightly dependent on the transistor beta, β .



Base-Emitter Loop

From Kirchhoff's voltage law:

$$V_{CC} - I'_C R_C - I_B R_B - V_{BE} - I_E R_E = 0$$

Where $I_B \ll I_C$:

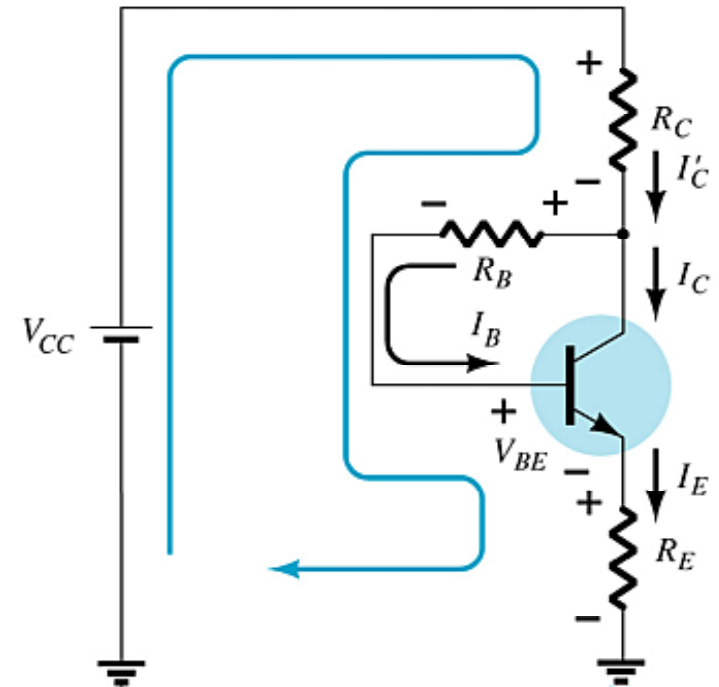
$$I'_C = I_C + I_B \cong I_C$$

Knowing $I_C = \beta I_B$ and $I_E \cong I_C$, the loop equation becomes:

$$V_{CC} - \beta I_B R_C - I_B R_B - V_{BE} - \beta I_B R_E = 0$$

Solving for I_B :

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)}$$



Collector-Emitter Loop

Applying Kirchoff's voltage law:

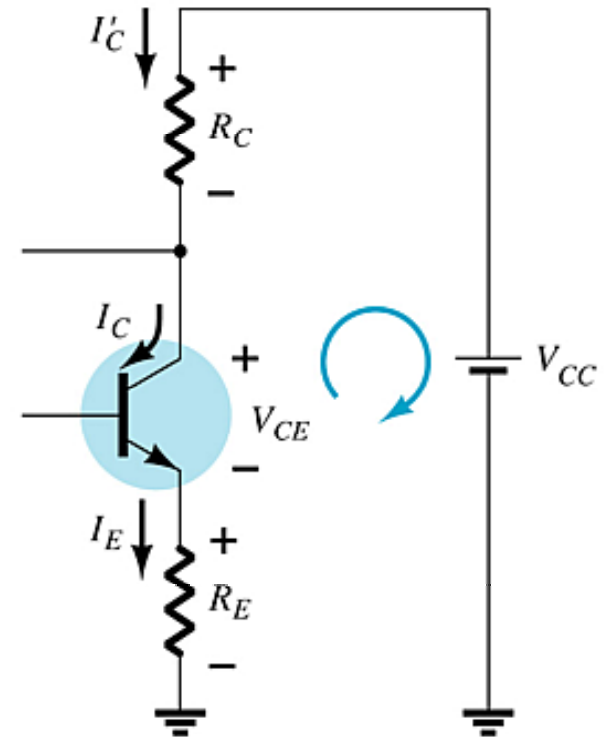
$$I_E + V_{CE} + I'_C R_C - V_{CC} = 0$$

Since $I'_C \cong I_C$ and $I_C = \beta I_B$:

$$I_C(R_C + R_E) + V_{CE} - V_{CC} = 0$$

Solving for V_{CE} :

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$



Base-Emitter Bias Analysis

Transistor Saturation Level

$$I_{C\text{sat}} = I_{C\text{max}} = \frac{V_{CC}}{R_C + R_E}$$

Load Line Analysis

Cutoff:

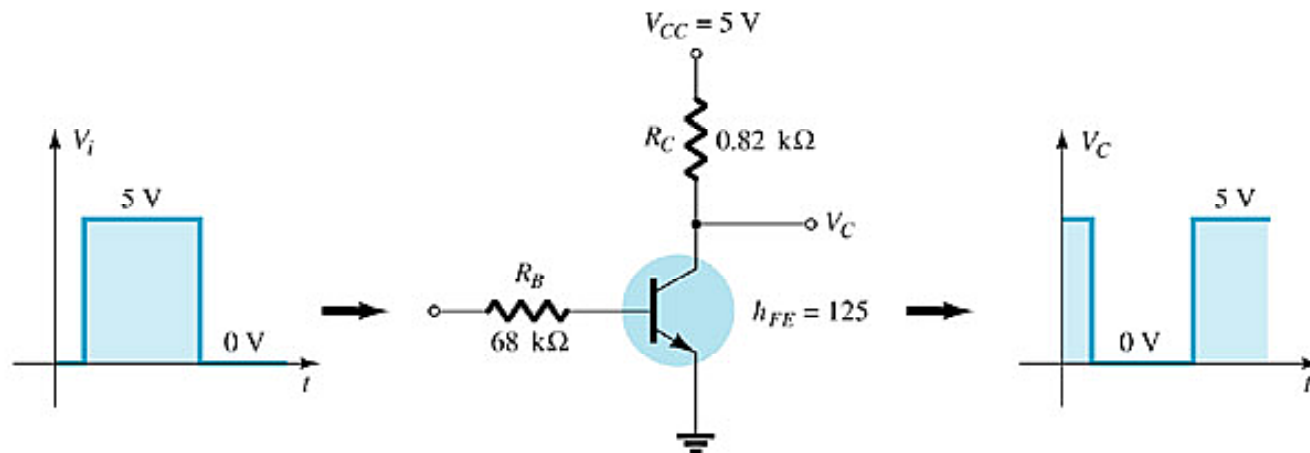
$$V_{CE} = V_{CC}$$
$$I_C = 0 \text{ mA}$$

Saturation:

$$I_C = \frac{V_{CC}}{R_C + R_E}$$
$$V_{CE} = 0 \text{ V}$$

Transistor Switching Networks

Transistors with only the DC source applied can be used as electronic switches.



Switching Circuit Calculations

Saturation current:

$$I_{C\text{sat}} = \frac{V_{CC}}{R_C}$$

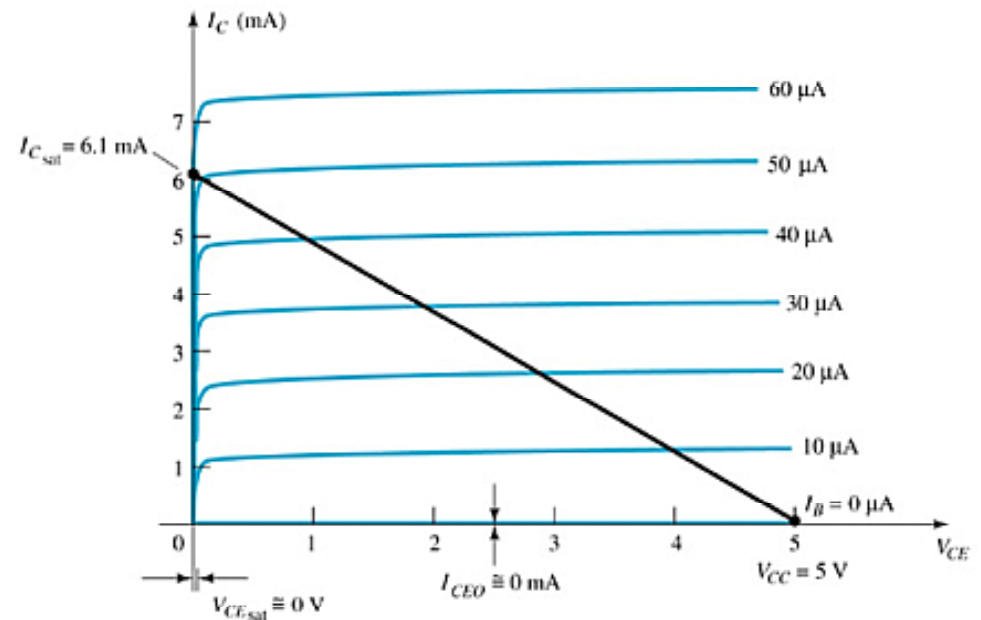
To ensure saturation:

$$I_B > \frac{I_{C\text{sat}}}{\beta_{dc}}$$

**Emitter-collector resistance
at saturation and cutoff:**

$$R_{\text{sat}} = \frac{V_{CE\text{sat}}}{I_{C\text{sat}}}$$

$$R_{\text{cutoff}} = \frac{V_{CC}}{I_{CEO}}$$

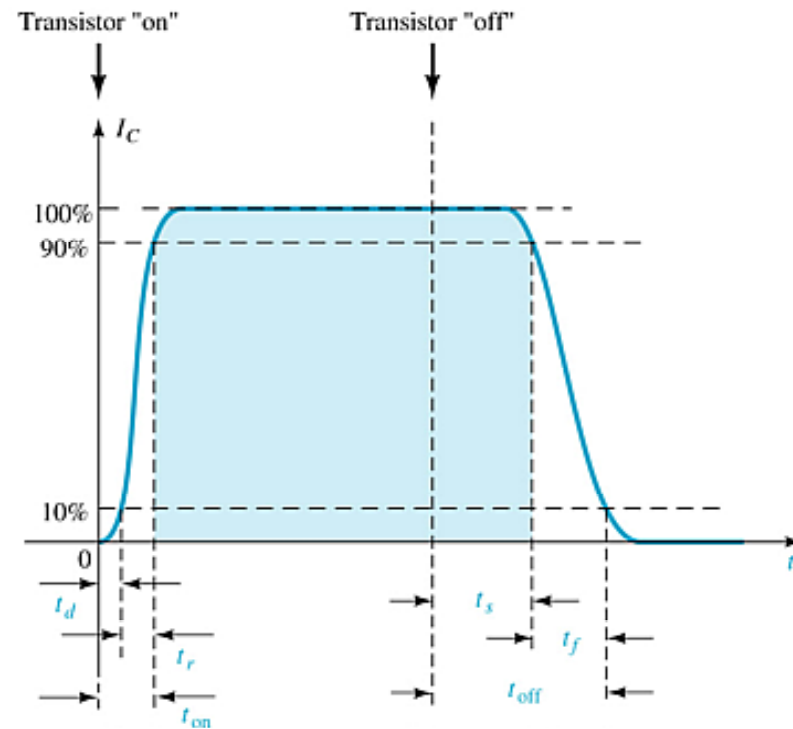


Switching Time

Transistor switching times:

$$t_{\text{on}} = t_r + t_d$$

$$t_{\text{off}} = t_s + t_f$$



Troubleshooting Hints

- **Approximate voltages**
 - $V_{BE} \cong .7 \text{ V}$ for silicon transistors
 - $V_{CE} \cong 25\% \text{ to } 75\% \text{ of } V_{CC}$
- **Test for opens and shorts with an ohmmeter.**
- **Test the solder joints.**
- **Test the transistor with a transistor tester or a curve tracer.**
- **Note that the load or the next stage affects the transistor operation.**

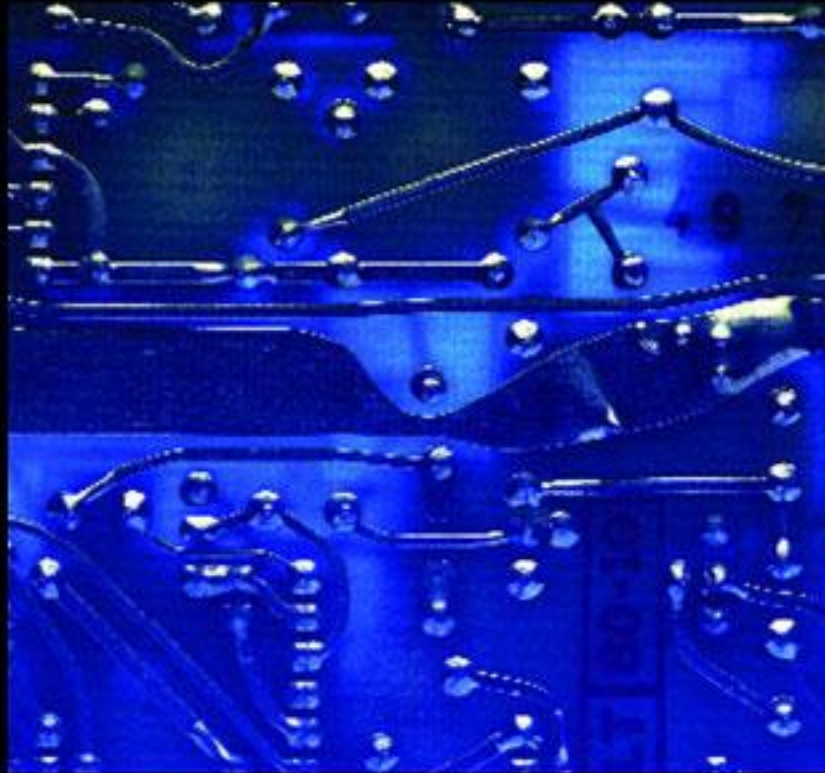
PNP Transistors

The analysis for *pnp* transistor biasing circuits is the same as that for *npn* transistor circuits. The only difference is that the currents are flowing in the opposite direction.

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Chapter 5: BJT AC Analysis

BJT Transistor Modeling

- A model is an equivalent circuit that represents the AC characteristics of the transistor.
- A model uses circuit elements that approximate the behavior of the transistor.
- There are two models commonly used in small signal AC analysis of a transistor:
 - r_e model
 - Hybrid equivalent model

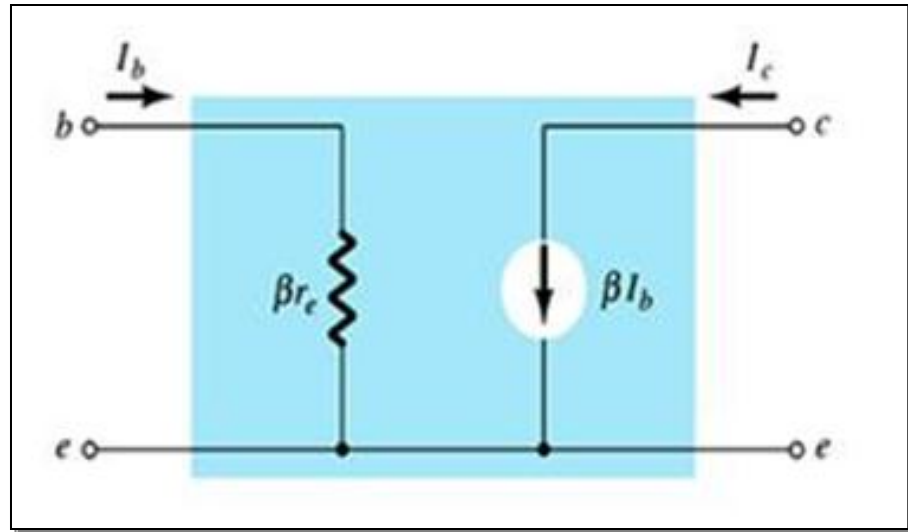
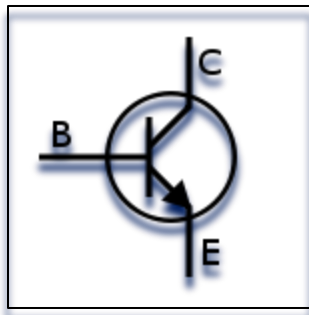
The r_e Transistor Model

- BJTs are basically **current-controlled** devices; therefore the r_e model uses a diode and a current source to duplicate the behavior of the transistor.
- One disadvantage to this model is its sensitivity to the DC level. This model is designed for specific circuit conditions.

The r_e Transistor Model

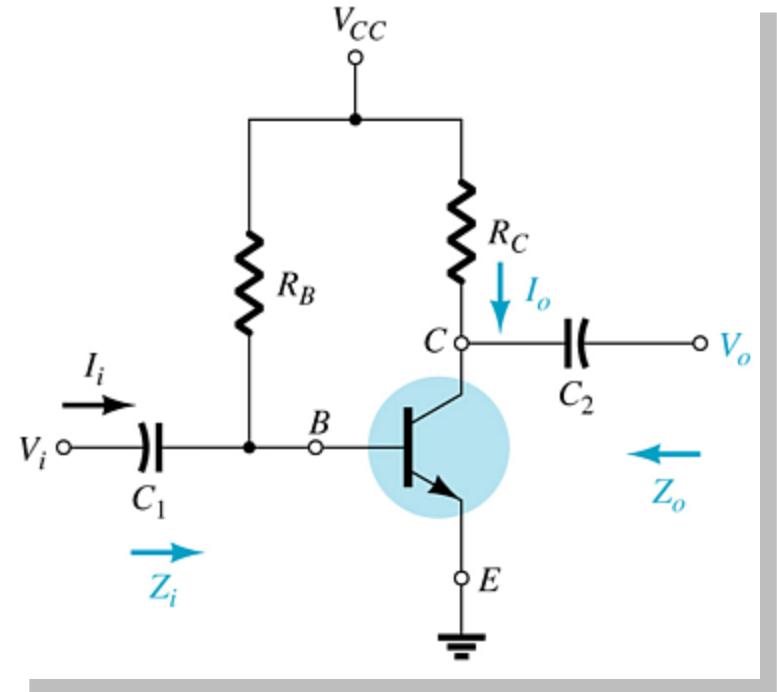
Common-Emitter Configuration

- The equivalent circuit of Fig above will be used throughout the analysis to follow for the common-emitter configuration.

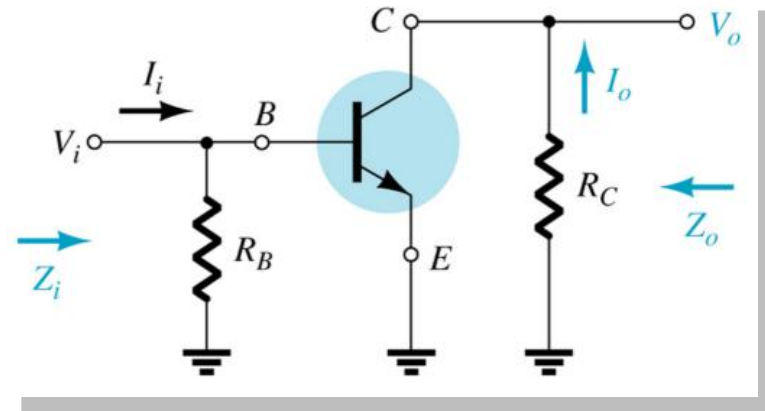
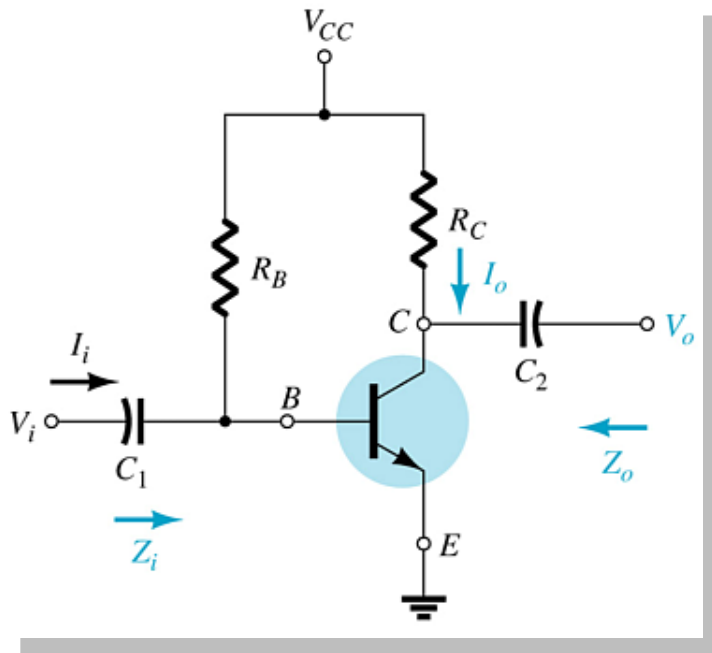


Common-Emitter Fixed-Bias Configuration

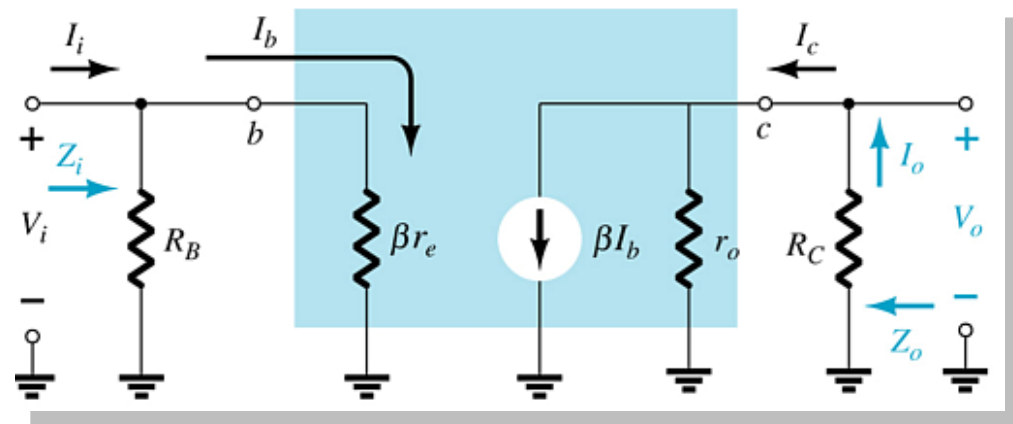
- The input is applied to the base
- The output is from the collector
- High input impedance
- Low output impedance
- High voltage and current gain
- Phase shift between input and output is 180°



Common-Emitter Fixed-Bias Configuration



AC equivalent



r_e model

Common-Emitter Fixed-Bias Calculations

Input impedance:

$$Z_i = R_B \parallel \beta r_e$$

$$Z_i \cong \beta r_e \mid R_E \geq 10\beta r_e$$

Output impedance:

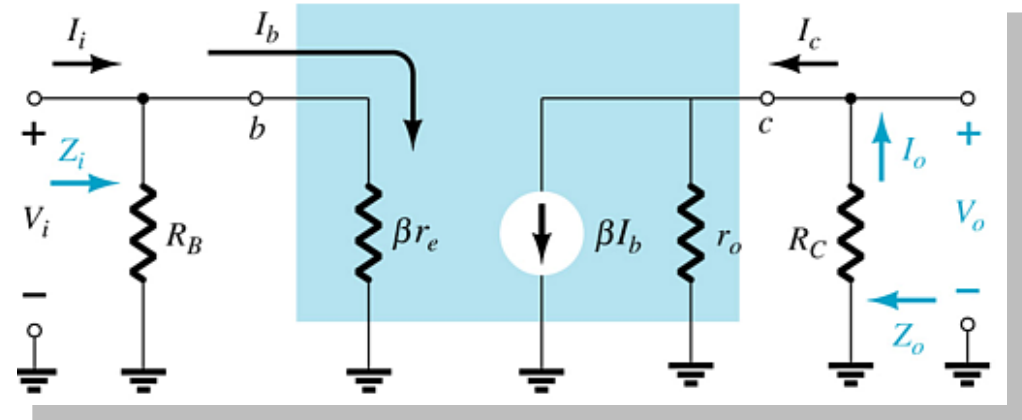
$$Z_o = R_C \parallel r_o$$

$$Z_o \cong R_C \mid r_o \geq 10R_C$$

Voltage gain:

$$A_v = \frac{V_o}{V_i} = -\frac{(R_C \parallel r_o)}{r_e}$$

$$A_v \cong -\frac{R_C}{r_e} \mid r_o \geq 10R_C$$



Current gain:

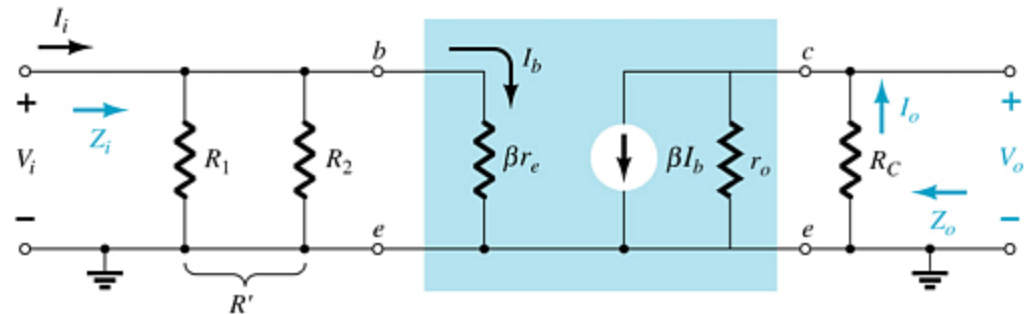
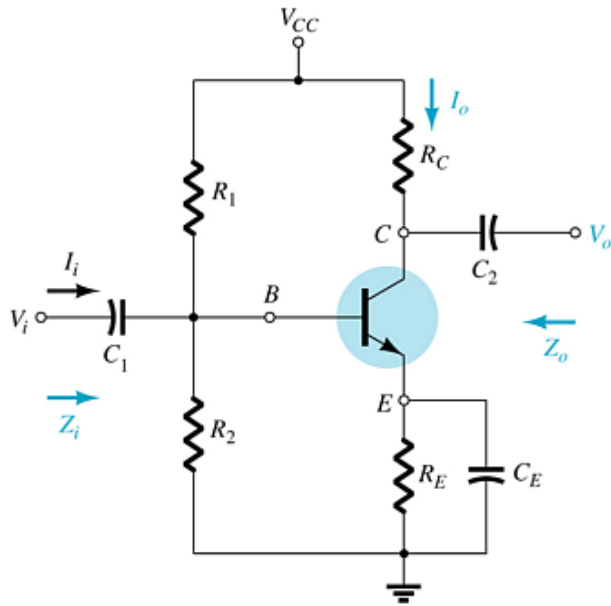
$$A_i = \frac{I_o}{I_i} = \frac{\beta R_B r_o}{(r_o + R_C)(R_B + \beta r_e)}$$

$$A_i \cong \beta \mid r_o \geq 10R_C, R_B \geq 10\beta r_e$$

Current gain from voltage gain:

$$A_i = -A_v \frac{Z_i}{R_C}$$

Common-Emitter Voltage-Divider Bias



r_e model requires you to determine β , r_e , and r_o .

Common-Emitter Voltage-Divider Bias Cal.

Input impedance:

$$R' = R_1 \parallel R_2$$

$$Z_i = R' \parallel \beta r_e$$

Output impedance:

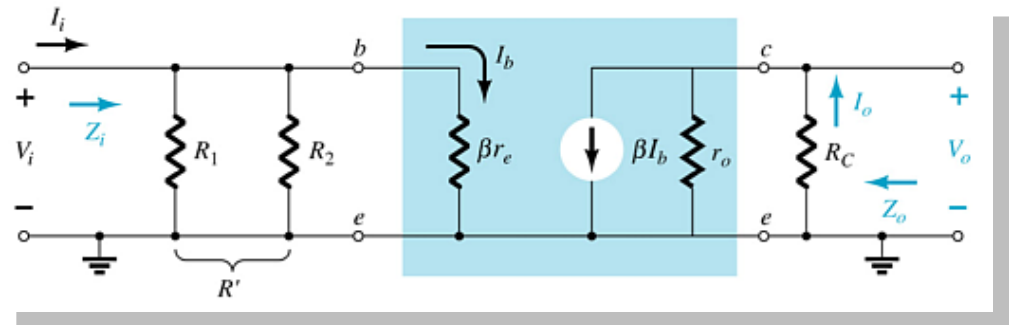
$$Z_o = R_C \parallel r_o$$

$$Z_o \cong R_C \Big|_{r_o \geq 10R_C}$$

Voltage gain:

$$A_v = \frac{V_o}{V_i} = \frac{-R_C \parallel r_o}{r_e}$$

$$A_v = \frac{V_o}{V_i} \cong -\frac{R_C}{r_e} \Big|_{r_o \geq 10R_C}$$



Current gain:

$$A_i = \frac{I_o}{I_i} = \frac{\beta R' r_o}{(r_o + R_C)(R' + \beta r_e)}$$

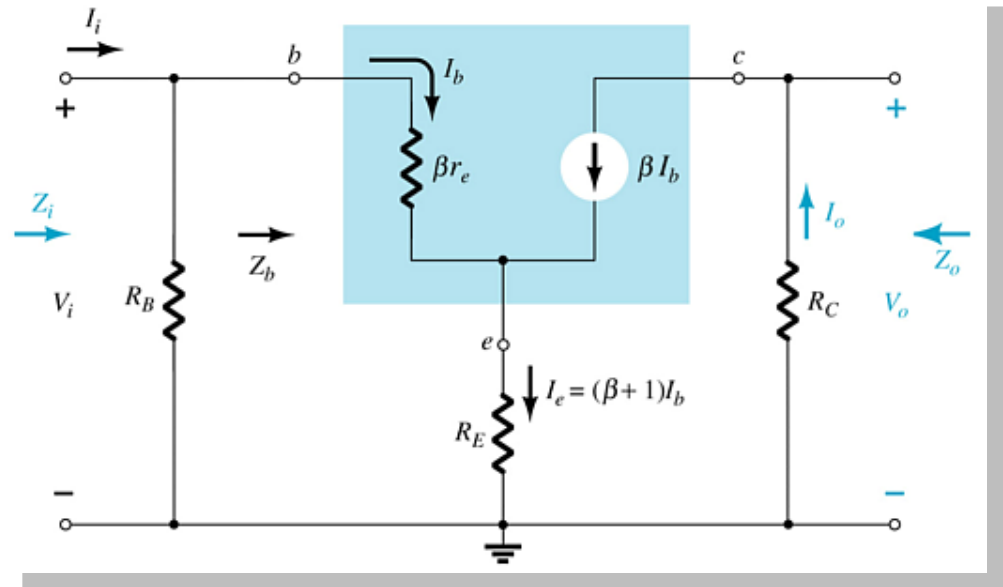
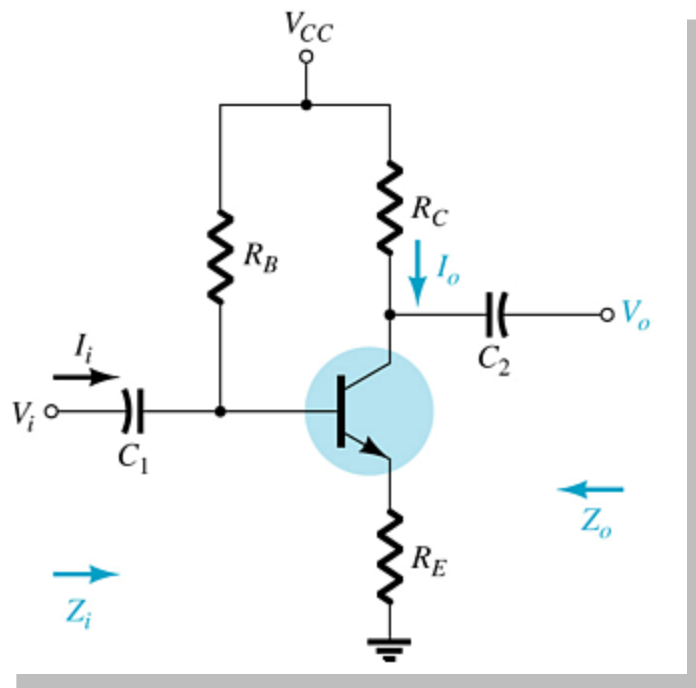
$$A_i = \frac{I_o}{I_i} \cong \frac{\beta R'}{R' + \beta r_e} \Big|_{r_o \geq 10R_C}$$

$$A_i = \frac{I_o}{I_i} \cong \beta \Big|_{r_o \geq 10R_C, R' \geq 10\beta r_e}$$

Current gain from voltage gain:

$$A_i = -A_v \frac{Z_i}{R_C}$$

Common-Emitter Emitter-Bias Config.



Impedance Calculations

Input impedance:

$$Z_i = R_B \parallel Z_b$$

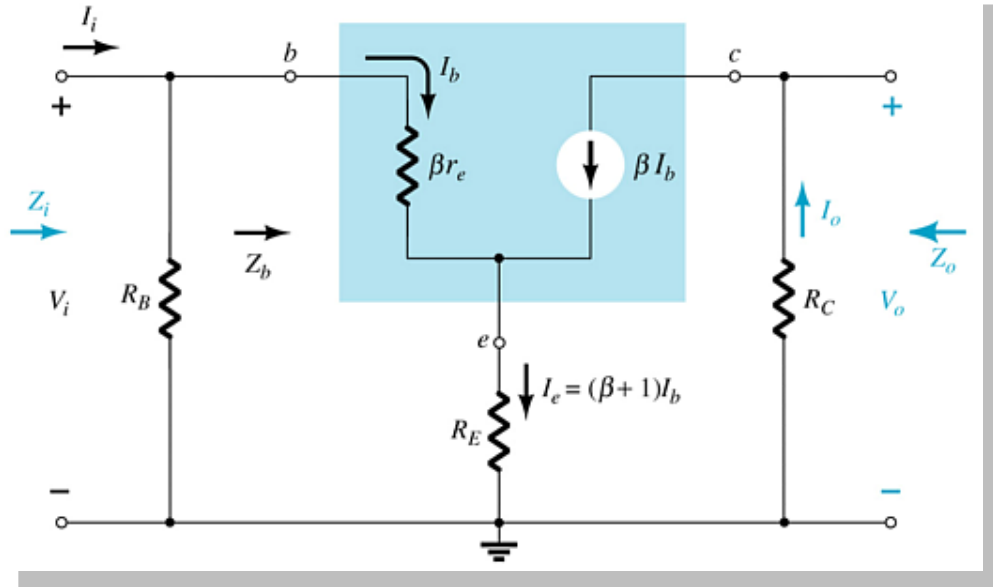
$$Z_b = \beta r_e + (\beta + 1)R_E$$

$$Z_b \cong \beta(r_e + R_E)$$

$$Z_b \cong \beta R_E$$

Output impedance:

$$Z_o = R_C$$



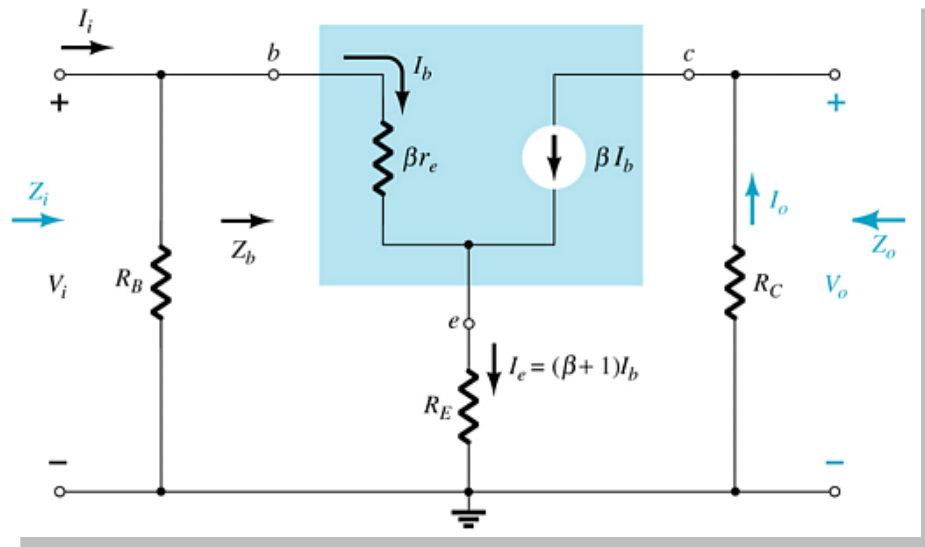
Gain Calculations

Voltage gain:

$$A_v = \frac{V_o}{V_i} = -\frac{\beta R_C}{Z_b}$$

$$A_v = \frac{V_o}{V_i} = -\frac{R_C}{r_e + R_E} \Big| Z_b = \beta(r_e + R_E)$$

$$A_v = \frac{V_o}{V_i} \cong -\frac{R_C}{R_E} \Big| Z_b \cong \beta R_E$$



Current gain:

$$A_i = \frac{I_o}{I_i} = \frac{\beta R_B}{R_B + Z_b}$$

Current gain from voltage gain:

$$A_i = -A_v \frac{Z_i}{R_C}$$

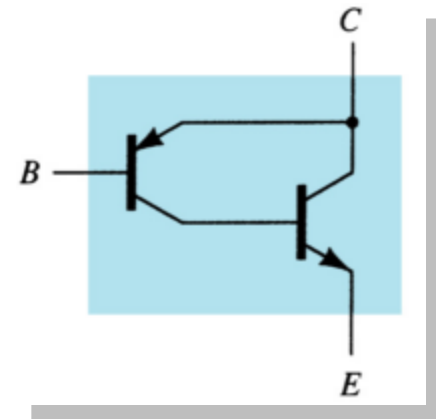
Feedback Pair

This is a two-transistor circuit that operates like a Darlington pair, *but it is not a Darlington pair.*

It has similar characteristics:

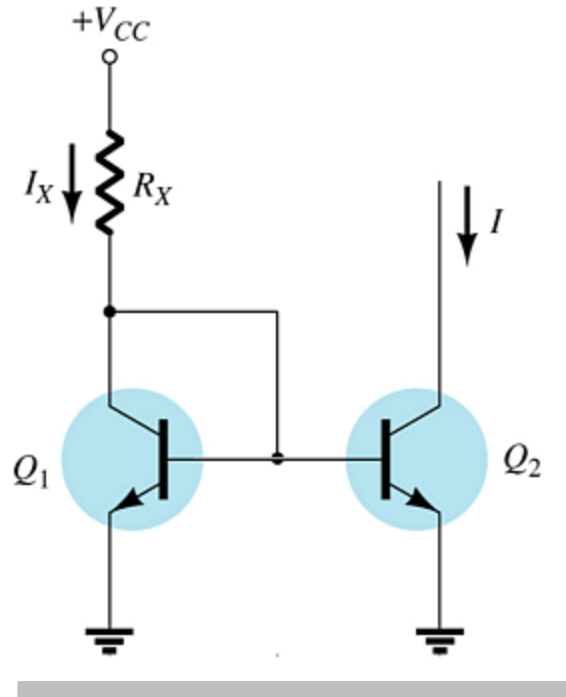
- **High current gain**
- **Voltage gain near unity**
- **Low output impedance**
- **High input impedance**

The difference is that a Darlington uses a pair of like transistors, whereas the feedback-pair configuration uses complementary transistors.



Current Mirror Circuits

Current mirror circuits provide constant current in integrated circuits.

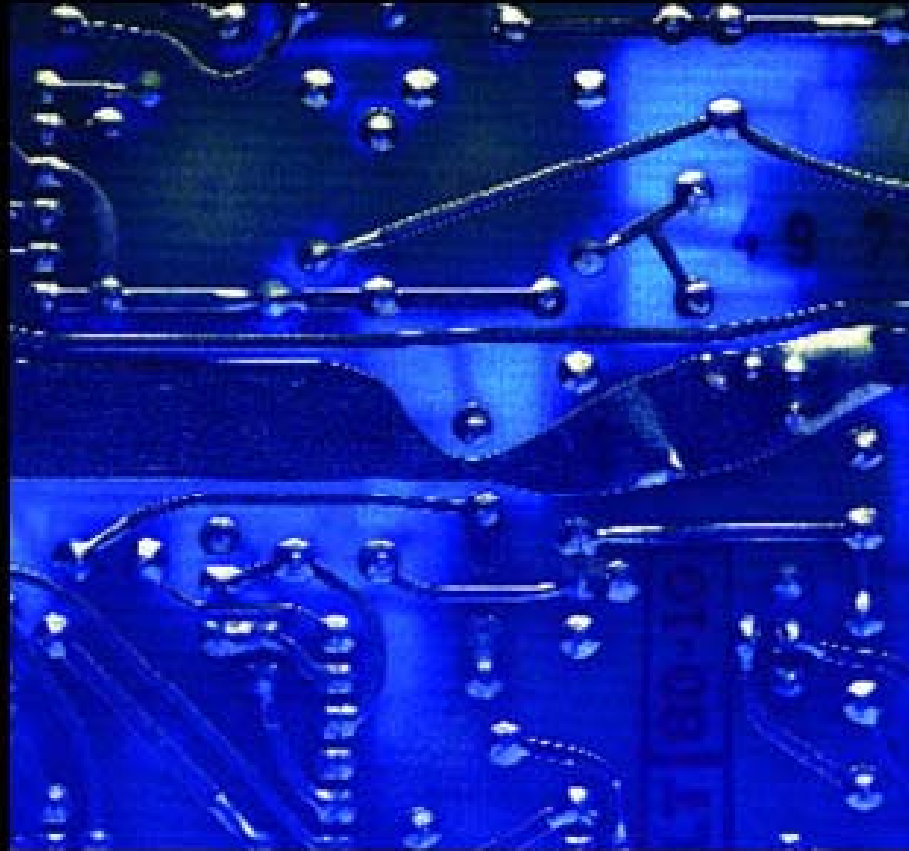


Example 4.26: Calculate the mirrored current I in the above circuit given, $R_x = 1.1 \text{ k}\Omega$ and $+V_{cc} = 12 \text{ V}$.

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Chapter 6: Field-Effect Transistors

FETs vs. BJTs

Similarities:

- Amplifiers
- Switching devices
- Impedance matching circuits

Differences:

- FETs are voltage controlled devices. BJTs are current controlled devices.
- FETs have a higher input impedance. BJTs have higher gains.
- FETs are less sensitive to temperature variations and are more easily integrated on ICs.
- FETs are generally more static sensitive than BJTs.

FET Types

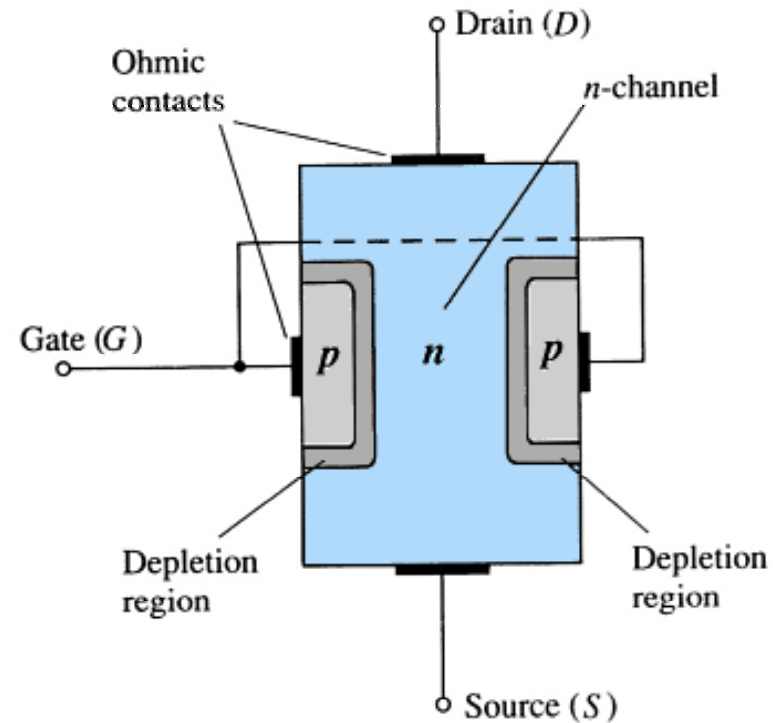
- **JFET:** Junction FET
- **MOSFET:** Metal–Oxide–Semiconductor FET
 - **D-MOSFET:** Depletion MOSFET
 - **E-MOSFET:** Enhancement MOSFET

JFET Construction

There are two types of JFETs

- ***n*-channel**
- ***p*-channel**

The *n*-channel is more widely used.



There are three terminals:

- **Drain (D)** and **Source (S)** are connected to the *n*-channel
- **Gate (G)** is connected to the *p*-type material

JFET Operation: The Basic Idea

JFET operation can be compared to a water spigot.

The source of water pressure is the accumulation of electrons at the negative pole of the drain-source voltage.

The drain of water is the electron deficiency (or holes) at the positive pole of the applied voltage.

The control of flow of water is the gate voltage that controls the width of the n-channel and, therefore, the flow of charges from source to drain.



JFET Operating Characteristics

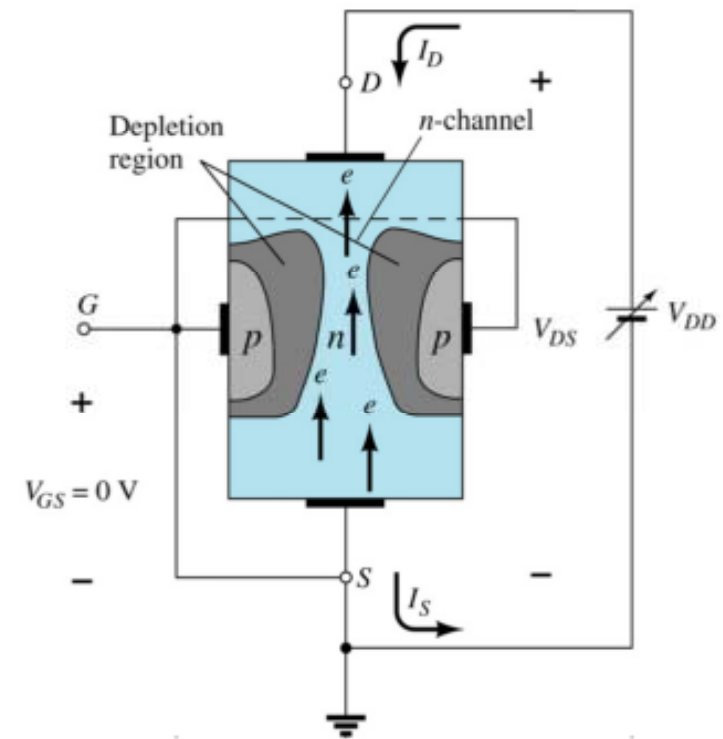
There are three basic operating conditions for a JFET:

- $V_{GS} = 0$, V_{DS} increasing to some positive value
- $V_{GS} < 0$, V_{DS} at some positive value
- **Voltage-controlled resistor**

JFET Operating Characteristics: $V_{GS} = 0 \text{ V}$

Three things happen when $V_{GS} = 0$ and V_{DS} is increased from 0 to a more positive voltage

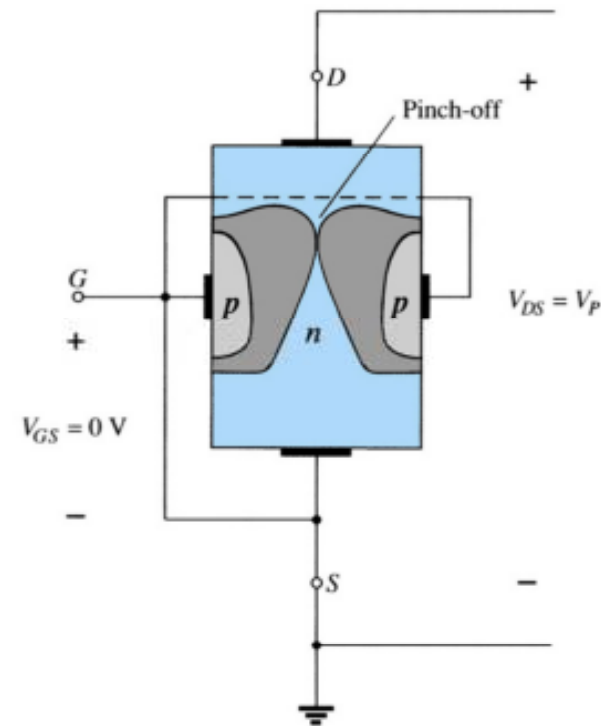
- The depletion region between p-gate and n-channel increases as electrons from n-channel combine with holes from p-gate.
- Increasing the depletion region, decreases the size of the n-channel which increases the resistance of the n-channel.
- Even though the n-channel resistance is increasing, the current (I_D) from source to drain through the n-channel is increasing. This is because V_{DS} is increasing.



JFET Operating Characteristics: Pinch Off

If $V_{GS} = 0$ and V_{DS} is further increased to a more positive voltage, then the depletion zone gets so large that it **pinches off** the n-channel.

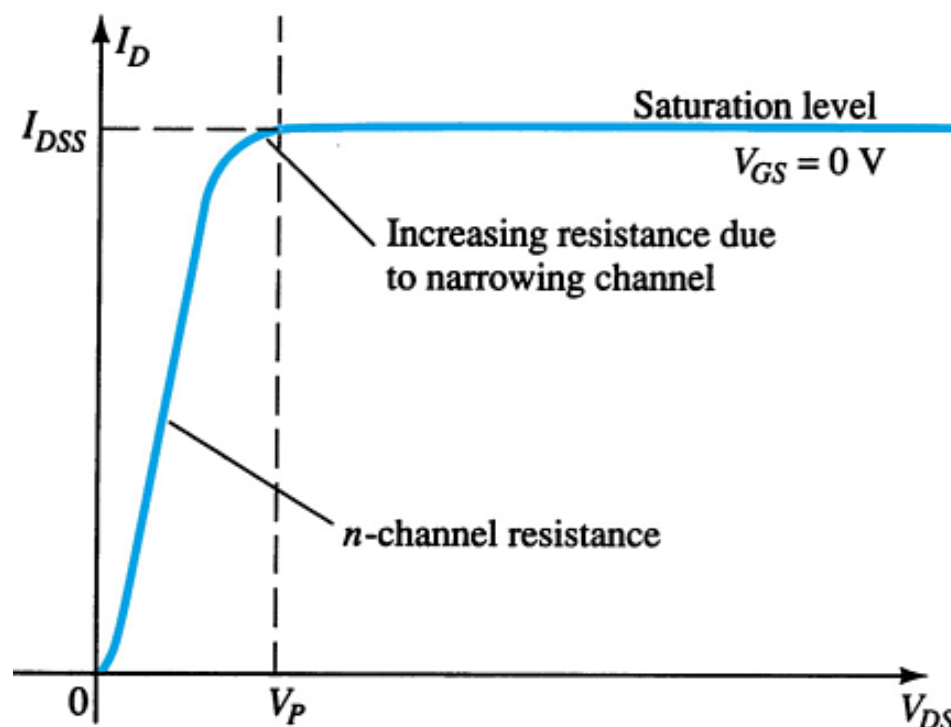
This suggests that the current in the n-channel (I_D) would drop to 0A, but it does just the opposite—as V_{DS} increases, so does I_D .



JFET Operating Characteristics: Saturation

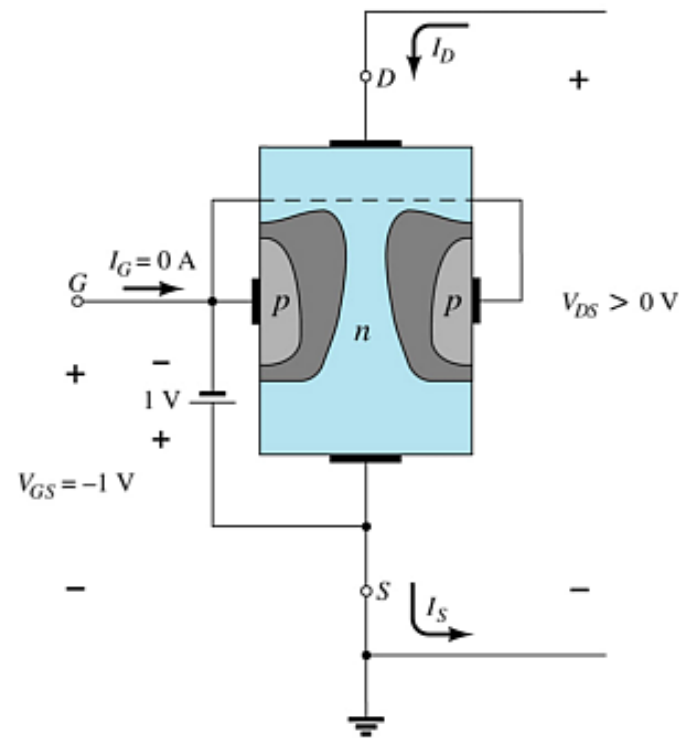
At the pinch-off point:

- Any further increase in V_{GS} does not produce any increase in I_D . V_{GS} at pinch-off is denoted as V_P .
- I_D is at saturation or maximum. It is referred to as I_{DSS} .
- The ohmic value of the channel is maximum.



JFET Operating Characteristics

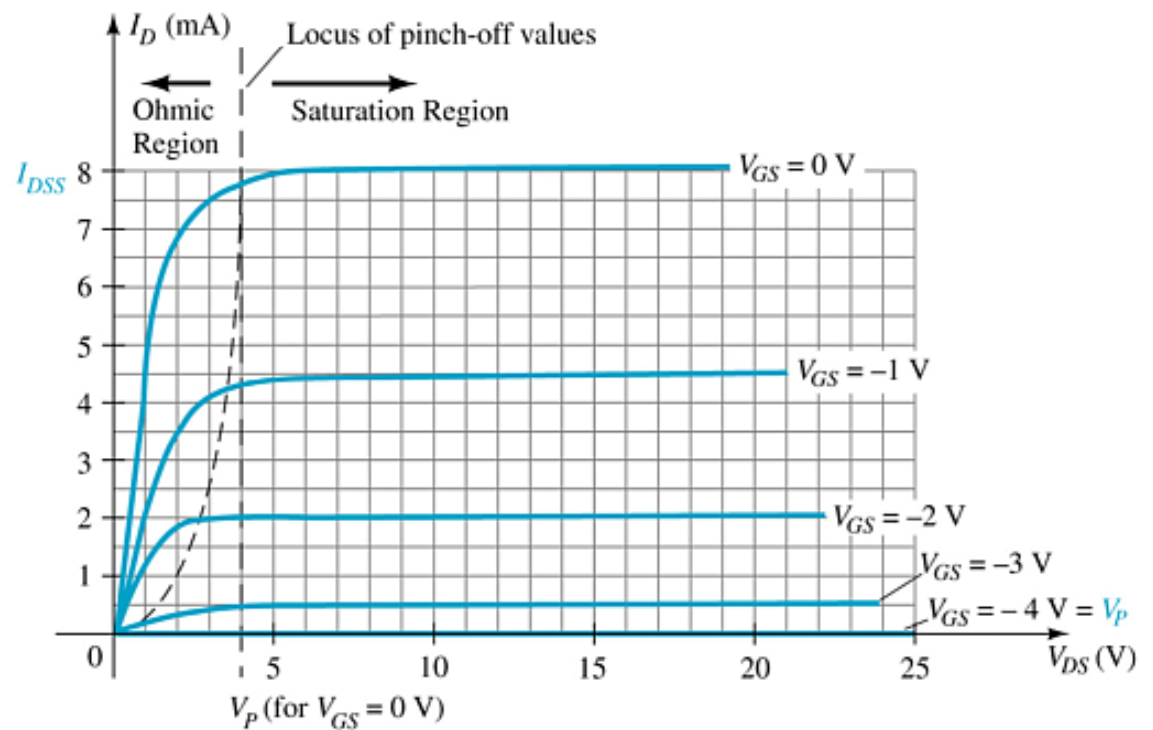
As V_{GS} becomes more negative, the depletion region increases.



JFET Operating Characteristics

As V_{GS} becomes more negative:

- The JFET experiences pinch-off at a lower voltage (V_P).
- I_D decreases ($I_D < I_{DSS}$) even though V_{DS} is increased.
- Eventually I_D reaches 0 A. V_{GS} at this point is called V_P or $V_{GS(off)}$.



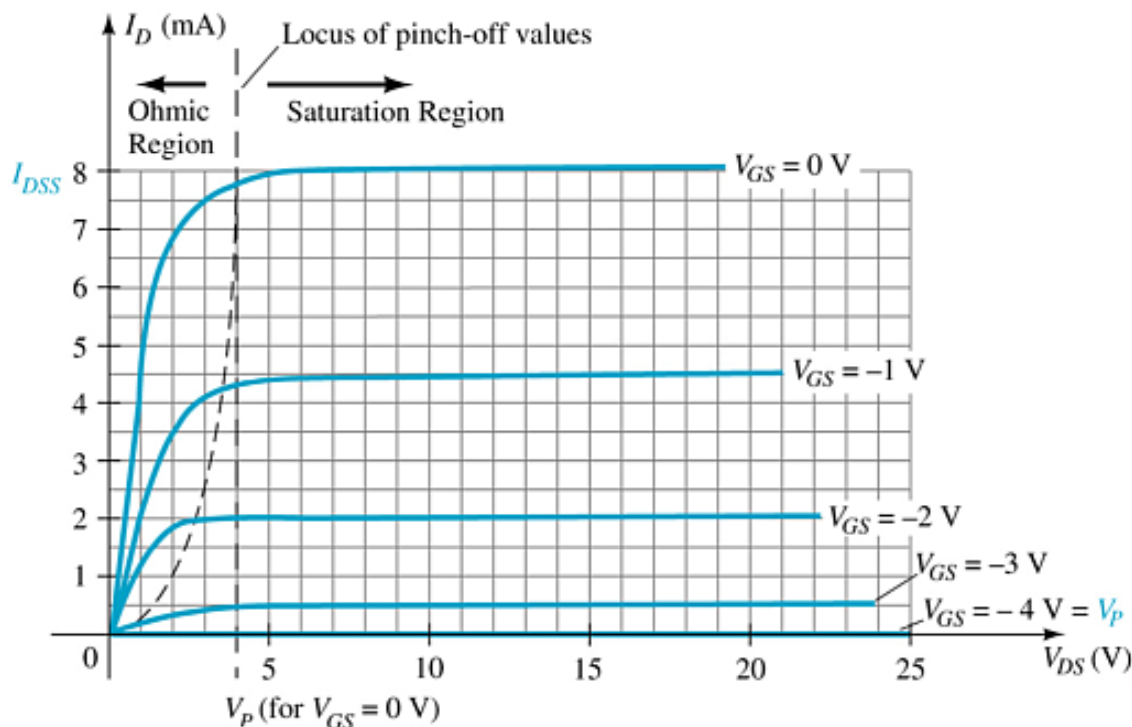
Also note that at high levels of V_{DS} the JFET reaches a breakdown situation. I_D increases uncontrollably if $V_{DS} > V_{DSmax}$.

JFET Operating Characteristics: Voltage-Controlled Resistor

The region to the left of the pinch-off point is called the **ohmic region**.

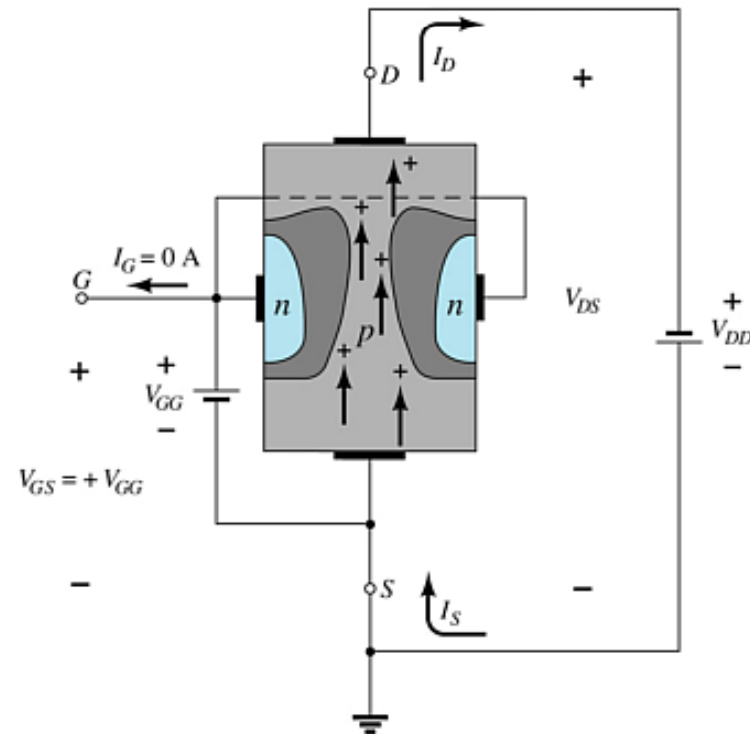
The JFET can be used as a variable resistor, where V_{GS} controls the drain-source resistance (r_d). As V_{GS} becomes more negative, the resistance (r_d) increases.

$$r_d = \frac{r_o}{\left(1 - \frac{V_{GS}}{V_P}\right)^2}$$



p-Channel JFETs

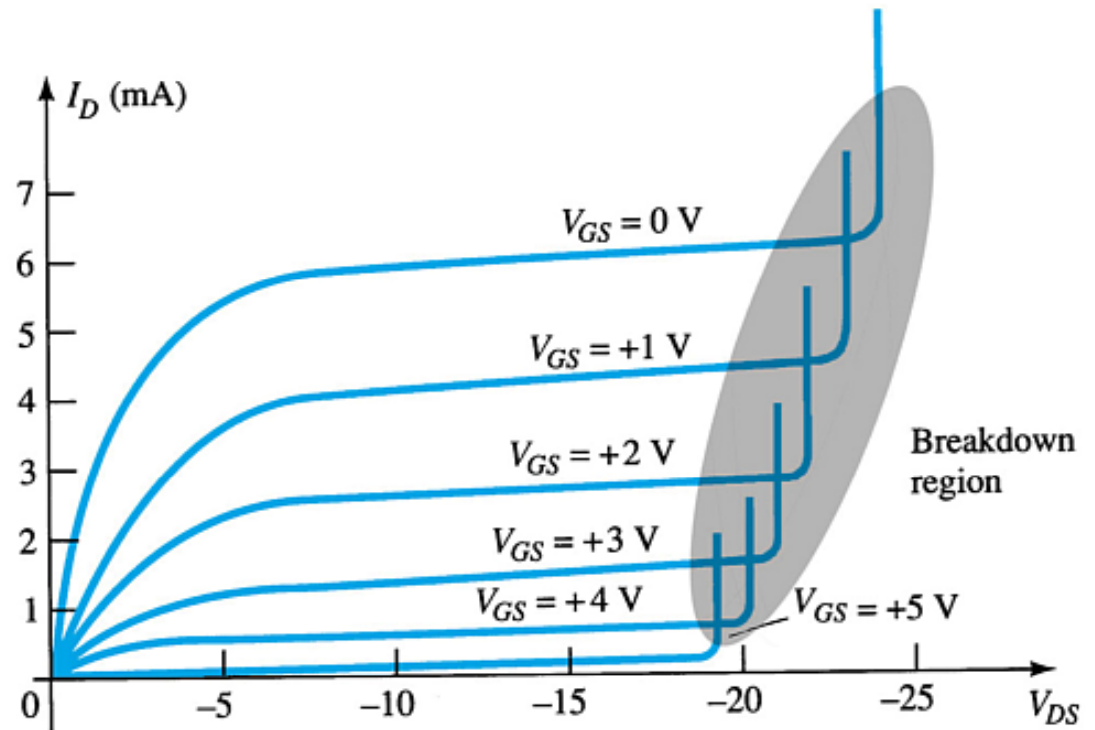
The *p*-channel JFET behaves the same as the *n*-channel JFET, except the voltage polarities and current directions are reversed.



p-Channel JFET Characteristics

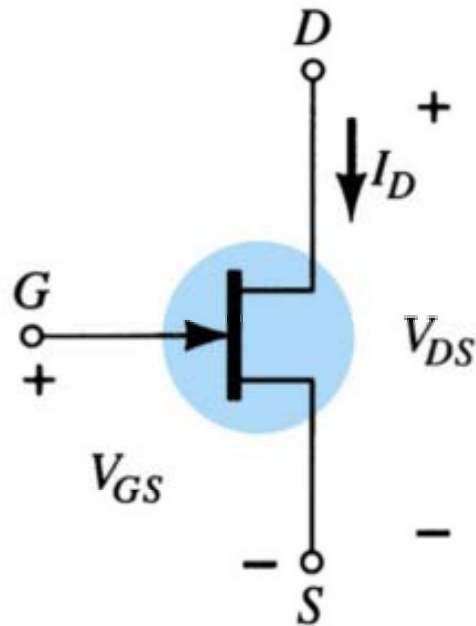
As V_{GS} increases more positively

- The depletion zone increases
- I_D decreases ($I_D < I_{DSS}$)
- Eventually $I_D = 0$ A



Also note that at high levels of V_{DS} the JFET reaches a breakdown situation: I_D increases uncontrollably if $V_{DS} > V_{DSmax}$.

N-Channel JFET Symbol



JFET Transfer Characteristics

The transfer characteristic of input-to-output is not as straightforward in a JFET as it is in a BJT.

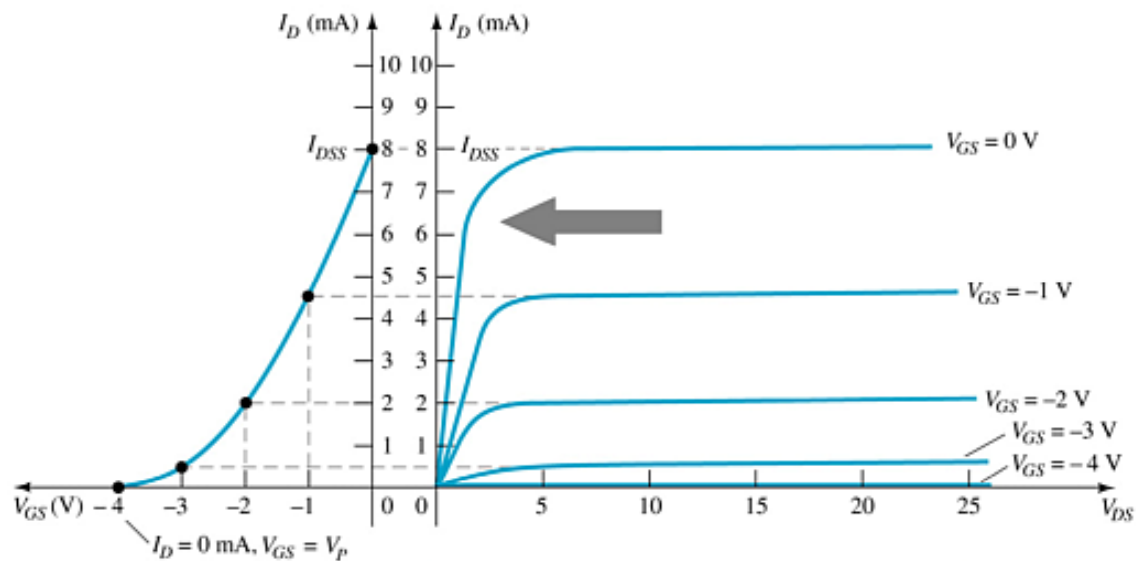
In a BJT, β indicates the relationship between I_B (input) and I_C (output).

In a JFET, the relationship of V_{GS} (input) and I_D (output) is a little more complicated:

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

JFET Transfer Curve

This graph shows the value of I_D for a given value of V_{GS} .



Plotting the JFET Transfer Curve

Using I_{DSS} and V_p ($V_{GS(off)}$) values found in a specification sheet, the transfer curve can be plotted according to these three steps:

Step 1

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

Solving for $V_{GS} = 0V$ $I_D = I_{DSS}$

Step 2

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

Solving for $V_{GS} = V_p$ ($V_{GS(off)}$) $I_D = 0A$

Step 3

Solving for $V_{GS} = 0V$ to V_p $I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$

JFET Specifications Sheet

Electrical Characteristics

ELECTRICAL CHARACTERISTICS ($T_A = 25^\circ\text{C}$ unless otherwise noted)

Characteristic	Symbol	Min	Typ	Max	Unit
----------------	--------	-----	-----	-----	------

OFF CHARACTERISTICS

Gate-Source Breakdown Voltage ($I_G = -10 \mu\text{A}$, $V_{DS} = 0$)	$V_{(BR)GS}$	-25	-	-	Vdc
Gate Reverse Current ($V_{GS} = -15 \text{ Vdc}$, $V_{DS} = 0$) ($V_{GS} = -15 \text{ Vdc}$, $V_{DS} = 0$, $T_A = 100^\circ\text{C}$)	I_{GSS}	-	-	-1.0 -200	nAdc
Gate Source Cutoff Voltage ($V_{DS} = 15 \text{ Vdc}$, $I_D = 10 \text{ nAdc}$)	$V_{GS(off)}$	-0.5	-	-6.0	Vdc
Gate Source Voltage ($V_{DS} = 15 \text{ Vdc}$, $I_D = 100 \mu\text{A}$)	V_{GS}	-	-2.5	-	Vdc

ON CHARACTERISTICS

Zero-Gate-Voltage Drain Current* ($V_{DS} = 15 \text{ Vdc}$, $V_{GS} = 0$)	I_{DSS}	1.0	3.0	5.0	mAdc
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SMALL-SIGNAL CHARACTERISTICS

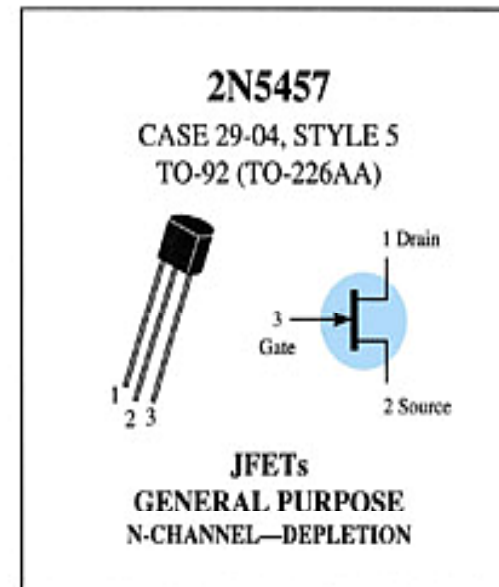
Forward Transfer Admittance Common Source* ($V_{DS} = 15 \text{ Vdc}$, $V_{GS} = 0$, $f = 1.0 \text{ kHz}$)	$ y_{fs} $	1000	-	5000	μmhos
Output Admittance Common Source* ($V_{DS} = 15 \text{ Vdc}$, $V_{GS} = 0$, $f = 1.0 \text{ kHz}$)	$ y_{os} $	-	10	50	μmhos
Input Capacitance ($V_{DS} = 15 \text{ Vdc}$, $V_{GS} = 0$, $f = 1.0 \text{ MHz}$)	C_{iss}	-	4.5	7.0	pF
Reverse Transfer Capacitance ($V_{DS} = 15 \text{ Vdc}$, $V_{GS} = 0$, $f = 1.0 \text{ MHz}$)	C_{rss}	-	1.5	3.0	pF

*Pulse Test: Pulse Width $\leq 630 \text{ ms}$; Duty Cycle $\leq 10\%$

JFET Specifications Sheet

Maximum Ratings

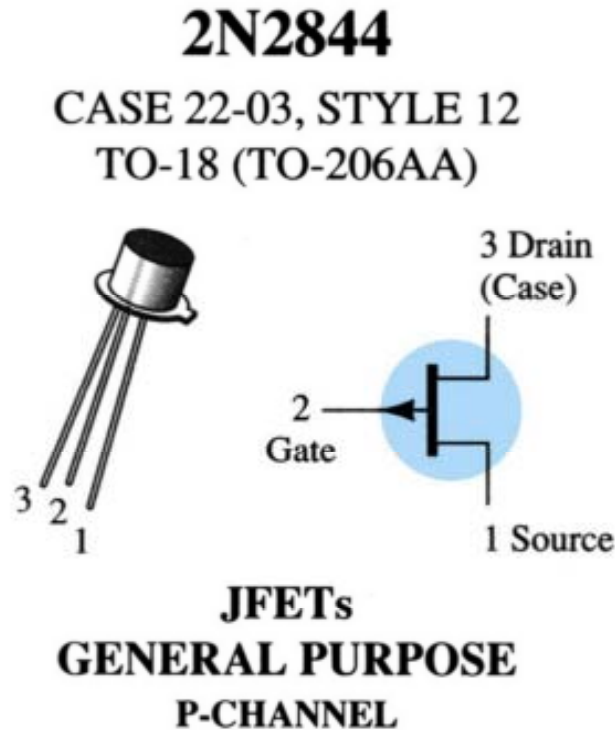
MAXIMUM RATINGS			
Rating	Symbol	Value	Unit
Drain-Source Voltage	V_{DS}	25	Vdc
Drain-Gate Voltage	V_{DG}	25	Vdc
Reverse Gate-Source Voltage	V_{GSR}	-25	Vdc
Gate Current	I_G	10	mAdc
Total Device Dissipation @ $T_A = 25^\circ\text{C}$ Derate above 25°C	P_D	310 2.82	mW mW/°C
Junction Temperature Range	T_J	125	°C
Storage Channel Temperature Range	T_{stg}	-65 to +150	°C



Refer to 2N4220 for graphs.

more...

Case and Terminal Identification



Testing JFETs

- **Curve Tracer**

A curve tracer displays the I_D versus V_{DS} graph for various levels of V_{GS} .

- **Specialized FET Testers**

These testers show I_{DSS} for the JFET under test.

MOSFETs

MOSFETs have characteristics similar to JFETs and additional characteristics that make them very useful.

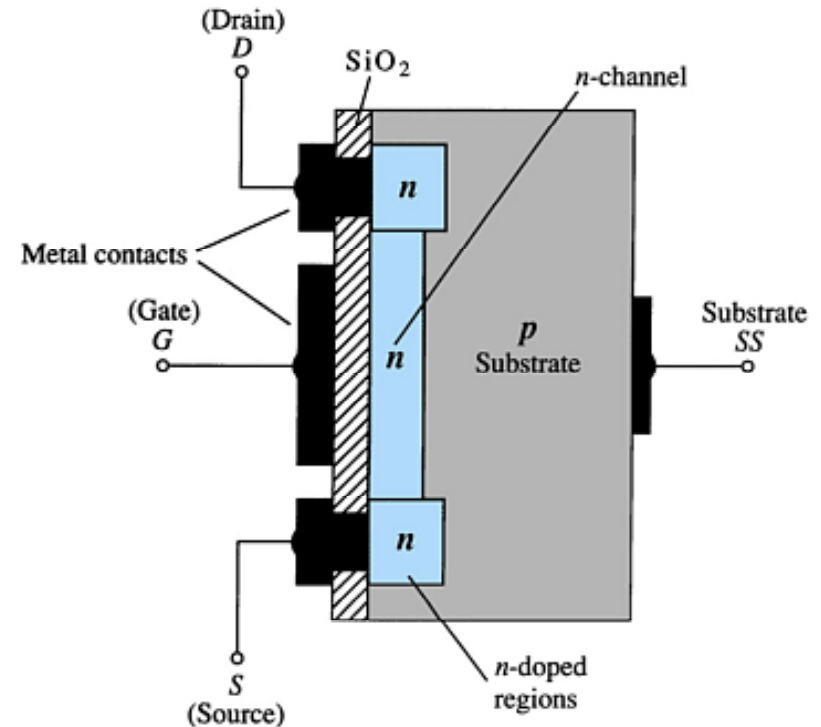
There are two types of MOSFETs:

- **Depletion-Type**
- **Enhancement-Type**

Depletion-Type MOSFET Construction

The **Drain** (D) and **Source** (S) connect to the n -doped regions. These n -doped regions are connected via an n -channel. This n -channel is connected to the **Gate** (G) via a thin insulating layer of SiO_2 .

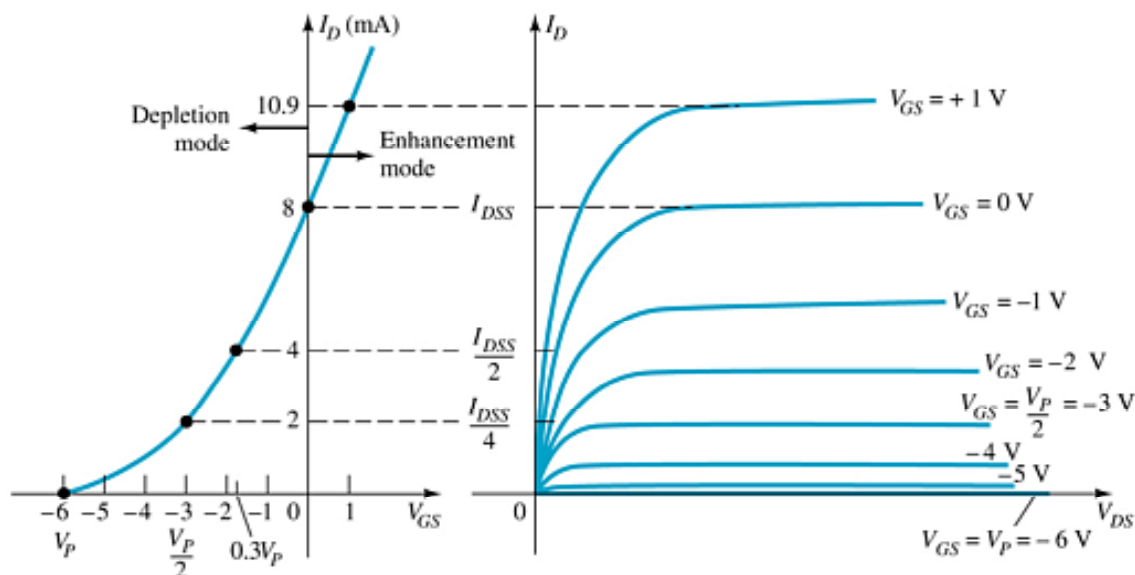
The n -doped material lies on a p -doped substrate that may have an additional terminal connection called **Substrate** (SS).



Basic MOSFET Operation

A depletion-type MOSFET can operate in two modes:

- Depletion mode
- Enhancement mode



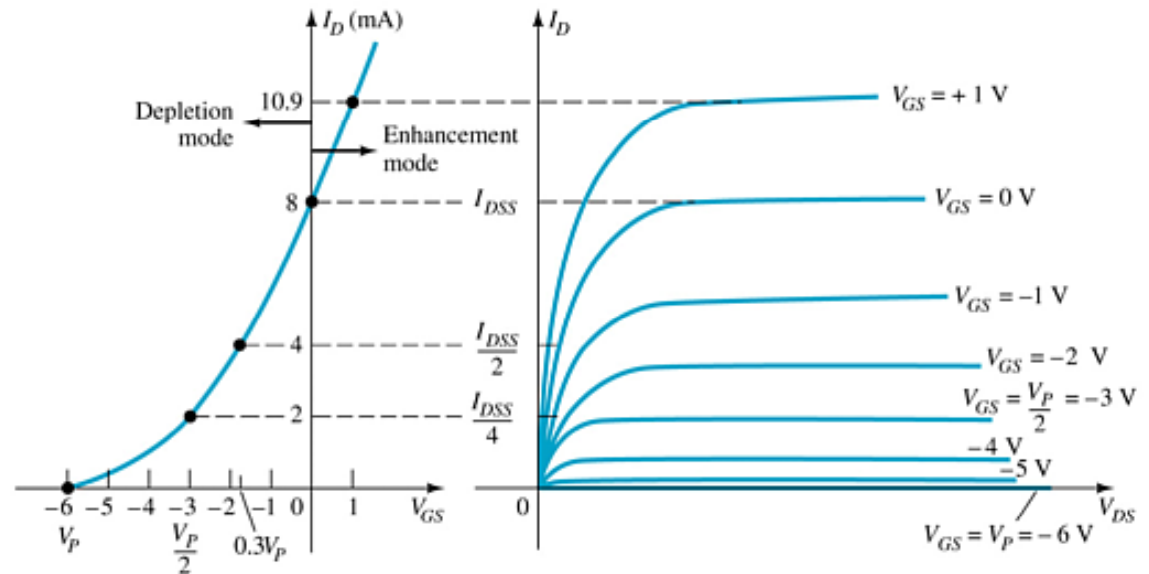
D-Type MOSFET in Depletion Mode

Depletion Mode

The characteristics are similar to a JFET.

- When $V_{GS} = 0 \text{ V}$, $I_D = I_{DSS}$
- When $V_{GS} < 0 \text{ V}$, $I_D < I_{DSS}$
- The formula used to plot the transfer curve still applies:

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

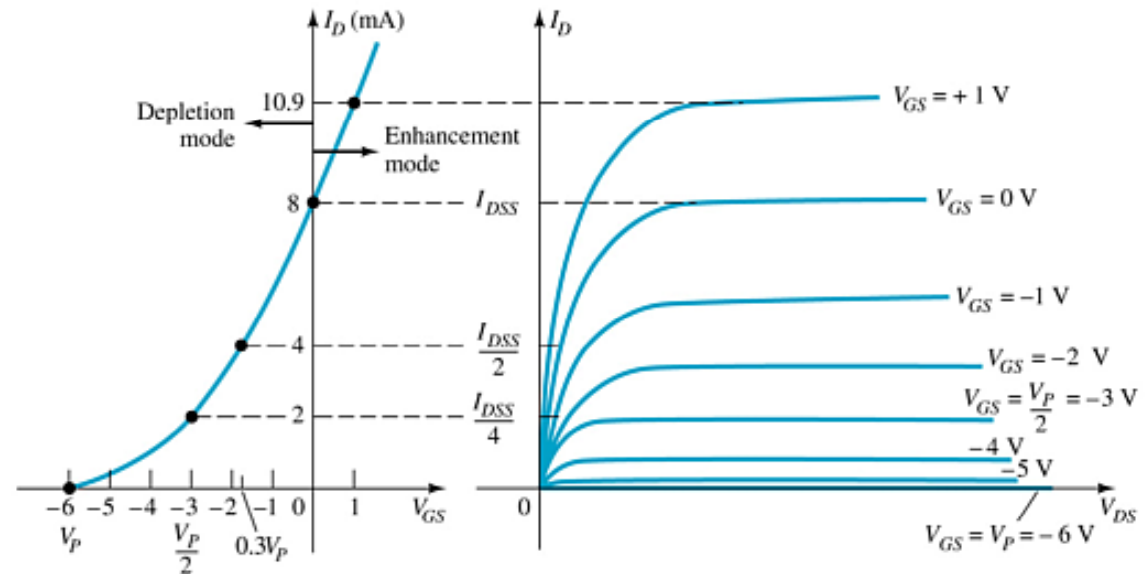


D-Type MOSFET in Enhancement Mode

Enhancement Mode

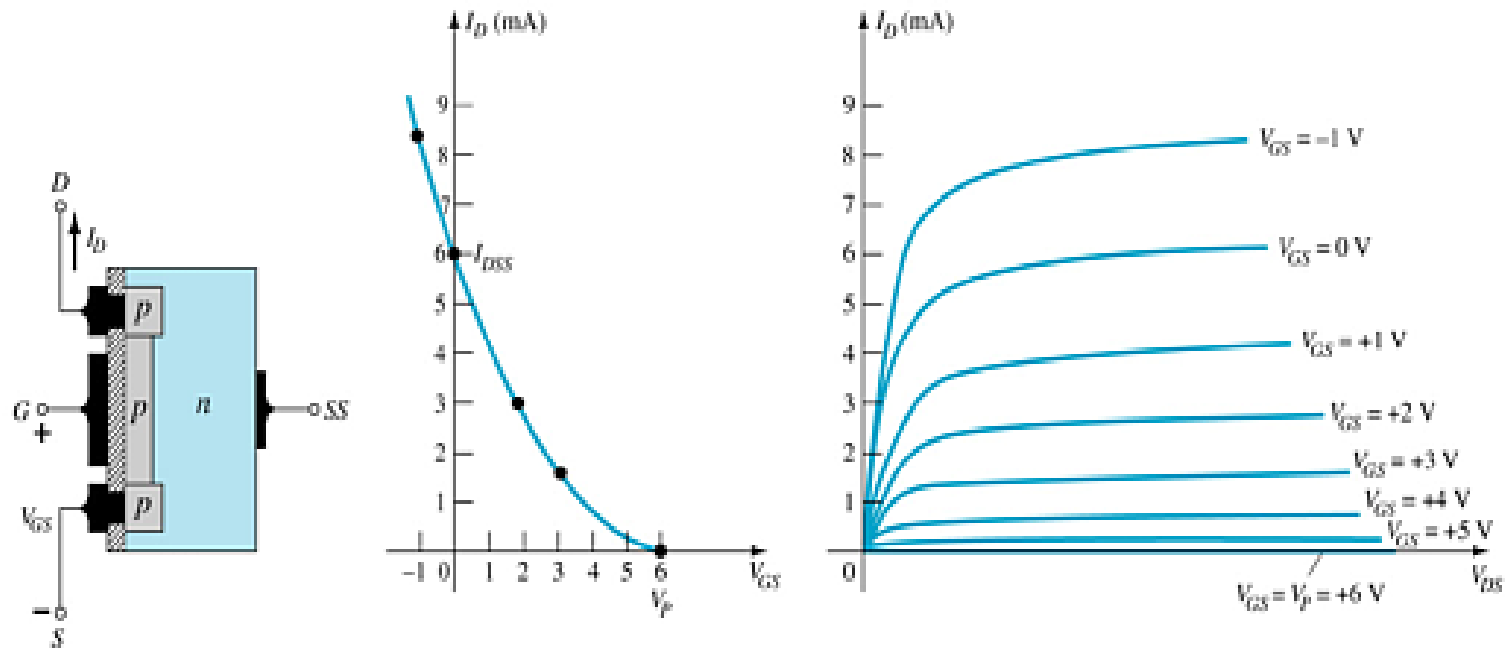
- $V_{GS} > 0 \text{ V}$
- I_D increases above I_{DSS}
- The formula used to plot the transfer curve still applies:

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

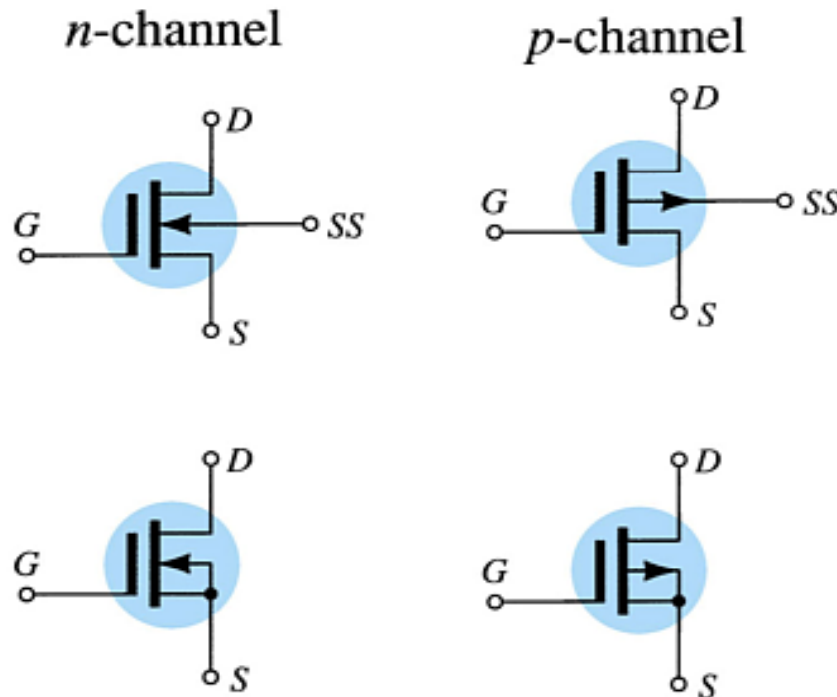


Note that V_{GS} is now a positive polarity

p-Channel D-Type MOSFET



D-Type MOSFET Symbols

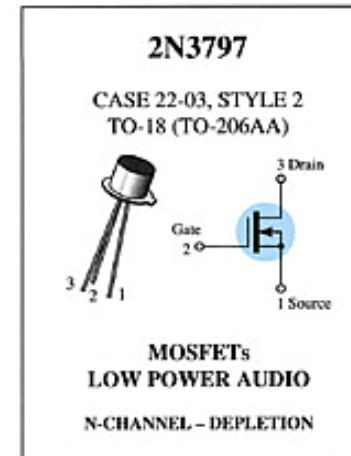


Specification Sheet

Maximum Ratings

MAXIMUM RATINGS

Rating	Symbol	Value	Unit
Drain-Source Voltage	V_{DS}	20	Vdc
Gate-Source Voltage	V_{GS}	± 10	Vdc
Drain Current	I_D	20	mA dc
Total Device Dissipation @ $T_A = 25^\circ\text{C}$ Derate above 25°C	P_D	200 1.14	mW mW/°C
Junction Temperature Range	T_J	+175	°C
Storage Channel Temperature Range	T_{UG}	-65 to +200	°C



more...

Specification Sheet

Electrical Characteristics

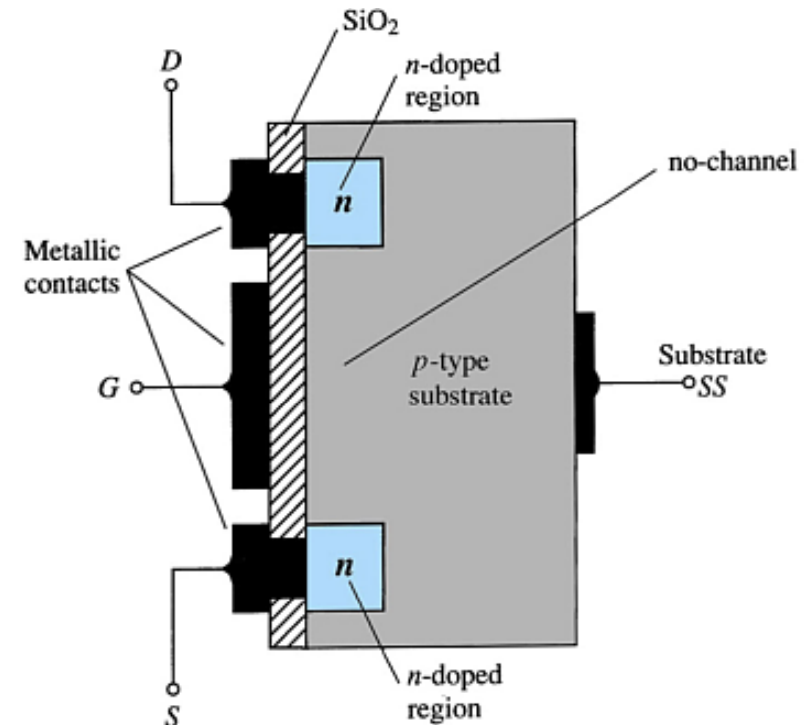
ELECTRICAL CHARACTERISTICS ($T_A = 25^\circ\text{C}$ unless otherwise noted)

Characteristic	Symbol	Min	Typ	Max	Unit
OFF CHARACTERISTICS					
Drain Source Breakdown Voltage ($V_{GS} = -7.0\text{ V}$, $I_D = 5.0\text{ }\mu\text{A}$)	$V_{(BR)DSX}$	20	25	—	Vdc
Gate Reverse Current (1) ($V_{GS} = -10\text{ V}$, $V_{DS} = 0$) ($V_{GS} = -10\text{ V}$, $V_{DS} = 0$, $T_A = 150^\circ\text{C}$)	I_{GSS}	—	—	1.0 200	pAdc
Gate Source Cutoff Voltage ($I_D = 2.0\text{ }\mu\text{A}$, $V_{DS} = 10\text{ V}$)	$V_{GS(off)}$	—	-5.0	-7.0	Vdc
Drain-Gate Reverse Current (1) ($V_{DG} = 10\text{ V}$, $I_S = 0$)	I_{DGO}	—	—	1.0	pAdc
ON CHARACTERISTICS					
Zero-Gate-Voltage Drain Current ($V_{DS} = 10\text{ V}$, $V_{GS} = 0$)	I_{DSS}	2.0	2.9	6.0	mAdc
On-State Drain Current ($V_{DS} = 10\text{ V}$, $V_{GS} = +3.5\text{ V}$)	$I_{D(on)}$	9.0	14	18	mAdc
SMALL-SIGNAL CHARACTERISTICS					
Forward Transfer Admittance ($V_{DS} = 10\text{ V}$, $V_{GS} = 0$, $f = 1.0\text{ kHz}$)	$ Y_{fs} $	1500	2300	3000	μmhos
($V_{DS} = 10\text{ V}$, $V_{GS} = 0$, $f = 1.0\text{ MHz}$)		1500	—	—	
Output Admittance ($I_{DS} = 10\text{ V}$, $V_{GS} = 0$, $f = 1.0\text{ kHz}$)	$ Y_{os} $	—	27	60	μmhos
Input Capacitance ($V_{DS} = 10\text{ V}$, $V_{GS} = 0$, $f = 1.0\text{ MHz}$)	C_{iss}	—	6.0	8.0	pF
Reverse Transfer Capacitance ($V_{DS} = 10\text{ V}$, $V_{GS} = 0$, $f = 1.0\text{ MHz}$)	C_{rss}	—	0.5	0.8	pF
FUNCTIONAL CHARACTERISTICS					
Noise Figure ($V_{DS} = 10\text{ V}$, $V_{GS} = 0$, $f = 1.0\text{ kHz}$, $R_G = 3\text{ megohms}$)	NF	—	3.8	—	dB

(1) This value of current includes both the FET leakage current as well as the leakage current associated with the test socket and fixture when measured under best attainable conditions.

E-Type MOSFET Construction

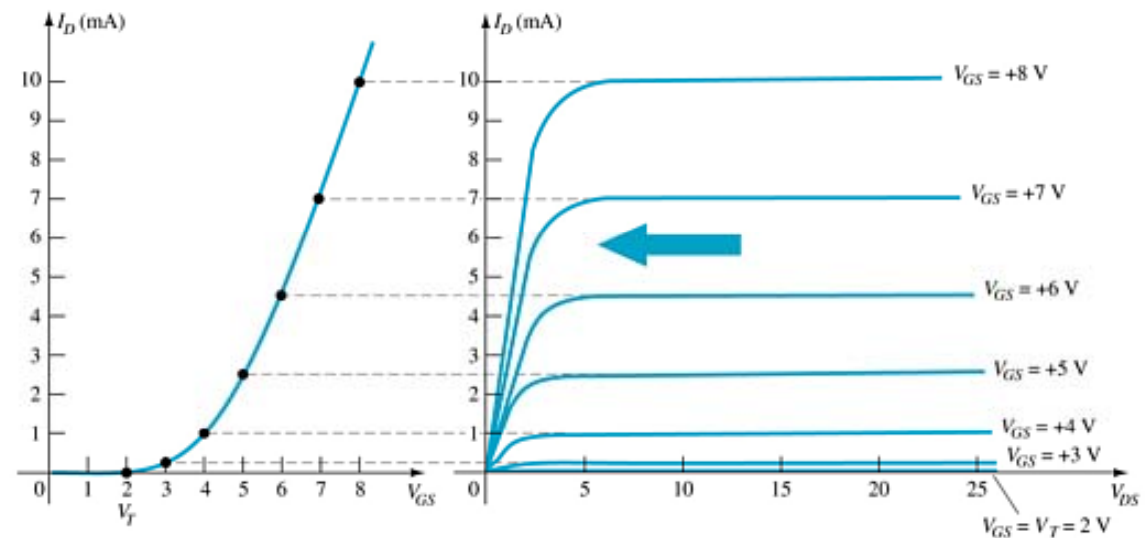
- The **Drain** (D) and **Source** (S) connect to the to n -doped regions. These n -doped regions are connected via an n -channel
- The **Gate** (G) connects to the p -doped substrate via a thin insulating layer of SiO_2
- There is no channel
- The n -doped material lies on a p -doped substrate that may have an additional terminal connection called the **Substrate** (SS)



Basic Operation of the E-Type MOSFET

The enhancement-type MOSFET operates only in the enhancement mode.

- V_{GS} is always positive
- As V_{GS} increases, I_D increases
- As V_{GS} is kept constant and V_{DS} is increased, then I_D saturates (I_{DSS}) and the saturation level, V_{DSsat} is reached



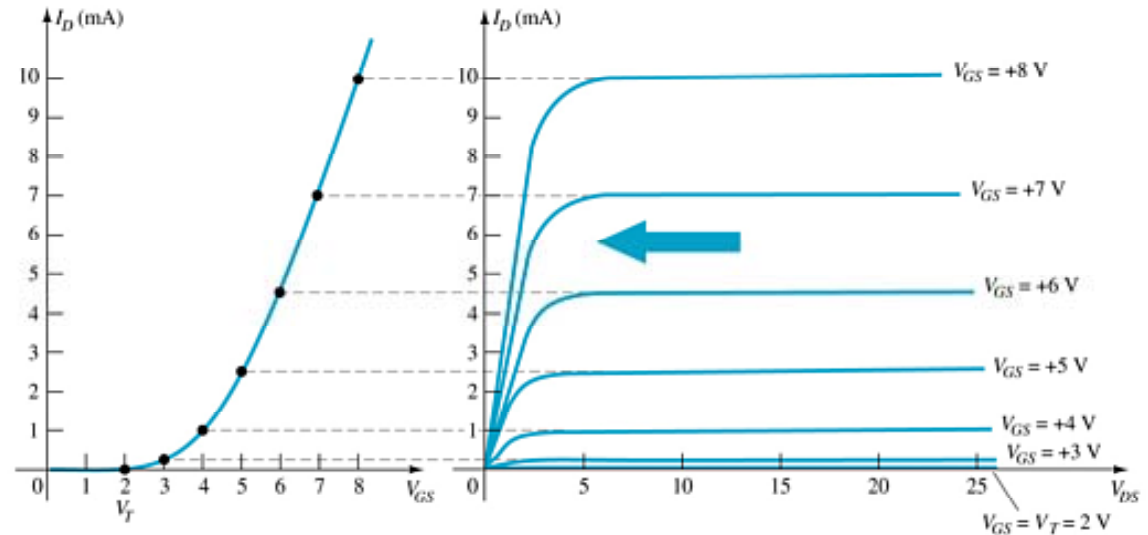
E-Type MOSFET Transfer Curve

To determine I_D given V_{GS} :

$$I_D = k(V_{GS} - V_T)^2$$

Where:

V_T = threshold voltage
or voltage at which the
MOSFET turns on



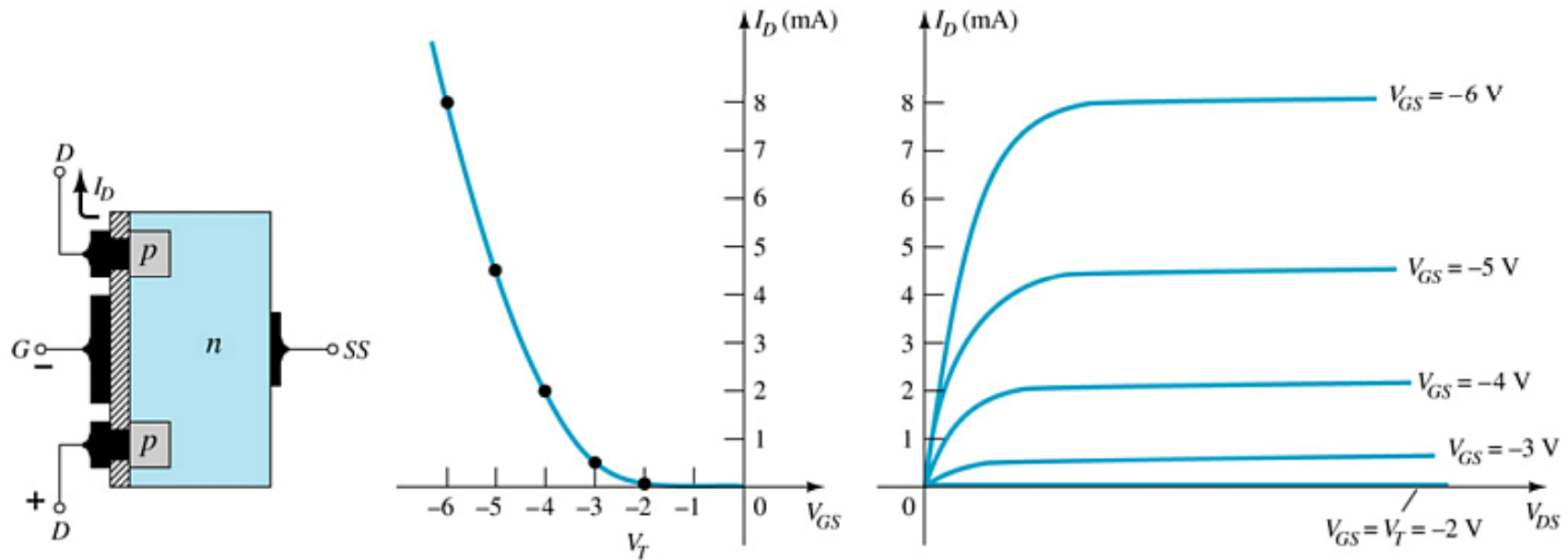
k , a constant, can be determined by using
values at a specific point and the formula:

$$k = \frac{I_{D(ON)}}{(V_{GS(ON)} - V_T)^2}$$

V_{DSsat} can be calculated by:

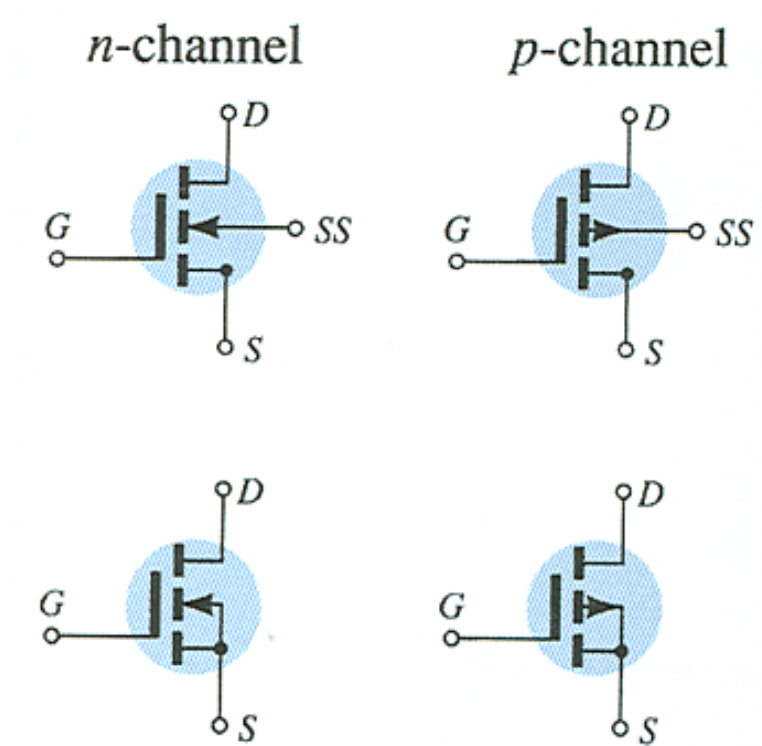
$$V_{DSsat} = V_{GS} - V_T$$

p-Channel E-Type MOSFETs



The *p*-channel enhancement-type MOSFET is similar to the *n*-channel, except that the voltage polarities and current directions are reversed.

MOSFET Symbols



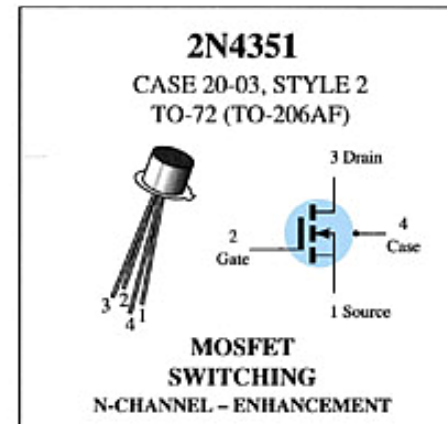
Specification Sheet

Maximum Ratings

MAXIMUM RATINGS

Rating	Symbol	Value	Unit
Drain-Source Voltage	V_{DS}	25	Vdc
Drain-Gate Voltage	V_{DG}	30	Vdc
Gate-Source Voltage*	V_{GS}	30	Vdc
Drain Current	I_D	30	mA dc
Total Device Dissipation @ $T_A = 25^\circ\text{C}$ Derate above 25°C	P_D	300 1.7	mW mW/ $^\circ\text{C}$
Junction Temperature Range	T_J	175	$^\circ\text{C}$
Storage Temperature Range	T_{stg}	-65 to +175	$^\circ\text{C}$

* Transient potentials of ± 75 Volt will not cause gate-oxide failure.



more...

Specification Sheet

Electrical Characteristics

ELECTRICAL CHARACTERISTICS ($T_A = 25^\circ\text{C}$ unless otherwise noted.)

Characteristic		Symbol	Min	Max	Unit
OFF CHARACTERISTICS					
Drain-Source Breakdown Voltage ($I_D = 10\ \mu\text{A}$, $V_{GS} = 0$)		$V_{(BR)DSX}$	25	—	Vdc
Zero-Gate-Voltage Drain Current ($V_{DS} = 10\ \text{V}$, $V_{GS} = 0$) $T_A = 25^\circ\text{C}$ $T_A = 150^\circ\text{C}$		I_{DSS}	— —	10 10	nAdc μAdc
Gate Reverse Current ($V_{GS} = \pm 15\ \text{Vdc}$, $V_{DS} = 0$)		I_{GSS}	—	± 10	pAdc
ON CHARACTERISTICS					
Gate Threshold Voltage ($V_{DS} = 10\ \text{V}$, $I_D = 10\ \mu\text{A}$)		$V_{GS(Th)}$	1.0	5	Vdc
Drain-Source On-Voltage ($I_D = 2.0\ \text{mA}$, $V_{GS} = 10\ \text{V}$)		$V_{DS(on)}$	—	1.0	V
On-State Drain Current ($V_{GS} = 10\ \text{V}$, $V_{DS} = 10\ \text{V}$)		$I_{D(on)}$	3.0	—	mAdc
SMALL-SIGNAL CHARACTERISTICS					
Forward Transfer Admittance ($V_{DS} = 10\ \text{V}$, $I_D = 2.0\ \text{mA}$, $f = 1.0\ \text{kHz}$)		$ y_{fs} $	1000	—	μmho
Input Capacitance ($V_{DS} = 10\ \text{V}$, $V_{GS} = 0$, $f = 140\ \text{kHz}$)		C_{iss}	—	5.0	pF
Reverse Transfer Capacitance ($V_{DS} = 0$, $V_{GS} = 0$, $f = 140\ \text{kHz}$)		C_{rss}	—	1.3	pF
Drain-Substrate Capacitance ($V_{DS(UB)} = 10\ \text{V}$, $f = 140\ \text{kHz}$)		$C_{d(sub)}$	—	5.0	pF
Drain-Source Resistance ($V_{GS} = 10\ \text{V}$, $I_D = 0$, $f = 1.0\ \text{kHz}$)		$r_{ds(on)}$	—	300	ohms
SWITCHING CHARACTERISTICS					
Turn-On Delay (Fig. 5)	$I_D = 2.0\ \text{mAdc}$, $V_{DS} = 10\ \text{Vdc}$, ($V_{GS} = 10\ \text{Vdc}$) (See Figure 9; Times Circuit Determined)	t_{d1}	—	45	ns
Rise Time (Fig. 6)		t_r	—	65	ns
Turn-Off Delay (Fig. 7)		t_{d2}	—	60	ns
Fall Time (Fig. 8)		t_f	—	100	ns

Handling MOSFETs

MOSFETs are very sensitive to static electricity. Because of the very thin SiO_2 layer between the external terminals and the layers of the device, any small electrical discharge can create an unwanted conduction.

Protection

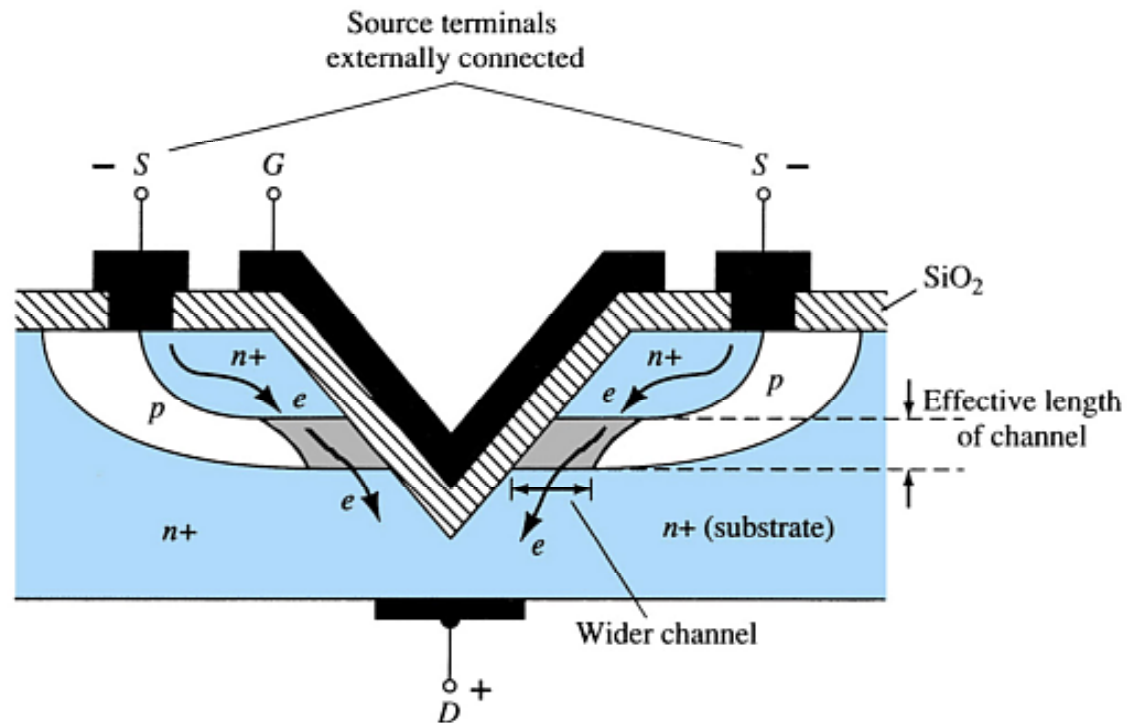
- **Always transport in a static sensitive bag**
- **Always wear a static strap when handling MOSFETs**
-
- **Apply voltage limiting devices between the gate and source, such as back-to-back Zeners to limit any transient voltage.**

VMOS Devices

VMOS (vertical MOSFET) increases the surface area of the device.

Advantages

- VMOS devices handle higher currents by providing more surface area to dissipate the heat.
- VMOS devices also have faster switching times.

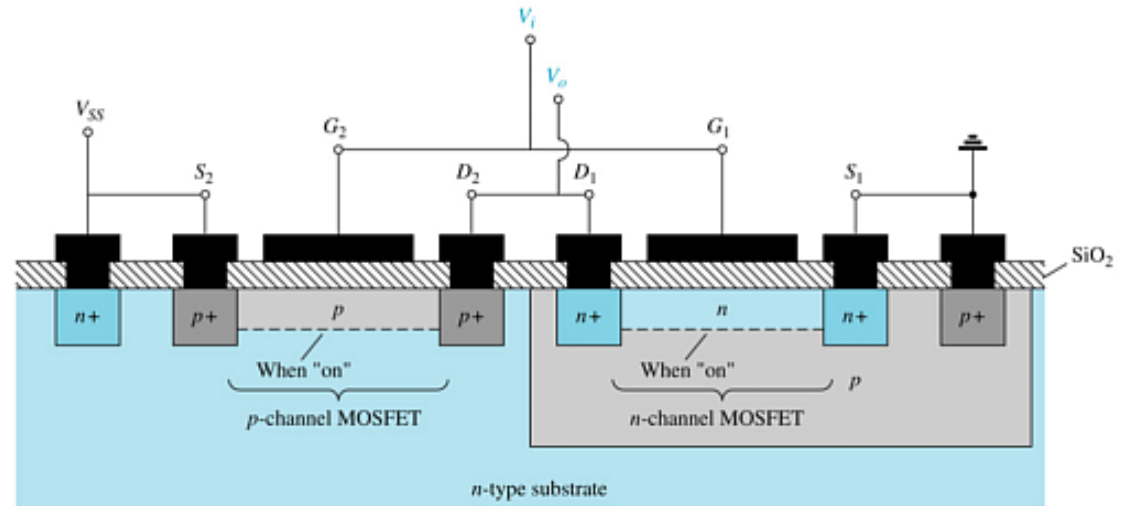


CMOS Devices

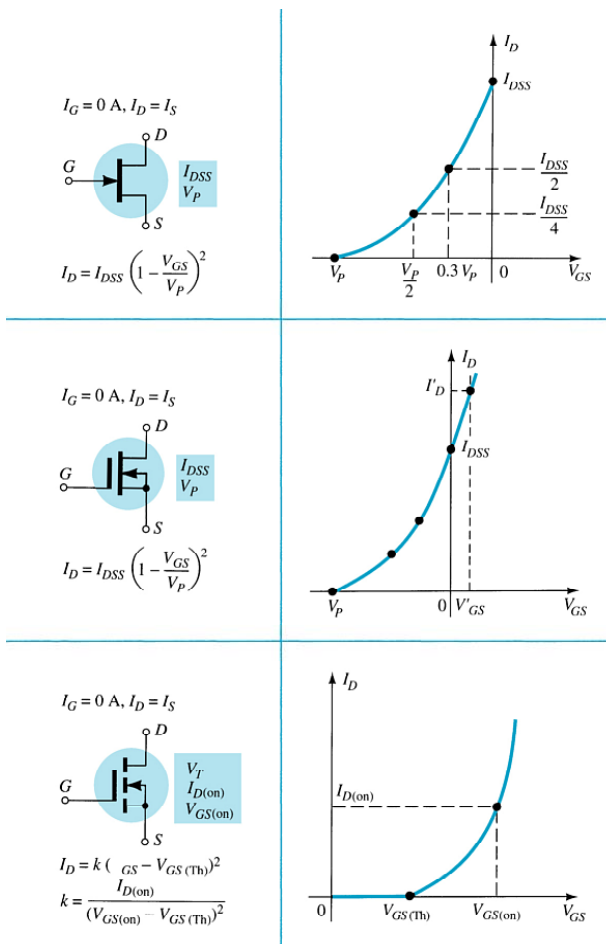
CMOS (complementary MOSFET) uses a *p*-channel and *n*-channel MOSFET; often on the same substrate as shown here.

Advantages

- Useful in logic circuit designs
- Higher input impedance
- Faster switching speeds
- Lower operating power levels



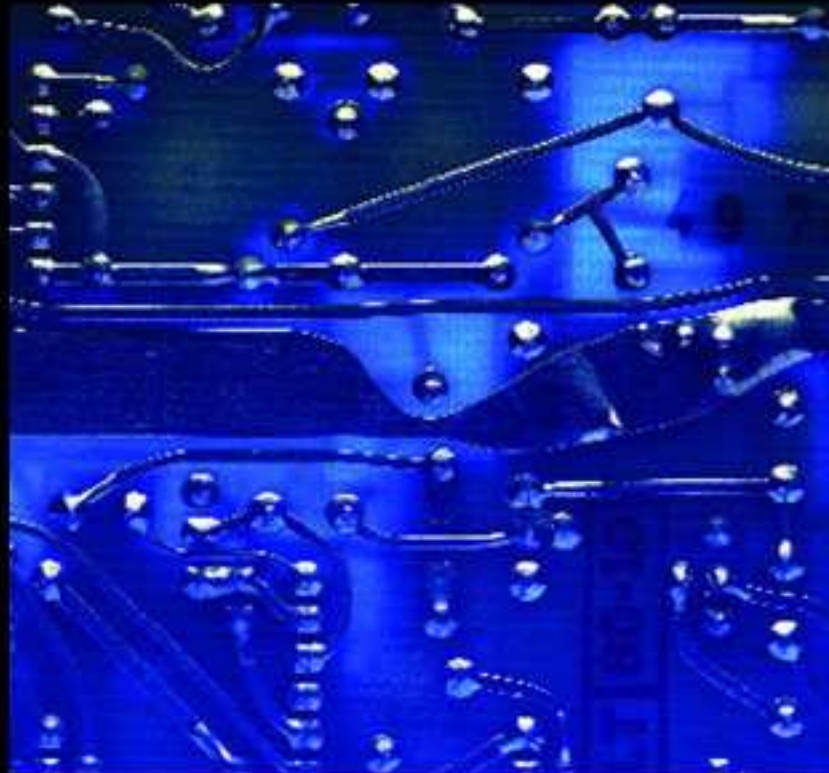
Summary Table



ELECTRONIC DEVICES AND CIRCUIT THEORY

TENTH EDITION

BOYLESTAD



PEARSON

Chapter 7: FET Biasing

Common FET Biasing Circuits

JFET Biasing Circuits

- Fixed – Bias
- Self-Bias
- Voltage-Divider Bias

D-Type MOSFET Biasing Circuits

- Self-Bias
- Voltage-Divider Bias

E-Type MOSFET Biasing Circuits

- Feedback Configuration
- Voltage-Divider Bias

Basic Current Relationships

For all FETs:

$$I_G \cong 0A$$

$$I_D = I_S$$

For JFETs and D-Type MOSFETs:

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

For E-Type MOSFETs:

$$I_D = k(V_{GS} - V_T)^2$$

Fixed-Bias Configuration

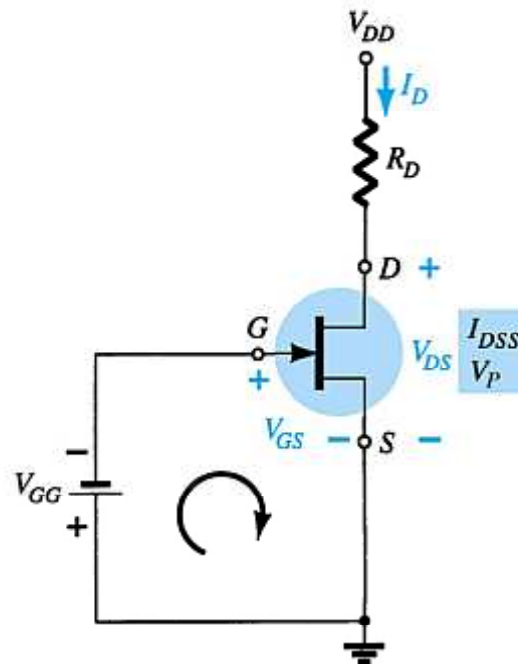
$$V_{DS} = V_{DD} - I_D R_D$$

$$V_S = 0V$$

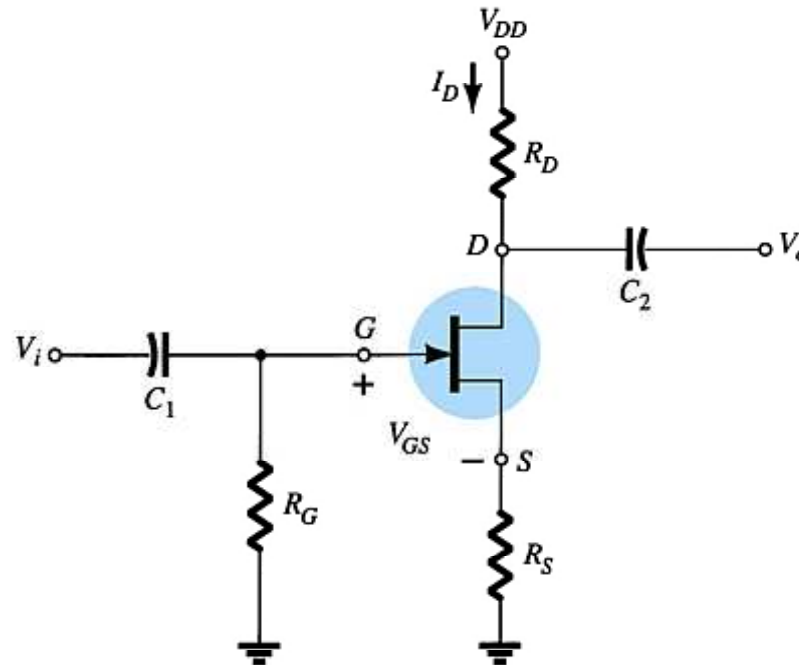
$$V_C = V_{DS}$$

$$V = V_{GS}$$

$$V_{GS} = -V_{GG}$$



Self-Bias Configuration



Self-Bias Calculations

For the indicated loop, $V_{GS} = -I_D R_S$

To solve this equation:

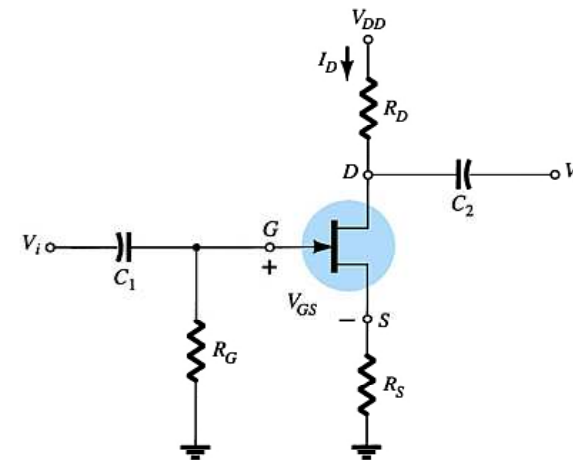
- Select an $I_D < I_{DSS}$ and use the component value of R_S to calculate V_{GS}
- Plot the point identified by I_D and V_{GS} . Draw a line from the origin of the axis to this point.
- Plot the transfer curve using I_{DSS} and V_P ($V_P = V_{GSoff}$ in specification sheets) and a few points such as $I_D = I_{DSS} / 4$ and $I_D = I_{DSS} / 2$ etc.

The Q-point is located where the first line intersects the transfer curve. Use the value of I_D at the Q-point (I_{DQ}) to solve for the other voltages:

$$V_{DS} = V_{DD} - I_D (R_S + R_D)$$

$$V_S = I_D R_S$$

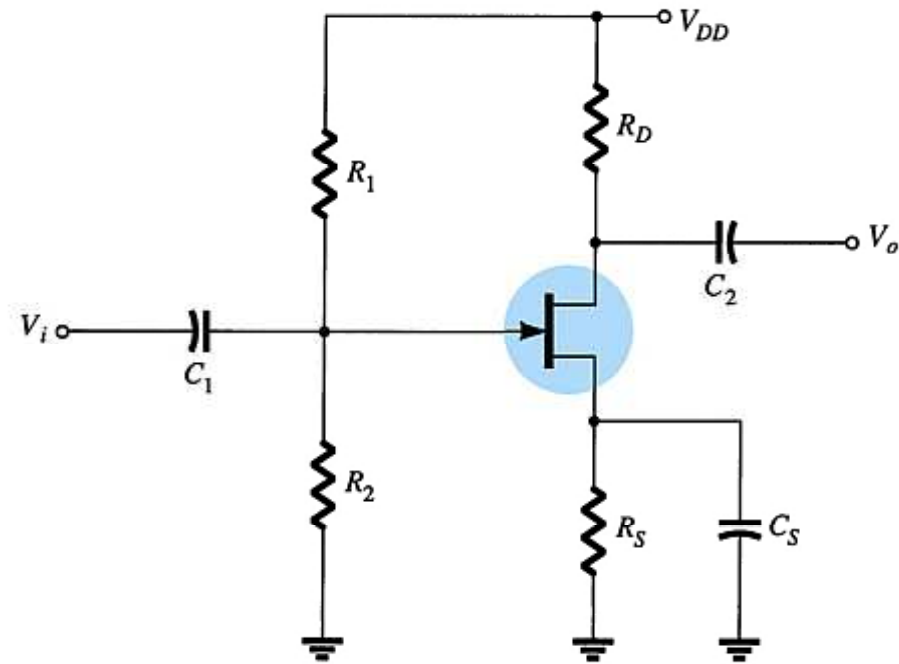
$$V_D = V_{DS} + V_S = V_{DD} - V_{RD}$$



Voltage-Divider Bias

$$I_G = 0 \text{ A}$$

I_D responds to changes in V_{GS} .



Voltage-Divider Bias Calculations

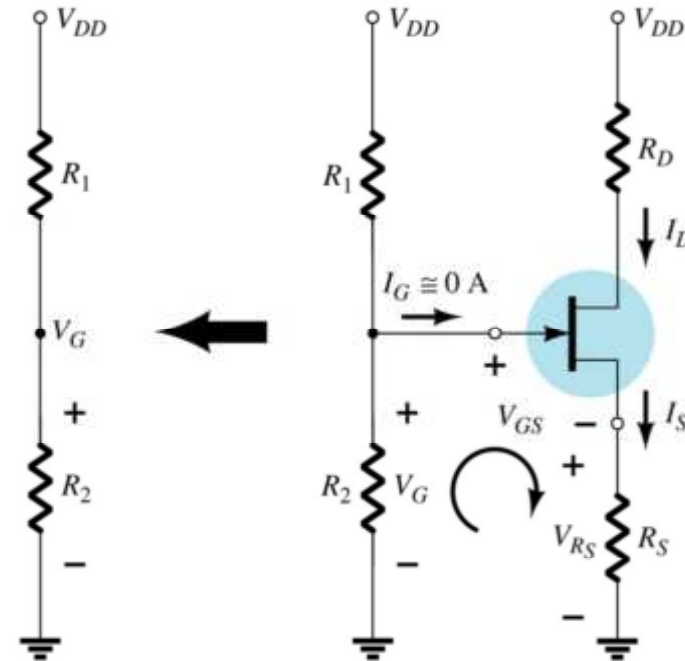
V_G is equal to the voltage across divider resistor R_2 :

$$V_G = \frac{R_2 V_{DD}}{R_1 + R_2}$$

Using Kirchhoff's Law:

$$V_{GS} = V_G - I_D R_S$$

The Q point is established by plotting a line that intersects the transfer curve.



Voltage-Divider Q-point

Step 1

Plot the line by plotting two points:

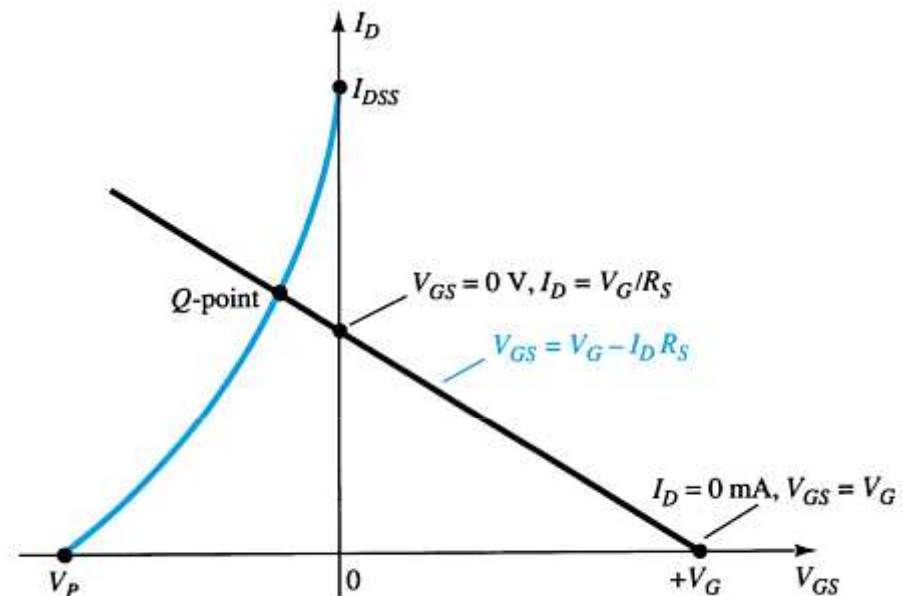
- $V_{GS} = V_G$, $I_D = 0$ A
- $V_{GS} = 0$ V, $I_D = V_G / R_S$

Step 2

Plot the transfer curve by plotting I_{DSS} , V_P and the calculated values of I_D

Step 3

The Q-point is located where the line intersects the transfer curve



Voltage-Divider Bias Calculations

Using the value of I_D at the Q-point, solve for the other variables in the voltage-divider bias circuit:

$$V_{DS} = V_{DD} - I_D(R_D + R_S)$$

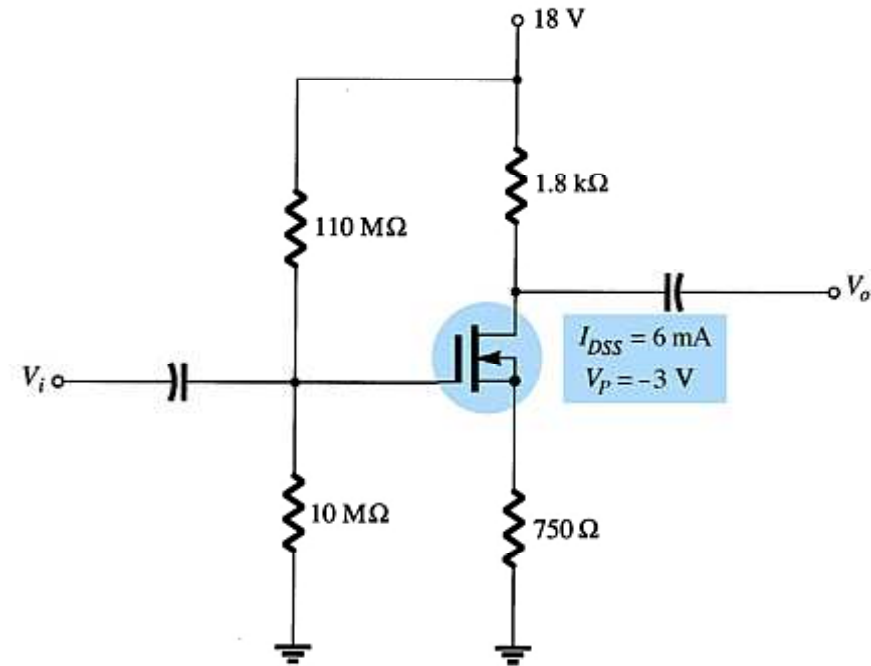
$$V_D = V_{DD} - I_D R_D$$

$$V_S = I_D R_S$$

$$I_{R1} = I_{R2} = \frac{V_{DD}}{R_1 + R_2}$$

D-Type MOSFET Bias Circuits

Depletion-type MOSFET bias circuits are similar to those used to bias JFETs. The only difference is that depletion-type MOSFETs can operate with positive values of V_{GS} and with I_D values that exceed I_{DSS} .



Self-Bias

Step 1

Plot line for

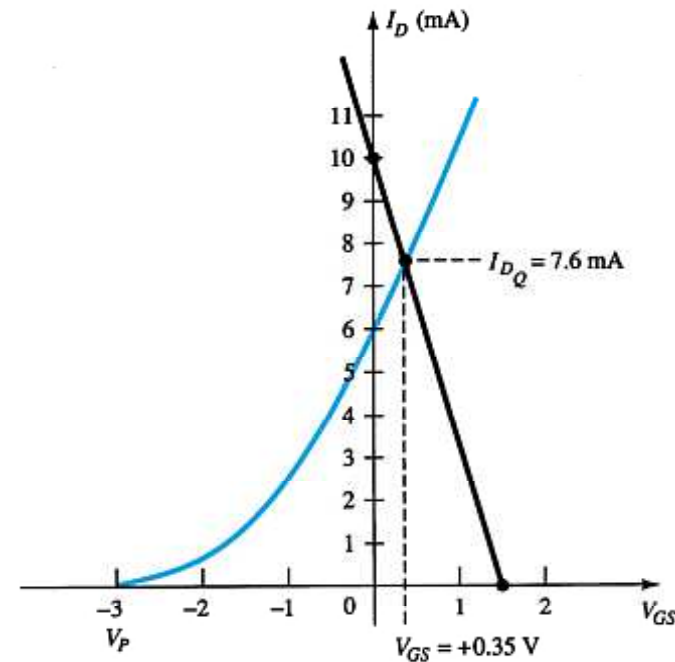
- $V_{GS} = V_G, I_D = 0 \text{ A}$
- $I_D = V_G/R_S, V_{GS} = 0 \text{ V}$

Step 2

Plot the transfer curve using I_{DSS} , V_P and calculated values of I_D

Step 3

The Q-point is located where the line intersects the transfer curve. Use the I_D at the Q-point to solve for the other variables in the voltage-divider bias circuit.



These are the same steps used to analyze JFET self-bias circuits.

Voltage-Divider Bias

Step 1

Plot the line for

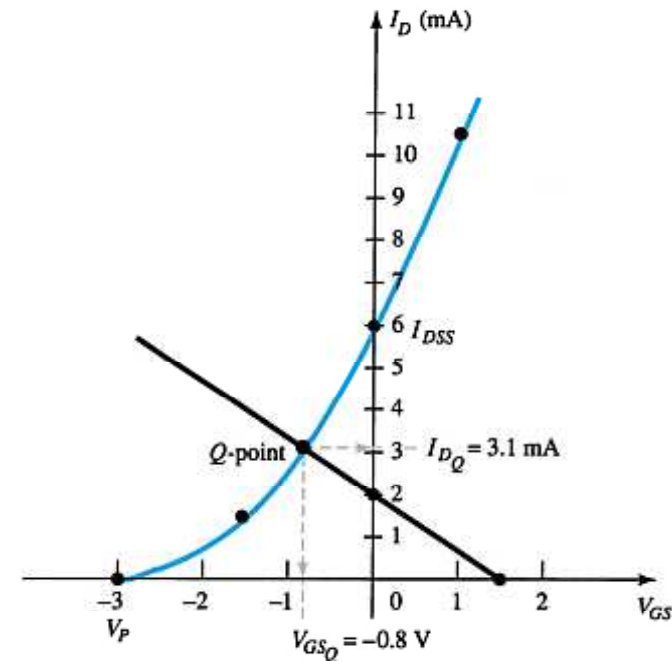
- $V_{GS} = V_G$, $I_D = 0$ A
- $I_D = V_G/R_S$, $V_{GS} = 0$ V

Step 2

Plot the transfer curve using I_{DSS} , V_P and calculated values of I_D .

Step 3

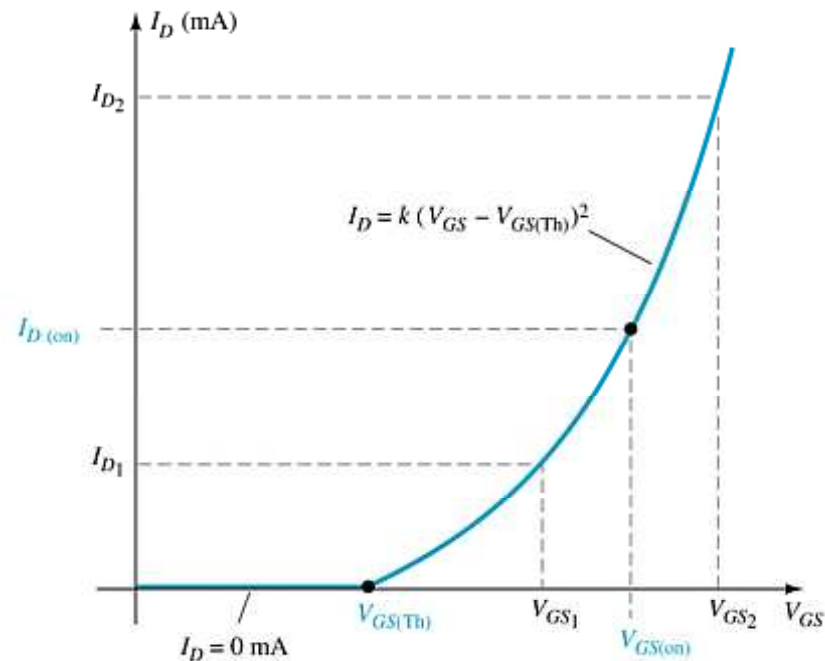
The Q-point is located where the line intersects the transfer curve. Use the I_D at the Q-point to solve for the other variables in the voltage-divider bias circuit.



These are the same steps used to analyze JFET voltage-divider bias circuits.

E-Type MOSFET Bias Circuits

The transfer characteristic for the e-type MOSFET is very different from that of a simple JFET or the d-type MOSFET.



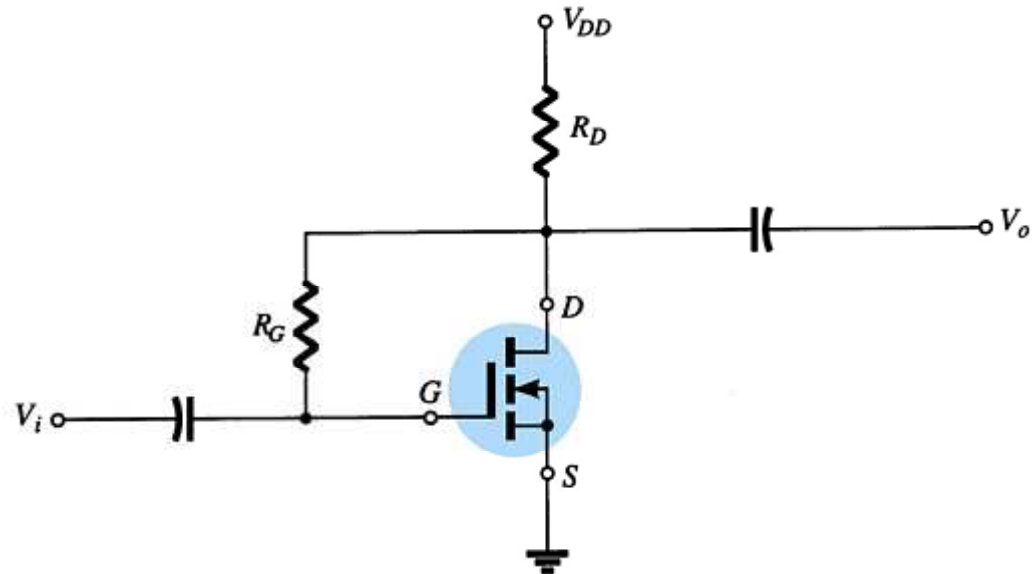
Feedback Bias Circuit

$$I_G = 0 \text{ A}$$

$$V_{RG} = 0 \text{ V}$$

$$V_{DS} = V_{GS}$$

$$V_{GS} = V_{DD} - I_D R_D$$



Feedback Bias Q-Point

Step 1

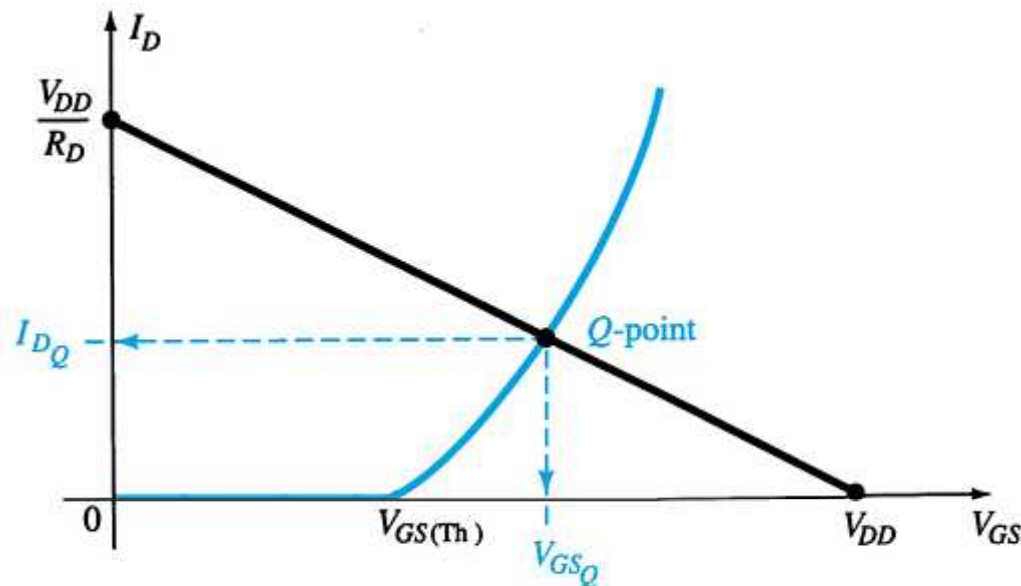
Plot the line using

- $V_{GS} = V_{DD}$, $I_D = 0$ A
- $I_D = V_{DD} / R_D$, $V_{GS} = 0$ V

Step 2

Using values from the specification sheet, plot the transfer curve with

- $V_{GS(Th)}$, $I_D = 0$ A
- $V_{GS(on)}$, $I_{D(on)}$



Step 3

The Q-point is located where the line and the transfer curve intersect

Step 4

Using the value of I_D at the Q-point, solve for the other variables in the bias circuit

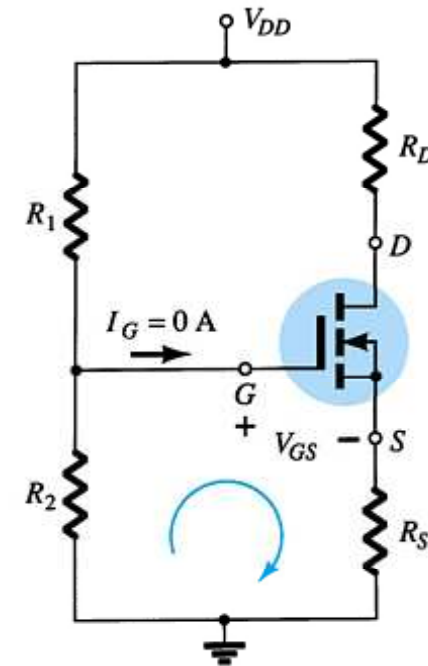
Voltage-Divider Biasing

Plot the line and the transfer curve to find the Q-point. Use these equations:

$$V_G = \frac{R_2 V_{DD}}{R_1 + R_2}$$

$$V_{GS} = V_G - I_D R_S$$

$$V_{DS} = V_{DD} - I_D (R_S + R_D)$$



Voltage-Divider Bias Q-Point

Step 1

Plot the line using

- $V_{GS} = V_G = (R_2 V_{DD}) / (R_1 + R_2)$, $I_D = 0$ A
- $I_D = V_G / R_S$, $V_{GS} = 0$ V

Step 2

Using values from the specification sheet, plot the transfer curve with

- V_{GSTh} , $I_D = 0$ A
- $V_{GS(on)}$, $I_{D(on)}$

Step 3

The point where the line and the transfer curve intersect is the Q-point.

Step 4

Using the value of I_D at the Q-point, solve for the other circuit values.

p -Channel FETs

For p -channel FETs the same calculations and graphs are used, except that the voltage polarities and current directions are reversed.

The graphs are mirror images of the n -channel graphs.

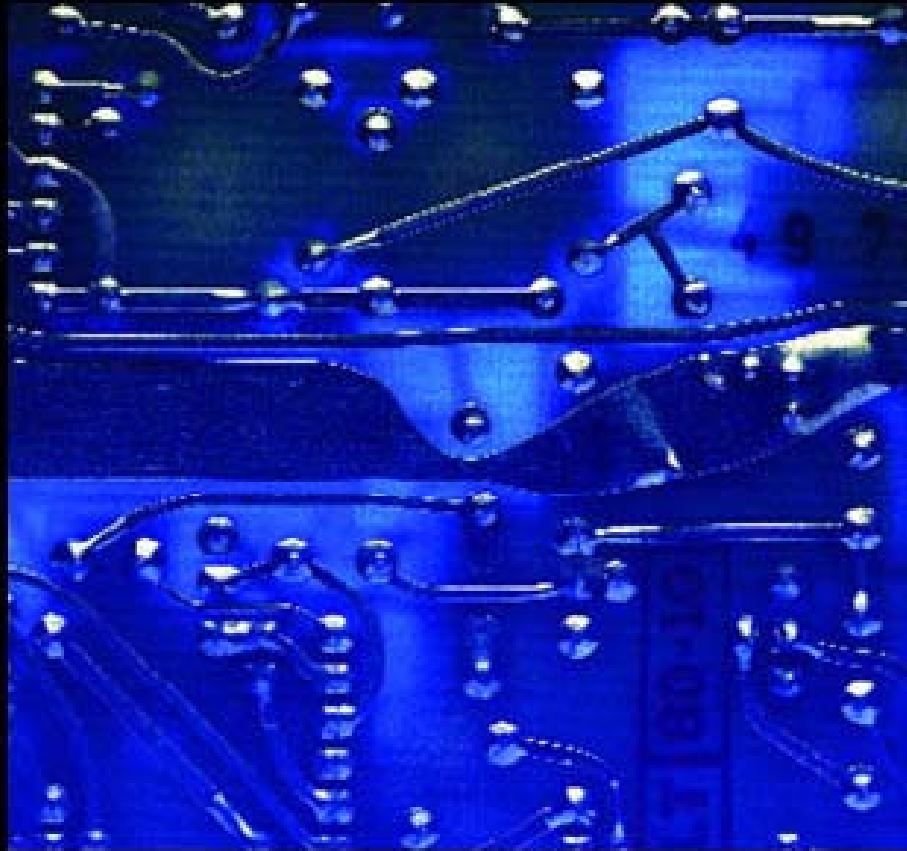
Applications

Voltage-controlled resistor
JFET voltmeter
Timer network
Fiber optic circuitry
MOSFET relay driver

ELECTRONIC DEVICES AND CIRCUIT THEORY

TENTH EDITION

BOYLESTAD



PEARSON

Chapter 8: FET Amplifiers

Introduction

FETs provide:

- **Excellent voltage gain**
- **High input impedance**
- **Low-power consumption**
- **Good frequency range**

FET Small-Signal Model

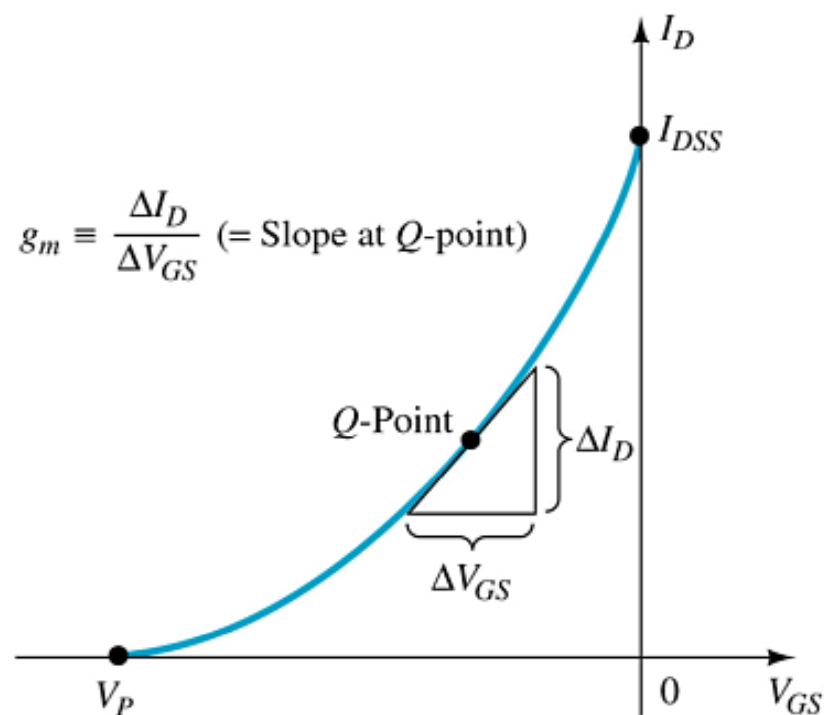
Transconductance

The relationship of a change in I_D to the corresponding change in V_{GS} is called **transconductance**

Transconductance is denoted g_m and given by:

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}}$$

Graphical Determination of g_m



Mathematical Definitions of g_m

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}}$$

$$g_m = \frac{2I_{DSS}}{|V_P|} \left[1 - \frac{V_{GS}}{V_P} \right]$$

Where $V_{GS} = 0V$ $g_{m0} = \frac{2I_{DSS}}{|V_P|}$

$$g_m = g_{m0} \left[1 - \frac{V_{GS}}{V_P} \right]$$

Where $1 - \frac{V_{GS}}{V_P} = \sqrt{\frac{I_D}{I_{DSS}}}$

$$g_m = g_{m0} \left(1 - \frac{V_{GS}}{V_P} \right) = g_{m0} \sqrt{\frac{I_D}{I_{DSS}}}$$

FET Impedance

Input impedance:

$$Z_i = \infty \Omega$$

Output Impedance:

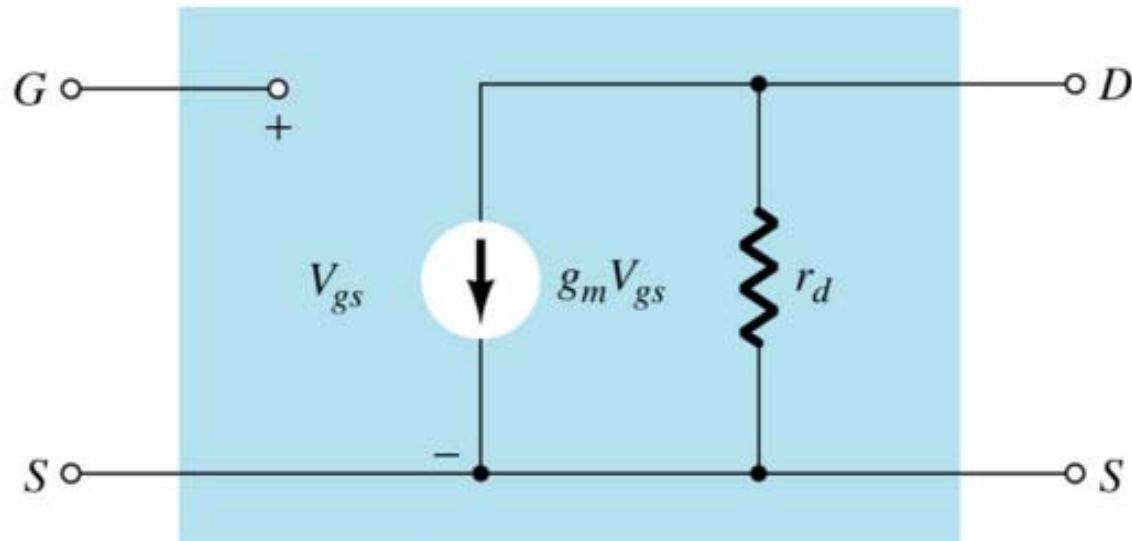
$$Z_o = r_d = \frac{1}{y_{os}}$$

where:

$$r_d = \left. \frac{\Delta V_{DS}}{\Delta I_D} \right|_{V_{GS} = \text{constant}}$$

y_{os} = admittance parameter listed on FET specification sheets.

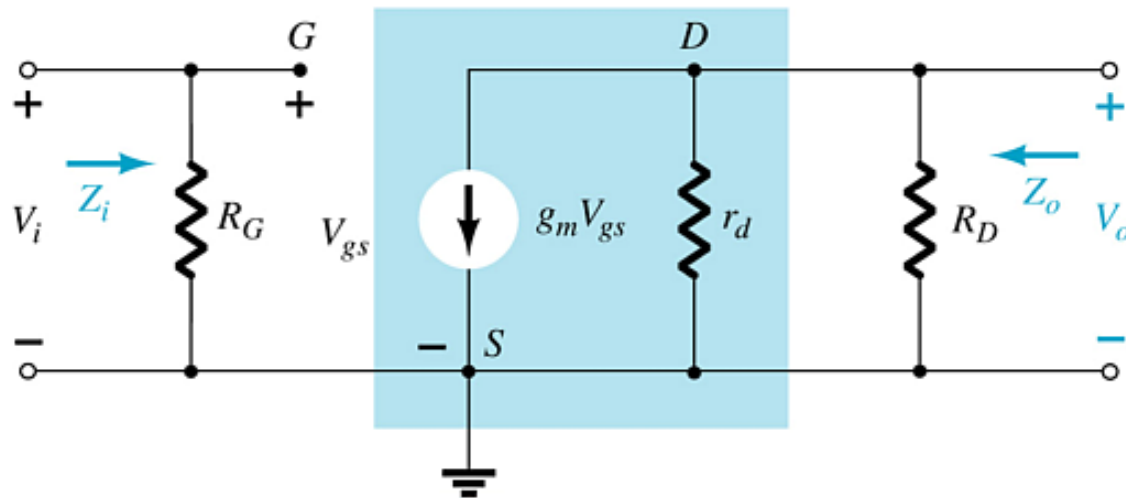
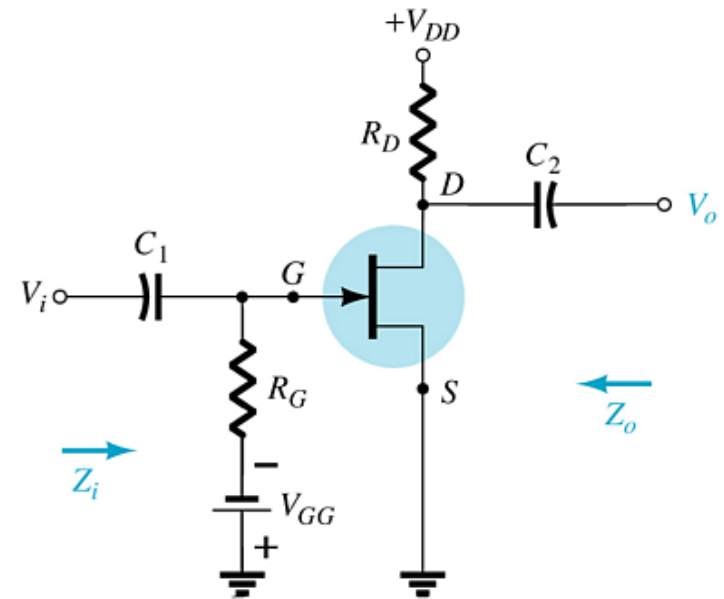
FET AC Equivalent Circuit



Common-Source (CS) Fixed-Bias Circuit

The input is on the gate and the output is on the drain

There is a 180° phase shift between input and output



Calculations

Input impedance:

$$Z_i = R_G$$

Output impedance:

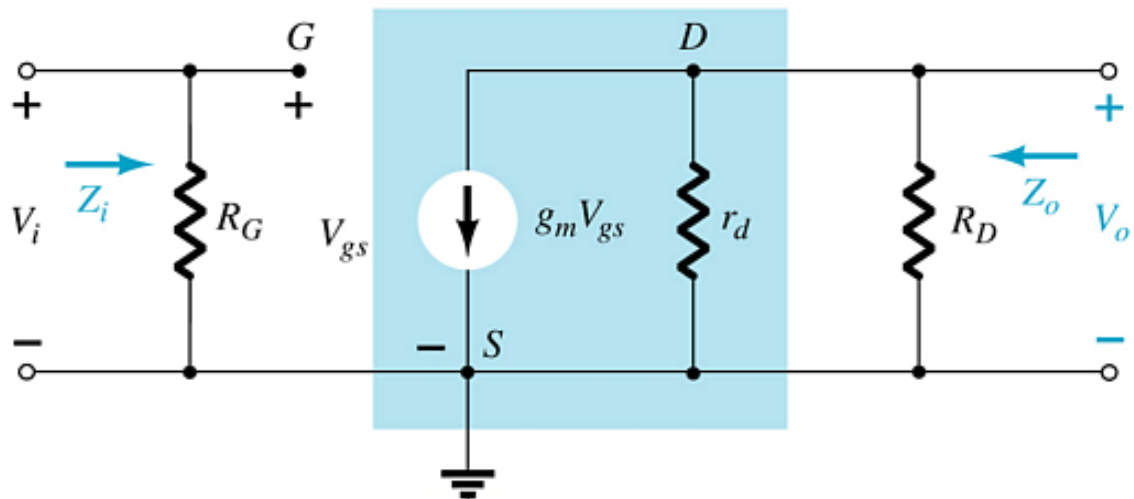
$$Z_o = R_D \parallel r_d$$

$$Z_o \cong R_D \quad \left| \quad r_d \geq 10R_D \right.$$

Voltage gain:

$$A_v = \frac{V_o}{V_i} = -g_m (r_d \parallel R_D)$$

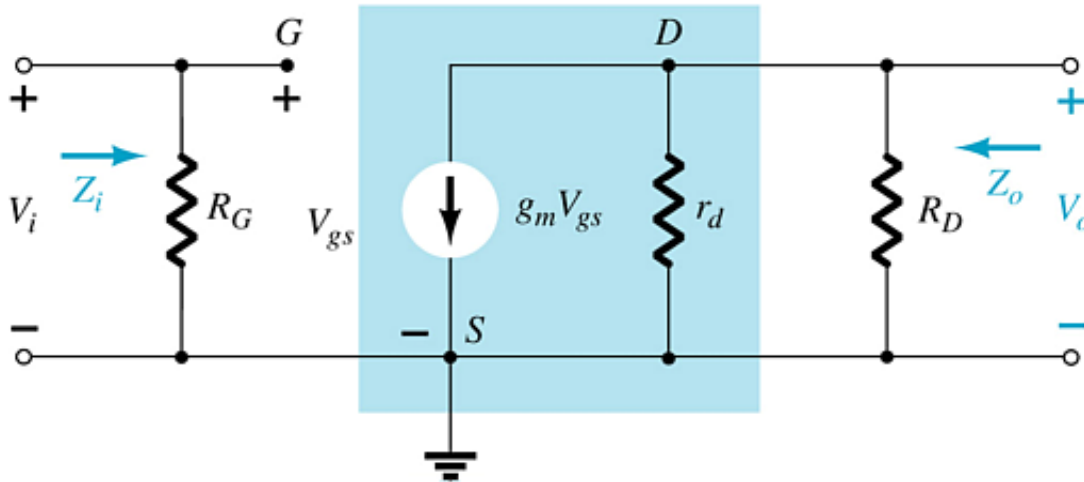
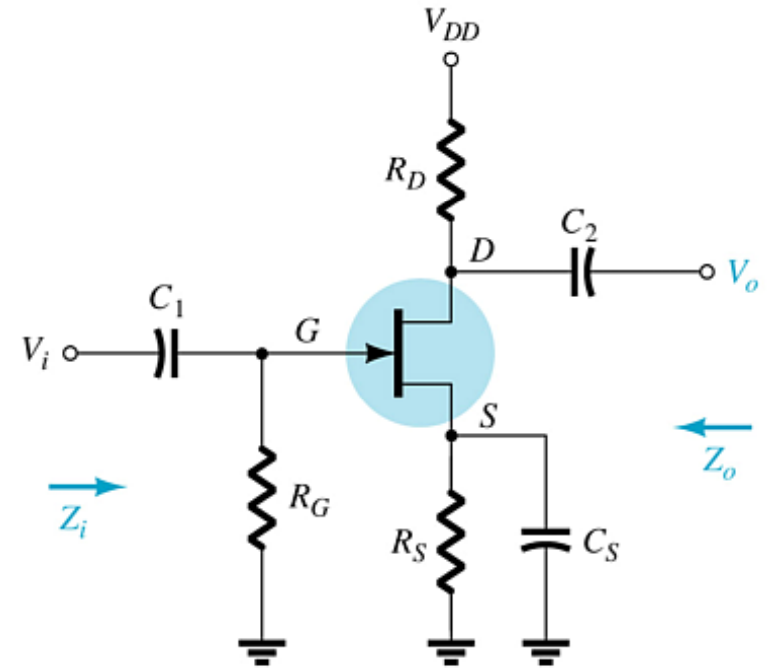
$$A_v = \frac{V_o}{V_i} = -g_m R_D \quad \left| \quad r_d \geq 10R_D \right.$$



Common-Source (CS) Self-Bias Circuit

This is a common-source amplifier configuration, so the input is on the gate and the output is on the drain

There is a 180° phase shift between input and output



Calculations

Input impedance:

$$Z_i = R_G$$

Output impedance:

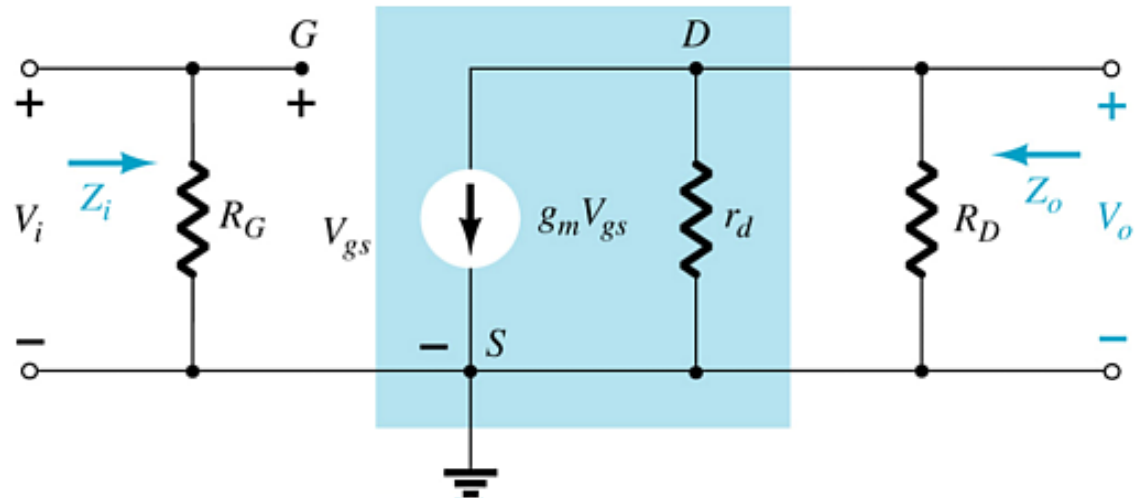
$$Z_o = r_d \parallel R_D$$

$$Z_o \cong R_D \quad \left| \quad r_d \geq 10R_D \right.$$

Voltage gain:

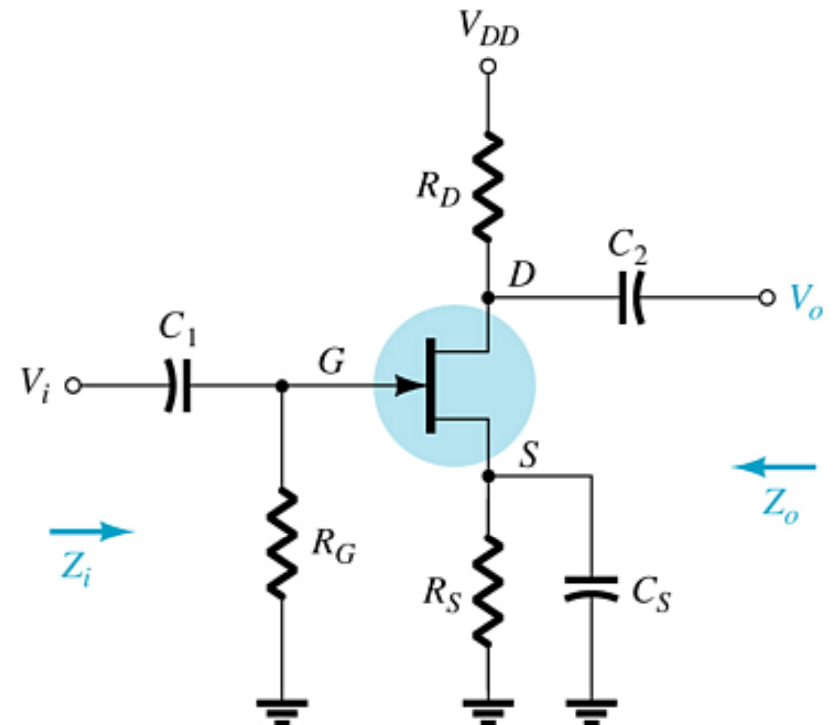
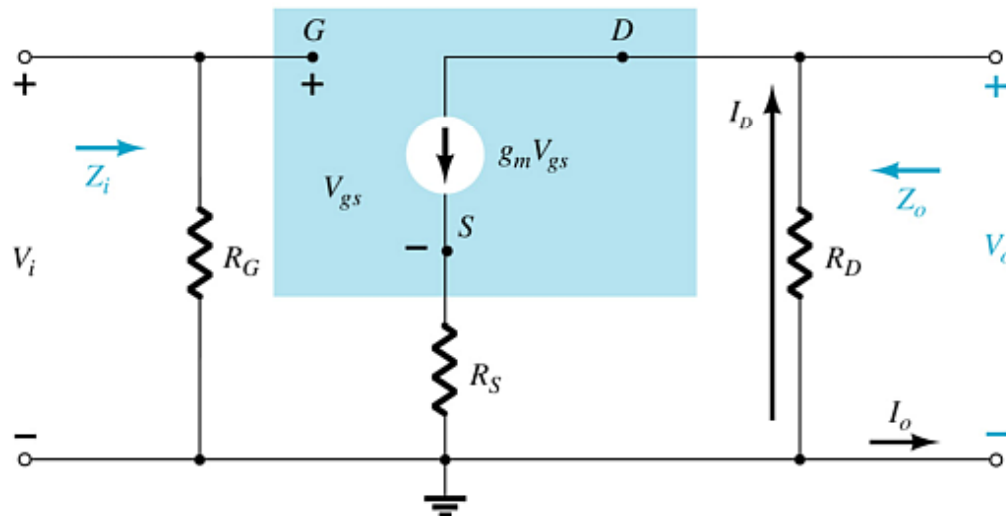
$$A_v = -g_m (r_d \parallel R_D)$$

$$A_v = -g_m R_D \quad \left| \quad r_d \geq 10R_D \right.$$



Common-Source (CS) Self-Bias Circuit

Removing C_s affects the gain of the circuit.



Calculations

Input impedance:

$$Z_i = R_G$$

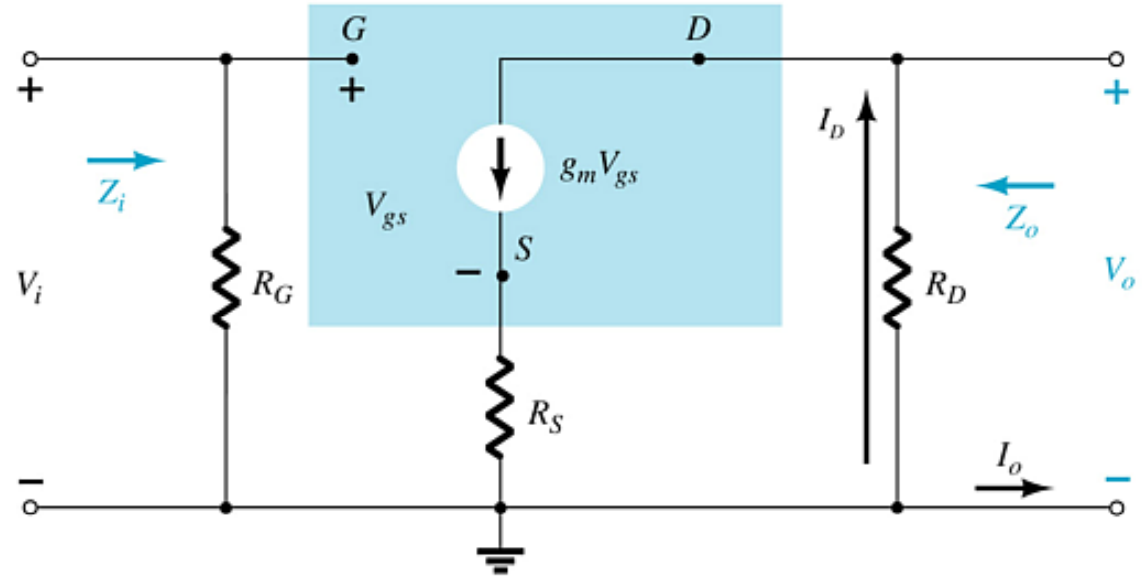
Output impedance:

$$Z_o \cong R_D \left| r_d \geq 10R_D \right.$$

Voltage gain:

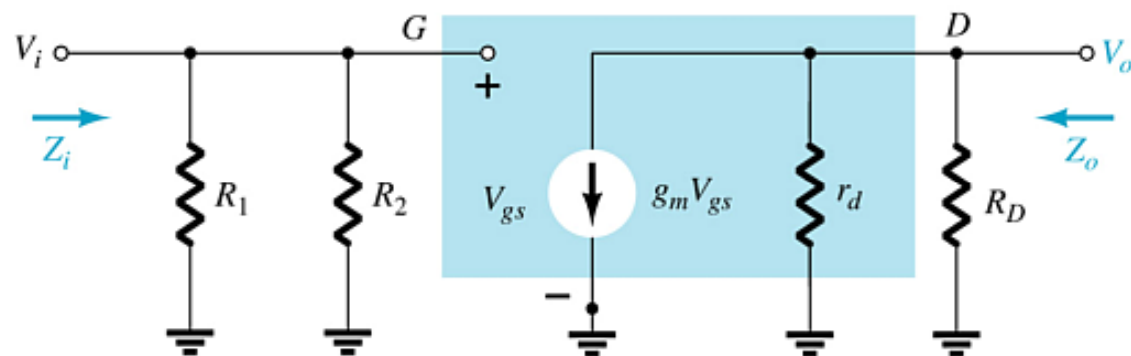
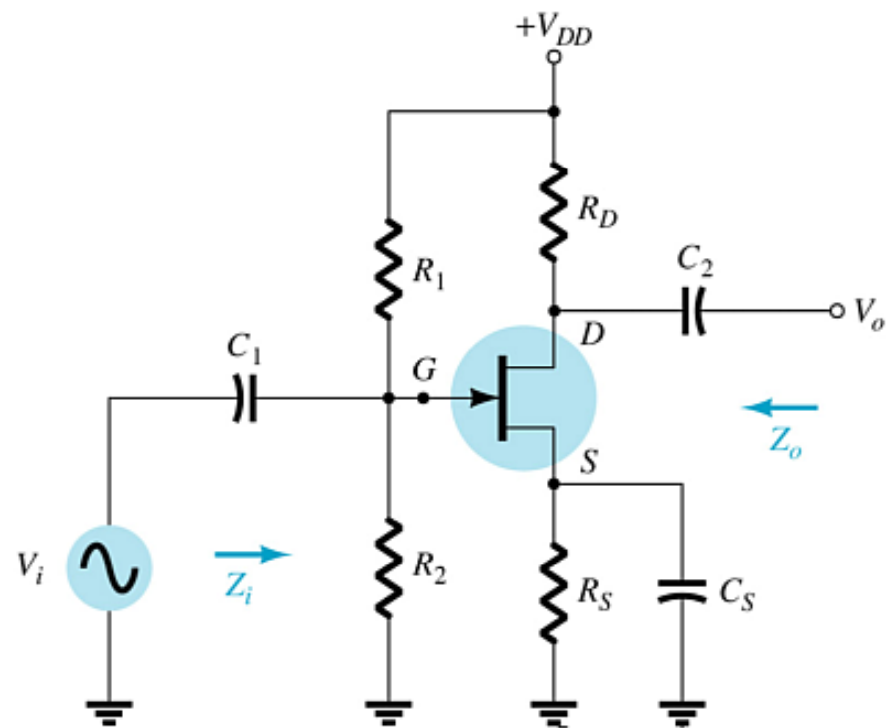
$$A_v = \frac{V_o}{V_i} = - \frac{g_m R_D}{1 + g_m R_S + \frac{R_D + R_S}{r_d}}$$

$$A_v = \frac{V_o}{V_i} = - \frac{g_m R_D}{1 + g_m R_S} \left| r_d \geq 10(R_D + R_S) \right.$$



Common-Source (CS) Voltage-Divider Bias

This is a common-source amplifier configuration, so the input is on the gate and the output is on the drain.



Impedances

Input impedance:

$$Z_i = R_1 \parallel R_2$$

Output impedance:

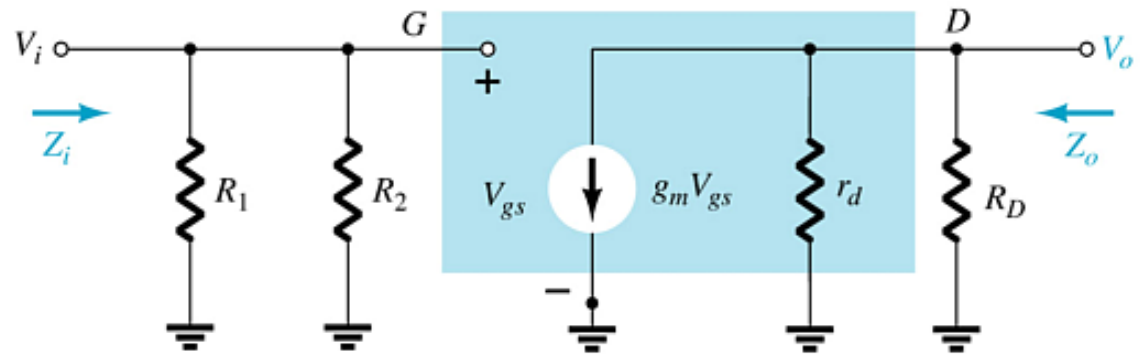
$$Z_o = r_d \parallel R_D$$

$$Z_o \cong R_D \left| \begin{array}{l} r_d \geq 10R_D \end{array} \right.$$

Voltage gain:

$$A_v = -g_m (r_d \parallel R_D)$$

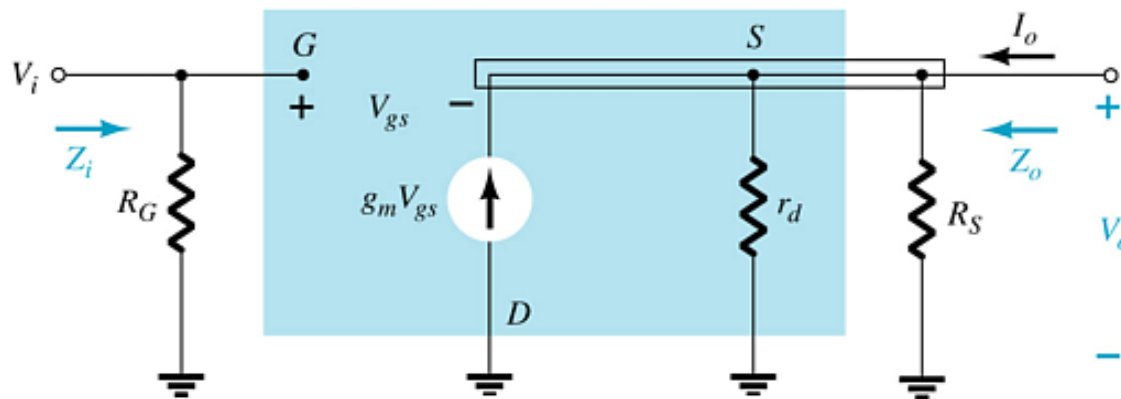
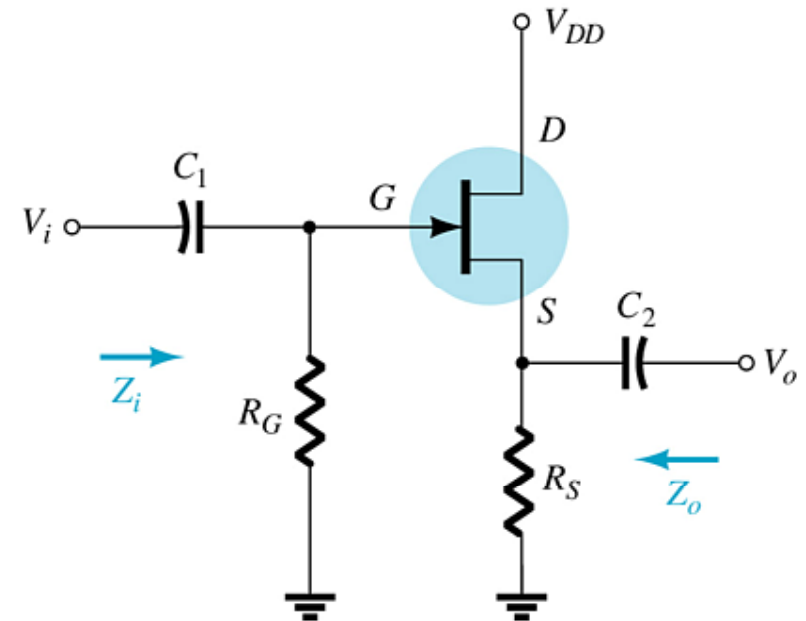
$$A_v = -g_m R_D \left| \begin{array}{l} r_d \geq 10R_D \end{array} \right.$$



Source Follower (Common-Drain) Circuit

In a common-drain amplifier configuration, the input is on the gate, but the output is from the source.

There is no phase shift between input and output.



Impedances

Input impedance:

$$Z_i = R_G$$

Output impedance:

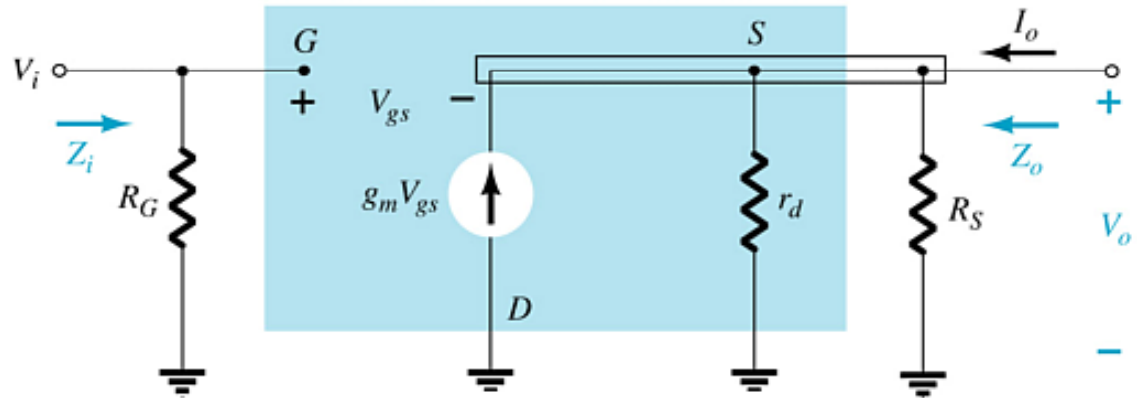
$$Z_o = r_d \parallel R_S \parallel \frac{1}{g_m}$$

$$Z_o \cong R_S \parallel \frac{1}{g_m} \Big|_{r_d \geq 10R_S}$$

Voltage gain:

$$A_v = \frac{V_o}{V_i} = \frac{g_m(r_d \parallel R_S)}{1 + g_m(r_d \parallel R_S)}$$

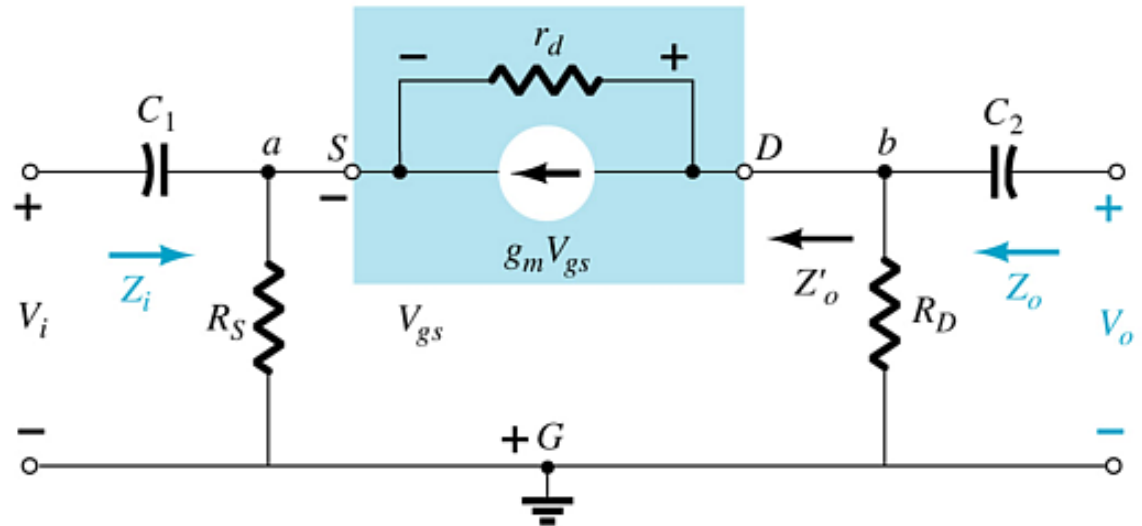
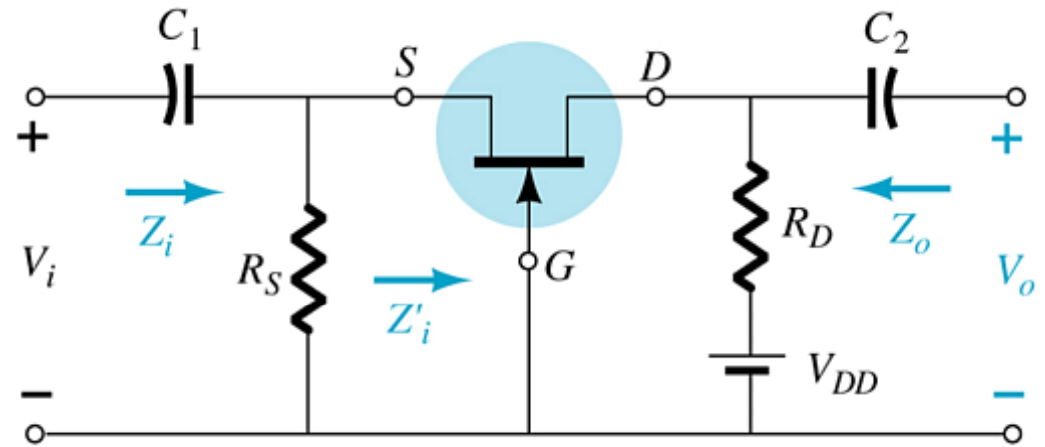
$$A_v = \frac{V_o}{V_i} = \frac{g_m R_S}{1 + g_m R_S} \Big|_{r_d \geq 10R_S}$$



Common-Gate (CG) Circuit

The input is on the source and the output is on the drain.

There is no phase shift between input and output.



Calculations

Input impedance:

$$Z_i = R_S \parallel \left[\frac{r_d + R_D}{1 + g_m r_d} \right]$$

$$Z_i \cong R_S \parallel \frac{1}{g_m} \Big|_{r_d \geq 10 R_D}$$

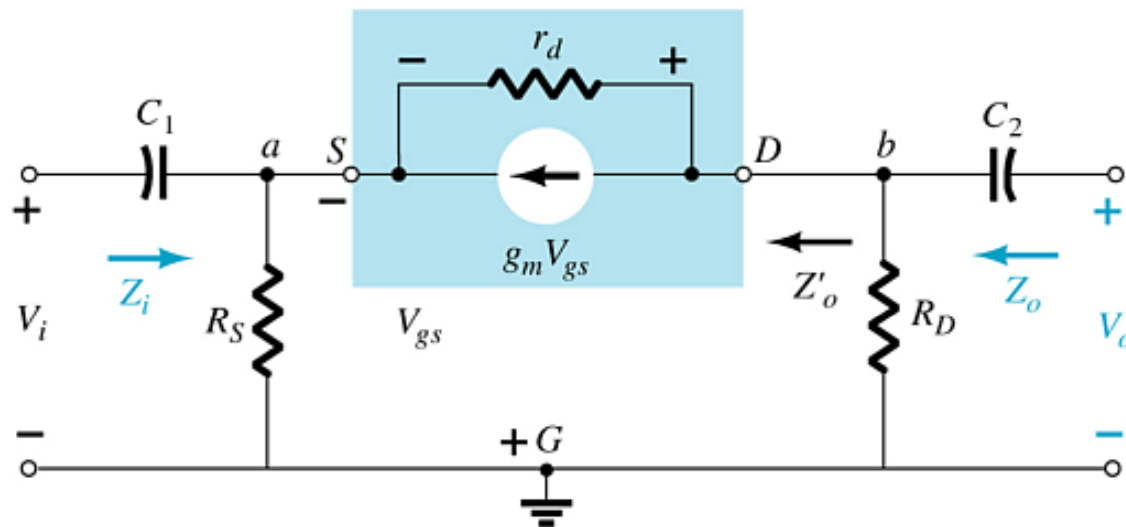
Output impedance:

$$Z_o = R_D \parallel r_d$$

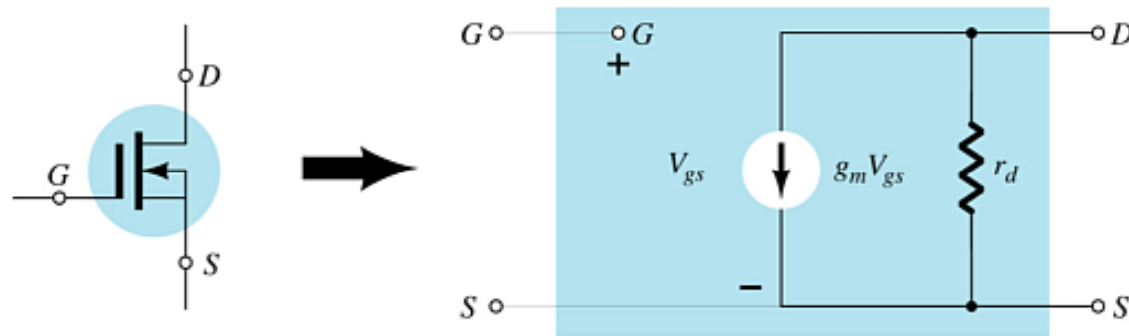
$$Z_o \cong R_D \Big|_{r_d \geq 10 R_D}$$

Voltage gain:

$$A_v = \frac{V_o}{V_i} = \frac{\left[g_m R_D + \frac{R_D}{r_d} \right]}{\left[1 + \frac{R_D}{r_d} \right]} \quad A_v = g_m R_D \Big|_{r_d \geq 10 R_D}$$

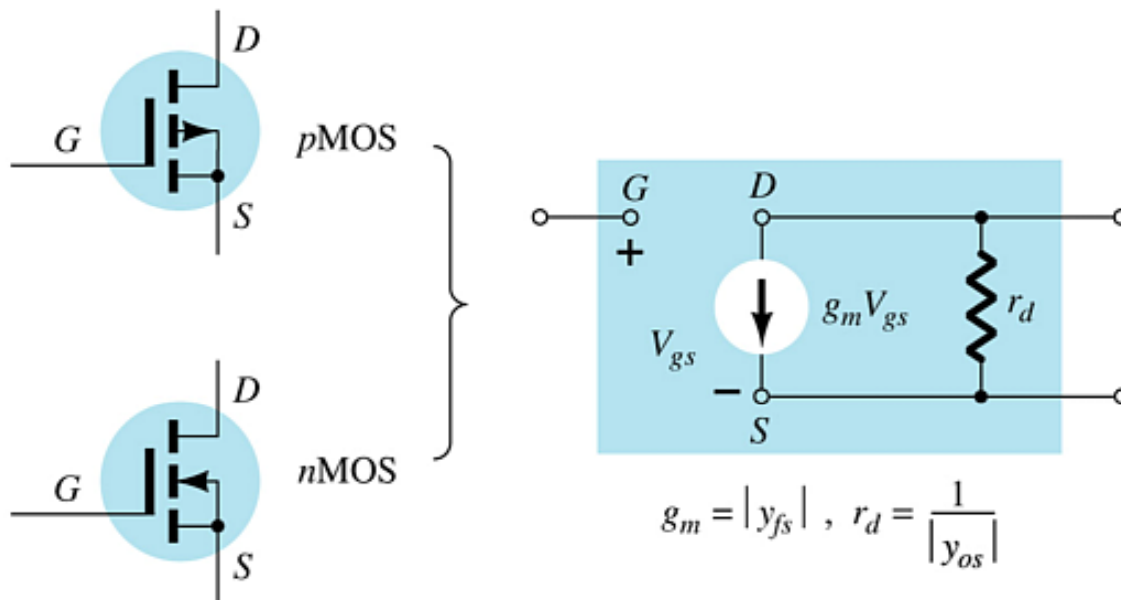


D-Type MOSFET AC Equivalent



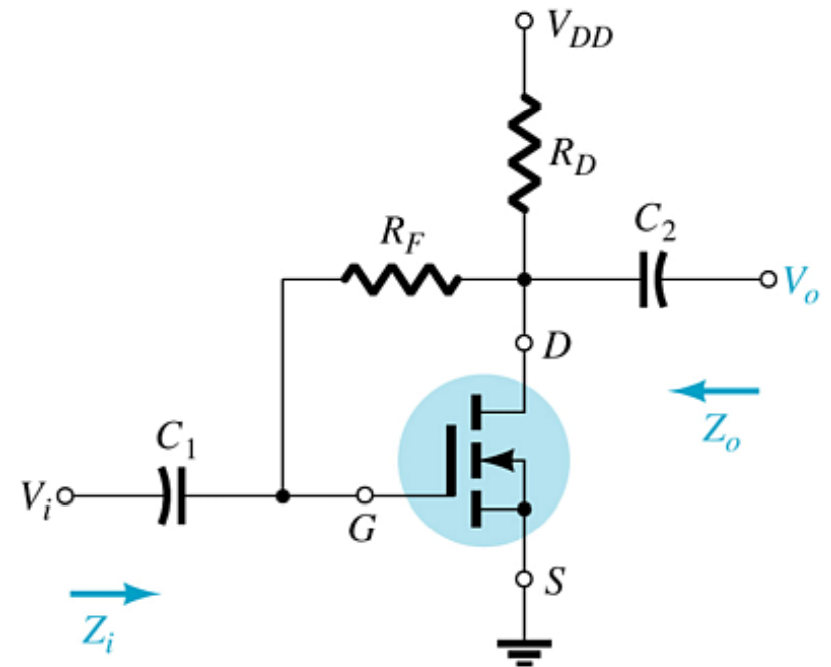
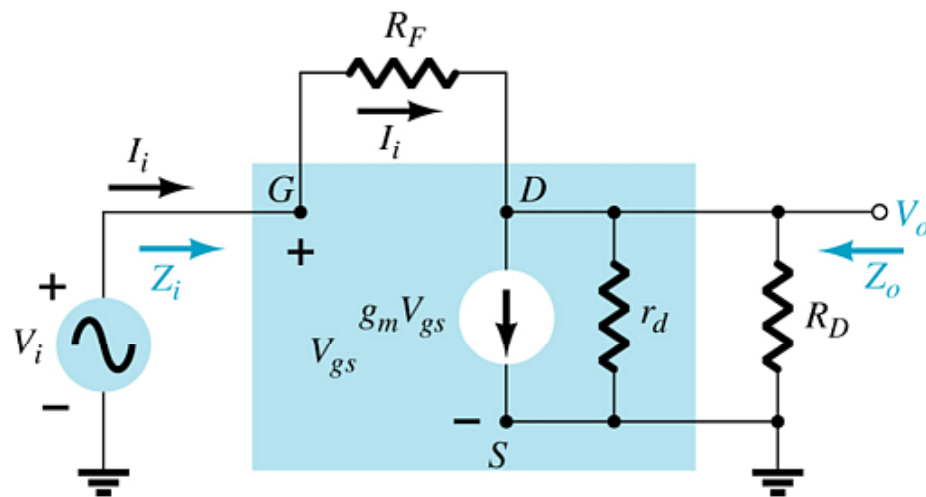
E-Type MOSFET AC Equivalent

g_m and r_d can be found in the specification sheet for the FET.



Common-Source Drain-Feedback

There is a 180° phase shift between input and output.



Calculations

Input impedance:

$$Z_i = \frac{R_F + r_d \parallel R_D}{1 + g_m(r_d \parallel R_D)}$$

$$Z_i \cong \frac{R_F}{1 + g_m R_D} \Big| R_F \gg r_d \parallel R_D, r_d \geq 10R_D$$

Output impedance:

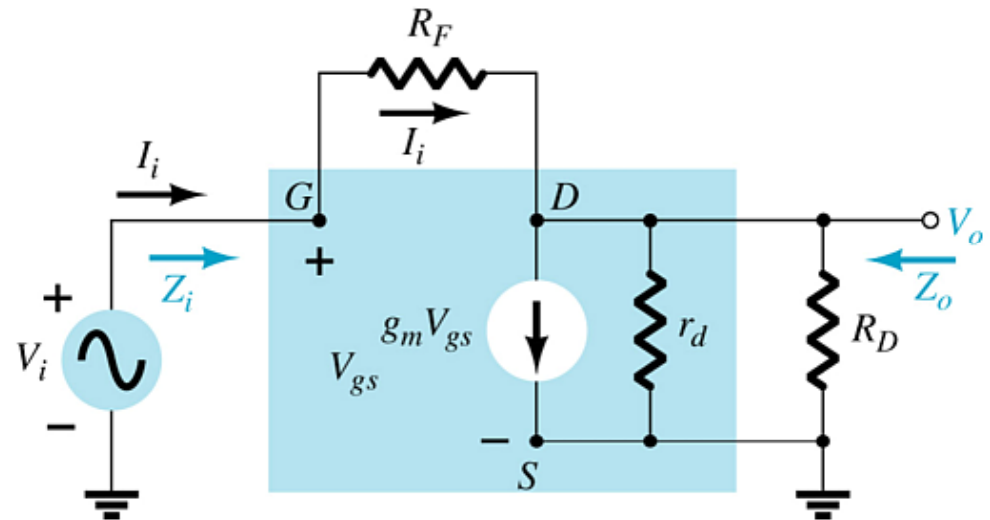
$$Z_o = R_F \parallel r_d \parallel R_D$$

$$Z_o \cong R_D \Big| R_F \gg r_d \parallel R_D, r_d \geq 10R_D$$

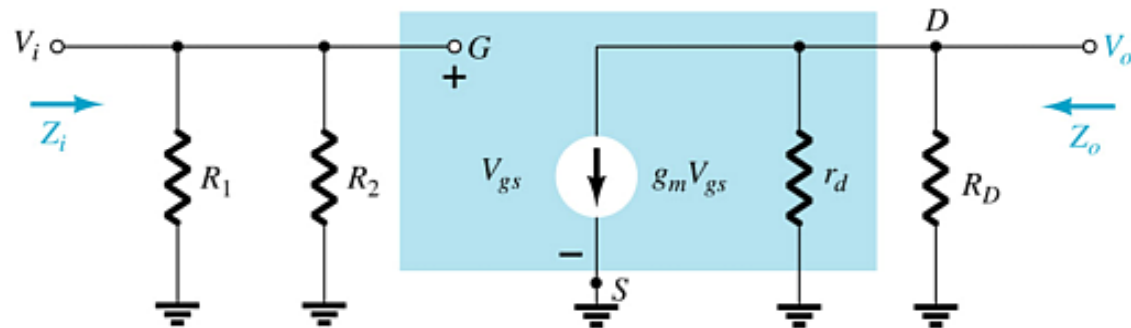
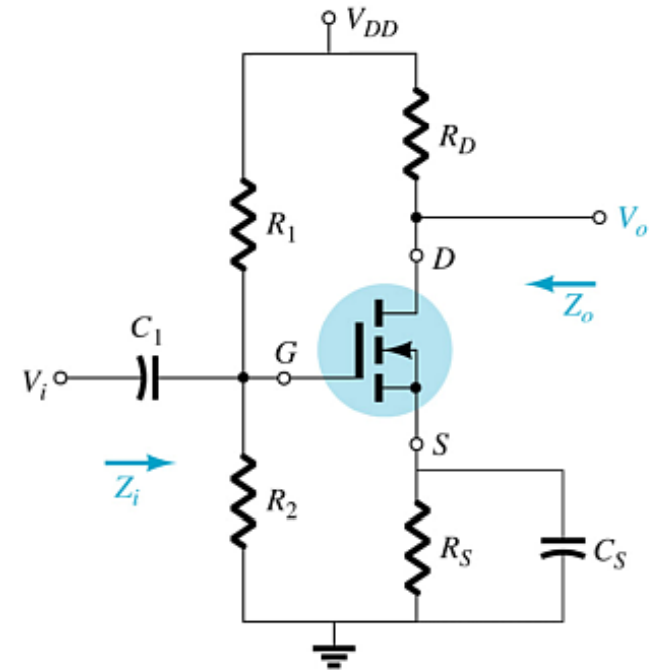
Voltage gain:

$$A_v = -g_m(R_F \parallel r_d \parallel R_D)$$

$$A_v \cong -g_m R_D \Big| R_F \gg r_d \parallel R_D, r_d \geq 10R_D$$



Common-Source Voltage-Divider Bias



Calculations

Input impedance:

$$Z_i = R_1 \parallel R_2$$

Output impedance:

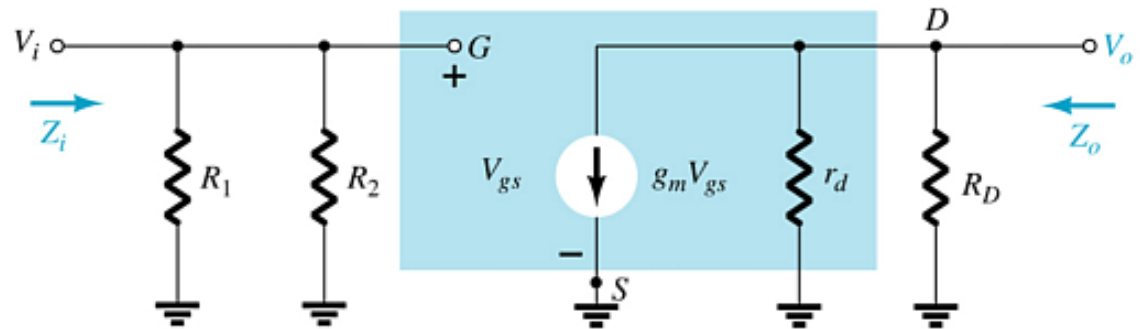
$$Z_o = r_d \parallel R_D$$

$$Z_o \cong R_D \big|_{r_d \geq 10R_D}$$

Voltage gain:

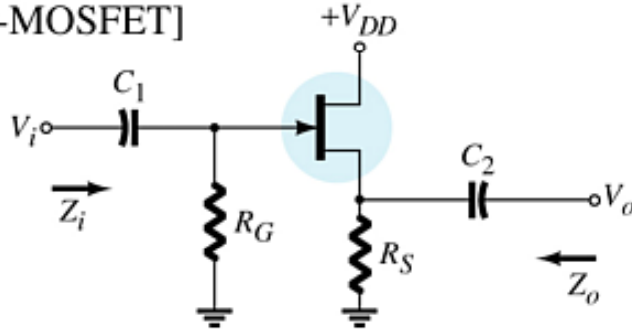
$$A_v = -g_m (r_d \parallel R_D)$$

$$A_v \cong -g_m R_D \big|_{r_d \geq 10R_D}$$

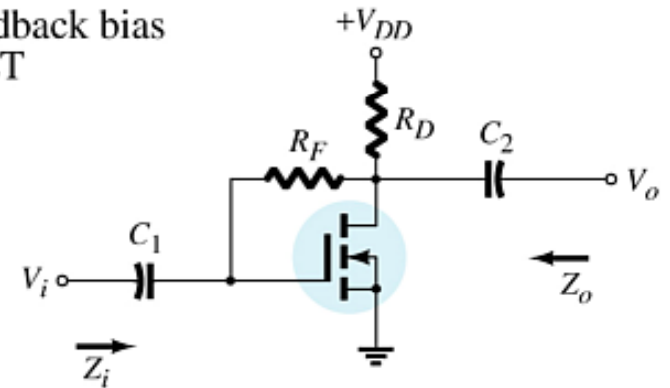


Summary Table

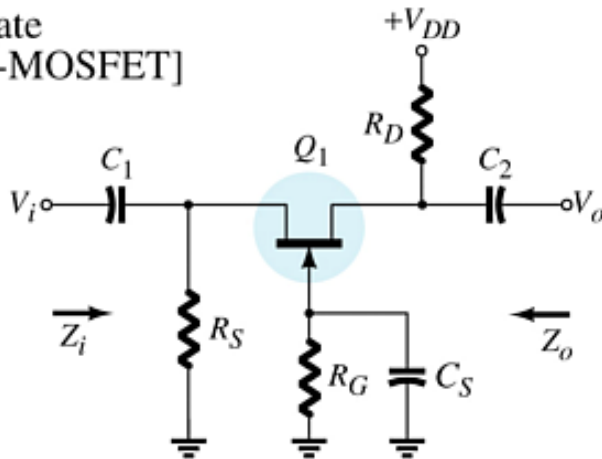
Source-follower
[JFET or D-MOSFET]



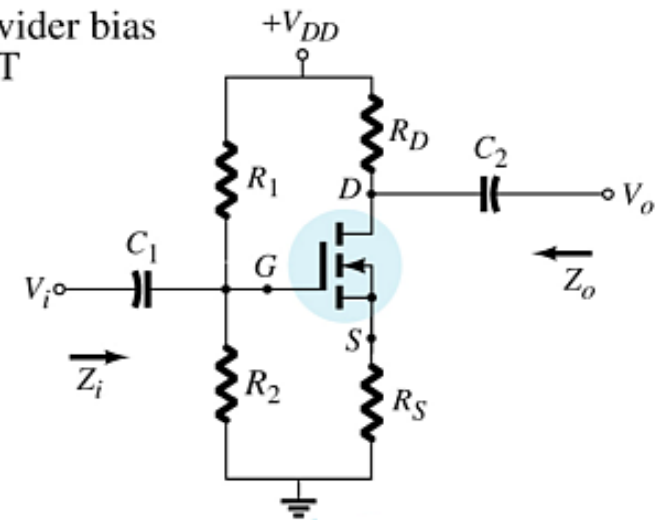
Drain-Feedback bias
E-MOSFET



Common-gate
[JFET or D-MOSFET]



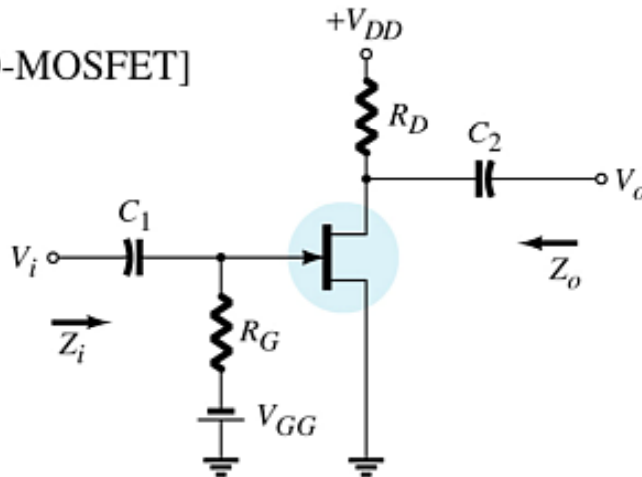
Voltage-divider bias
E-MOSFET



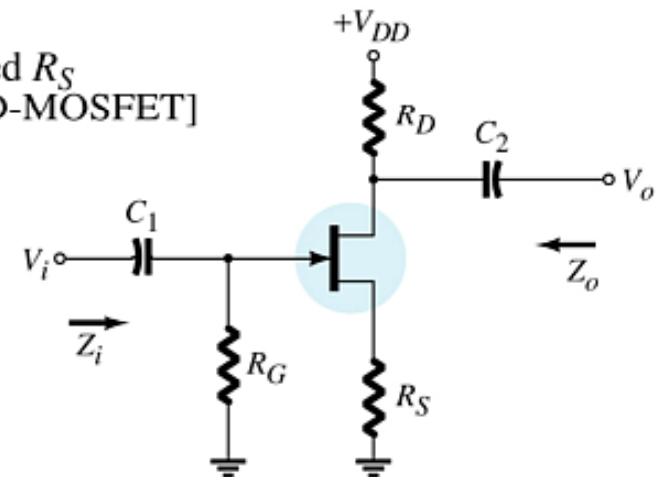
more...

Summary Table

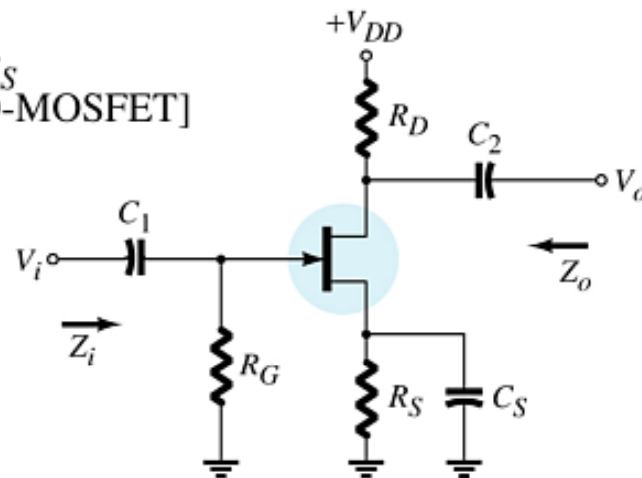
Fixed-bias
[JFET or D-MOSFET]



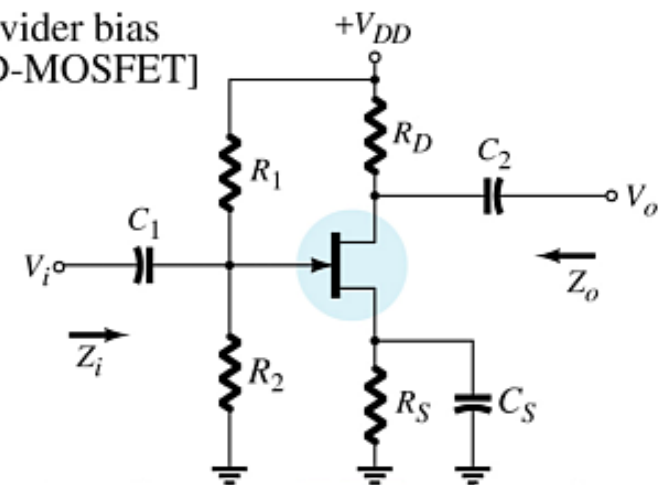
Self-bias
Unbypassed R_S
[JFET or D-MOSFET]



Self-bias
bypassed R_S
[JFET or D-MOSFET]



Voltage-divider bias
[JFET or D-MOSFET]



Troubleshooting

Check the DC bias voltages:

If not correct check power supply, resistors, FET. Also check to ensure that the coupling capacitor between amplifier stages is OK.

Check the AC voltages:

If not correct check FET, capacitors and the loading effect of the next stage

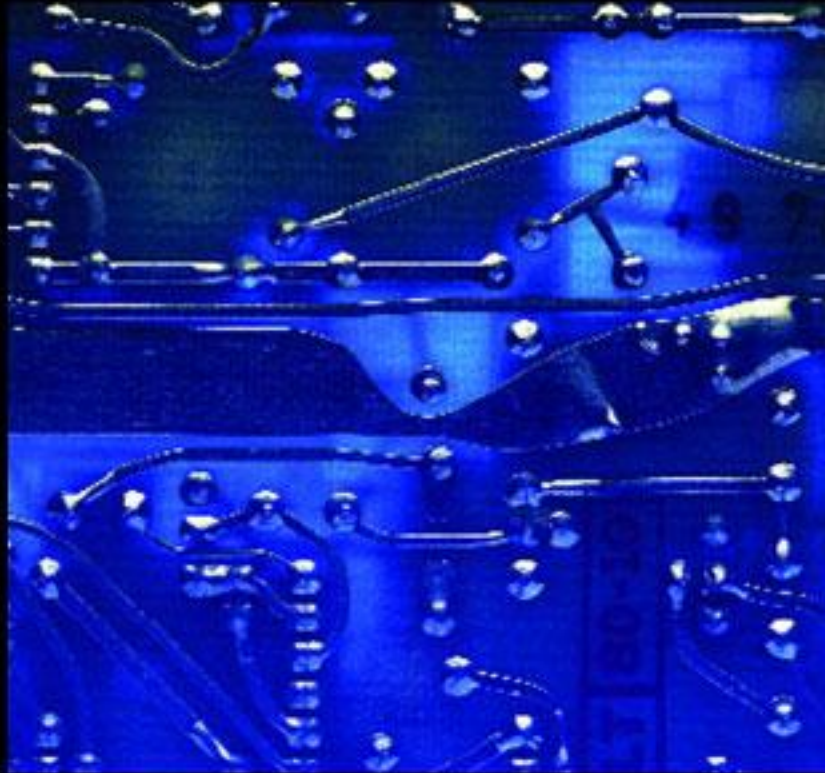
Practical Applications

Three-Channel Audio Mixer
Silent Switching
Phase Shift Networks
Motion Detection System

ELECTRONIC DEVICES AND CIRCUIT THEORY

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PEARSON

Chapter 9: BJT and FET Frequency Response

General Frequency Considerations

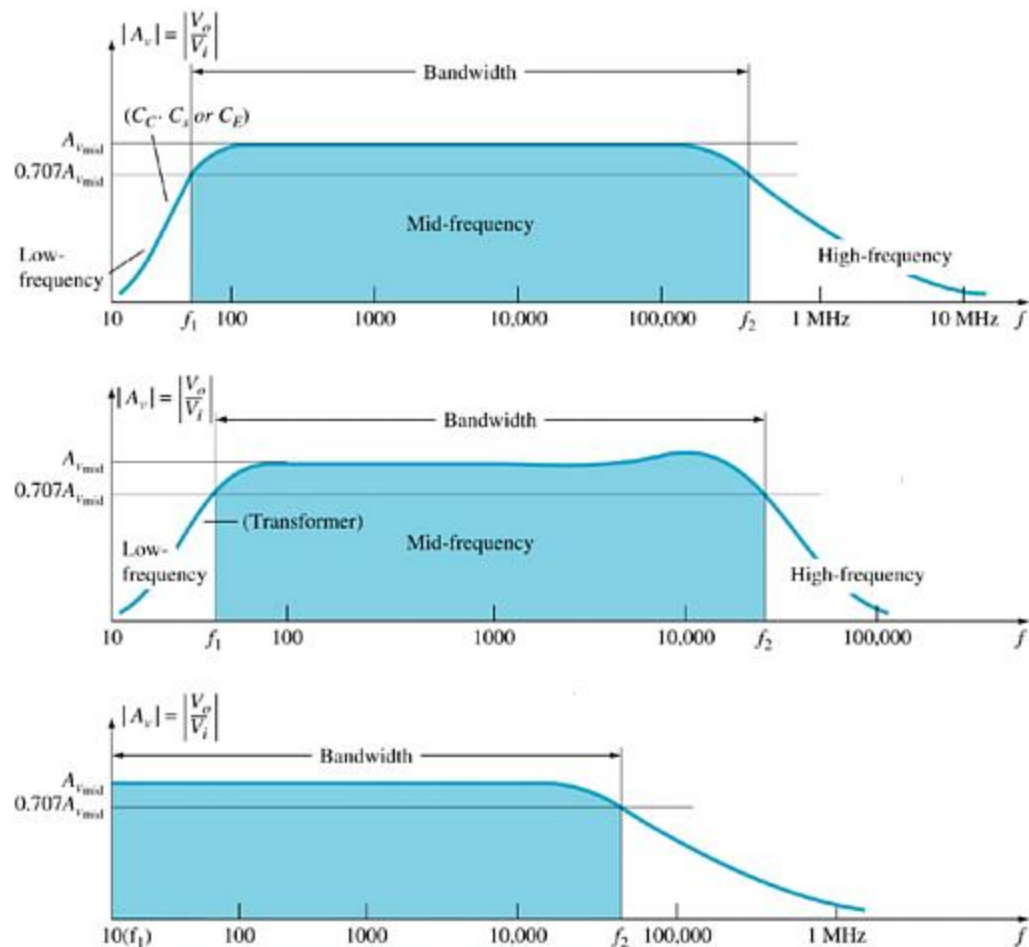
The **frequency response** of an amplifier refers to the frequency range in which the amplifier will operate with negligible effects from capacitors and device internal capacitance. This range of frequencies can be called the **mid-range**.

- At frequencies above and below the midrange, capacitance and any inductance will affect the gain of the amplifier.
- At low frequencies the coupling and bypass capacitors lower the gain.
- At high frequencies stray capacitances associated with the active device lower the gain.
- Also, cascading amplifiers limits the gain at high and low frequencies.

Bode Plot

A Bode plot indicates the frequency response of an amplifier.

The horizontal scale indicates the frequency (in Hz) and the vertical scale indicates the gain (in dB).

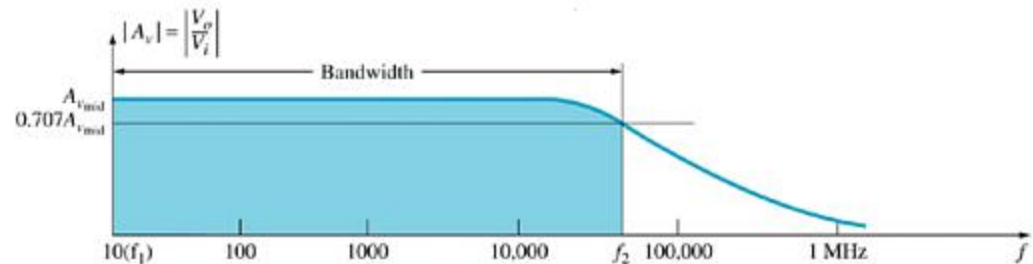
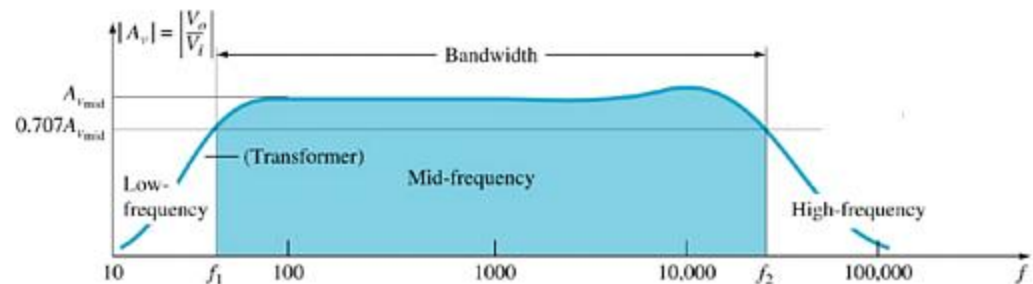
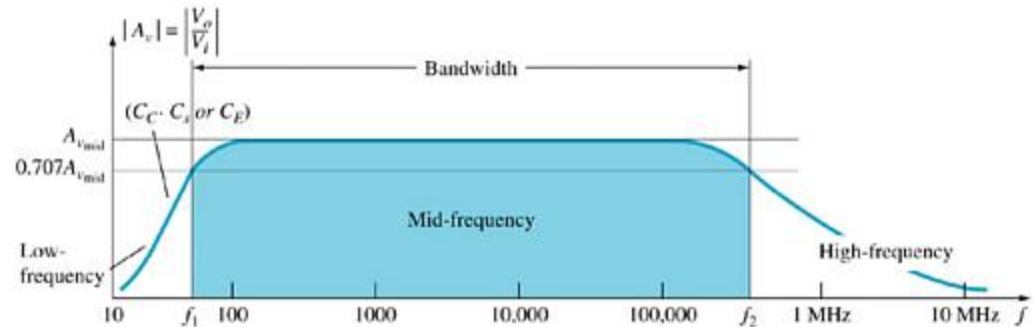


Cutoff Frequencies

The mid-range frequency range of an amplifier is called the bandwidth of the amplifier.

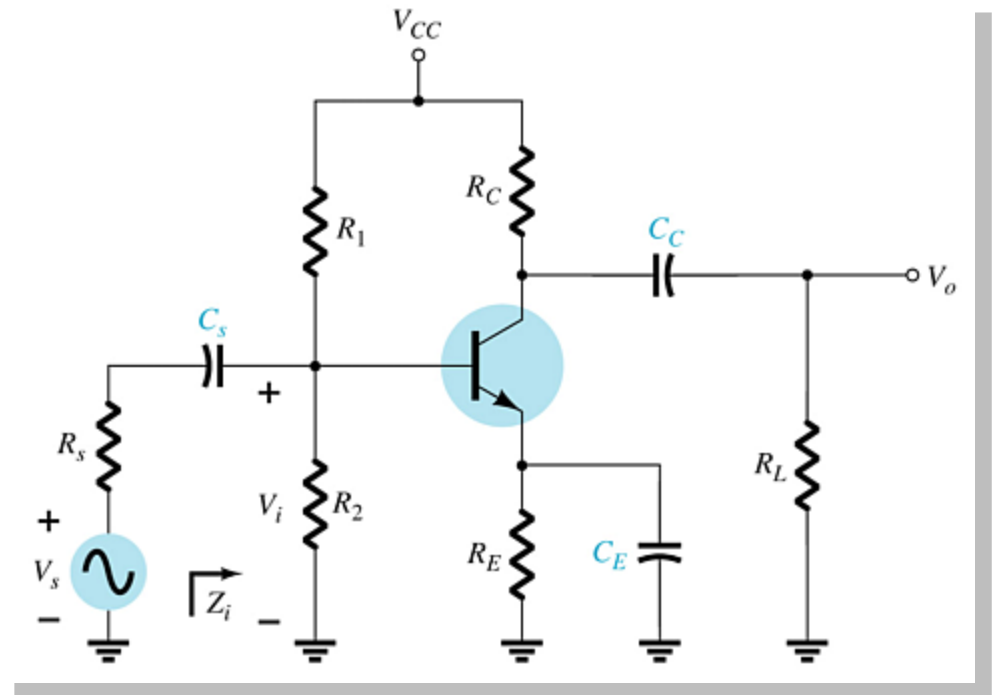
The **bandwidth** is defined by the lower and upper cutoff frequencies.

Cutoff – any frequency at which the gain has dropped by 3 dB.



BJT Amplifier Low-Frequency Response

At low frequencies, coupling capacitor (C_S , C_C) and bypass capacitor (C_E) reactances affect the circuit impedances.



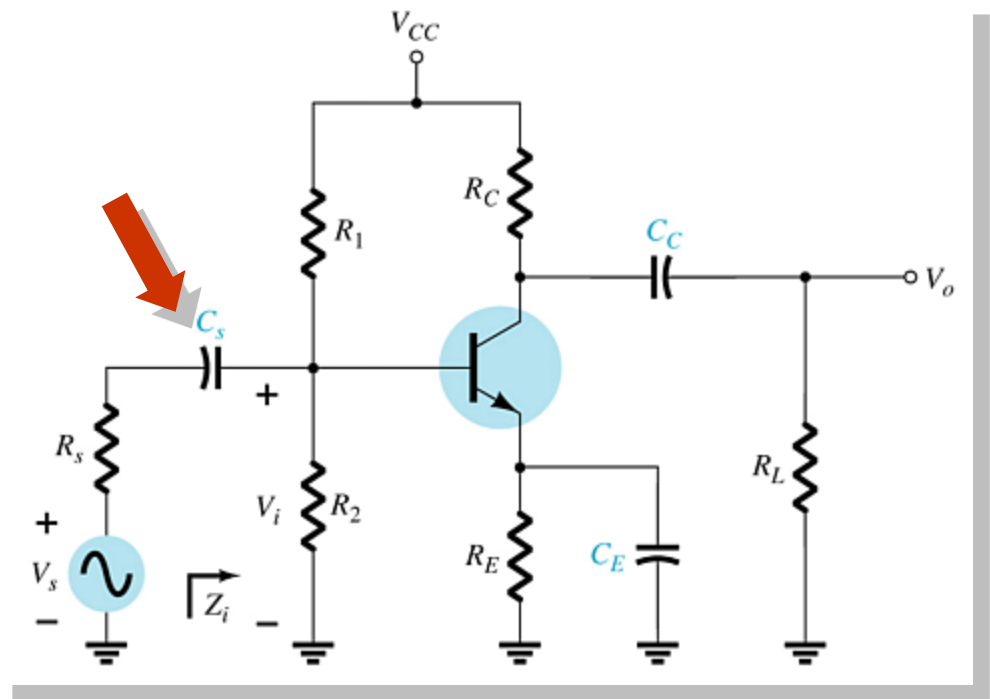
Coupling Capacitor (C_s)

The cutoff frequency due to C_s can be calculated by

$$f_{Ls} = \frac{1}{2\pi(R_s + R_i)C_s}$$

where

$$R_i = R_1 \parallel R_2 \parallel \beta r_e$$



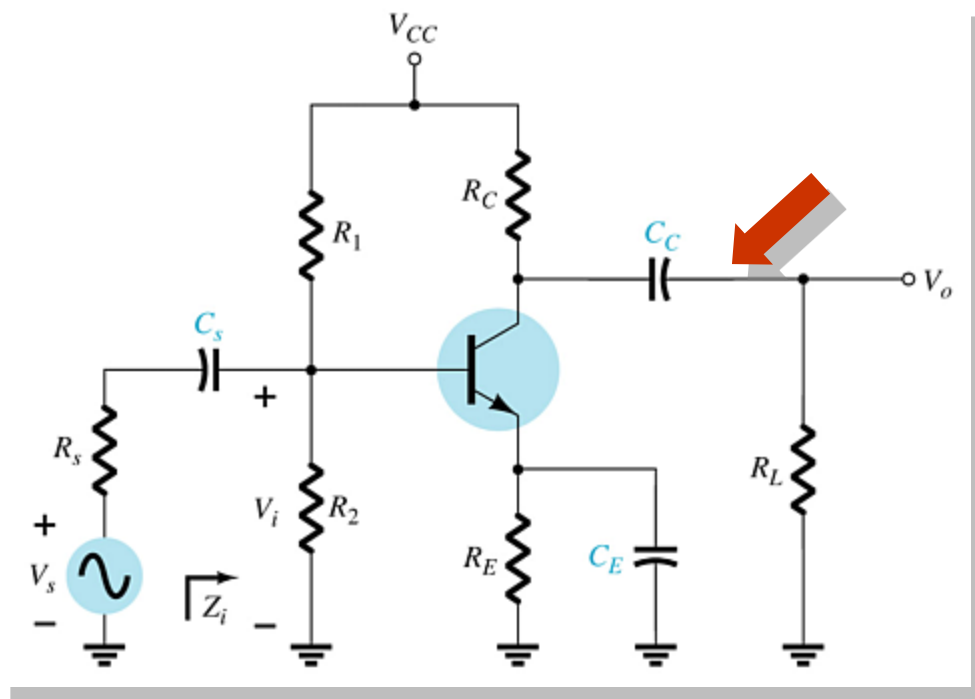
Coupling Capacitor (C_C)

The cutoff frequency due to C_C can be calculated with

$$f_{LC} = \frac{1}{2\pi(R_o + R_L)C_C}$$

where

$$R_o = R_C \parallel r_o$$



Bypass Capacitor (C_E)

The cutoff frequency due to C_E can be calculated with

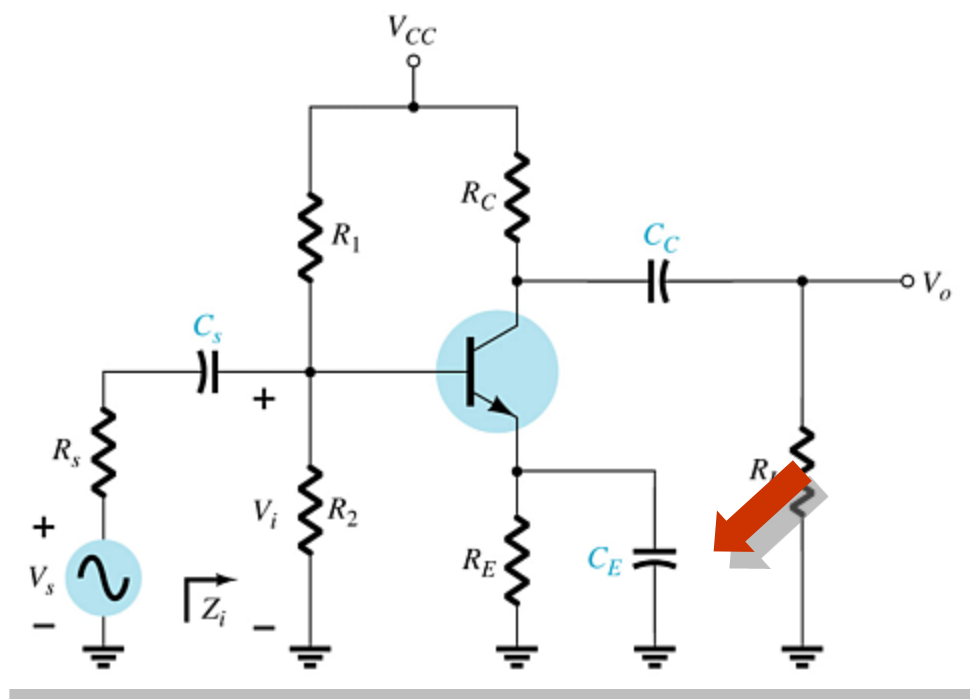
$$f_{LE} = \frac{1}{2\pi R_e C_E}$$

where

$$R_e = R_E \parallel \left(\frac{R'_s}{\beta} + r_e \right)$$

and

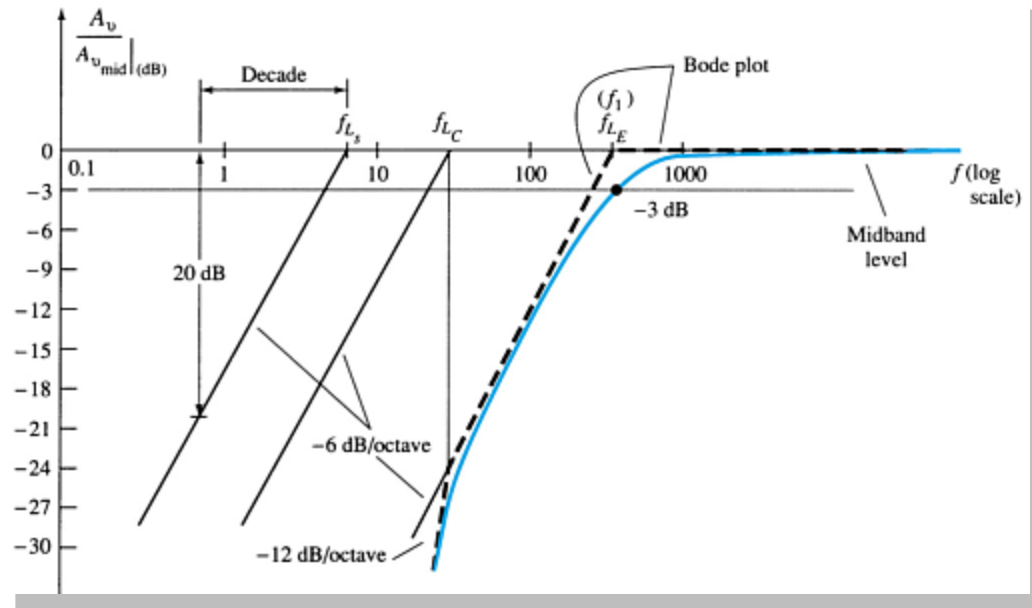
$$R'_s = R_s \parallel R_1 \parallel R_2$$



BJT Amplifier Low-Frequency Response

The Bode plot indicates that each capacitor may have a different cutoff frequency.

It is the device that has the *highest* lower cutoff frequency (f_L) that dominates the overall frequency response of the amplifier.

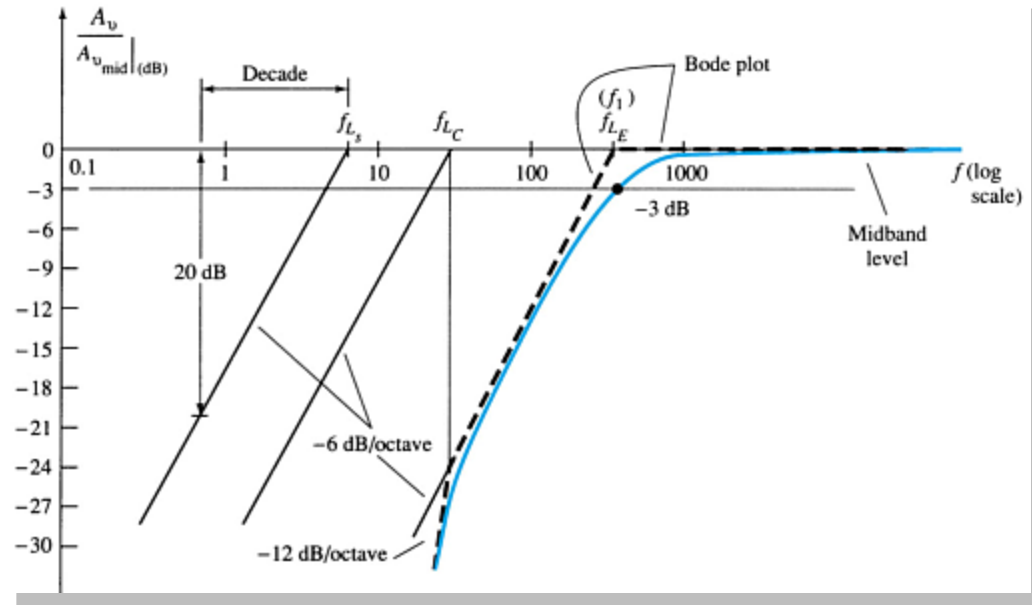


Roll-Off of Gain in the Bode Plot

The Bode plot not only indicates the cutoff frequencies of the various capacitors it also indicates the amount of attenuation (loss in gain) at these frequencies.

The amount of attenuation is sometimes referred to as roll-off.

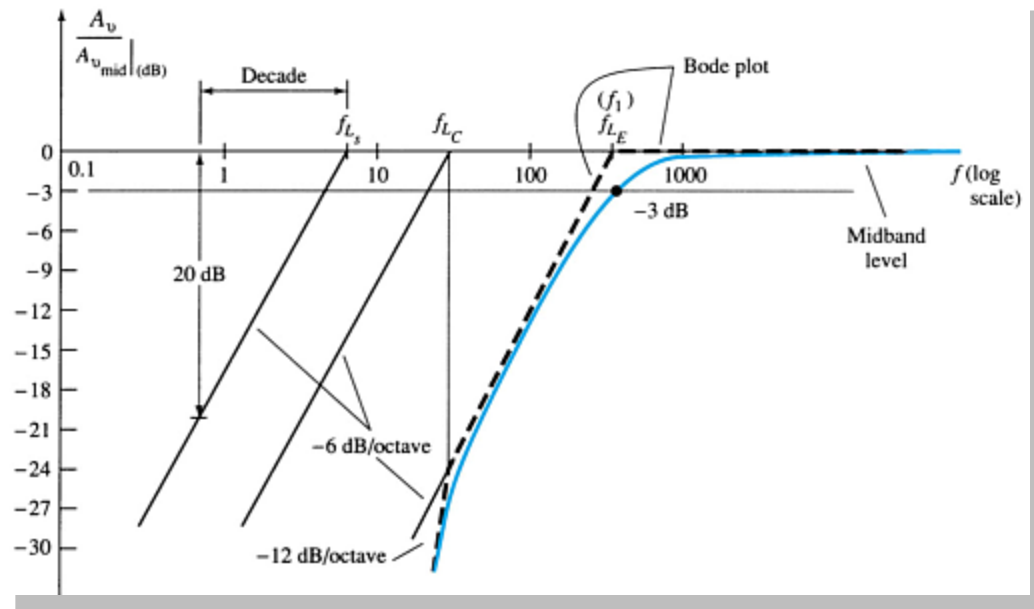
The roll-off is described as dB loss-per-octave or dB loss-per-decade.



Roll-off Rate (-dB/Decade)

-dB/decade refers to the attenuation for every 10-fold change in frequency.

For attenuations at the low-frequency end, it refers to the loss in gain from the lower cutoff frequency to a frequency that is one-tenth the cutoff value.



In this example:

$f_{LS} = 9\text{kHz}$ gain is 0dB

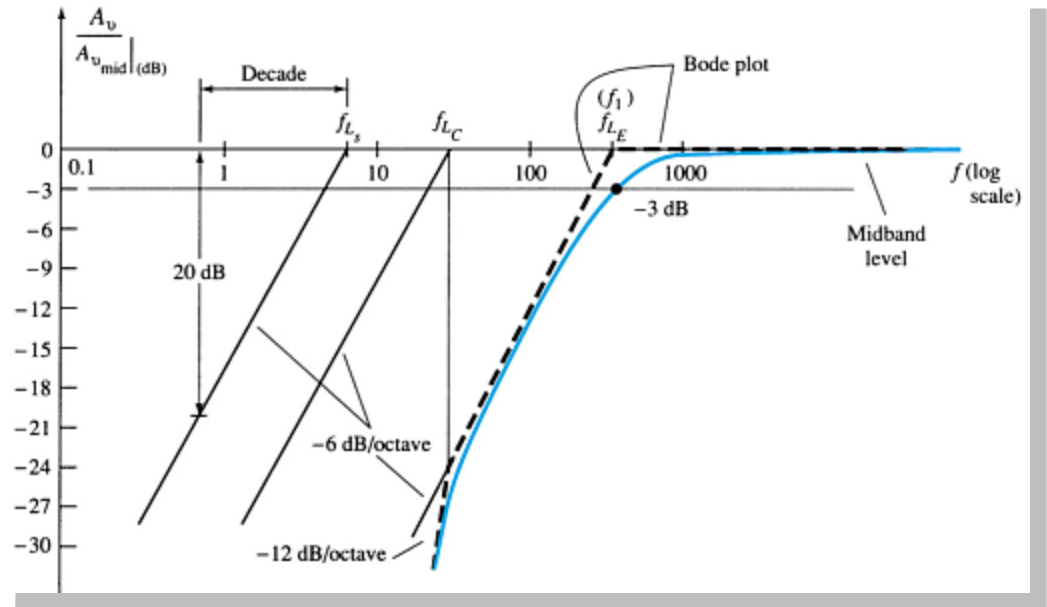
$f_{LS}/10 = .9\text{kHz}$ gain is -20dB

Thus the roll-off is 20dB/decade

The gain decreases by -20dB/decade

Roll-Off Rate (–dB/Octave)

–dB/octave refers to the attenuation for every 2-fold change in frequency. For attenuations at the low-frequency end, it refers to the loss in gain from the lower cutoff frequency to a frequency one-half the cutoff value.



In this example:

$f_{LS} = 9\text{kHz}$ gain is 0dB

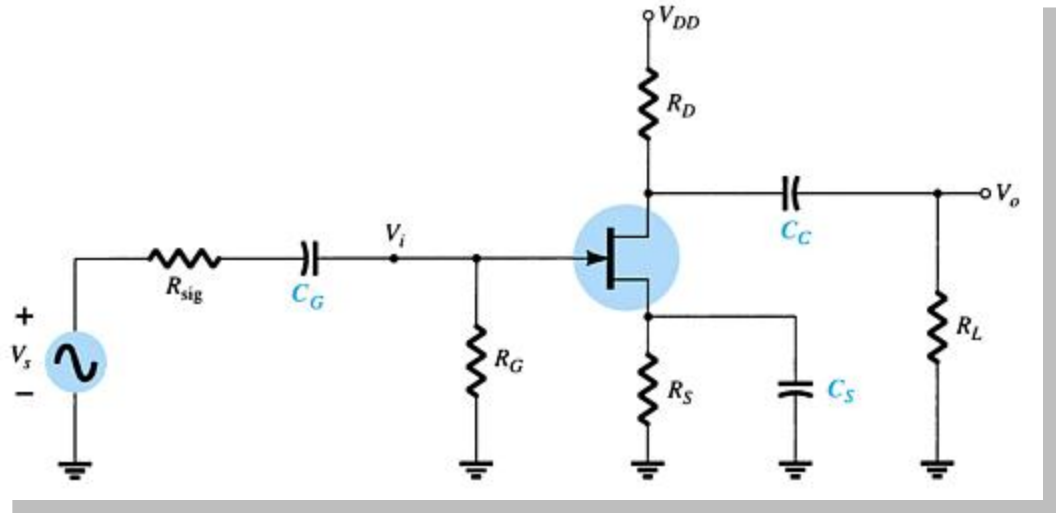
$f_{LS} / 2 = 4.5\text{kHz}$ gain is –6dB

Therefore the roll-off is 6dB/octave.

This is a little difficult to see on this graph because the horizontal scale is a logarithmic scale.

FET Amplifier Low-Frequency Response

At low frequencies, coupling capacitor (C_G , C_C) and bypass capacitor (C_S) reactances affect the circuit impedances.



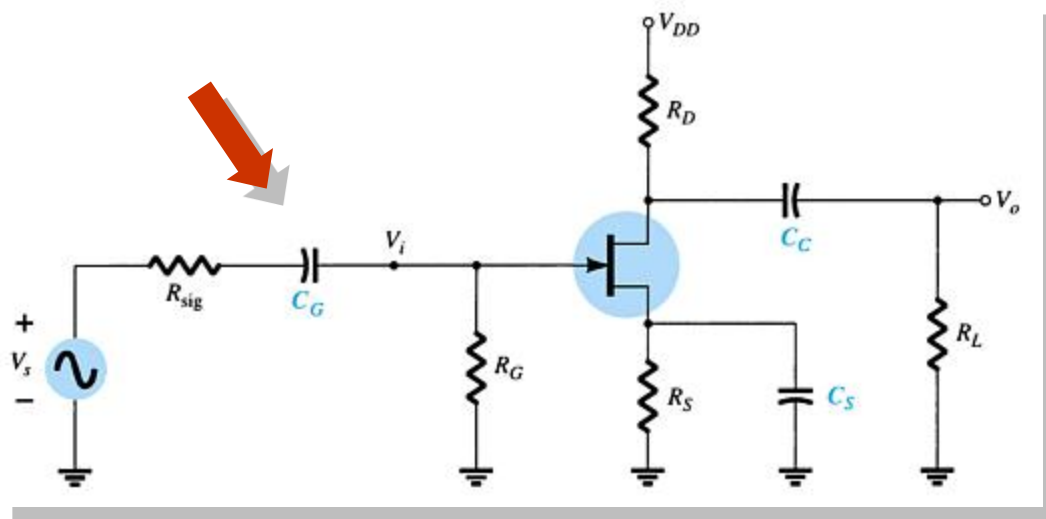
Coupling Capacitor (C_G)

The cutoff frequency due to C_G can be calculated with

$$f_{LC} = \frac{1}{2\pi(R_{sig} + R_i)C_G}$$

where

$$R_i = R_G$$



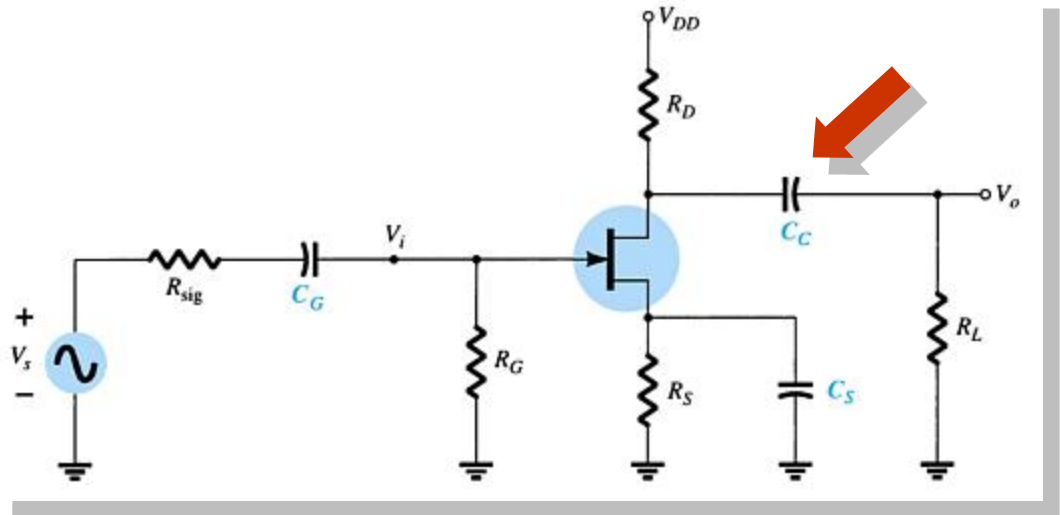
Coupling Capacitor (C_C)

The cutoff frequency due to C_C can be calculated with

$$f_{LC} = \frac{1}{2\pi(R_o + R_L)C_C}$$

where

$$R_o = R_D \parallel r_d$$



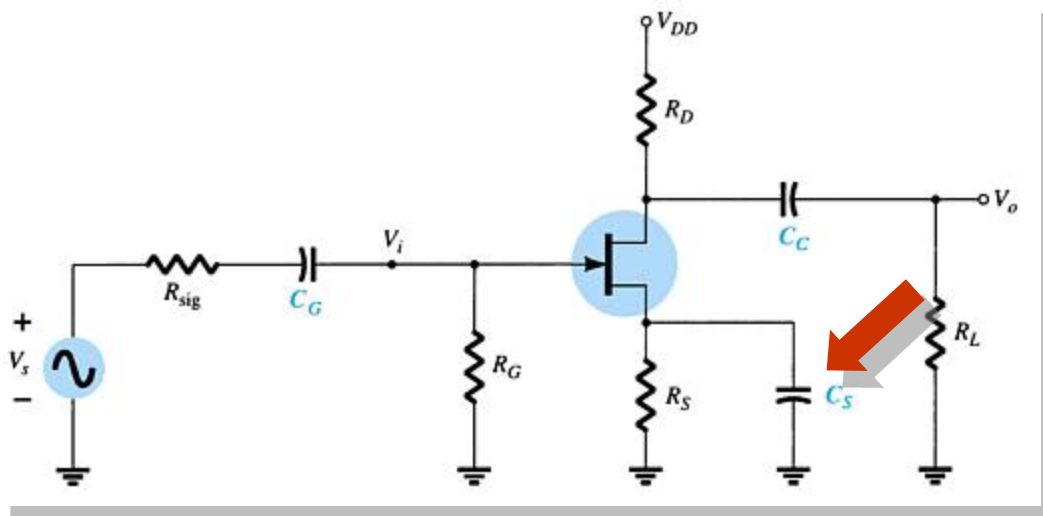
Bypass Capacitor (C_S)

The cutoff frequency due to C_S can be calculated with

$$f_{LS} = \frac{1}{2\pi R_{eq} C_S}$$

where

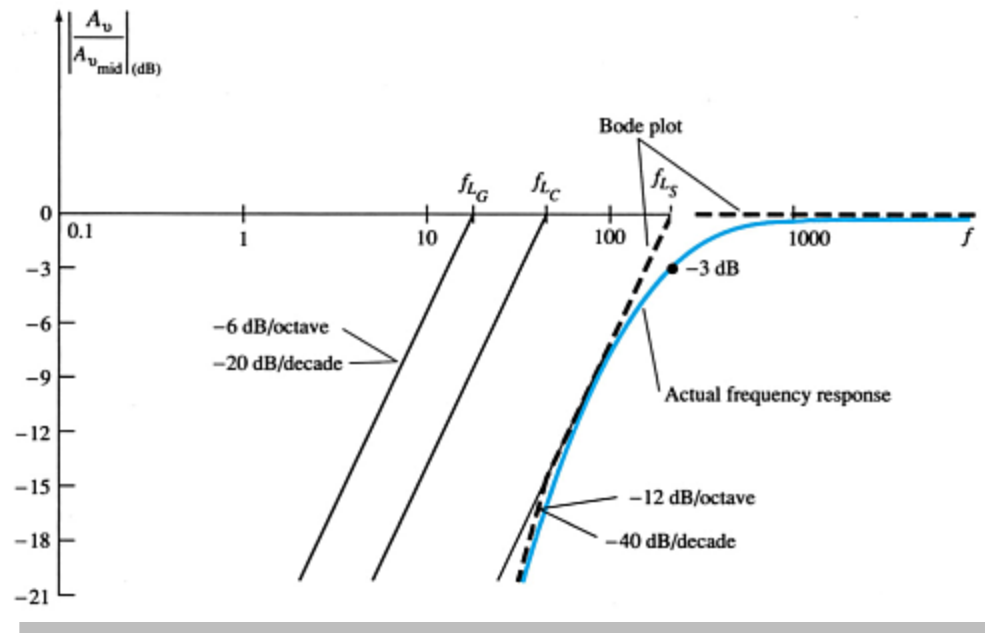
$$R_{eq} = R_S \parallel \left. \frac{1}{g_m} \right|_{r_d \cong \infty \Omega}$$



FET Amplifier Low-Frequency Response

The Bode plot indicates that each capacitor may have a different cutoff frequency.

The capacitor that has the *highest* lower cutoff frequency (f_L) is closest to the actual cutoff frequency of the amplifier.



Miller Capacitance

Any p - n junction can develop capacitance. In a BJT amplifier, this capacitance becomes noticeable across:

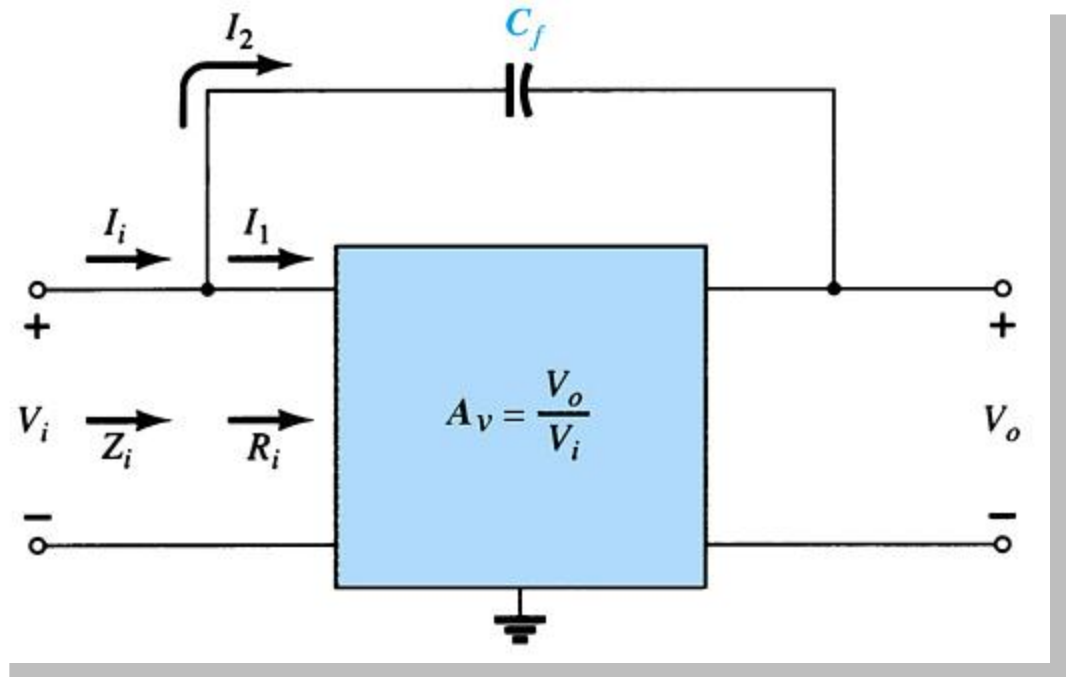
- **The base-collector junction at high frequencies in common-emitter BJT amplifier configurations**
- **The gate-drain junction at high frequencies in common-source FET amplifier configurations.**

These capacitances are represented as separate input and output capacitances, called the **Miller Capacitances.**

Miller Input Capacitance (C_{Mi})

$$C_{Mi} = (1 - A_v)C_f$$

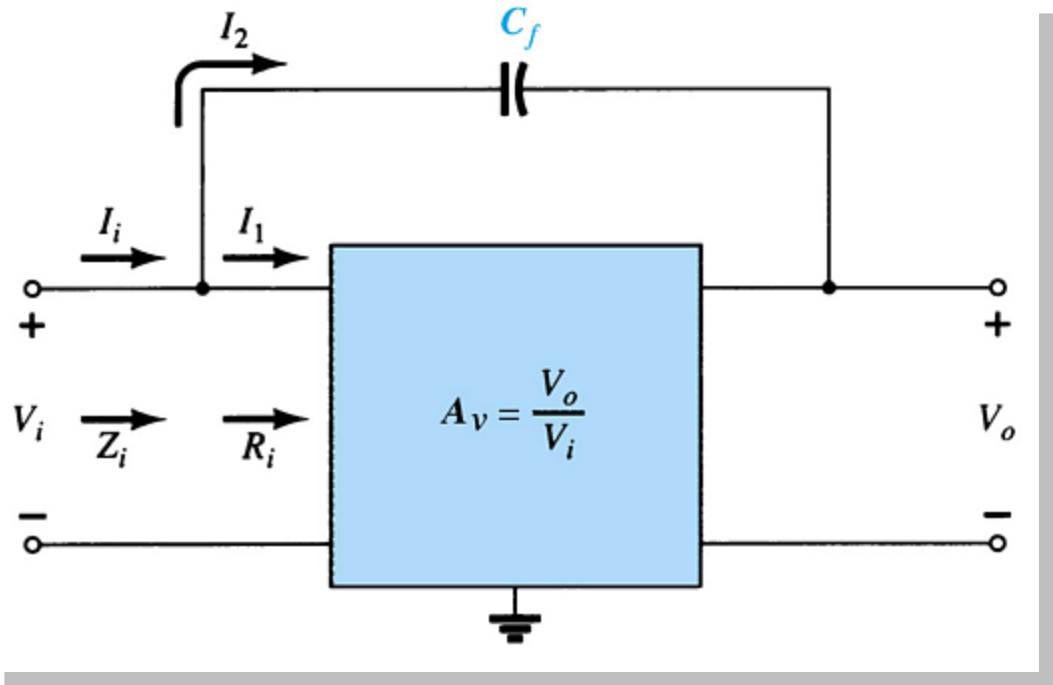
Note that the amount of Miller capacitance is dependent on inter-electrode capacitance from input to output (C_f) and the gain (A_v).



Miller Output Capacitance (C_{Mo})

If the gain (A_v) is considerably greater than 1, then

$$C_{Mo} \cong C_f$$



BJT Amplifier High-Frequency Response

Capacitances that affect the high-frequency response are

- Junction capacitances

$$C_{be}, C_{bc}, C_{ce}$$

- Wiring capacitances

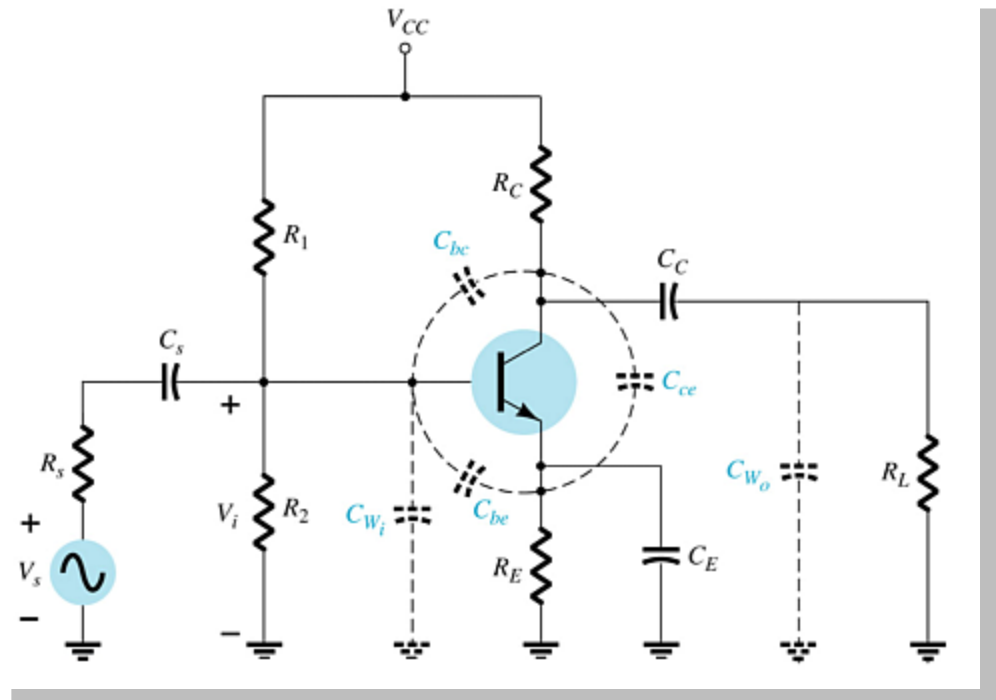
$$C_{wi}, C_{wo}$$

- Coupling capacitors

$$C_S, C_C$$

- Bypass capacitor

$$C_E$$



Input Network (f_{Hi}) High-Frequency Cutoff

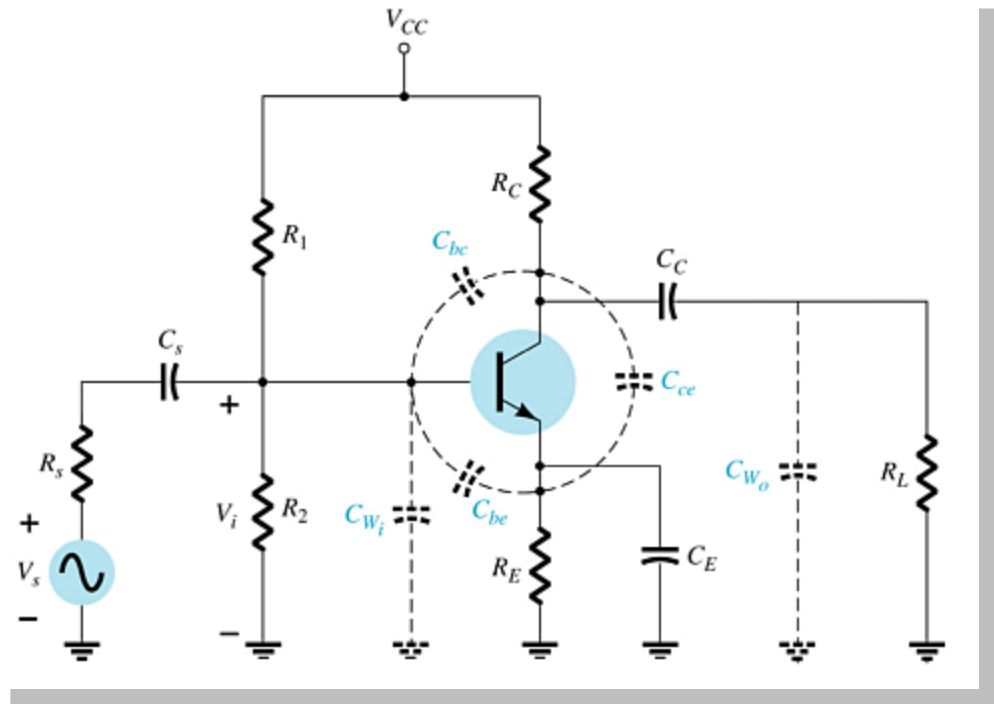
$$f_{Hi} = \frac{1}{2\pi R_{Thi} C_i}$$

where

$$R_{Thi} = R_s \parallel R_1 \parallel R_2 \parallel R_i$$

and

$$\begin{aligned} C_i &= C_{Wi} + C_{be} + C_{Mi} \\ &= C_{Wi} + C_{be} + (1 - A_v) C_{bc} \end{aligned}$$



Output Network (f_{H_o}) High-Frequency Cutoff

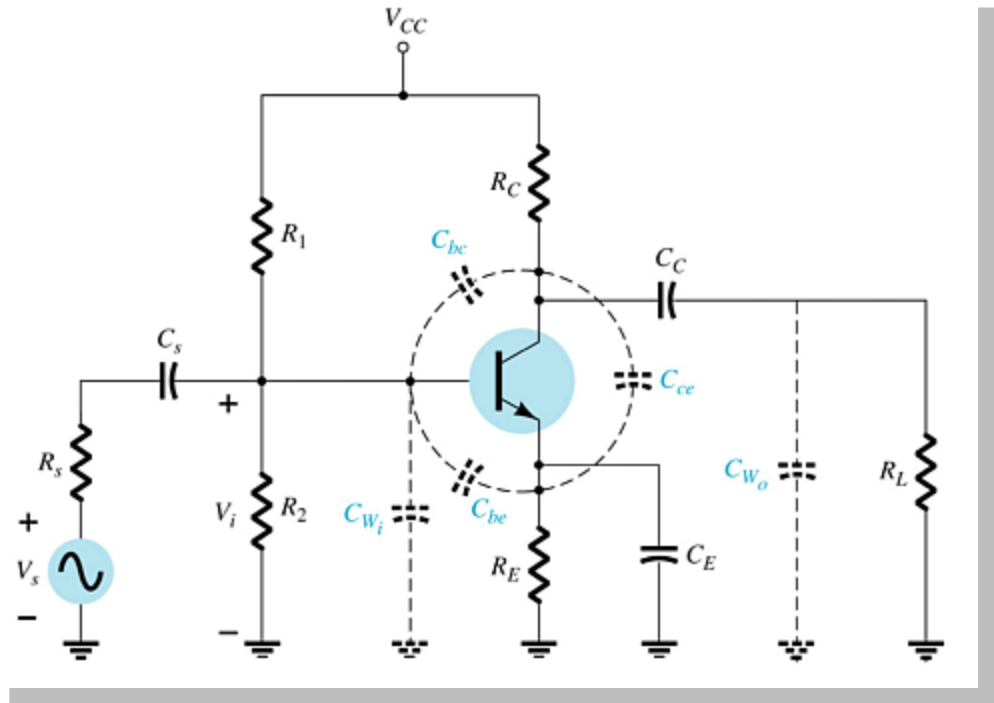
$$f_{H_o} = \frac{1}{2\pi R_{Th_o} C_o}$$

where

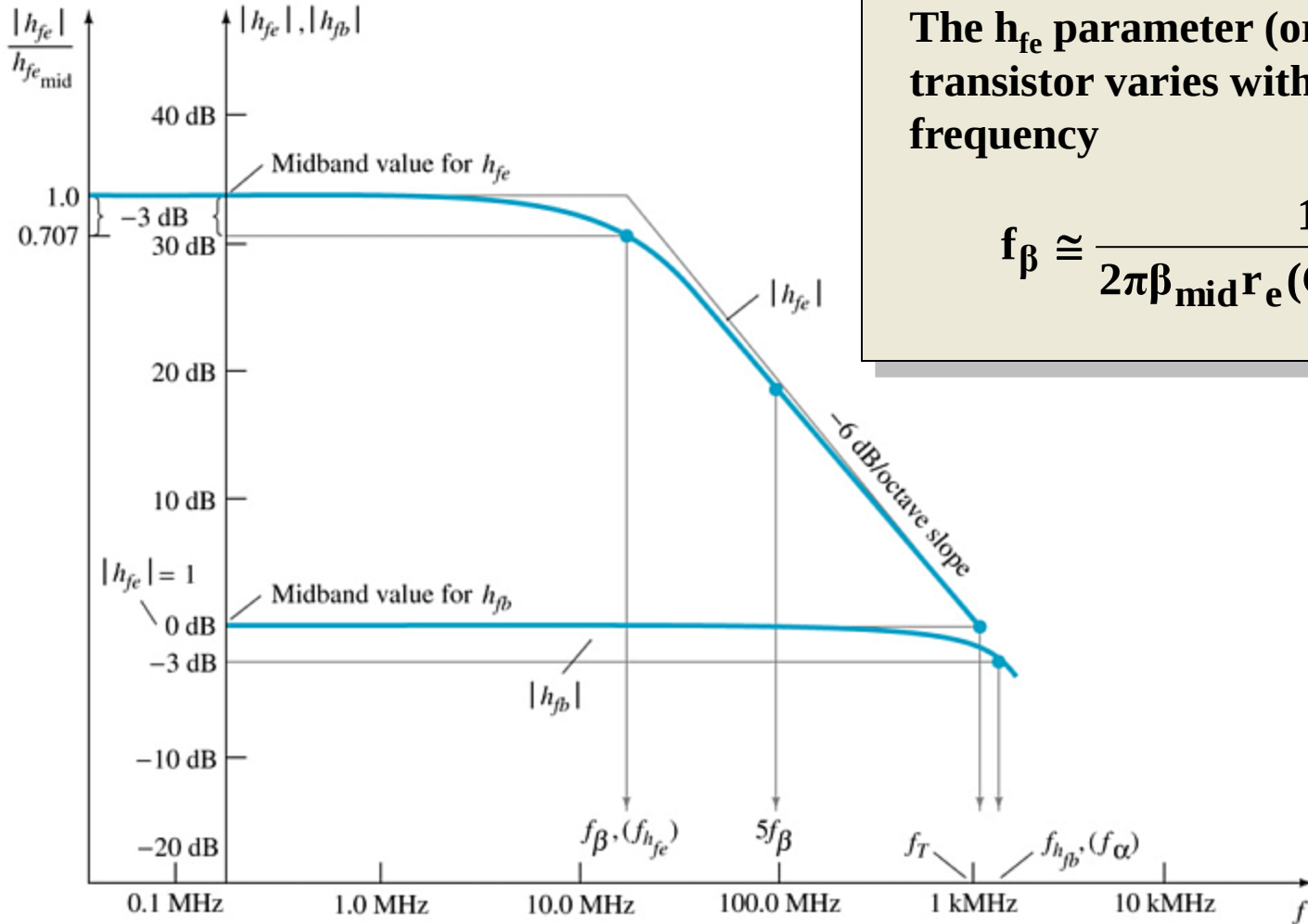
$$R_{Th_o} = R_C \parallel R_L \parallel r_o$$

and

$$C_o = C_{W_o} + C_{ce} + C_{M_o}$$



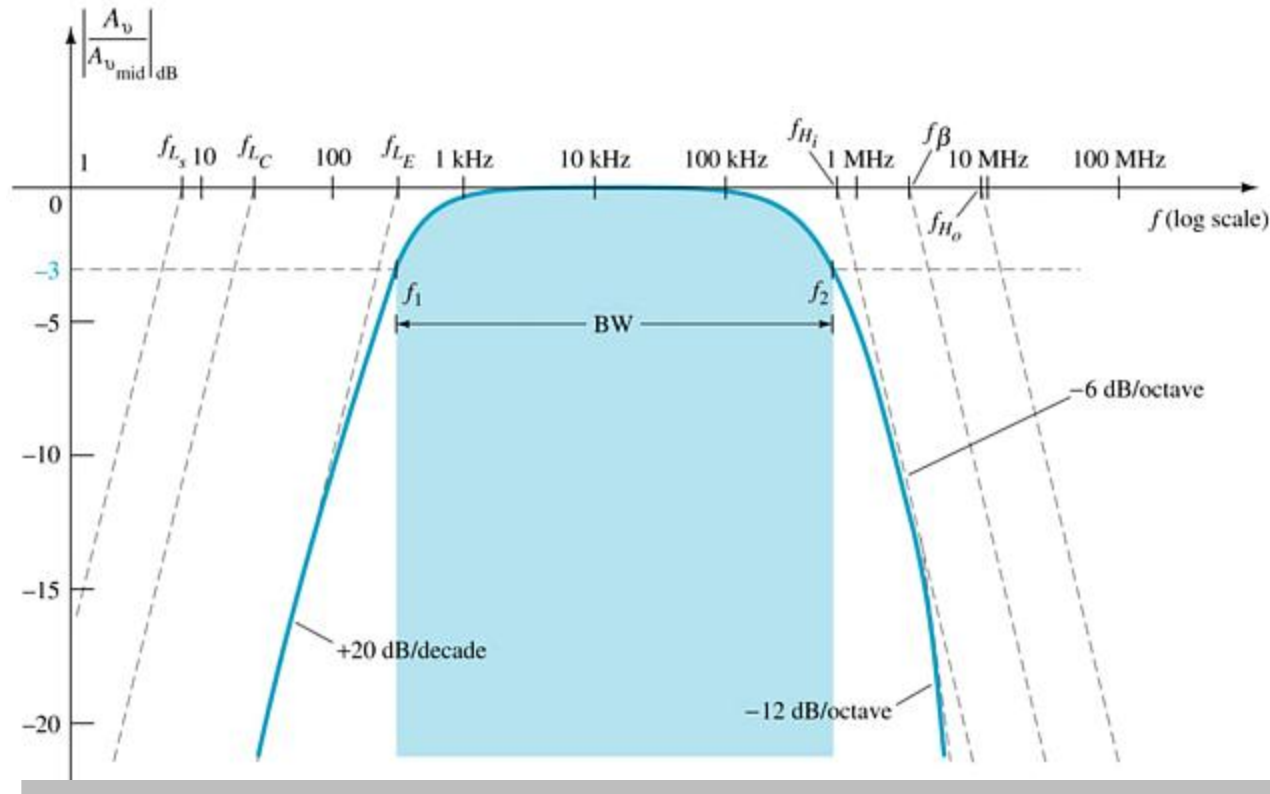
h_{fe} (or β) Variation



The h_{fe} parameter (or β) of a transistor varies with frequency

$$f_{\beta} \cong \frac{1}{2\pi\beta_{mid}r_e(C_{be} + C_{bc})}$$

BJT Amplifier Frequency Response



Note the *highest* lower cutoff frequency (f_L) and the *lowest* upper cutoff frequency (f_H) are closest to the actual response of the amplifier.

FET Amplifier High-Frequency Response

Capacitances that affect the high-frequency response are

- Junction capacitances

$$C_{gs}, C_{gd}, C_{ds}$$

- Wiring capacitances

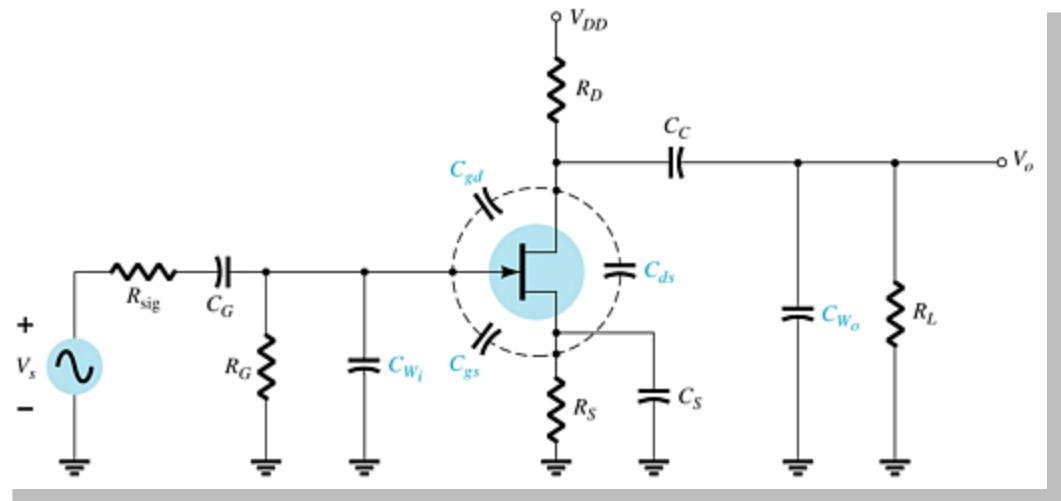
$$C_{wi}, C_{wo}$$

- Coupling capacitors

$$C_G, C_C$$

- Bypass capacitor

$$C_S$$



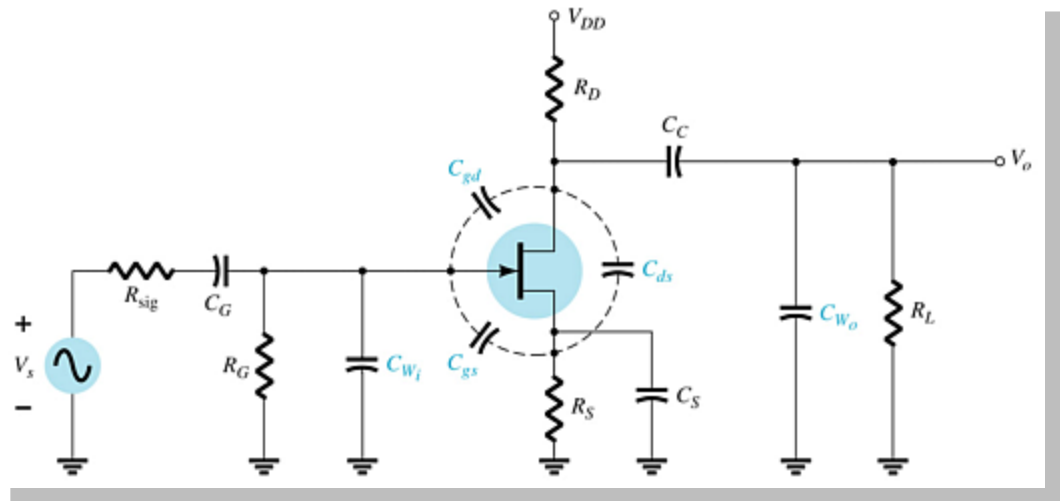
Input Network (f_{Hi}) High-Frequency Cutoff

$$f_{Hi} = \frac{1}{2\pi R_{Thi} C_i}$$

$$C_i = C_{Wi} + C_{gs} + C_{Mi}$$

$$C_{Mi} = (1 - A_v) C_{gd}$$

$$R_{Thi} = R_{sig} \parallel R_G$$



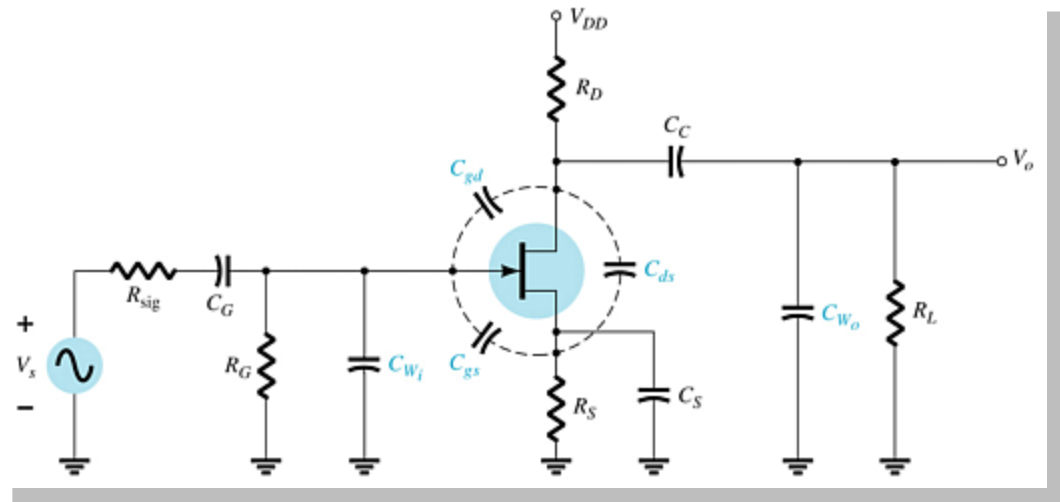
Output Network (f_{H_o}) High-Frequency Cutoff

$$f_{H_o} = \frac{1}{2\pi R_{Th_o} C_o}$$

$$C_o = C_{W_o} + C_{ds} + C_{M_o}$$

$$C_{M_o} = \left(1 - \frac{1}{A_v}\right) C_{gd}$$

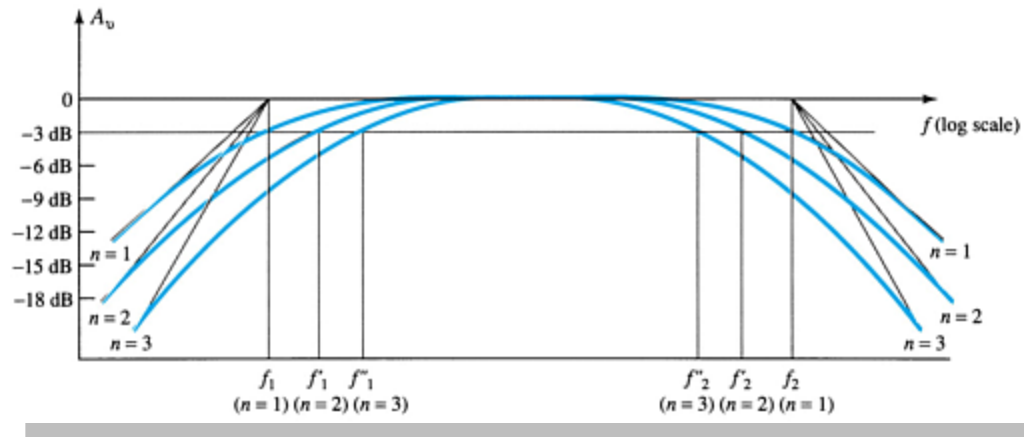
$$R_{Th_o} = R_D \parallel R_L \parallel r_d$$



Multistage Frequency Effects

Each stage will have its own frequency response, but the output of one stage will be affected by capacitances in the subsequent stage. This is especially so when determining the high frequency response. For example, the output capacitance (C_o) will be affected by the input Miller Capacitance (C_{Mi}) of the next stage.

Multistage Amplifier Frequency Response



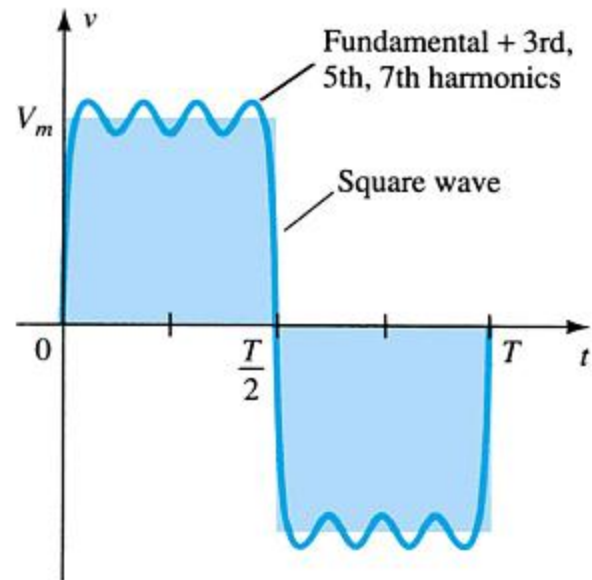
Once the cutoff frequencies have been determined for each stage (taking into account the shared capacitances), they can be plotted.

Note the *highest* lower cutoff frequency (f_L) and the *lowest* upper cutoff frequency (f_H) are closest to the actual response of the amplifier.

Square Wave Testing

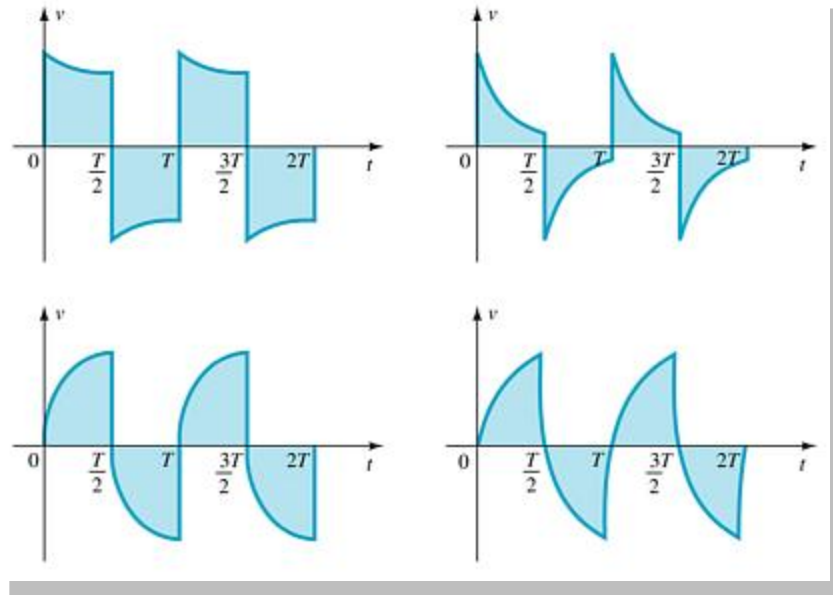
In order to determine the frequency response of an amplifier by experimentation, you must apply a wide range of frequencies to the amplifier.

One way to accomplish this is to apply a square wave. A square wave consists of multiple frequencies (by Fourier analysis: it consists of odd harmonics).



Square Wave Response Waveforms

If the output of the amplifier is not a perfect square wave then the amplifier is 'cutting' off certain frequency components of the square wave.



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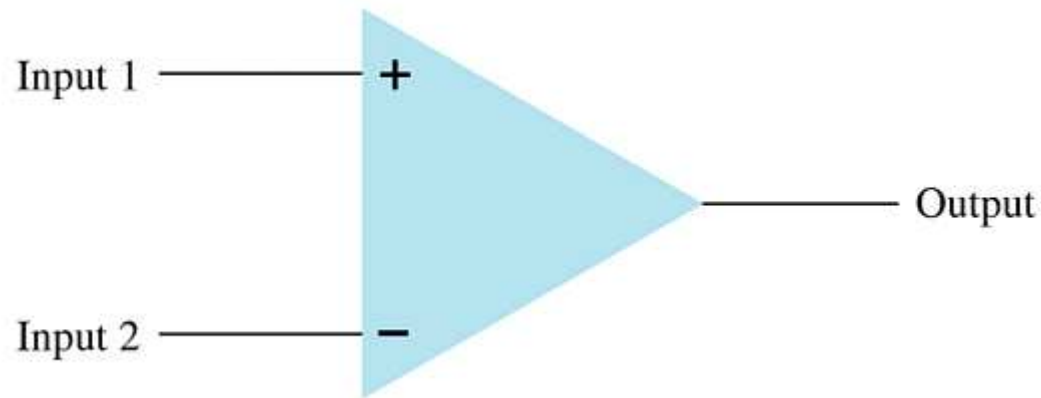
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Chapter 10: Operational Amplifiers

Basic Op-Amp



Operational amplifier or op-amp, is a very high gain differential amplifier with a high input impedance (typically a few meg-Ohms) and low output impedance (less than 100 Ω).

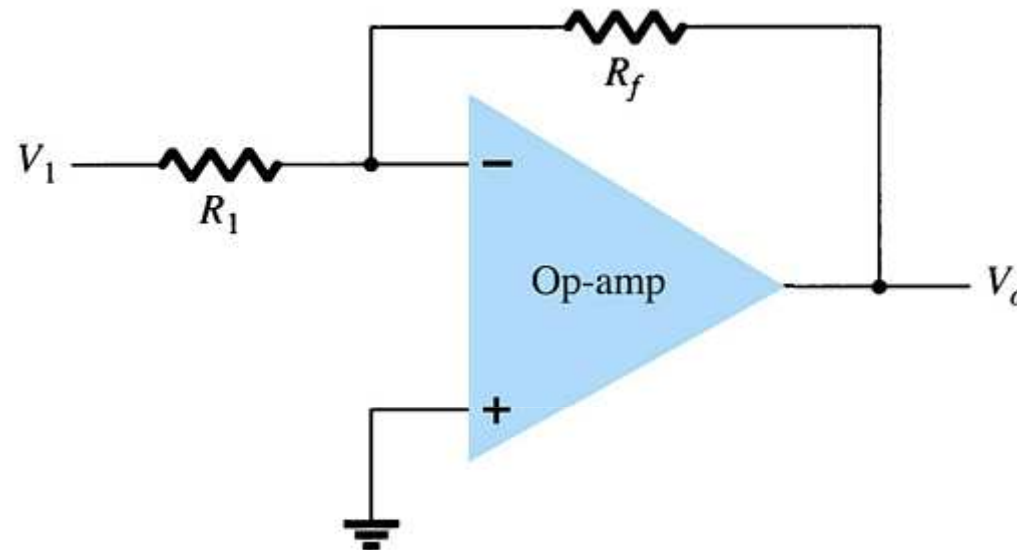
Note the op-amp has two inputs and one output.

Op-Amp Gain

Op-Amps have a very high gain. They can be connected open-loop or closed-loop.

- **Open-loop** refers to a configuration where there is no feedback from output back to the input. In the open-loop configuration the gain can exceed 10,000.
- **Closed-loop** configuration reduces the gain. In order to control the gain of an op-amp it must have feedback. This feedback is a negative feedback. A **negative feedback** reduces the gain and improves many characteristics of the op-amp.

Inverting Op-Amp



- The signal input is applied to the **inverting (-) input**
- The **non-inverting input (+)** is grounded
- The resistor R_f is the **feedback resistor**. It is connected from the output to the negative (inverting) input. This is **negative feedback**.

Inverting Op-Amp Gain

Gain can be determined from external resistors: R_f and R_1

$$A_v = \frac{V_o}{V_i} = \frac{R_f}{R_1}$$

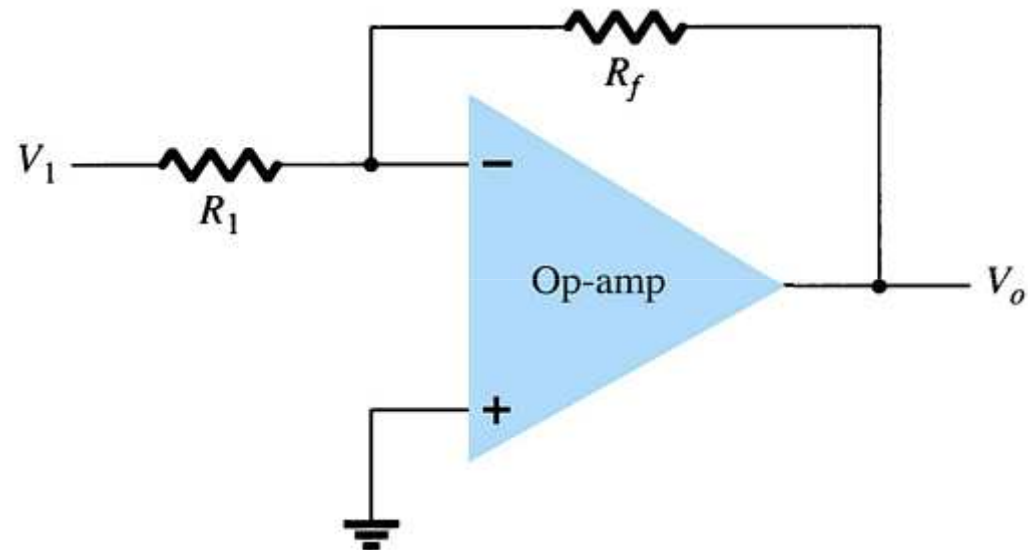
Unity gain—voltage gain is 1

$$R_f = R_1$$

$$A_v = \frac{-R_f}{R_1} = -1$$

The negative sign denotes a 180° phase shift between input and output.

Constant Gain— R_f is a multiple of R_1

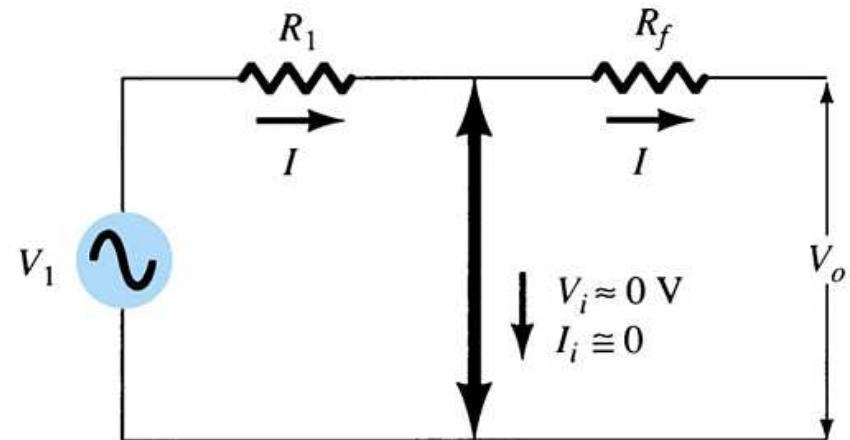
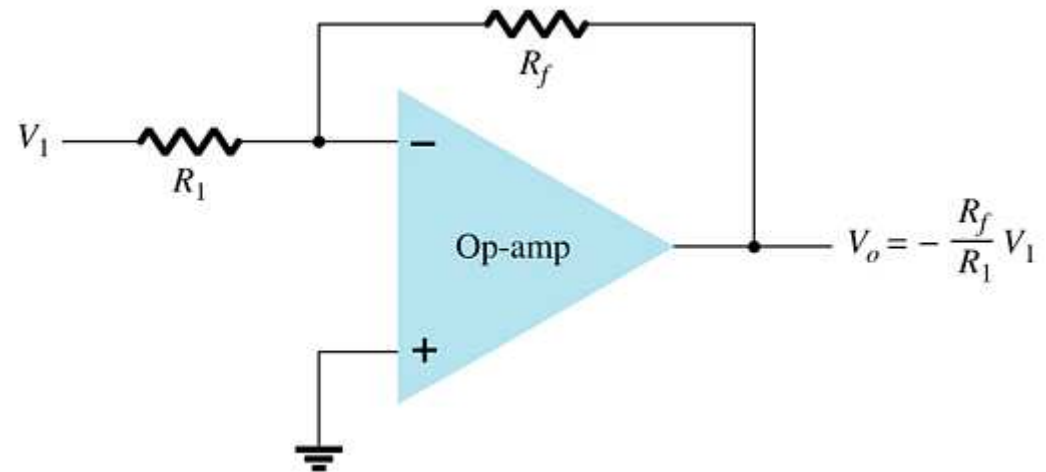


Virtual Ground

An understanding of the concept of **virtual ground** provides a better understanding of how an op-amp operates.

The *non-inverting* input pin is at ground. The *inverting* input pin is also at 0 V for an AC signal.

The op-amp has such high input impedance that even with a high gain there is no current from inverting input pin, therefore there is no voltage from inverting pin to ground—all of the current is through R_f .



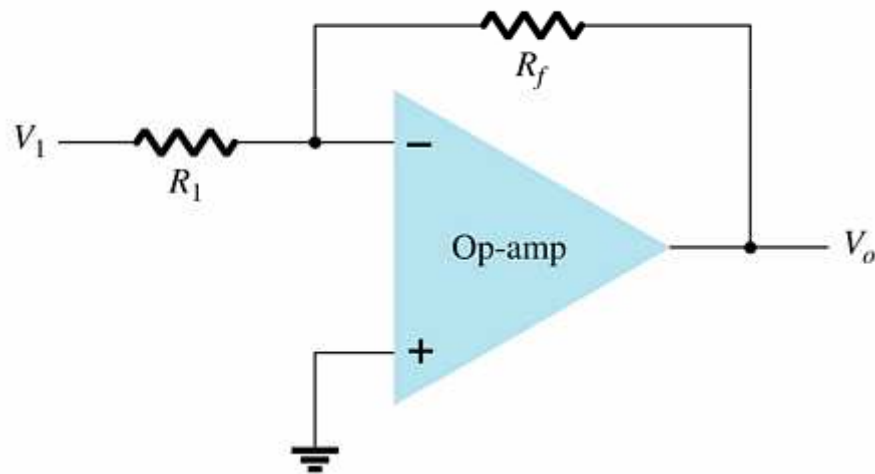
Practical Op–Amp Circuits

Inverting amplifier
Noninverting amplifier
Unity follower
Summing amplifier
Integrator
Differentiator

Inverting/Noninverting Op-Amps

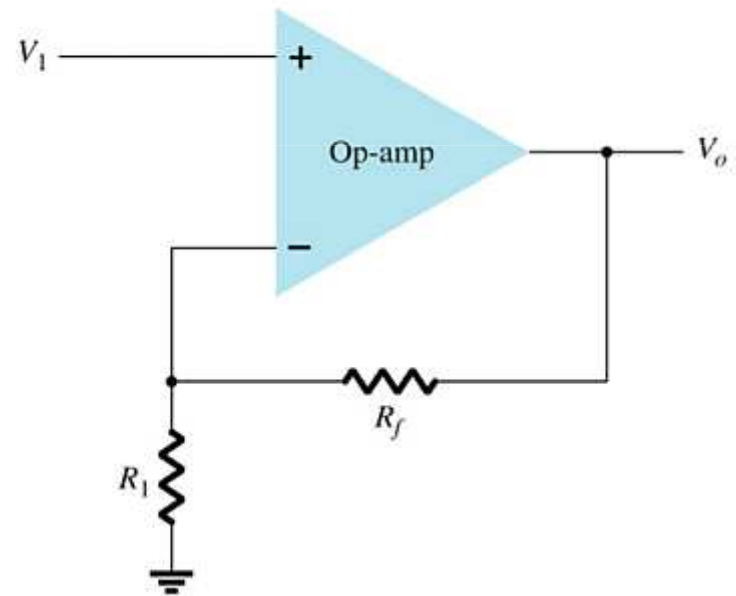
Inverting Amplifier

$$V_o = -\frac{R_f}{R_1} V_1$$



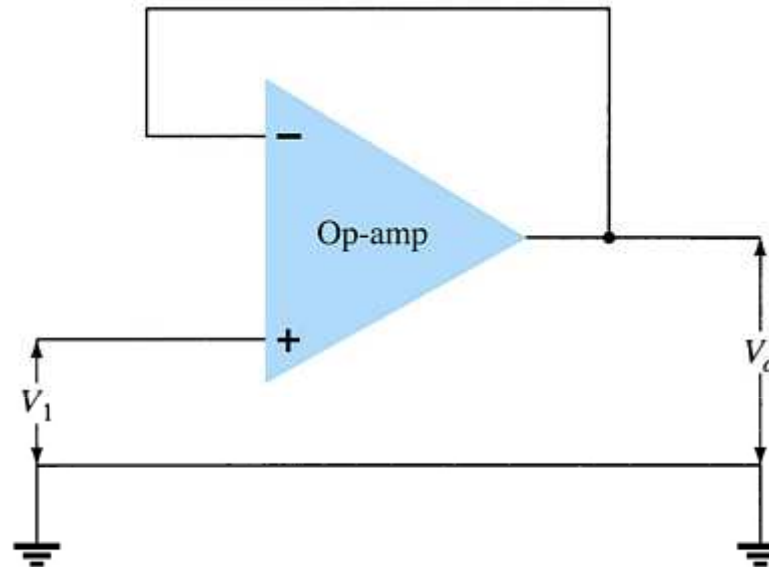
Noninverting Amplifier

$$V_o = \left(1 + \frac{R_f}{R_1}\right) V_1$$



Unity Follower

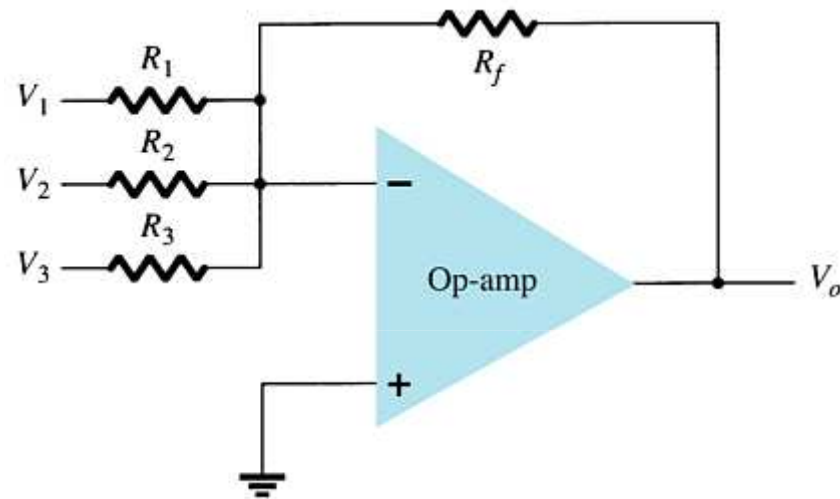
$$V_o = V_1$$



Summing Amplifier

Because the op-amp has a high input impedance, the multiple inputs are treated as separate inputs.

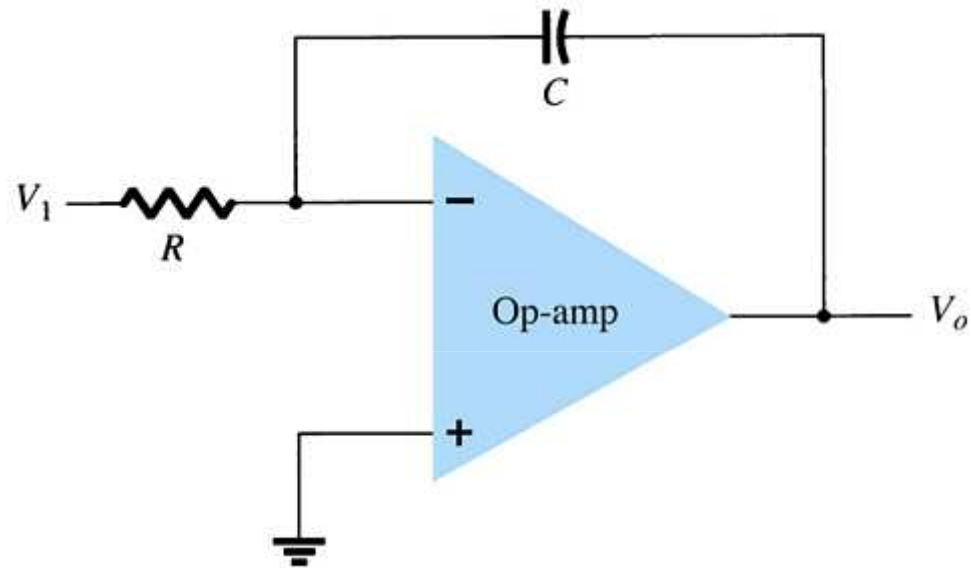
$$V_o = -\left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3\right)$$



Integrator

The output is the integral of the input. Integration is the operation of summing the area under a waveform or curve over a period of time. This circuit is useful in low-pass filter circuits and sensor conditioning circuits.

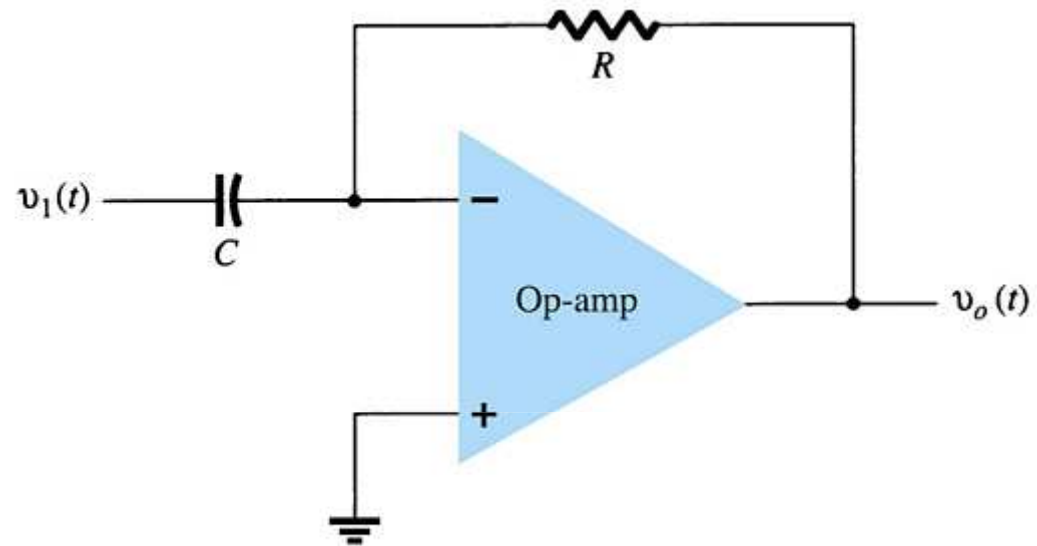
$$v_o(t) = -\frac{1}{RC} \int v_1(t) dt$$



Differentiator

The differentiator takes the derivative of the input. This circuit is useful in high-pass filter circuits.

$$v_o(t) = -RC \frac{dv_1(t)}{dt}$$



Op-Amp Specifications—DC Offset Parameters

Even when the input voltage is zero, there can be an output **offset**. The following can cause this offset:

- Input offset voltage
- Input offset current
- Input offset voltage *and* input offset current
- Input bias current

Input Offset Voltage (V_{IO})

The specification sheet for an op-amp indicate an input offset voltage (V_{IO}).

The effect of this input offset voltage on the output can be calculated with

$$V_{o(\text{offset})} = V_{IO} \frac{R_1 + R_f}{R_1}$$

Output Offset Voltage Due to Input Offset Current (I_{IO})

If there is a difference between the dc bias currents for the same applied input, then this also causes an output offset voltage:

- The input offset Current (I_{IO}) is specified in the specifications for the op-amp.
- The effect on the output can be calculated using:

$$V_{o(\text{offset due to } I_{IO})} = I_{IO} R_f$$

Total Offset Due to V_{IO} and I_{IO}

Op-amps may have an output offset voltage due to both factors V_{IO} and I_{IO} . The total output offset voltage will be the sum of the effects of both:

$$V_o(\text{offset}) = V_o(\text{offset due to } V_{IO}) + V_o(\text{offset due to } I_{IO})$$

Input Bias Current (I_{IB})

A parameter that is related to input offset current (I_{IO}) is called **input bias current** (I_{IB})

The separate input bias currents are:

$$I_{IB}^{-} = I_{IB} - \frac{I_{IO}}{2}$$

$$I_{IB}^{+} = I_{IB} + \frac{I_{IO}}{2}$$

The total input bias current is the average:

$$I_{IB} = \frac{I_{IB}^{-} + I_{IB}^{+}}{2}$$

Frequency Parameters

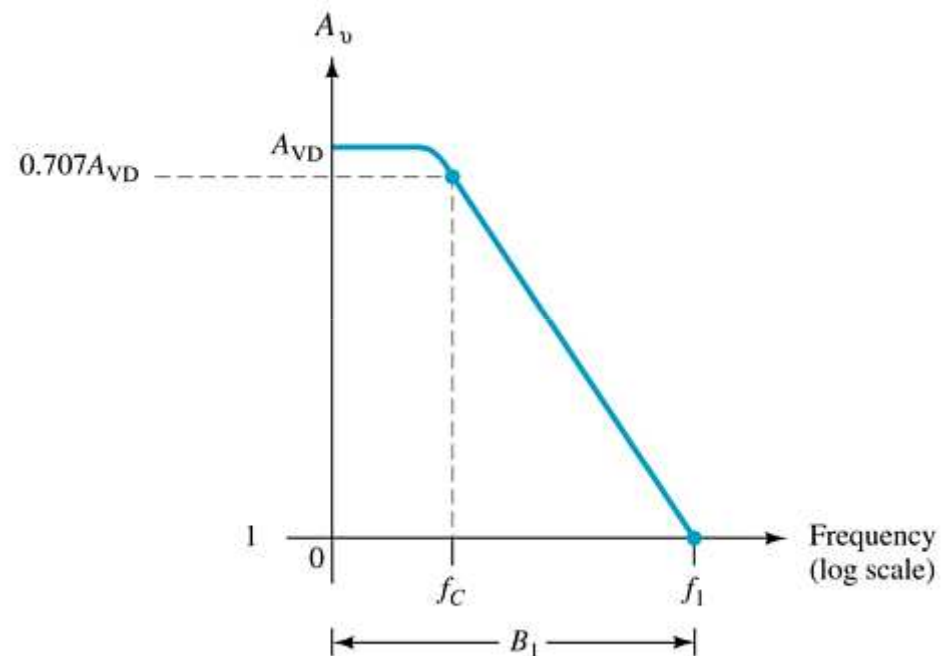
An op-amp is a wide-bandwidth amplifier. The following affect the bandwidth of the op-amp:

- **Gain**
- **Slew rate**

Gain and Bandwidth

The op-amp's high frequency response is limited by internal circuitry. The plot shown is for an open loop gain (A_{OL} or A_{VD}). This means that the op-amp is operating at the highest possible gain with no feedback resistor.

In the open loop, the op-amp has a narrow bandwidth. The bandwidth widens in closed-loop operation, but then the gain is lower.



Slew Rate (SR)

Slew rate (SR) is the maximum rate at which an op-amp can change output without distortion.

$$SR = \frac{\Delta V_o}{\Delta t} \quad (\text{in } V/\mu s)$$

The SR rating is given in the specification sheets as V/μs rating.

Maximum Signal Frequency

The slew rate determines the highest frequency of the op-amp without distortion.

$$f \leq \frac{SR}{2\pi V_p}$$

where V_p is the peak voltage

General Op–Amp Specifications

Other ratings for op-amp found on specification sheets are:

- **Absolute Ratings**
- **Electrical Characteristics**
- **Performance**

Absolute Ratings

**These are common
maximum ratings
for the op-amp.**

Absolute Maximum Ratings	
Supply voltage	6 22 V
Internal power dissipation	500 mW
Differential input voltage	6 30 V
Input voltage	6 15 V

Electrical Characteristics

TABLE 13.2 mA741 Electrical Characteristics: $V_{CC} = \pm 15 \text{ V}$, $T_A = 25^\circ\text{C}$

Characteristic	MIN	TYP	MAX	Unit
V_{IO} Input offset voltage		1	6	mV
I_{IO} Input offset current		20	200	nA
I_{IB} Input bias current		80	500	nA
V_{ICR} Common-mode input voltage range	± 12	± 13		V
V_{OM} Maximum peak output voltage swing	± 12	± 14		V
A_{VD} Large-signal differential voltage amplification	20	200		V/mV
r_i Input resistance	0.3	2		M Ω
r_o Output resistance		75		Ω
C_i Input capacitance		1.4		pF
CMRR Common-mode rejection ratio	70	90		dB
I_{CC} Supply current		1.7	2.8	mA
P_D Total power dissipation		50	85	mW

Note: These ratings are for specific circuit conditions, and they often include minimum, maximum and typical values.

CMRR

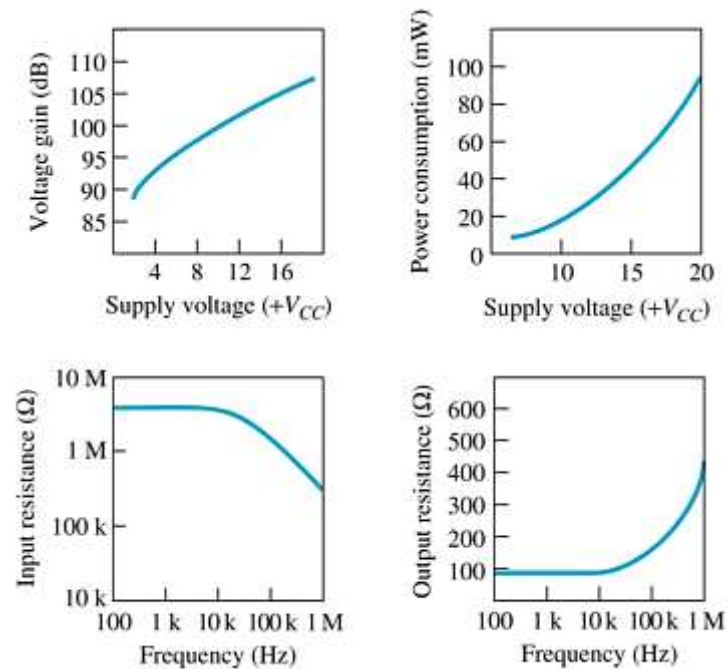
One rating that is unique to op-amps is CMRR or **common-mode rejection ratio**.

Because the op-amp has two inputs that are opposite in phase (inverting input and the non-inverting input) any signal that is common to both inputs will be cancelled.

Op-amp CMRR is a measure of the ability to cancel out common-mode signals.

Op-Amp Performance

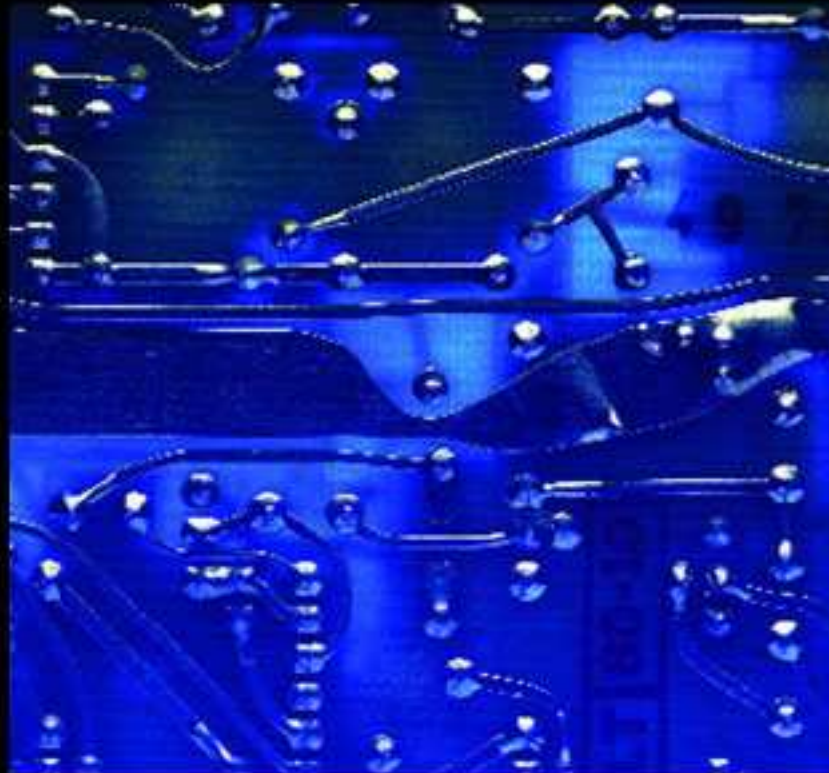
The specification sheets will also include graphs that indicate the performance of the op-amp over a wide range of conditions.



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Chapter 11 Op-Amp Applications

Op-Amp Applications

Constant-gain multiplier

Voltage summing

Voltage buffer

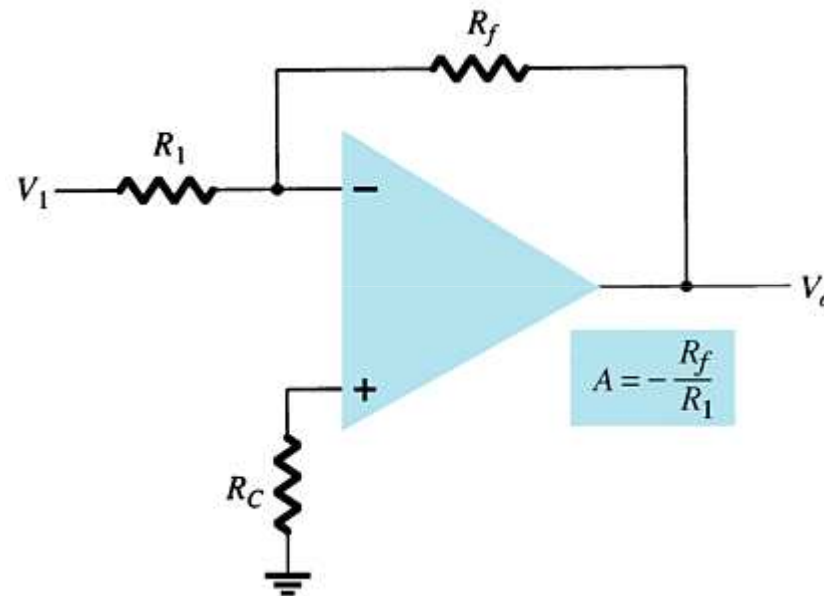
Controlled sources

Instrumentation circuits

Active filters

Constant-Gain Amplifier

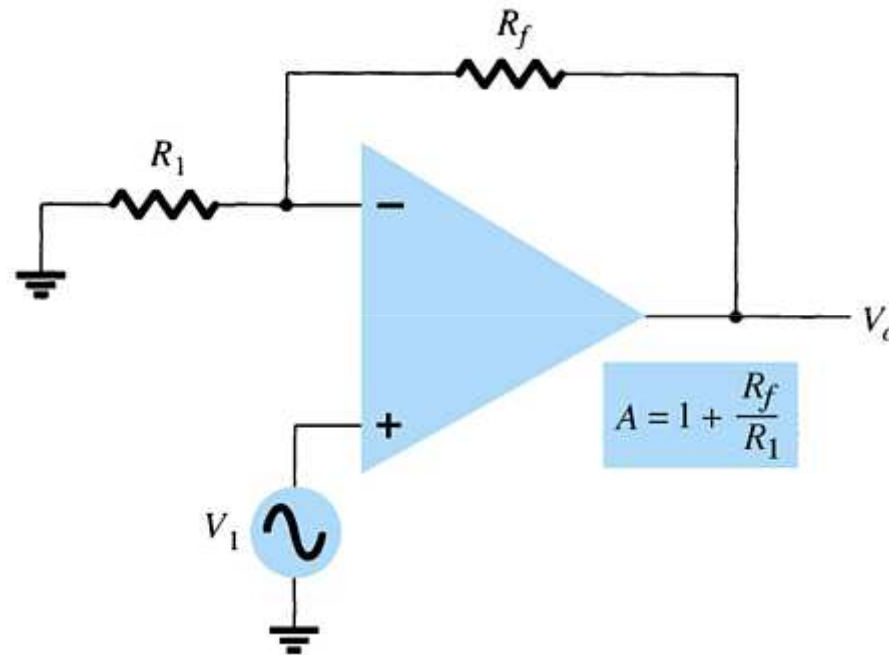
Inverting Version



[more...](#)

Constant-Gain Amplifier

Noninverting Version



Multiple-Stage Gains

The total gain (3-stages) is given by:

$$\mathbf{A} = \mathbf{A}_1 \mathbf{A}_2 \mathbf{A}_3$$

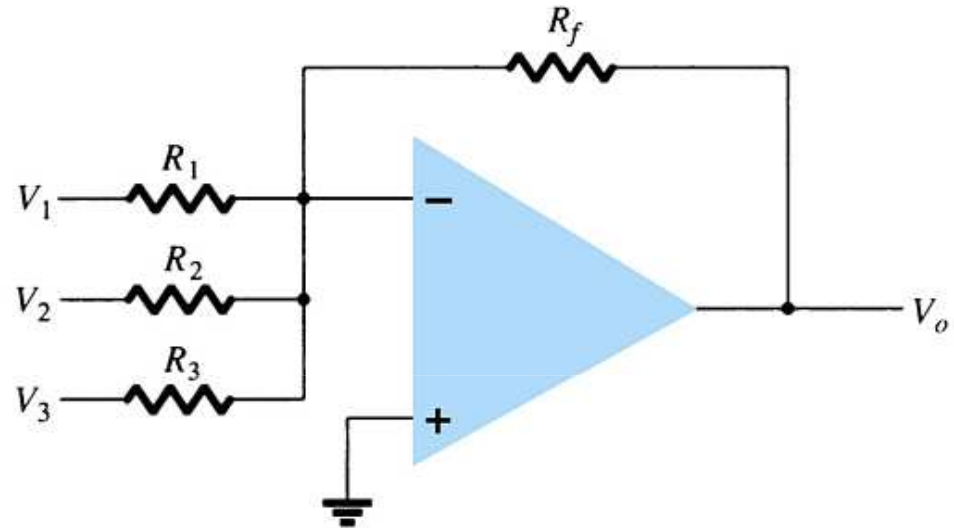
or

$$\mathbf{A} = \left(1 + \frac{\mathbf{R}_f}{\mathbf{R}_1} \right) \left(-\frac{\mathbf{R}_f}{\mathbf{R}_2} \right) \left(-\frac{\mathbf{R}_f}{\mathbf{R}_3} \right)$$

Voltage Summing

The output is the sum of individual signals times the gain:

$$V_o = -\left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3\right)$$



[Formula 14.3]

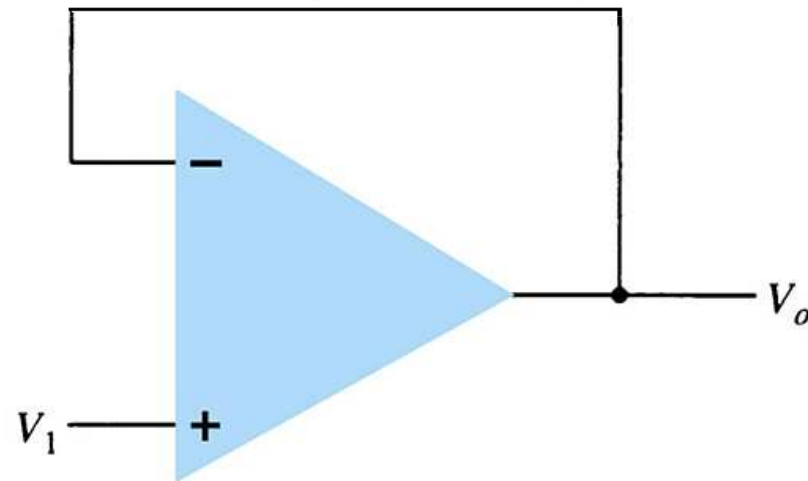
Voltage Buffer

Any amplifier with no gain or loss is called a **unity gain amplifier**.

The advantages of using a unity gain amplifier:

- Very high input impedance
- Very low output impedance

Realistically these circuits are designed using equal resistors ($R_1 = R_f$) to avoid problems with offset voltages.



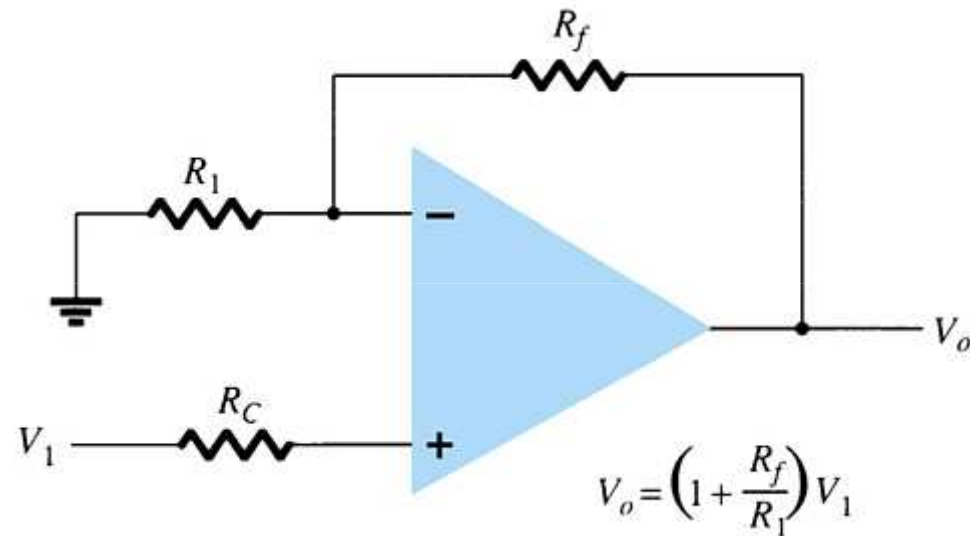
Controlled Sources

Voltage-controlled voltage source
Voltage-controlled current source
Current-controlled voltage source
Current-controlled current source

Voltage-Controlled Voltage Source

The output voltage is the gain times the input voltage. What makes an op-amp different from other amplifiers is its impedance characteristics and gain calculations that depend solely on external resistors.

Noninverting Amplifier Version

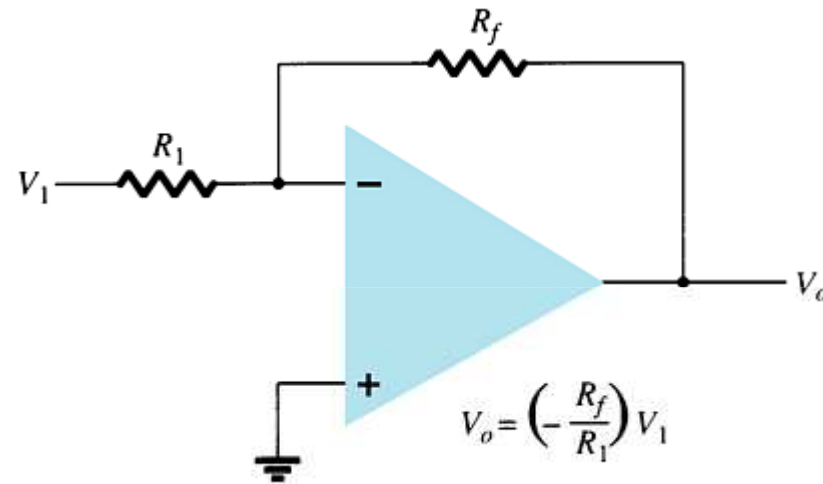


more...

Voltage-Controlled Voltage Source

The output voltage is the gain times the input voltage. What makes an op-amp different from other amplifiers is its impedance characteristics and gain calculations that depend solely on external resistors.

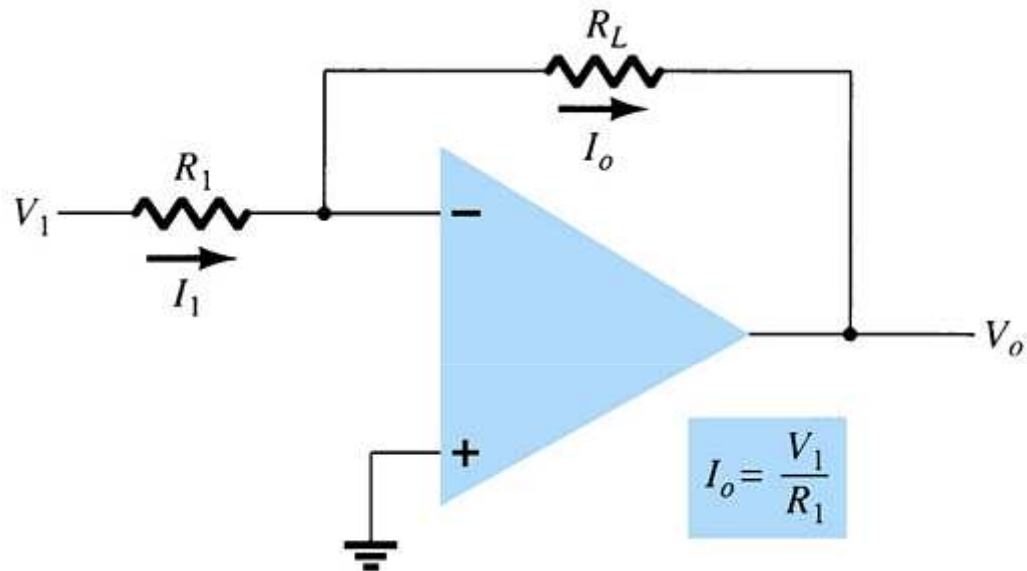
Inverting Amplifier Version



Voltage-Controlled Current Source

The output current is:

$$I_o = \frac{V_1}{R_1} = kV_1$$



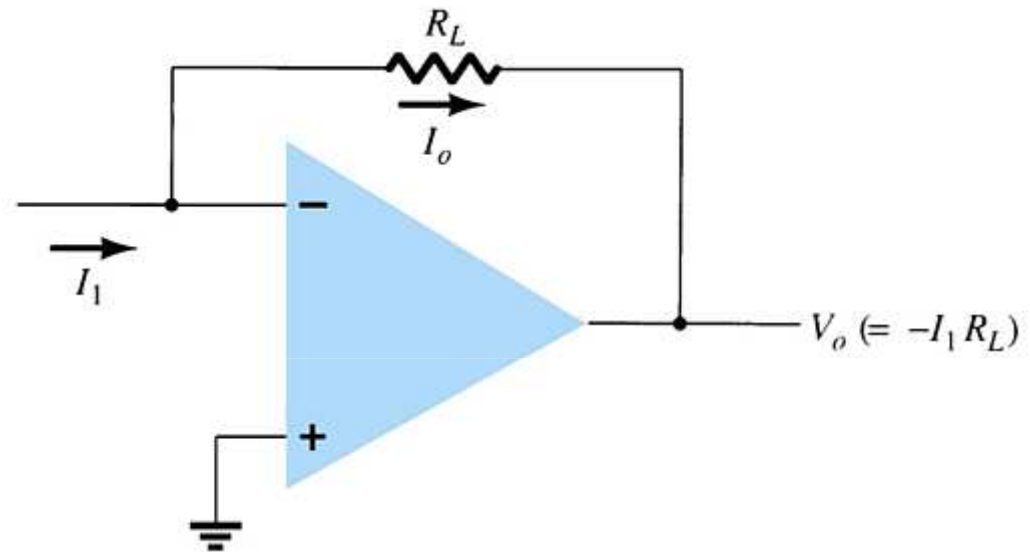
Current-Controlled Voltage Source

This is simply another way of applying the op-amp operation. Whether the input is a current determined by V_{in}/R_1 or as I_1 :

$$V_{out} = \frac{-R_f}{R_1} V_{in}$$

or

$$V_{out} = -I_1 R_L$$



Current-Controlled Current Source

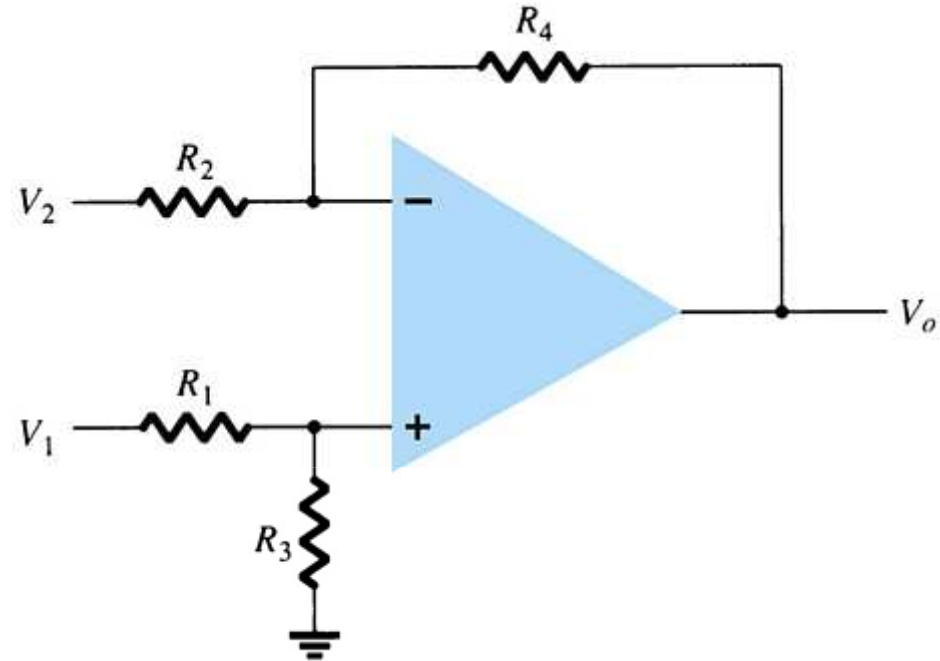
This circuit may appear more complicated than the others but it is really the same thing.

$$V_{out} = -\left(\frac{R_f}{R_{in}}\right)V_{in} \quad I_o = -\frac{V_{in}}{R_1 \parallel R_2}$$

$$\frac{V_{out}}{R_f} = -\frac{V_{in}}{R_1 \parallel R_2} \quad I_o = -V_{in}\left(\frac{R_1 + R_2}{R_1 \times R_2}\right)$$

$$\frac{V_{out}}{R_f} = -\frac{V_{in}}{R_{in}} \quad I_o = -\frac{V_{in}}{R_1}\left(\frac{R_1 + R_2}{R_2}\right)$$

$$I_o = -I\left(1 + \frac{R_1}{R_2}\right) = kI$$

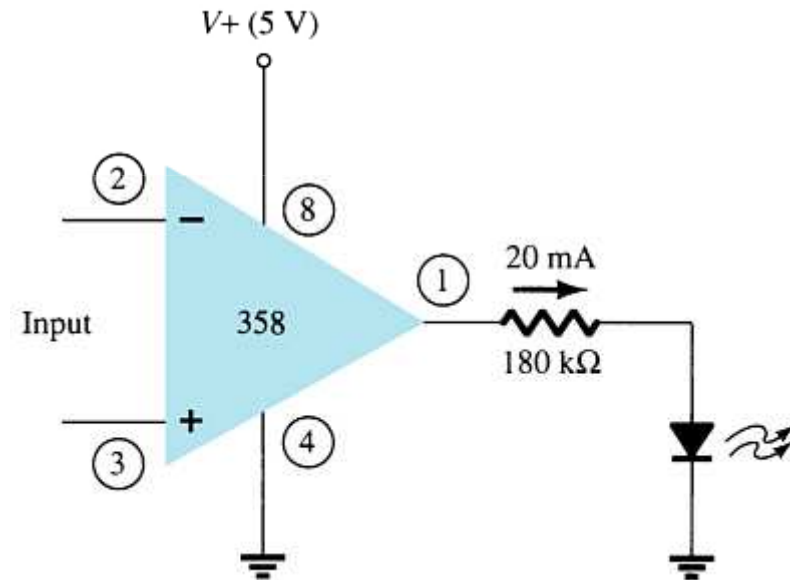
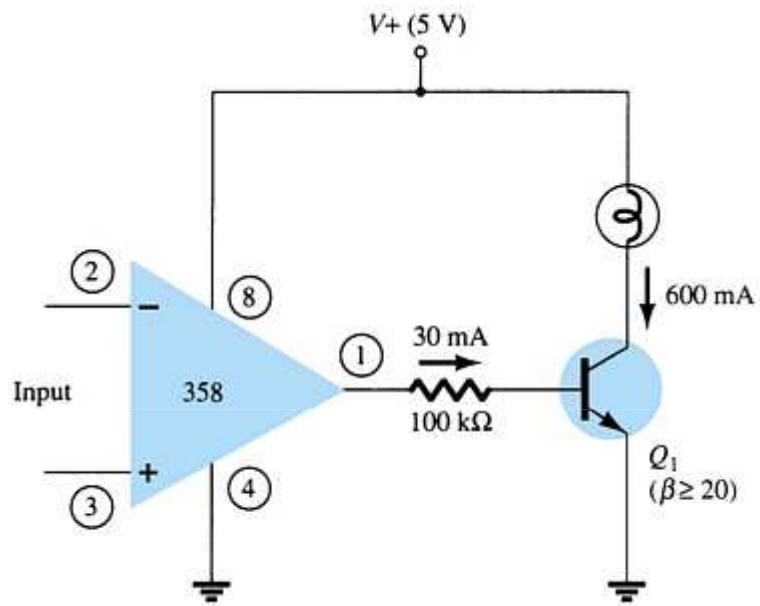


Instrumentation Circuits

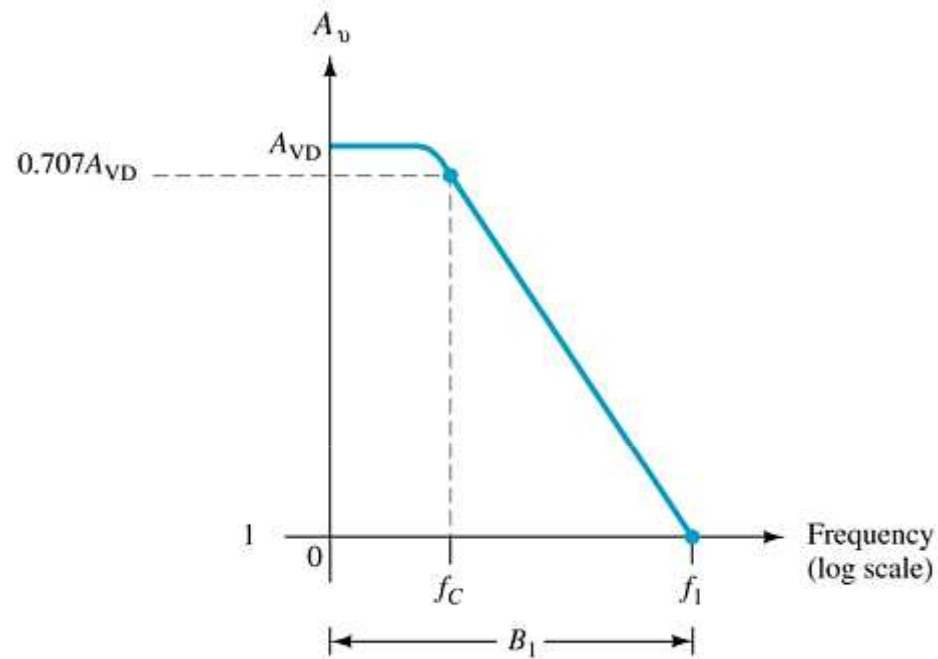
Some examples of instrumentation circuits using op-amps:

- **Display driver**
- **Instrumentation amplifier**

Display Driver



Instrumentation Amplifier



For all R_s at the same value (except R_p):

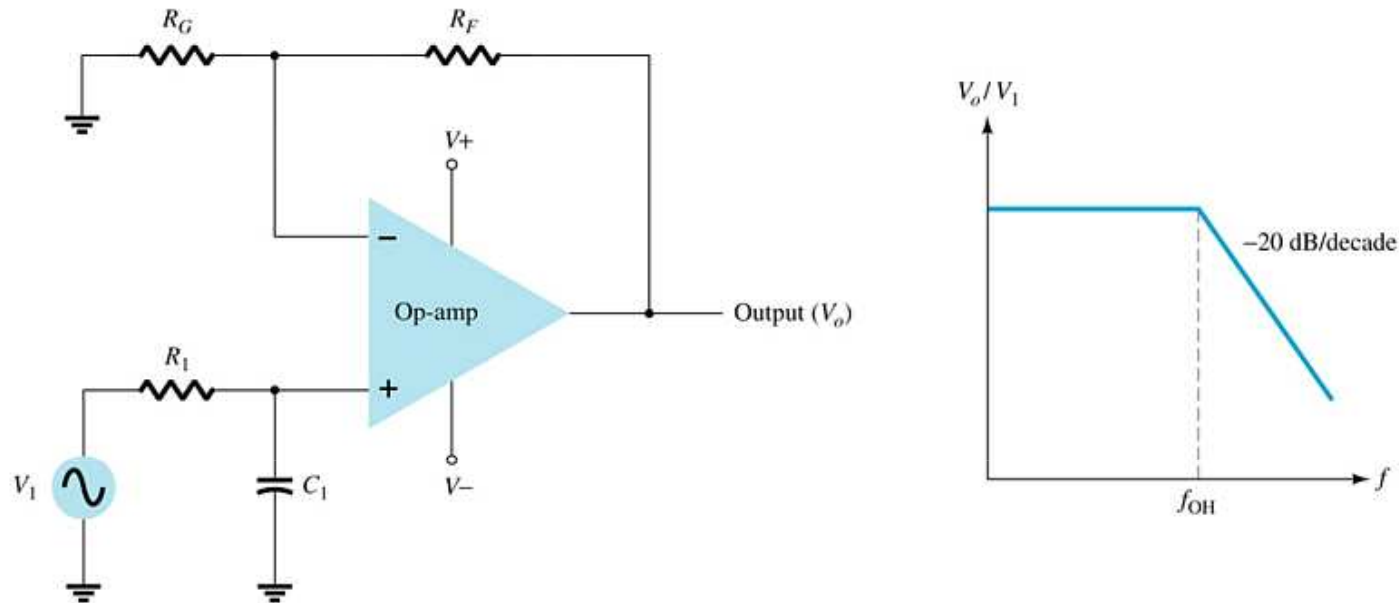
$$V_o = \left(1 + \frac{2R}{R_p}\right)(V_1 - V_2) = k(V_1 - V_2)$$

Active Filters

Adding capacitors to op-amp circuits provides external control of the cutoff frequencies. The op-amp active filter provides controllable cutoff frequencies and controllable gain.

- **Low-pass filter**
- **High-pass filter**
- **Bandpass filter**

Low-Pass Filter—First-Order

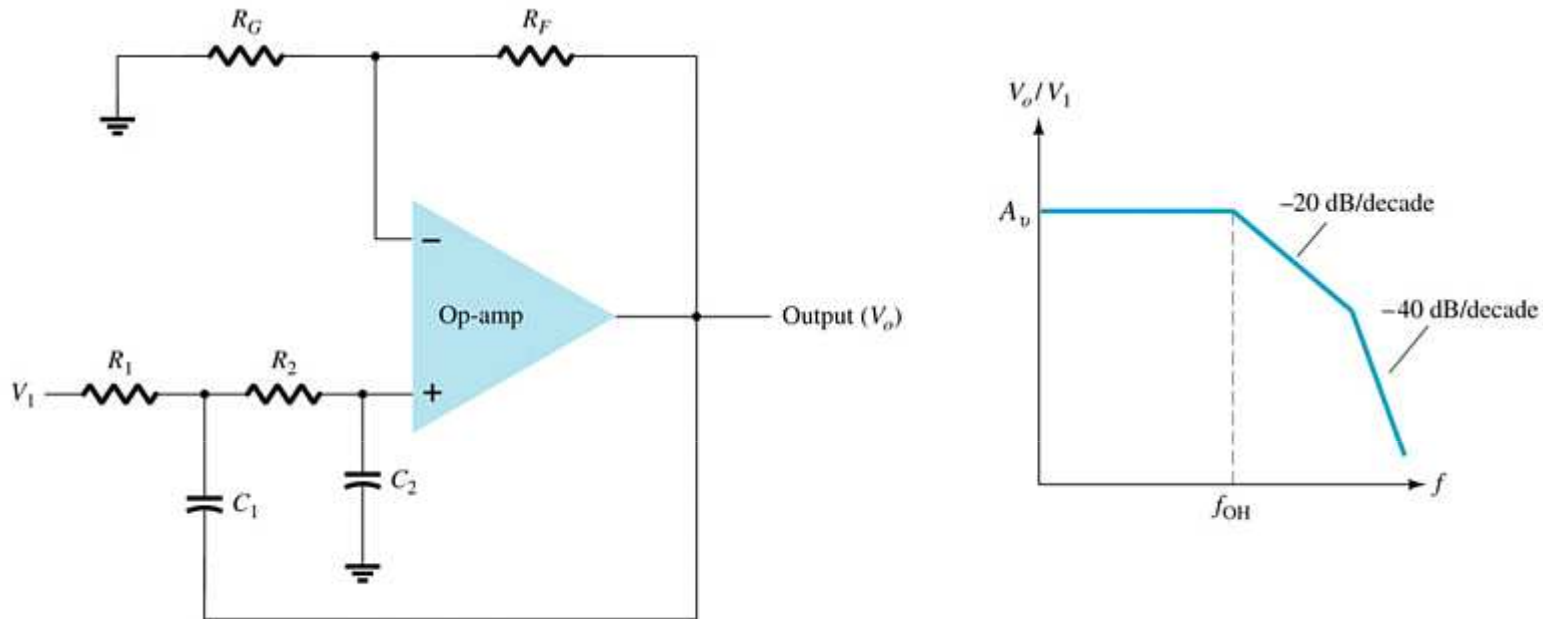


The upper cutoff frequency and voltage gain are given by:

$$f_{OH} = \frac{1}{2\pi R_1 C_1}$$

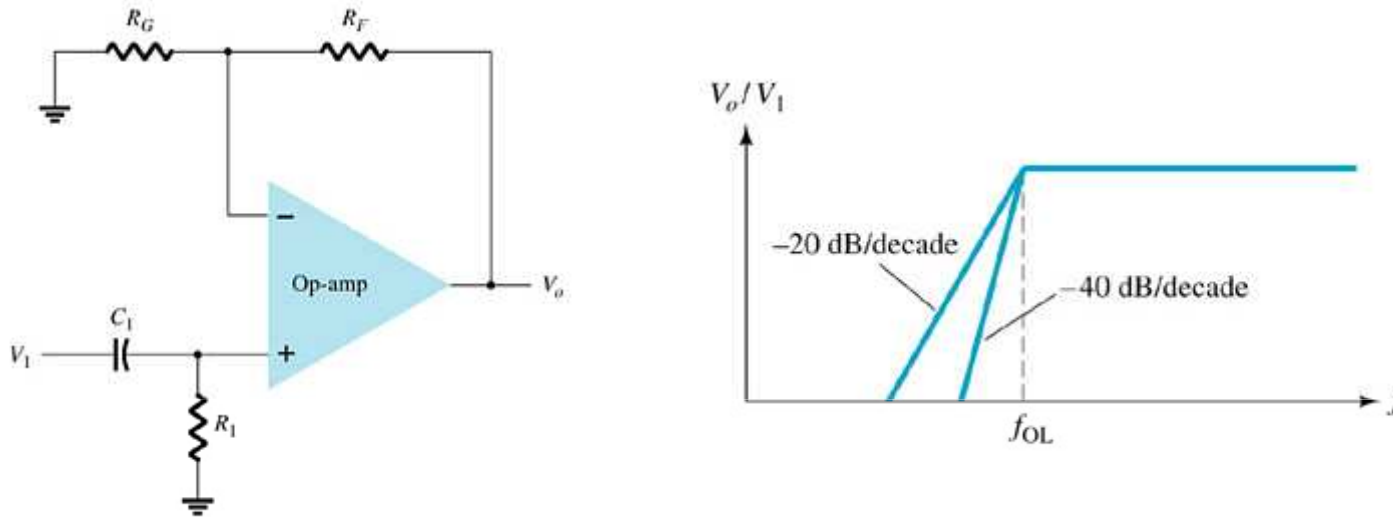
$$A_v = 1 + \frac{R_f}{R_1}$$

Low-Pass Filter—Second-Order



The roll-off can be made steeper by adding more RC networks.

High-Pass Filter



The cutoff frequency is determined by:

$$f_{OL} = \frac{1}{2\pi R_1 C_1}$$

Bandpass Filter

There are two cutoff frequencies: upper and lower. They can be calculated using the same low-pass cutoff and high-pass cutoff frequency formulas in the appropriate sections.

